Electrical Machines

Detailed Solutions • 1991–2026

GATE ELECTRICAL ENGINEERING

(EE)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book works, in full, every question GATE has actually asked in Electrical Machines (EE), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, and every solution is taken step by step rather than asserted.

Each solution carries the same identifier as the question it answers in the companion questions volume, so you can move between practice and explanation without a lookup table. Where the examining authority revised an answer or awarded marks to all (MTA), that is noted with the question it affects.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

○○○ ●○○ Easy ●●○ Medium ●●● Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EE	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

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Contents

1. **Magnetic circuits** 1
2. **Transformers** 19
3. **DC Machines** 113
4. **Three Phase Induction Machines** 185
5. **Synchronous Machines** 252
6. **Miscellaneous topics** 320

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SOLUTIONS

Magnetic circuits

Topics

1. Magnetically Coupled Circuits, Dot Convention
2. Energy in Coupled circuits
3. Electric and Magnetic Materials

Unit Snapshot*

Stream: EE	**Questions:** 17	**MCQ:** 9	**MSQ:** 0	**NAT:** 8
1M: 4	**2M:** 10	**Descriptive:** 3	**Avg Weightage** ~2M	**Range:** 1M(2021)–2M(2025)

**See preface for how Avg Weightage and Range are calculated.*

Q1.25.1 MCQ | 2025 | 2M | Ans: A,D

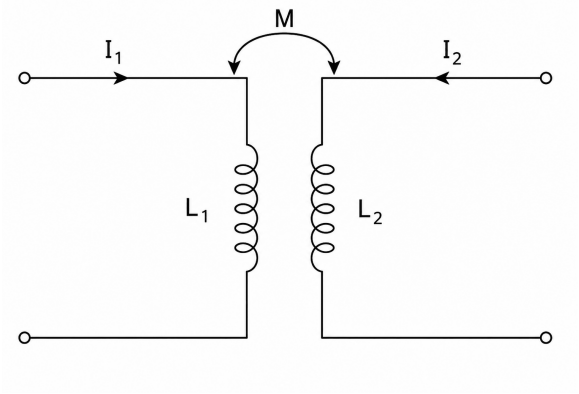


Fig. Solution Q1.25.1

The magnetic energy stored in two coupled inductors is

$$E = \frac{1}{2}L_1 I_1^2 + \frac{1}{2}L_2 I_2^2 + M I_1 I_2$$

For minimum stored magnetic energy,

$$\frac{\partial E}{\partial I_1} = 0 \quad \text{and} \quad \frac{\partial E}{\partial I_2} = 0$$

From

$$\frac{\partial E}{\partial I_1} = L_1 I_1 + M I_2 = 0$$

we get

$$\frac{I_1}{I_2} = -\frac{M}{L_1}$$

Also,

$$\frac{\partial E}{\partial I_2} = L_2 I_2 + M I_1 = 0$$

which gives

$$\frac{I_1}{I_2} = -\frac{L_2}{M}$$

Hence, for minimum stored magnetic energy,

$$\frac{I_1}{I_2} = -\frac{M}{L_1} \quad \text{or} \quad -\frac{L_2}{M}$$

Therefore, the correct options are

$$\boxed{A, D}$$

Key Points

Energy stored in coupled inductors is

$$E = \frac{1}{2}L_1 I_1^2 + \frac{1}{2}L_2 I_2^2 \pm M I_1 I_2$$

depending on the dot convention.
Minimum energy condition is obtained by setting the first partial derivatives equal to zero.
Mutual inductance couples the energies of the two coils through the term MI_1I_2.

Mistakes to be avoided

Do not differentiate with respect to only one current and stop.
Use the correct sign of the mutual energy term according to the given current directions and coupling.

Q1.21.1 MCQ | 2021 | 1M | Ans: A

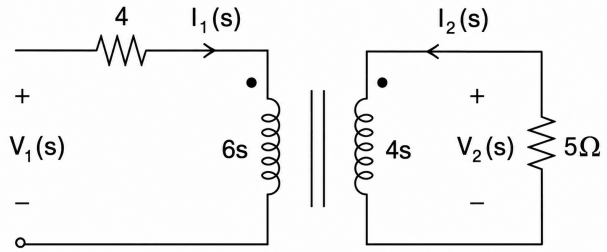


Fig. Solution Q1.21.1

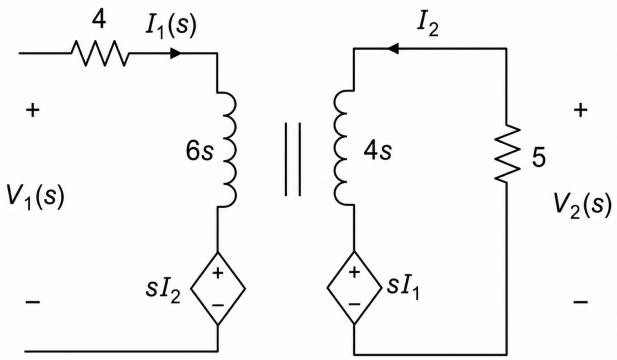


Fig. Solution Q1.21.1

Let the input current be $I_1(s)$ and the current in the secondary loop be $I_2(s)$.
The coupled inductors have

$$L_1 = 6\,\text{H}, \qquad L_2 = 4\,\text{H}, \qquad M = 1\,\text{H}$$

Since both currents enter the dotted terminals, the mutual terms are positive.
Applying KVL to the primary side,

$$V_1(s) = (4 + 6s)I_1(s) + sI_2(s) \tag{1}$$

Applying KVL to the secondary loop,

$$V_2(s) = 4sI_2(s) + sI_1(s)$$

Also,

$$I_2(s) = -\frac{V_2(s)}{5}$$

Hence,

$$5I_2(s) = -(4sI_2(s) + sI_1(s))$$

$$(4s+5)I_2(s) = -sI_1(s)$$

$$I_2(s) = -\frac{sI_1(s)}{4s+5} \tag{2}$$

Substituting (2) into (1),

$$V_1(s) = (4+6s)I_1(s) - \frac{s^2 I_1(s)}{4s+5}$$

Therefore,

$$\frac{V_1(s)}{I_1(s)} = \frac{(4+6s)(4s+5) - s^2}{4s+5}$$

$$= \frac{16s + 20 + 24s^2 + 30s - s^2}{4s+5}$$

$$= \frac{23s^2 + 46s + 20}{4s+5}$$

Hence,

$$Z_{in}(s) = \frac{23s^2 + 46s + 20}{4s+5}$$

Therefore, the correct option is

$$\boxed{A}$$

Key Points

For coupled inductors, use positive mutual terms when both currents enter the dotted terminals.
Input impedance is obtained from

$$Z_{in}(s) = \frac{V_1(s)}{I_1(s)}.$$

Mistakes to be avoided

Do not use the wrong sign for the mutual inductance term. The sign depends on the dot convention and current directions.
Do not forget to eliminate the secondary current before evaluating the input impedance.
Substitute the expression for $I_2(s)$ carefully to avoid algebraic errors while simplifying $Z_{in}(s)$.

Q1.19.1 NAT | 2019 | 2M | Ans: 0.83 to 0.84 | ● ● ○

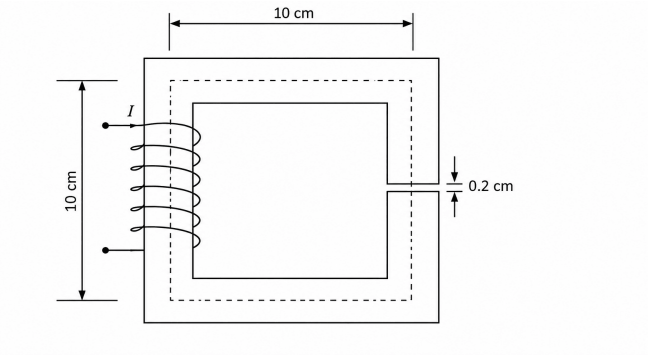


Fig. Solution Q1.19.1

Initial reluctance calculation for the magnetic circuit:

$$\mathcal{R}_{core} = \frac{l}{\mu_r \mu_0 A} = 0 \quad (\text{since } \mu_r = \infty)$$

$$\mathcal{R}_{gap} = \frac{0.2 \times 10^{-2}}{\mu_0 \times A} = \frac{2 \times 10^{-3}}{\mu_0 A}$$

The total initial reluctance $\mathcal{R}_1$ is:

$$\mathcal{R}_1 = \mathcal{R}_{core} + \mathcal{R}_{gap} = \frac{2 \times 10^{-3}}{\mu_0 A}$$

Final reluctance calculation after changes in core properties:

$$\mathcal{R}_{core} = \frac{l_{core}}{\mu_0 \mu_r A} = \frac{40 \times 10^{-2}}{\mu_0 \times 1000 \times A} = \frac{4 \times 10^{-4}}{\mu_0 A}$$

$$\mathcal{R}_{gap} = \frac{2 \times 10^{-3}}{\mu_0 A}$$

The total final reluctance $\mathcal{R}_2$ is:

$$\mathcal{R}_2 = \mathcal{R}_{core} + \mathcal{R}_{gap} = \frac{2.4 \times 10^{-3}}{\mu_0 A}$$

Given that the current I is constant, the Magnetomotive Force (MMF) remains constant (NI = constant). Since $\phi = \frac{\text{MMF}}{\mathcal{R}}$, we have:

$$\phi \propto \frac{1}{\mathcal{R}}$$

The flux density B is proportional to the flux ϕ for a constant area A, so:

$$\frac{\phi_2}{\phi_1} = \frac{B_2}{B_1} = \frac{\mathcal{R}_1}{\mathcal{R}_2}$$

Substituting the calculated values:

$$B_2 = B_1 \times \frac{2.0}{2.4} = 1 \times \frac{2.0}{2.4} = 0.833\,\text{T}$$

$$\boxed{0.83}$$

Key Points

- Reluctance ($\mathcal{R}$) of a magnetic path is given by $\frac{l}{\mu_0 \mu_r A}$.
- For an ideal core ($\mu_r \to \infty$), the reluctance of the core is treated as zero.
- Total reluctance in a series magnetic circuit is the sum of individual reluctances: $\mathcal{R}_{eq} = \mathcal{R}_{core} + \mathcal{R}_{gap}$.
- Flux ϕ and Flux Density B are inversely proportional to total reluctance when MMF is held constant.

Mistakes to be avoided

- **Unit Conversion:** Ensure lengths are converted to meters (m). For example, $0.2\,\text{cm} = 0.2 \times 10^{-2}\,\text{m}$.
- **Reluctance vs. Resistance:** Do not confuse the symbol R for resistance with $\mathcal{R}$ for reluctance, though they follow similar mathematical laws (Ohm's Law for magnetic circuits).
- **Inverse Proportionality:** When calculating the ratio, remember that $B_2/B_1 = \mathcal{R}_1/\mathcal{R}_2$. Swapping the reluctances will lead to an incorrect increase in flux density.
- **Neglecting Air Gap:** Even if the core permeability is very high, the air gap reluctance usually dominates the circuit and cannot be ignored.

Q1.18.1 MCQ 2018 2M Ans: D ●●○

For a toroidal core,

$$L = \frac{\mu N_p^2 A}{l}$$

where

$$A = \pi r^2, \qquad l = 2\pi R$$

$$L = \frac{\mu N_p^2 \pi r^2}{2\pi R} = \frac{\mu N_p^2 r^2}{2R}$$

Given,

$$i_p = I \sin \omega t$$

$$v_p = L\frac{di_p}{dt} = LI\omega \cos \omega t$$

Comparing with

$$v_p = V \cos \omega t$$

$$V = LI\omega$$

$$V = \frac{\mu N_p^2 r^2}{2R} I\omega$$

$$R = \frac{\mu I r^2 N_p^2 \omega}{2V}$$

Hence, the correct option is

$\boxed{D}$

Q1.16.1 NAT | 2016 | 2M | Ans: 188.56 | ● ● ○

Energy in a magnetic system is given by

$$E = \int i(\lambda)\, d\lambda$$

From the relation given,

$$\lambda = \frac{\sqrt{i}}{g}$$

$$i = \lambda^2 g^2$$

Therefore,

$$E = \int \lambda^2 g^2\, d\lambda = g^2 \frac{\lambda^3}{3}$$

Mechanical force is given by

$$F = \frac{dE}{dg}$$

$$F = \frac{d}{dg}\left(\frac{g^2\lambda^3}{3}\right) = \frac{2g\lambda^3}{3}$$

Using

$$\lambda = \frac{\sqrt{i}}{g}$$

$$F = \frac{2g}{3}\left(\frac{\sqrt{i}}{g}\right)^3$$

$$F = \frac{2i^{3/2}}{3g^2}$$

Substituting $i = 2\,\text{A}$ and $g = 0.1\,\text{m}$,

$$F = \frac{2 \times 2^{3/2}}{3(0.1)^2}$$

$$F = 188.56\,\text{N}$$

$\boxed{188.56}$

Key Points

- Magnetic field energy is obtained using $E = \int i\, d\lambda$.
- Mechanical force is calculated from energy as $F = \frac{dE}{dg}$.
- Air-gap length g must be expressed in meters before substitution.

Mistakes to be avoided

- Do not use $E = \frac{1}{2}Li^2$ directly since inductance is not explicitly given.
- Convert 10 cm to 0.1 m before calculation.
- Differentiate energy with respect to gap length g while keeping flux linkage relation intact.

Q1.15.1 NAT 2015 2M Ans: 0 ●●○

Let the mutual inductance between the two identical coils be M.
Since both coils are wound on the same core,

$$L_1 = L_2 = L$$

The equivalent inductance for series connection is

$$L_{eq} = L_1 + L_2 \pm 2M$$

$$L_{eq} = 2L \pm 2M$$

For the minimum possible inductance, the coils must be connected in series opposition:

$$L_{min} = 2L - 2M$$

For two coils on the same core, the maximum possible mutual inductance is

$$M = \sqrt{L_1 L_2} = L$$

Hence,

$$L_{min} = 2L - 2L = 0$$

Given

$$L_{eq} = \alpha L$$

therefore,

$$\alpha = \frac{L_{eq}}{L} = \frac{0}{L} = 0$$

$$\boxed{\alpha = 0}$$

Mistakes to be avoided

- Do not use parallel connection formulas; the minimum value is obtained from series-opposing connection.
- Do not assume $M < L$ when the coils are perfectly coupled.
- Remember that the question asks for the minimum value of α, not the maximum.

Q1.07.1 MCQ 2007 2M Ans: D ● ○ ○

For an iron core with negligible iron reluctance, the inductance is

$$L = \frac{\mu_0 N^2 A}{l}$$

Substituting the given values,

$$L = \frac{4\pi \times 10^{-7} \times (400)^2 \times 16 \times 10^{-4}}{1 \times 10^{-3}} = 0.3217\,\text{H}$$

The inductive reactance is

$$X_L = 2\pi f L$$

$$X_L = 2\pi \times 50 \times 0.3217 = 101.07\,\Omega$$

Hence,

$$I = \frac{V}{X_L} = \frac{230}{101.07} = 2.275\,\text{A}$$

Therefore,

$$\boxed{D}$$

Key Points

- When iron reluctance is neglected, only the air-gap reluctance determines the inductance.
- For a pure inductor, current is given by $I = V/X_L$ where $X_L = 2\pi f L$.
- Air-gap length has a strong influence on inductance since $L \propto 1/l$.

Q1.07.2 MCQ 2007 2M Ans: A ● ○ ○

The energy stored in the magnetic field is

$$W = \frac{1}{2} L I^2$$

Using

$$L = 0.3217\,\text{H}, \qquad I = 2.275\,\text{A}$$

$$W = \frac{1}{2} \times 0.3217 \times (2.275)^2 = 0.83249\,\text{J}$$

The average force required to close the air gap is obtained from the energy method,

$$F = \frac{W}{d}$$

where

$$d = 1\,\text{mm} = 10^{-3}\,\text{m}$$

Therefore,

$$F = \frac{0.83249}{10^{-3}} = 832.49\,\text{N}$$

$$F \approx 832.29\,\text{N}$$

Hence,

$$\boxed{A}$$

Q1.04.1 MCQ | 2004 | 1M | Ans: A | ● ○ ○

For a linear electromagnetic circuit,

$$W_f = \int_0^i \lambda\, di$$

and the co-energy is

$$W'_f = \int_0^\lambda i\, d\lambda$$

For a linear magnetic circuit,

$$\lambda = Li$$

Hence,

$$W_f = \int_0^i Li\, di = \frac{1}{2}Li^2$$

and

$$W'_f = \int_0^{Li} \frac{\lambda}{L}\, d\lambda = \frac{1}{2}Li^2$$

Therefore,

$$W_f = W'_f$$

Thus, field energy is equal to co-energy.

$$\boxed{A}$$

Q1.01.1 NAT | 2001 | 5M | Ans: 530.52, 10 | ● ● ●

Let i_1 be the current flowing clockwise in the primary loop and i_2 be the current flowing clockwise in the short-circuited secondary loop. Both currents enter the dotted terminals of their respective inductors.
Applying Kirchhoff's Voltage Law (KVL) to the secondary loop:

$$j\omega L_2 i_2 + j\omega M i_1 = 0$$

Expressing i_2 in terms of i_1:

$$i_2 = -\frac{M}{L_2} i_1$$

Applying KVL to the primary loop:

$$V = i_1 R + \frac{i_1}{j\omega C} + j\omega L_1 i_1 + j\omega M i_2$$

Substituting i_2 into the primary loop equation:

$$V = i_1 \left(R + \frac{1}{j\omega C} + j\omega L_1 - j\omega \frac{M^2}{L_2} \right)$$

$$V = i_1 \left(R + j \left(\omega \left(L_1 - \frac{M^2}{L_2} \right) - \frac{1}{\omega C} \right) \right)$$

Thus, the net equivalent inductance L_{eq} of the coupled circuit is:

$$L_{eq} = L_1 - \frac{M^2}{L_2}$$

Given data: $L_1 = 40\,\text{mH}$, $L_2 = 10\,\text{mH}$, and $M = 10\,\text{mH}$.

$$L_{eq} = 40\,\text{mH} - \frac{(10\,\text{mH})^2}{10\,\text{mH}} = 30\,\text{mH} = 30 \times 10^{-3}\,\text{H}$$

At resonance, the imaginary part of the total impedance is zero ($\omega_r L_{eq} = \frac{1}{\omega_r C}$):

$$\omega_r = \frac{1}{\sqrt{L_{eq} C}}$$

Given $C = 3\,\mu\text{F} = 3 \times 10^{-6}\,\text{F}$:

$$\omega_r = \frac{1}{\sqrt{30 \times 10^{-3} \times 3 \times 10^{-6}}} = \frac{1}{\sqrt{90 \times 10^{-9}}} = \frac{1}{3 \times 10^{-4}} = 3333.33\,\text{rad/s}$$

The resonance frequency in Hertz (f_r) is:

$$f_r = \frac{\omega_r}{2\pi} = \frac{3333.33}{2\pi} \approx 530.52\,\text{Hz}$$

The Quality factor (Q-factor) for the series resonant circuit is given by:

$$Q = \frac{\omega_r L_{eq}}{R}$$

Given $R = 10\,\Omega$:

$$Q = \frac{3333.33 \times 30 \times 10^{-3}}{10} = 10$$

$$\boxed{530.52,\ 10}$$

Key Points

- When a secondary coupled inductor is short-circuited, the reflected impedance modifies the primary self-inductance to an equivalent value: $L_{eq} = L_1 - \frac{M^2}{L_2}$.
- The resonance frequency of a series $R - L - C$ network occurs when inductive reactance equals capacitive reactance: $\omega_r = \frac{1}{\sqrt{L_{eq} C}}$.
- The quality factor Q measures the sharpness of resonance and is defined as $Q = \frac{\omega_r L_{eq}}{R}$.

Mistakes to be avoided

- Do not ignore the short-circuited secondary loop; the mutual coupling directly reduces the effective inductance of the primary loop.

Q1.01.2 MCQ 2001 2M Ans: C ● ○ ○

The area of the hysteresis loop represents the energy loss per unit volume per cycle. According to the given scales, the hysteresis energy loss per cycle is calculated as:

$$\text{Energy loss per cycle} = \text{Area of the loop} \times (\text{Scale of X-axis} \times \text{Scale of Y-axis})$$

Given: Area = $5\,\text{cm}^2$ Scale of X-axis (Magnetizing force/Current) = 2 AT Scale of Y-axis (Flux) = 50 mWb

$$\text{Energy loss per cycle} = 5 \times (2\,\text{AT} \times 50\,\text{mWb}) = 500\,\text{mJ}$$

The total hysteresis power loss (P_h) at a frequency (f) of 50 Hz is:

$$P_h = \text{Energy loss per cycle} \times f$$

$$P_h = 500\,\text{mJ} \times 50 = 25000\,\text{mW} = 25\,\text{W}$$

C

Key Points

- Hysteresis loss is the energy dissipated in the form of heat due to the reversal of magnetization of the core.
- Power loss formula: $P_h = E_{cycle} \times f$, where E_{cycle} is the energy loss per cycle (area of the B-H loop) and f is the frequency.
- Always account for axis scaling factors when calculating energy from a geometric area.

Mistakes to be avoided

- Forgetting to multiply the area by the frequency to find the power loss (Watts).
- Incorrectly handling units like milli (10^{-3}) which can lead to a result of 25000 instead of 25.

Q1.97.1 NAT 1997 5M Ans: 0.0628 ● ● ○

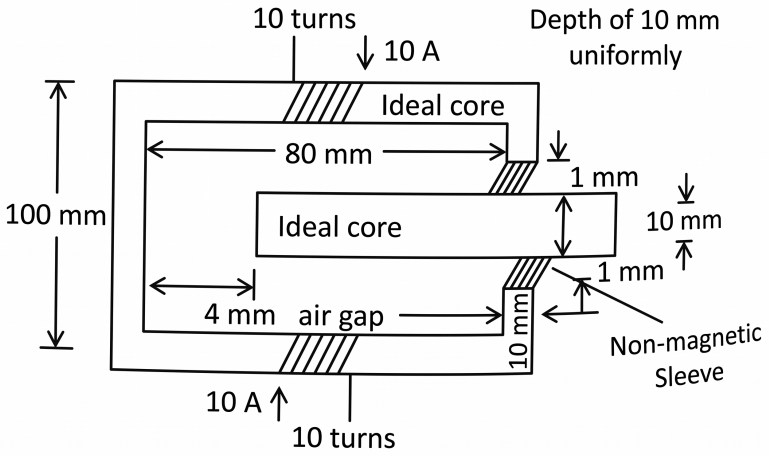


Fig. Solution Q1.97.1

Method 1

To determine the magnetic flux density(B) in the air gap, we use ampere's law for magnetic circuits. Given the magnetic circuit with two identical coils aiding each other, we first calculate the total Magnetomotive Force (MMF).

$$\text{Total MMF} = 2 \times (N \times I) = 2 \times (10 \times 10) = 200\,\text{AT}$$

For an ideal core, the reluctance of the core material is zero ($\mathcal{R}_{core} = 0$). Therefore, the entire MMF is dropped across the air gap.

$$\text{MMF} = H \times g$$

$$H = \frac{\text{MMF}}{g} = \frac{200}{4 \times 10^{-3}} = 50,000 \text{ A/m}$$

The magnetic flux density (B) in the air gap is:

$$B = \mu_0 \times H = (4\pi \times 10^{-7}) \times 50,000$$

$$B = 0.02\pi \approx 0.0628 \text{ T} = 62.83 \text{ mT}$$

Method 2

Alternatively, we can solve using the concept of magnetic reluctance ($\mathcal{R}$). The MMF of each winding is $10 \times 10 = 100$ AT. Since the core is ideal, the only reluctance is offered by the air gap:

$$\mathcal{R}_{ag} = \frac{l_{ag}}{\mu_0 A} = \frac{4 \times 10^{-3}}{4\pi \times 10^{-7} \times (10^{-2} \times 10^{-2})} \approx 0.3183 \times 10^8 \text{ AT/Wb}$$

Total flux (ϕ) passing through the air gap (sum of fluxes from both coils):

$$\phi_{total} = \phi_1 + \phi_2 = 2\phi = 2 \times \frac{100}{0.3183 \times 10^8} = 6.283 \, \mu\text{Wb}$$

Flux density (B) in the air gap:

$$B = \frac{\phi_{total}}{A} = \frac{6.283 \times 10^{-6}}{10^{-2} \times 10^{-2}} = 0.06283 \text{ Wb/m}^2$$

0.0628

Key Points

- **Ideal Core:** Means permeability $\mu_r \to \infty$, thus reluctance $\mathcal{R}_{core} = 0$.
- **Aiding Coils:** When currents are in directions that produce flux in the same direction, MMFs are additive.
- **Flux Density:** $B = \mu_0 H$ in air or $B = \phi / A$.

Mistakes to be avoided

- Ensure the cross-sectional area A is converted correctly to m^2.
- Do not forget to account for both coils if they are aiding each other.
- The dimensions of the core and the "non-magnetic sleeve are not required for this specific calculation because core is ideal and the flux density is primarily determined by the air gap reluctance.

Q1.97.2 NAT 1997 2M Ans: 0.11 ● ○ ○

Let the self-inductance of each coil be L and mutual inductance be M.
When connected in series aiding,

$$L_{\text{aiding}} = 2L + 2M$$

Given,

$$V = 200 \text{ V}, \qquad f = 50 \text{ Hz}, \qquad I = 10 \text{ A}$$

$$X_{\text{aiding}} = \frac{V}{I} = \frac{200}{10} = 20\,\Omega$$

$$\omega L_{\text{aiding}} = 20$$

$$2L + 2M = \frac{20}{\omega}$$

When one coil terminals are reversed (series opposing),

$$L_{\text{opposing}} = 2L - 2M$$

Given current is 8 A,

$$X_{\text{opposing}} = \frac{200}{8} = 25\,\Omega$$

$$\omega L_{\text{opposing}} = 25$$

$$2L - 2M = \frac{25}{\omega}$$

Dividing,

$$\frac{2L - 2M}{2L + 2M} = \frac{25}{20}$$

$$\frac{L - M}{L + M} = \frac{5}{4}$$

Since opposing connection must have lower inductance, taking correct ratio,

$$\frac{L + M}{L - M} = \frac{25}{20} = \frac{5}{4}$$

$$4L + 4M = 5L - 5M$$

$$L = 9M$$

Coefficient of coupling,

$$k = \frac{M}{\sqrt{LL}} = \frac{M}{L} = \frac{1}{9}$$

$$\boxed{0.11}$$

Q1.95.1 NAT | 1995 | 5M | Ans: 303.3 | ● ● ●

The given problem describes an electromagnetic circuit with an E-shaped core and an armature. The total reluctance of the magnetic circuit depends entirely on the air gaps because the core material has infinite permeability, $\mu_r = \infty$.

Area of the central limb air gap is

$$A_c = 25\,\text{mm} \times 25\,\text{mm} = 625 \times 10^{-6}\,\text{m}^2$$

Length of central air gap is

$$g_c = 0.5\,\text{mm} = 0.5 \times 10^{-3}\,\text{m}$$

Reluctance of central air gap is

$$\mathcal{R}_{g,c} = \frac{g_c}{\mu_0 A_c} = \frac{0.5 \times 10^{-3}}{(4\pi \times 10^{-7})(625 \times 10^{-6})} = \frac{2 \times 10^6}{\pi} \text{ AT/Wb}$$

Each side limb has

$$A_s = 12.5 \text{ mm} \times 25 \text{ mm} = 312.5 \times 10^{-6} \text{ m}^2 = \frac{A_c}{2}$$

Hence,

$$\mathcal{R}_{g,s} = \frac{g_c}{\mu_0 A_s} = 2\mathcal{R}_{g,c}$$

The two side gaps are in parallel, therefore

$$\mathcal{R}_{g,\text{side,eq}} = \frac{\mathcal{R}_{g,s}}{2} = \mathcal{R}_{g,c}$$

Total equivalent reluctance is

$$\mathcal{R}_{eq} = \mathcal{R}_{g,c} + \mathcal{R}_{g,\text{side,eq}} = 2\mathcal{R}_{g,c} = \frac{4 \times 10^6}{\pi} \text{ AT/Wb}$$

Since winding resistance and core losses are neglected,

$$V = 2\pi f N \phi_{\text{rms}}$$

$$\phi_{\text{rms}} = \frac{V}{2\pi f N} = \frac{230}{2\pi \times 50 \times 1500} = 4.8808 \times 10^{-4} \text{ Wb}$$

The current drawn is

$$I = \frac{\phi_{\text{rms}} \mathcal{R}_{eq}}{N}$$

$$I = \frac{(4.8808 \times 10^{-4}) \left(\frac{4 \times 10^6}{\pi} \right)}{1500} = 0.414 \text{ A}$$

For the central air gap,

$$F_c = \frac{\phi^2}{2\mu_0 A_c}$$

For each side gap, flux is $\frac{\phi}{2}$ and area is $\frac{A_c}{2}$, so

$$F_s = \frac{\left(\frac{\phi}{2} \right)^2}{2\mu_0 \left(\frac{A_c}{2} \right)} = \frac{\phi^2}{4\mu_0 A_c} = \frac{F_c}{2}$$

Total instantaneous force is

$$F = F_c + 2F_s = \frac{\phi^2}{\mu_0 A_c}$$

Using RMS flux for average force,

$$F_{\text{avg}} = \frac{\phi_{\text{rms}}^2}{\mu_0 A_c}$$

$$F_{\text{avg}} = \frac{(4.8808 \times 10^{-4})^2}{(4\pi \times 10^{-7})(625 \times 10^{-6})} = 303.3 \text{ N}$$

$$\boxed{0.414}$$

Key Points

- Since $\mu_r = \infty$, reluctance of the core is neglected.
- The central air gap is in series with the parallel combination of the two side air gaps.
- For sinusoidal flux, $V = 2\pi f N \phi_{\text{rms}}$.
- Average force is obtained using RMS flux.

Q1.92.1 MCQ 1992 1M Ans: B ● ● ●

In electromechanical energy conversion systems, the developed force or torque acts in a direction that minimizes the stored magnetic energy when the flux is held constant.
For a magnetic system,

$$F = \left(\frac{\partial W'_m}{\partial x}\right)_i$$

where W'_m is the magnetic co-energy.
Similarly,

$$F = -\left(\frac{\partial W_m}{\partial x}\right)_\lambda$$

where W_m is the stored magnetic energy and λ is the flux linkage.
Hence, the force acts in a direction that tends to

decrease the stored energy at constant flux.

Therefore, the correct option is

$$\boxed{B}$$

Key Points

- At constant flux (or flux linkage), electromagnetic systems move toward minimum stored magnetic energy.
- At constant current (or mmf), electromagnetic systems move toward maximum co-energy.
- Energy and co-energy methods are widely used to derive force and torque expressions in magnetic systems.

Mistakes to be avoided

- Do not confuse stored energy with co-energy.
- Remember that the criterion changes depending on whether flux or current (mmf) is kept constant.
- Option (D) is incorrect because it refers to constant mmf, whereas decrease in stored energy applies to constant flux.

Q1.91.1 NAT 1991 2M Ans: 100, 1.44×10^7, 0.69, 0.345 ● ● ○

Given data: Area of core, $A_{core} = 50\,\text{mm}^2$ Mean magnetic path length, $L_{core} = 100\,\text{mm}$ Air-gap length, $L_{ag} = 1\,\text{mm}$ Number of turns, $N = 100$ Exciting current, $I = 1\,\text{A}$
Due to the fringing effect, the effective area of the air-gap increases by 10%:

$$A_{ag} = A_{core} + (10\% \times A_{core}) = 50 + 5 = 55\,\text{mm}^2$$

(a) The exciting mmf of the coil:

$$\text{MMF} = N \times I = 100 \times 1 = 100\,\text{AT}$$

(b) The effective reluctance of the magnetic circuit: Since the iron is a perfect magnetic material ($\mu_r = \infty$), the reluctance of the core $R_{core} = 0$. The net reluctance is equal to the air-gap reluctance R_{ag}:

$$R_{net} = R_{ag} = \frac{L_{ag}}{\mu_0 A_{ag}}$$

$$R_{net} = \frac{1 \times 10^{-3}}{4\pi \times 10^{-7} \times 55 \times 10^{-6}} = 1.44 \times 10^7 \text{ AT/Wb}$$

(c) The inductance of the coil:

$$L = \frac{N^2}{R_{net}} = \frac{100^2}{1.44 \times 10^7} = 0.69 \times 10^{-3} \text{ H} = 0.69 \text{ mH}$$

(d) The energy stored in the magnetic field:

$$E = \frac{1}{2}LI^2 = \frac{1}{2} \times 0.69 \times 10^{-3} \times (1)^2 = 0.345 \text{ mJ}$$

$$\boxed{100, 1.44 \times 10^7, 0.69, 0.345}$$

Key Points

- MMF is the product of turns and current: MMF = NI.
- Reluctance $R = \frac{l}{\mu A}$. For ideal magnetic materials, $\mu_r \to \infty$ and $R \to 0$.
- Inductance can be expressed in terms of reluctance as $L = \frac{N^2}{R}$.
- Magnetic energy stored in an inductor is $W = \frac{1}{2}LI^2$.

Mistakes to be avoided

- Ensure the area is converted from mm^2 to m^2 ($1 \text{ mm}^2 = 10^{-6} \text{ m}^2$) before calculation.
- Do not ignore the fringing effect; it specifically changes the effective area of the air gap, not the core.

Q1.91.2 MCQ 1991 1M Ans: B ●○○

For a solenoid, the inductance is

$$L = \frac{\mu_0 N^2 A}{l}$$

Given,

$$l = 30 \text{ cm} = 0.3 \text{ m}, \quad d = 3 \text{ cm} = 0.03 \text{ m}$$

$$r = \frac{d}{2} = 0.015 \text{ m}$$

$$A = \pi r^2 = \pi (0.015)^2 = 7.068 \times 10^{-4} \text{ m}^2$$

$$L = \frac{4\pi \times 10^{-7} \times (1000)^2 \times 7.068 \times 10^{-4}}{0.3}$$

$$L = 2.96 \times 10^{-3} \text{ H}$$

Energy stored in magnetic field is

$$W = \frac{1}{2}LI^2$$

$$W = \frac{1}{2} \times 2.96 \times 10^{-3} \times (10)^2$$

$$W = 0.148 \text{ J} \approx 0.15 \text{ J}$$

$$\boxed{B}$$

Key Points

- Inductance of a solenoid is $L = \dfrac{\mu_0 N^2 A}{l}$.
- Energy stored in magnetic field is $W = \dfrac{1}{2} L I^2$.

PrepFusion

SOLUTIONS

11

Transformers

Topics

1. Electromechanical Energy Conversion Principles
2. Single-Phase Transformer - Equivalent Circuit and Phasor Diagram
3. Construction of Transformer
4. Effect of Frequency in Transformer
5. Open Circuit and Short Circuit Tests
6. Voltage Regulation and Efficiency of Transformer
7. Parallel Operation of Transformers
8. Auto-Transformer
9. Three-Phase Transformer - Connections and Vector Groups
10. Effect of Frequency in Transformer
11. Scott - T transformers
12. Harmonics in Transformer - Open Delta Connection

Unit Snapshot*

Stream: EE	**Questions:** 109	**MCQ:** 82	**MSQ:** 1	**NAT:** 26
1M: 48	**2M:** 56	**Descriptive:** 5	**Avg Weightage** ~4M	**Range:** 1M(2024)–7M(2015)

**See preface for how Avg Weightage and Range are calculated.*

Q2.26.1 MCQ | 2026 | 1M | Ans: A | ● ● ●

The fundamental relationship for a transformer is derived from the EMF equation:

$$V \approx 4.44 f N \phi_m$$

Since

$$\phi_m = B_m A$$

we get:

$$B_m \propto \frac{V}{f}$$

Comparing the operating conditions:

$$\text{India:} \quad V_1 = 220\,\text{V}, \qquad f_1 = 50\,\text{Hz}$$
$$\frac{V_1}{f_1} = \frac{220}{50} = 4.4$$

$$\text{USA:} \quad V_2 = 110\,\text{V}, \qquad f_2 = 60\,\text{Hz}$$
$$\frac{V_2}{f_2} = \frac{110}{60} \approx 1.83$$

- **A. No-load current drawn would be smaller in the USA: Correct**
  Since the $\frac{V}{f}$ ratio (and hence B_m) is significantly lower in the USA, the core operates much farther from saturation. Therefore, the magnetizing current required to establish the flux is much smaller.
- **B. For the same load current, the losses would be higher in the USA: Incorrect**
  Copper losses:
  $$P_{cu} = I^2 R$$
  remain unchanged for the same current.
  Core losses generally decrease because the reduction in flux density dominates the effect of increased frequency.
- **C. Peak magnetic flux density (B_m) would be higher in the USA: Incorrect**
  Since:
  $$\left(\frac{V}{f}\right)_{\text{USA}} < \left(\frac{V}{f}\right)_{\text{India}}$$
  the peak flux density in the USA is lower.
- **D. Eddy current losses would be approximately 44% higher in the USA: Incorrect**
  Eddy current loss:
  $$P_e \propto f^2 B_m^2$$
  Since:
  $$B_m \propto \frac{V}{f}$$
  we obtain:
  $$P_e \propto V^2$$
  Therefore:
  $$\frac{P_{e(\text{USA})}}{P_{e(\text{India})}} = \left(\frac{110}{220}\right)^2 = \frac{1}{4}$$
  Hence, eddy current losses reduce to nearly 25% of the original value.

Therefore, the only correct statement is:

A

Key Points

- Peak flux density in a transformer core is proportional to the $\frac{V}{f}$ ratio.
- Magnetizing current decreases significantly when the transformer operates at lower flux density.
- Eddy current loss:

$$P_e \propto f^2 B_m^2$$

- Hysteresis loss:

$$P_h \propto f B_m^x$$

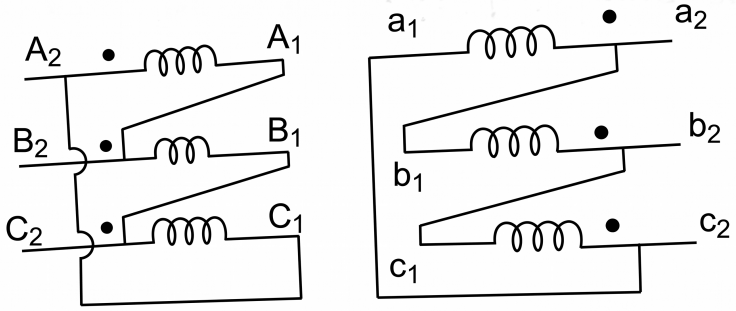
where x is the Steinmetz exponent.

Mistakes to be avoided

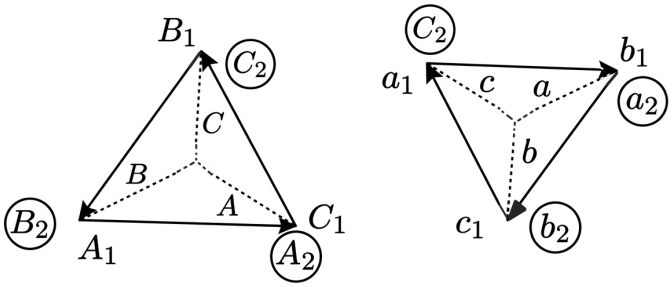
- Do not assume that higher frequency always increases total core losses.
- Avoid comparing only voltage values; transformer flux depends on the $\frac{V}{f}$ ratio.
- Do not confuse magnetic flux ϕ_m with flux density B_m.

Q2.26.2 MCQ 2026 2M Ans: D ●●○

The given transformer bank consists of three single-phase transformers connected in a Delta-Delta (Δ-Δ) configuration. To determine the vector group, we analyze the phase displacement between the High Voltage (HV) and Low Voltage (LV) sides.



Phasor diagram for the primary and secondary sides of the given three phase transformer.

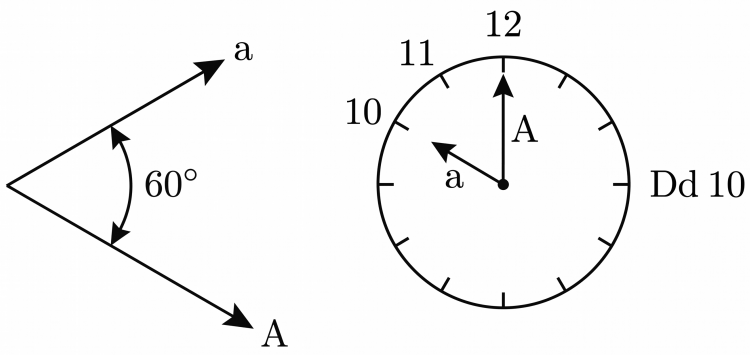


Based on the phasor relationship provided, the LV phase voltage V_a leads the HV phase voltage V_A by 60°.

3. Clock Convention:

A lead of 60° corresponds to moving two positions counter-clockwise from 12 o'clock.

$$+60^\circ \implies 12 - 2 = 10 \text{ o'clock}$$



Therefore, the vector group is identified as **Dd10**.

D

Key Points

- In the clock convention, each hour represents a 30° phase shift.
- Lagging angles are measured clockwise (e.g., −30° = 1 o'clock), and leading angles are measured counter-clockwise (e.g., +30° = 11 o'clock).
- For Delta-Delta connections, the possible phase shifts are even multiples of 30° (0°, 60°, 120°, etc.), resulting in even clock numbers (0, 2, 4, 6, 8, 10).

Mistakes to be avoided

- **Subscript Confusion:** Ensure A_1, A_2 and a_1, a_2 are correctly mapped to the dot convention; reversing them results in a 180° error.
- **Direction of Rotation:** Always assume anti-clockwise rotation for lead angles when mapping to the clock face.

Q2.26.3 MCQ | 2026 | 1M | Ans: D | ● ○ ○

Given a two-winding transformer with:

- Power rating $S_{2W} = 15\,\text{kVA}$
- Voltage ratings $V_{HV} = 1100\,\text{V}$, $V_{LV} = 220\,\text{V}$
- Reconfigured autotransformer ratio: $1.32\,\text{kV}/1.1\,\text{kV}$

Method 1

The autotransformer rating S_{AT} is related to the two-winding rating S_{2W} by the formula:

$$S_{AT} = S_{2W} \times \left(\frac{V_{High,AT}}{V_{series}}\right)$$

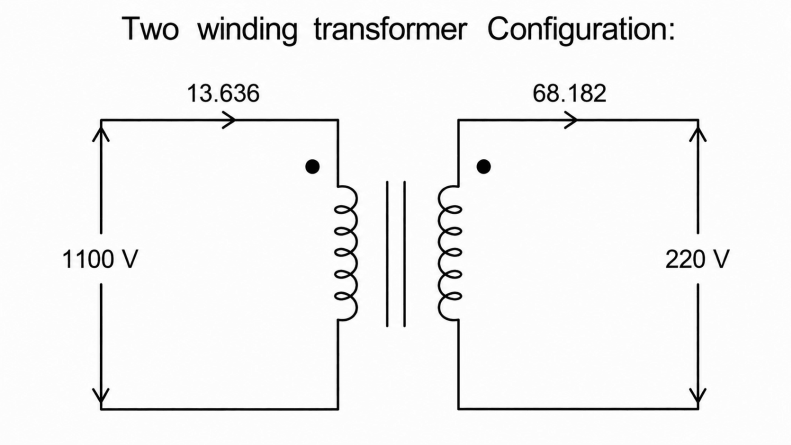
The series winding voltage (V_{series}) is the difference between the high and low voltages of the autotransformer:

$$V_{series} = 1.32\,\text{kV} - 1.1\,\text{kV} = 0.22\,\text{kV} = 220\,\text{V}$$

Substituting the values:

$$S_{AT} = 15\,\text{kVA} \times \left(\frac{1.32\,\text{kV}}{0.22\,\text{kV}}\right) = 15 \times 6 = 90\,\text{kVA}$$

Method 2

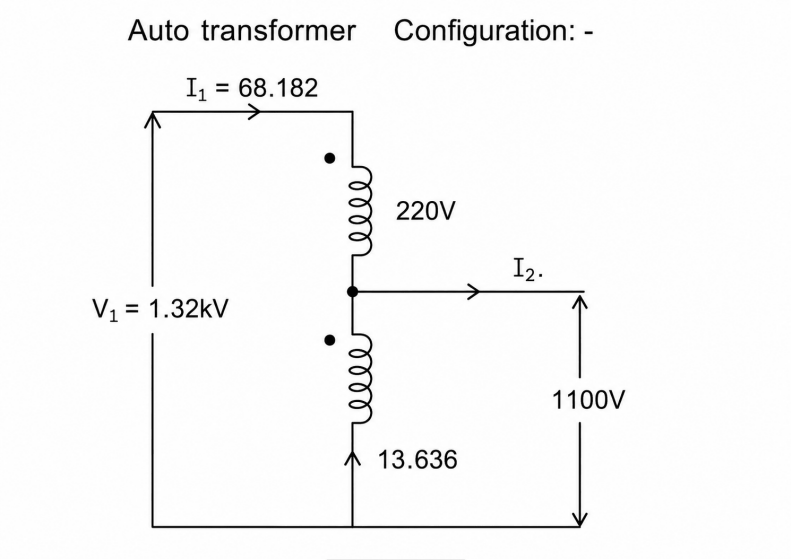


The rated currents for the two-winding transformer are:

$$I_{HV} = \frac{15 \times 10^3}{1100} = 13.636\,\text{A}$$

$$I_{LV} = \frac{15 \times 10^3}{220} = 68.182\,\text{A}$$

In the $1.32\,\text{kV}/1.1\,\text{kV}$ configuration, the 220 V winding is the series winding, and the 1100 V winding is the common winding. The current I_1 (input current at the 1.32 kV side) must be limited by the series winding current.



$$S_{auto} = V_1 \times I_{series} = 1320\,\text{V} \times 68.182\,\text{A} = 90000\,\text{VA} = 90\,\text{kVA}$$

$\boxed{D}$

Key Points

- The kVA rating of an autotransformer is always higher than its two-winding counterpart because power is transferred both conductively and inductively.
- $S_{AT} = \frac{1}{1-k} S_{2W}$, where k is the transformation ratio (V_{low}/V_{high}).
- The series winding must be able to carry the full rated current of the winding used for that section.

Mistakes to be avoided

- Do not confuse V_{series} with the low-voltage rating of the autotransformer; V_{series} is specifically the voltage across the winding not common to both sides.
- Ensure units are consistent (converting kV to V) before performing division or multiplication.

Q2.25.1 MCQ 2025 2M Ans: A ●●○

Given:

$$\frac{N_p}{N_s} = \frac{2}{1}$$

and

$$\bar{I}_a + \bar{I}_b + \bar{I}_c = 0$$

The primary and secondary phase currents of the same phase are drawn parallel to each other in the phasor diagram.

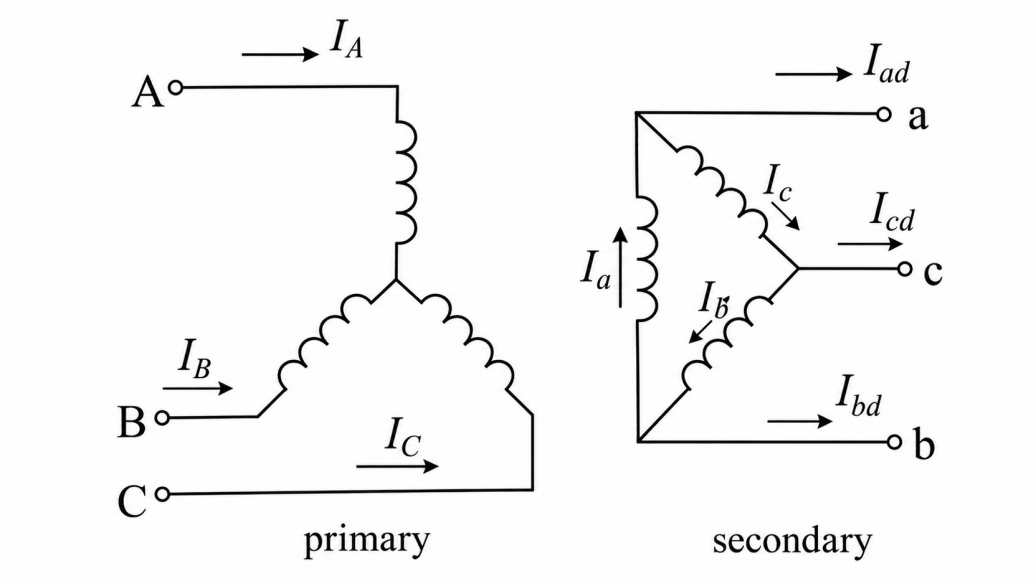


Fig. Solution Q2.25.1

From the phasor relationship:

$$\bar{I}_{ad} = \bar{I}_a - \bar{I}_c$$

Thus, I_{ad} lags I_A by 30°.
Using the transformer current ratio:

$$\frac{I_A}{I_a} = \frac{N_s}{N_p} = \frac{1}{2}$$

Therefore,

$$I_a = 2I_A$$

For a delta connection, the line current is:

$$I_{ad} = \sqrt{3}\, I_a$$

Substituting:

$$I_{ad} = \sqrt{3}(2I_A) = 2\sqrt{3}\, I_A$$

Hence,

$$\frac{I_A}{I_{ad}} = \frac{1}{2\sqrt{3}}$$

$\boxed{A}$

Key Points

- In a transformer, current ratio is inversely proportional to turns ratio:

$$\frac{I_p}{I_s} = \frac{N_s}{N_p}$$

- In a delta connection:

$$I_L = \sqrt{3}\, I_{ph}$$

- The line current in delta connection leads or lags the phase current by 30°.

Mistakes to be avoided

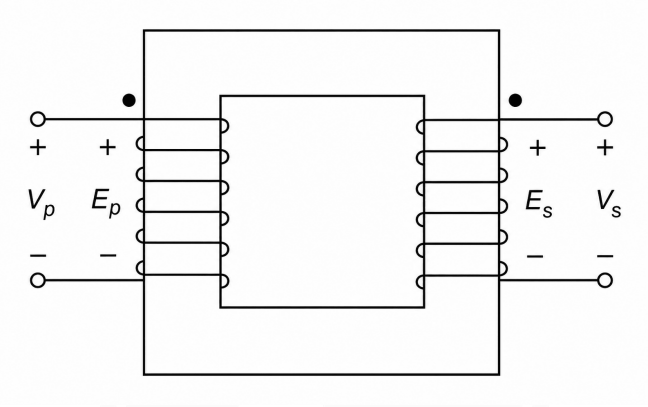
- Do not confuse phase current with line current in delta connection.
- Ensure proper interpretation of phasor subtraction:

$$\bar{I}_{ad} = \bar{I}_a - \bar{I}_c$$

- Always use inverse proportionality between current and turns ratio in transformers.

Q2.24.1 MCQ 2024 1M Ans: MTA ●●○

The basic principle of a transformer states that the dots on both the primary and secondary sides must have the same polarity of voltage.This condition is satisfied in option B, but in option B the dots are wrongly placed.The correct diagram should be:



But this does not match any of the given options.Therefore, the correct answer is:

MTA

Key Points

- Dot convention indicates the relative polarity of transformer windings.
- If current enters the dotted terminal of the primary winding, the induced voltage at the dotted terminal of the secondary winding becomes positive.
- Primary and secondary induced voltages must maintain identical instantaneous polarity at dotted terminals.

Mistakes to be avoided

- Do not place dots arbitrarily on opposite polarity ends.
- Always verify induced voltage polarity before deciding transformer connections.
- Do not confuse current direction with voltage polarity in dot convention problems.

Q2.23.1 NAT 2023 2M Ans: 2150 to 2250 ●●○

Given:

- Open circuit voltage across terminals a-b is 200 V (rms)
- Frequency:

$$f = 50\,\text{Hz}$$

- Maximum flux density relation:

$$\phi_m = 5000 \times 10^{-6}\,\text{Wb}$$

- Relative permeability:

$$\mu_r = 5000$$

- Leakage inductance of primary:

$$L_{\ell p} = 0.16\,\text{H}$$

Using the emf equation of transformer:

$$E = 4.44 f N \phi_m$$

Substituting the given values:

$$200 = 4.44 \times 50 \times N_1 \times 5000 \times 10^{-6}$$

$$N_1 = 180 \text{ turns}$$

Reluctance of flux path inside the core is:

$$\mathcal{R} = \frac{\ell}{\mu_0 \mu_r A}$$

Substituting the values:

$$\mathcal{R} = \frac{0.5}{4\pi \times 10^{-7} \times 5000 \times 5000 \times 10^{-6}} = 15915.494\,\text{A-T/Wb}$$

Mutual inductance referred to primary:

$$M = \frac{N_1^2}{\mathcal{R}}$$

$$M = \frac{(180)^2}{15915.494} = 2.0357\,\text{H}$$

Self inductance of primary:

$$L_p = M + L_{\ell p}$$

$$L_p = 2.0357 + 0.16 = 2.1957\,\text{H}$$

$$L_p = 2195.7\,\text{mH}$$

$$\boxed{2195.7}$$

Key Points

- Transformer emf equation:

$$E = 4.44 f N \phi_m$$

- Mutual inductance is related to reluctance by:

$$M = \frac{N^2}{\mathcal{R}}$$

- Self inductance includes both mutual and leakage inductance components.

Q2.21.1 NAT | 2021 | 2M | Ans: 50 to 52 | ● ● ○

Peak voltage across resistor is

$$\hat{V}_r = \frac{5}{2} = 2.5\,\text{V}$$

Peak current is

$$I_m = \frac{\hat{V}_r}{R} = \frac{2.5}{22}\,\text{A}$$

Assume

$$i_p = I_m \sin \omega t$$

Induced voltage across secondary is

$$v_s = M \frac{di_p}{dt}$$

$$v_s = M \frac{d}{dt} (I_m \sin \omega t) = \omega M I_m \cos \omega t$$

Hence,

$$\hat{V}_s = \omega M I_m$$

Substituting the values,

$$\frac{7.3}{2} = \left(2\pi \times 10^5\right) M \left(\frac{2.5}{22}\right)$$

$$M = 51.12\,\mu\text{H}$$

$$\boxed{51.12}$$

Key Points

- For sinusoidal current, induced voltage due to mutual inductance is $v_s = M \frac{di_p}{dt}$.
- Peak value relation is $\hat{V}_s = \omega M I_m$.
- Use peak voltage values consistently throughout the calculation.

Q2.21.2 MCQ | 2021 | 2M | Ans: B | ● ○ ○

Given:

- Wattmeter reading:

$$W_{i1} = 2500\,\text{W} \quad \text{at } 440\,\text{V}, 50\,\text{Hz}$$

- Wattmeter reading:

$$W_{i2} = 850\,\text{W} \quad \text{at } 220\,\text{V}, 25\,\text{Hz}$$

- Total core loss:

$$W_{i3} = P_{e3} + P_{h3}$$

- Since

$$\frac{V}{f} = \text{constant}$$

Total iron loss is given by:

$$P_i = P_h + P_e$$

where

$$P_h = Af$$

and

$$P_e = Bf^2$$

Hence,

$$2500 = Af + Bf^2$$

Dividing by f:

$$\frac{2500}{f} = A + Bf$$

For 50 Hz:

$$\frac{2500}{50} = A + 50B$$

$$50 = A + 50B \quad \text{(i)}$$

For 25 Hz:

$$\frac{850}{25} = A + 25B$$

$$34 = A + 25B \quad \text{(ii)}$$

Subtracting (ii) from (i):

$$25B = 50 - 34$$

$$25B = 16$$

$$B = \frac{16}{25}$$

Substituting into (i):

$$A = 50 - \frac{16}{25} \times 50$$

$$A = 50 - 32 = 18$$

Therefore hysteresis loss at 50 Hz is:

$$P_h = Af$$

$$P_h = 18 \times 50 = 900$$

Eddy current loss:

$$P_e = Bf^2$$

$$P_e = \left(\frac{16}{25}\right)(50)^2 = 1600\,\text{W}$$

$$\boxed{B}$$

Q2.20.1 MCQ | 2020 | 2M | Ans: MTA | ● ● ●

Given:

- Number of turns:

$$N_A = 20, \quad N_B = 20, \quad N_C = 20, \quad N_X = 2$$

- Applied balanced three-phase voltages:

$$\bar{V}_A = 230\angle 0^\circ\ \text{V}$$

$$\bar{V}_B = 230\angle(-120^\circ)\ \text{V}$$

$$\bar{V}_C = 230\angle 120^\circ\ \text{V}$$

Assuming winding polarity from the figure:

$$\bar{V}_{\text{net}} = \bar{V}_A - \bar{V}_B - \bar{V}_C$$

First calculate:

$$\bar{V}_B + \bar{V}_C = 230\angle(-120^\circ) + 230\angle 120^\circ$$

$$\bar{V}_B + \bar{V}_C = 230\,[(-0.5 - j0.866) + (-0.5 + j0.866)]$$

$$\bar{V}_B + \bar{V}_C = -230\angle 0^\circ \text{ V}$$

Hence,

$$\bar{V}_{\text{net}} = 230\angle 0^\circ - (-230\angle 0^\circ)$$

$$\bar{V}_{\text{net}} = 460\angle 0^\circ \text{ V}$$

Using transformer turns ratio:

$$\frac{N_A}{N_X} = \frac{20}{2} = 10$$

Therefore,

$$V_X = \frac{460}{10} = 46 \text{ V}$$

Technical flaw:

Windings A, B, and C are placed on the same magnetic limb and therefore share the same mutual flux:

$$\phi(t)$$

For ideal coupled windings on the same magnetic path, induced EMFs must either be in phase or exactly 180° apart.
Applying three independent sources displaced by 120° across windings on the same magnetic limb violates magnetic consistency and Kirchhoff's Voltage Law.
Hence, the configuration becomes physically inconsistent and the question was officially awarded Marks To All (MTA).

MTA

Key Points

- Windings on the same magnetic limb share identical mutual flux.
- Transformer induced voltages depend on common magnetic flux linkage.
- Three independent phase-shifted sources cannot simultaneously impose incompatible flux conditions.

Mistakes to be avoided

- Do not directly superimpose voltages without checking magnetic feasibility.
- Do not assume ideal transformer constraints can be violated.
- Always verify whether winding placement causes shared flux coupling.

Q2.20.2 NAT 2020 1M Ans: 162.5 ● ○ ○

Given,

$$200 \text{ V},\ 50 \text{ Hz}, \quad P_c = 450 \text{ W}$$

$$160 \text{ V},\ 40 \text{ Hz}, \quad P_c = 320 \text{ W}$$

$$100 \text{ V},\ 25 \text{ Hz}, \quad P_c = ?$$

Since

$$\frac{V}{f} = \text{constant}$$

$$\frac{200}{50} = \frac{160}{40} = \frac{100}{25}$$

So, core loss is given by

$$P_c = Af + Bf^2$$

For 50 Hz,

$$450 = A(50) + B(50)^2$$

$$\frac{450}{50} = A + B(50)$$

$$9 = A + 50B \qquad \cdots (i)$$

For 40 Hz,

$$320 = A(40) + B(40)^2$$

$$\frac{320}{40} = A + B(40)$$

$$8 = A + 40B \qquad \cdots (ii)$$

Subtracting equation (ii) from equation (i),

$$9 - 8 = 50B - 40B$$

$$1 = 10B$$

$$B = \frac{1}{10}$$

Substituting in equation (i),

$$A = 9 - \frac{1}{10} \times 50 = 4$$

Now at 100 V, 25 Hz,

$$P_c = 4 \times 25 + \frac{1}{10} \times (25)^2 = 162.5$$

$$\boxed{162.50\,\text{W}}$$

Q2.19.1 NAT | 2019 | 2M | Ans: 0.83 to 0.84 | ● ○ ○

Initial reluctance:

$$R_{\text{core}} = \frac{l}{\mu_0 \mu_r A} = 0 \qquad (\mu_r = \infty)$$

$$R_{\text{gap}} = \frac{0.2 \times 10^{-2}}{\mu_0 A} = \frac{2 \times 10^{-3}}{\mu_0 A}$$

Hence,

$$R_1 = R_{\text{core}} + R_{\text{gap}} = \frac{2 \times 10^{-3}}{\mu_0 A}$$

Final reluctance:

$$R_{\text{core}} = \frac{l_{\text{core}}}{\mu_0 \mu_r A}$$

$$= \frac{40 \times 10^{-2}}{\mu_0 \times 1000 \times A}$$

$$= \frac{4 \times 10^{-4}}{\mu_0 A}$$

$$R_{gap} = \frac{2 \times 10^{-3}}{\mu_0 A}$$

Hence,

$$R_2 = R_{core} + R_{gap}$$

$$R_2 = \frac{2.4 \times 10^{-3}}{\mu_0 A}$$

Since current is constant,

$$I = \text{constant}$$

Therefore MMF is constant:

$$NI = \text{constant}$$

Hence,

$$\phi \propto \frac{1}{R}$$

$$\frac{\phi_2}{\phi_1} = \frac{B_2}{B_1} = \frac{R_1}{R_2}$$

$$B_2 = B_1 \left(\frac{2.0}{2.4}\right)$$

$$B_2 = 1 \times 0.833$$

$$B_2 = 0.83\,\text{T}$$

PrepFusion

$$\boxed{0.83}$$

Q2.19.2 NAT | 2019 | 2M | Ans: 7.10 to 7.30 | ● ○ ○

Given load:

$$P_L = 12\,\text{kW}$$

Power factor:

$$\cos\phi_L = 0.6$$

Reactive power of load:

$$Q_L = P_L \tan\phi_L$$

$$Q_L = 12 \tan(\cos^{-1}(0.6))$$

$$Q_L = 12 \times \frac{0.8}{0.6} = 16\,\text{kVAr}$$

Transformer rating:

$$S = 25\,\text{kVA}$$

Using power triangle relation:

$$S^2 = P^2 + Q^2$$

Since additional load has unity power factor, it consumes only active power.
Hence total reactive power remains:

$$Q = 16\,\text{kVAr}$$

If additional active power is P,

$$(P + 12)^2 + 16^2 = 25^2$$

$$(P + 12)^2 = 625 - 256 = 369$$

$$P + 12 = 19.21$$

$$P = 7.21\,\text{kW}$$

$$\boxed{7.21}$$

Q2.19.3 MCQ 2019 1M Ans: C ●○○

Voltage regulation is given by:

$$\text{Voltage Regulation} = \frac{V_{NL} - V_{FL}}{V_{NL}} \times 100\%$$

Substituting the given values,

$$\text{Voltage Regulation} = \frac{100 - 95}{100} \times 100\%$$

$$= \frac{5}{100} \times 100$$

$$= 5\%$$

$$\boxed{C}$$

Key Points

- Voltage regulation indicates the change in terminal voltage from no-load to full-load conditions.
- Standard expression:

$$\text{VR} = \frac{V_{NL} - V_{FL}}{V_{NL}} \times 100\%$$

- Positive voltage regulation means terminal voltage decreases with loading.

Q2.18.1 NAT 2018 2M Ans: 97.36 ● ○ ○

Given:

$$S = 900\,\text{kVA}, \qquad V_{\text{line}}(\Delta) = 3\,\text{kV}, \qquad V_{\text{line}}(Y) = \sqrt{3}\,\text{kV}$$

For delta connection:

$$I_{\text{ph}}(\Delta) = \frac{S}{3V_{\text{ph}}} = \frac{900}{3 \times 3} = 100\,\text{A}$$

For star connection:

$$I_{\text{ph}}(Y) = I_{\text{line}} = \frac{S}{\sqrt{3}V_{\text{line}}} = \frac{900}{\sqrt{3} \times \sqrt{3}} = 300\,\text{A}$$

Copper loss:

$$P_{cu} = 3I^2_{\text{ph}_{\text{pri}}} R_p + 3I^2_{\text{ph}_{\text{sec}}} R_{\text{sec}}$$

$$P_{cu} = 3(100)^2(0.3) + 3(300)^2(0.02) = 14400\,\text{W} = 14.4\,\text{kW}$$

Core loss:

$$P_{\text{core}} = 10\,\text{kW}$$

Total losses:

$$P_{\text{loss}} = 14.4 + 10 = 24.4\,\text{kW}$$

Efficiency:

$$\eta = \frac{\text{Output}}{\text{Output + Losses}} \times 100 = \frac{900}{900 + 24.4} \times 100 = 97.36\%$$

$$\boxed{97.36}$$

Key Points

- For delta connection, $V_{\text{ph}} = V_{\text{line}}$.
- For star connection, $I_{\text{ph}} = I_{\text{line}}$.
- Copper loss is $P_{cu} = 3I^2R$.

Mistakes to be avoided

- Do not confuse phase and line quantities in star and delta connections.
- Include both copper loss and core loss while calculating efficiency.

Q2.18.2 MCQ 2018 1M Ans: A ● ○ ○

Impedance:

$$Z = 5\% = 0.05\,\text{pu}$$

Resistance:

$$R = 3\% = 0.03\,\text{pu}$$

Reactance:

$$X = \sqrt{Z^2 - R^2} = \sqrt{(0.05)^2 - (0.03)^2} = 0.04\,\text{pu}$$

Voltage regulation at 0.8 lagging p.f.:

$$VR = R\cos\phi + X\sin\phi$$

$$= 0.03(0.8) + 0.04(0.6) = 0.024 + 0.024 = 0.048$$

$$VR = 4.8\%$$

$$\boxed{A}$$

Q2.17.1 NAT 2017 2M Ans: 625.089 ● ○ ○

Given:

$$\bar{S}_1 = 900\angle 36.86^\circ \text{ kVA}, \qquad \bar{S}_2 = 300\angle 53.13^\circ \text{ kVA}$$

Total power drawn:

$$\bar{S} = 900(0.8 + j0.6) + 300(0.6 + j0.8)$$

$$= (720 + j540) + (180 + j240)$$

$$= 900 + j780 = 1190.95\angle 40.9^\circ \text{ kVA}$$

Using

$$S = \sqrt{3} V_L I_L$$

$$I_L = \frac{1190.95}{\sqrt{3} \times 1.1} = 625.1 \text{ A}$$

$$\boxed{625.1}$$

Q2.17.2 NAT 2017 2M Ans: 37.878 ● ○ ○

For a $Y - \Delta$ transformer, line-to-line voltage ratio is:

$$\frac{V_{\text{sec}}}{V_{\text{pri}}} = \frac{N_s}{\sqrt{3} N_p} = \frac{1}{5\sqrt{3}}$$

Hence secondary line voltage:

$$V_{\text{sec}} = \frac{3.3}{5\sqrt{3}} \text{ kV}$$

Since kVA rating remains same on both sides of transformer,

$$S = 25 \text{ kVA}$$

Secondary line current:

$$I_L = \frac{S}{\sqrt{3} V_L} = \frac{25}{\sqrt{3} \times \frac{3.3}{5\sqrt{3}}} = \frac{125}{3.3} = 37.878 \text{ A}$$

$$\boxed{37.878}$$

Q2.16.1 MCQ 2016 1M Ans: B ● ○ ○

In the given problem,

$$n = \frac{N_p}{N_s}$$

When inductance is referred to the primary side,

$$L_p = \left(\frac{N_p}{N_s}\right)^2 L = n^2 L$$

$$\boxed{B}$$

Key Points

- Impedance referred from secondary to primary scales as:

$$Z_p = n^2 Z$$

- Inductance referred to primary also scales as:

$$L_p = n^2 L$$

Q2.16.2 MCQ 2016 2M Ans: C ● ● ●

Method 1

Using phasor analysis,
For negative sequence excitation,

$$V_{AN} = V\angle 0^\circ, \qquad V_{BN} = V\angle 120^\circ, \qquad V_{CN} = V\angle -120^\circ$$

Star-side line voltage:

$$V_{AB} = V_{AN} - V_{BN}$$

$$V_{AB} = \sqrt{3}V\angle(-30^\circ)$$

From dot convention,

$$V_{AN} \text{ is in phase with } V_{ac}$$

Hence,

$$V_{ac} = V\angle 0^\circ$$

For delta connection under negative sequence,

$$V_{ac} \text{ leads } V_{ab} \text{ by } 60^\circ$$

$$V_{ab} = V\angle(-60^\circ)$$

Comparing phase angles,

$$V_{AB} = \angle(-30^\circ), \qquad V_{ab} = \angle(-60^\circ)$$

Hence,

$$V_{AB} \text{ leads } V_{ab} \text{ by } 30^\circ$$

Method 2

Using transformer phase-shift property,
For a star-delta transformer, line voltages possess an inherent phase shift of:

$$30^\circ$$

For positive sequence, the phase shift occurs in one direction.
For negative sequence excitation, the phase displacement reverses.
Hence,

$$V_{AB} \text{ leads } V_{ab} \text{ by } 30^\circ$$

$$\boxed{C}$$

Key Points

- Negative sequence phase order:
$$ABC \rightarrow ACB$$
- For negative sequence:
$$V_{AB} = \sqrt{3}V\angle(-30^\circ)$$
- Star-delta transformers introduce a phase shift of 30°.
- Under negative sequence excitation, transformer phase displacement reverses.

Mistakes to be avoided

- Do not use positive-sequence phase relations directly.
- Do not confuse phase voltage and line voltage.
- Carefully follow the dot convention while relating primary and secondary voltages.

Q2.16.3 NAT 2016 1M Ans: 10 ● ○ ○

Referring the secondary circuit to the primary side,

$$R_p = \left(\frac{N_p}{N_s}\right)^2 R_s = \frac{80 \times 10^3}{10^4} = 8\,\Omega$$

$$X_p = \left(\frac{N_p}{N_s}\right)^2 X_s = \frac{40 \times 10^3}{10^4} = 4\,\Omega$$

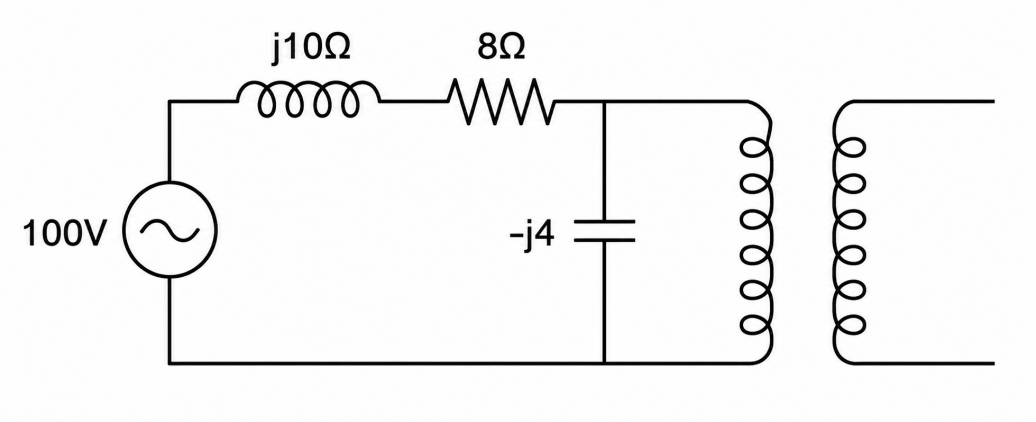


Fig. Solution Q2.16.3

Equivalent impedance:

$$Z_{eq} = 8 + j10 - j4$$

$$= 8 + j6 = 10\angle 36.87^\circ\ \Omega$$

Current:

$$I = \frac{V}{Z_{eq}} = \frac{100\angle 0^\circ}{10\angle 36.87^\circ}$$

$$= 10\angle(-36.87^\circ)\ \text{A}$$

Magnitude of current:

$$|I| = 10\ \text{A}$$

10

Q2.16.4 MCQ 2016 2M Ans: A

Given,
Transformer kVA rating remains same on both sides:

$$\sqrt{3}V_{LP}I_{LP} = \sqrt{3}V_{LS}I_{LS}$$

$$\sqrt{3} \times 110 \times I_{LP} = \sqrt{3} \times 11 \times I_{LS}$$

$$\frac{I_{LP}}{I_{LS}} = \frac{1}{10}$$

Since the primary is delta connected and the secondary is star connected,

$$I_{LP} = \sqrt{3}\, I_{\text{ph},P}$$

$$I_{LS} = I_{\text{ph},S}$$

Substituting,

$$\frac{\sqrt{3}I_{ph,P}}{I_{ph,S}} = \frac{1}{10}$$

$$\frac{I_{ph,P}}{I_{ph,S}} = \frac{1}{10\sqrt{3}}$$

$\boxed{A}$

Q2.16.5 NAT | 2016 | 2M | Ans: 6 | ● ○ ○

Given,

$$(\text{kVA})_{\text{auto}} = \frac{(\text{kVA})_{\text{2-wdg}}}{1-k}$$

where,

$$k = \frac{\text{Low voltage}}{\text{High voltage}}$$

Possible additive-polarity configurations:

$$k_1 = \frac{100}{300} = \frac{1}{3}$$

$$k_2 = \frac{200}{300} = \frac{2}{3}$$

For first configuration,

$$(\text{kVA})_1 = \frac{2}{1-\frac{1}{3}} = 3\,\text{kVA}$$

For the second configuration,

$$(\text{kVA})_2 = \frac{2}{1-\frac{2}{3}} = 6\,\text{kVA}$$

Since

$$(\text{kVA})_2 > (\text{kVA})_1$$

Maximum rating:

$\boxed{6}$

Key Points

- Autotransformer rating:

$$(\text{kVA})_{\text{auto}} = \frac{(\text{kVA})_{\text{2-wdg}}}{1-k}$$

- Voltage ratio:

$$k = \frac{V_L}{V_H}$$

- Larger value of k gives higher autotransformer rating.

Mistakes to be avoided

- Do not use subtractive polarity here; additive polarity is specified.
- Maintain:

$$k < 1$$

- Use lower-voltage to higher-voltage ratio only.

Q2.16.6 NAT | 2016 | 2M | Ans: 1.4423 | ● ○ ○

Given,

$$V_i = 400\,\text{V}, \qquad f_i = 50\,\text{Hz}$$

$$V_f = 200\,\text{V}, \qquad f_f = 25\,\text{Hz}$$

Since

$$\frac{V}{f} = \frac{400}{50} = \frac{200}{25} = 8$$

Hence flux remains constant.
Core-loss components:

$$P_h = K_h f, \qquad P_e = K_e f^2$$

At rated condition:

$$5000 = 50K_h + 2500K_e \tag{i}$$

At 25 Hz:

$$2000 = 25K_h + 625K_e \tag{ii}$$

From

$$(i) - 2(ii)$$

$$1000 = 1250K_e$$

$$K_e = 0.8$$

Substituting in (i),

$$5000 = 50K_h + 0.8(2500)$$

$$K_h = 60$$

Thus,

$$P_i = 60f + 0.8f^2$$

At

$$f = 52\,\text{Hz}$$

$$P_h = 60(52) = 3120\,\text{W}$$

$$P_e = 0.8(52)^2 = 2163.2\,\text{W}$$

Required ratio:

$$\frac{P_h}{P_e} = \frac{3120}{2163.2}$$

$$= 1.4423$$

$$\boxed{1.4423}$$

Q2.16.7 NAT | 2016 | 2M | Ans: 242 | ● ○ ○

Since output voltage is the sum of two voltages, the autotransformer is connected in additive polarity.

Method 1

Configuration: 220/2420

$$K = \frac{\text{Low voltage}}{\text{High voltage}}$$

$$K = \frac{220}{2420} = \frac{1}{11}$$

Autotransformer rating:

$$(\text{kVA})_{\text{auto}} = \frac{(\text{kVA})_{\text{2-wdg}}}{1 - K}$$

$$(\text{kVA})_{\text{auto}} = \frac{22}{1 - \frac{1}{11}}$$

$$= \frac{22 \times 11}{10}$$

$$= 24.2\,\text{kVA}$$

Method 2

Configuration: 2200/2420

$$K = \frac{2200}{2420} = \frac{10}{11}$$

$$(\text{kVA})_{\text{auto}} = \frac{22}{1 - \frac{10}{11}}$$

$$= 22 \times 11$$

$$= 242\,\text{kVA}$$

Since,

$$242 > 24.2$$

Maximum rating:

$$(\text{kVA})_{\max} = 242\,\text{kVA}$$

$$\boxed{242}$$

Q2.15.1 NAT | 2015 | 2M | Ans: 1.33 | ● ○ ○

Method 1

Using power balance
Load current on high-voltage side:

$$I_L = \frac{600}{12} = 50\,\text{A}$$

For an ideal transformer,

$$P_{\text{in}} = P_{\text{out}}$$

$$200 I_p = 600 \times 50$$

$$I_p = 150\,\text{A}$$

Equivalent load resistance as seen from the low-voltage side:

$$R_{\text{eq}} = \frac{V_{\text{in}}}{I_p} = \frac{200}{150} = 1.33\,\Omega$$

Method 2

Using impedance transformation
Given autotransformer voltage rating:

$$V_L = 200\,\text{V}, \qquad V_H = 600\,\text{V}$$

Impedance referred from high-voltage side to low-voltage side:

$$R_{\text{eq}} = \left(\frac{V_L}{V_H}\right)^2 R_L$$

$$R_{\text{eq}} = \left(\frac{200}{600}\right)^2 (12) = \frac{1}{9} \times 12 = 1.33\,\Omega$$

$$\boxed{1.33}$$

Key Points

- For ideal transformer:

$$P_{\text{in}} = P_{\text{out}}$$

- Impedance transformation relation:

$$R_{eq} = \left(\frac{V_L}{V_H}\right)^2 R_L$$

- Resistance referred to lower-voltage side reduces by the square of voltage ratio.

Mistakes to be avoided

- Do not use the original two-winding voltage rating 200/400 V for impedance reflection.
- Use the autotransformer voltage rating 200/600 V.
- Do not forget that impedance transforms as square of voltage ratio.

Q2.15.2 NAT | 2015 | 2M | Ans: 555 to 556 | ● ○ ○

Method 1

Using impedance ratio
Given,

$$Z_1 = 1 + j6$$

$$Z_2 = 0.8 + j4.8$$

Since

$$Z_2 = 0.8Z_1$$

the two impedances have the same angle. Hence load sharing is inversely proportional to impedance magnitude.

$$S_1 : S_2 = \frac{1}{Z_1} : \frac{1}{Z_2}$$

$$S_1 : S_2 = Z_2 : Z_1 = 0.8 : 1$$

Total load:

$$S_T = 1000\,\text{kVA}$$

Therefore,

$$S_2 = \frac{1}{1 + 0.8} \times 1000$$

$$S_2 = 555.56\,\text{kVA}$$

Method 2

Using current sharing
Since both transformers are in parallel,

$$\frac{I_1}{I_2} = \frac{Z_2}{Z_1}$$

$$\frac{I_1}{I_2} = \frac{0.8 + j4.8}{1 + j6} = 0.8$$

So,

$$I_1 = 0.8I_2$$

Since terminal voltage is same, kVA sharing follows current sharing:

$$S_1 = 0.8S_2$$

$$S_1 + S_2 = 1000$$

$$0.8S_2 + S_2 = 1000$$

$$S_2 = 555.56\,\text{kVA}$$

555.56

Key Points

- In parallel transformers, current sharing is inversely proportional to leakage impedance:

$$\frac{I_1}{I_2} = \frac{Z_2}{Z_1}$$

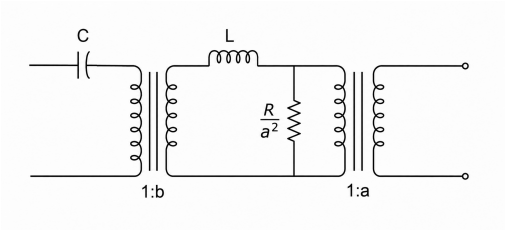
- If impedance angles are equal, kVA sharing is directly obtained from impedance magnitudes.
- Here, $Z_2 = 0.8Z_1$, so T_2 carries more load.

Mistakes to be avoided

- Do not divide load equally just because both transformers have equal kVA rating.
- Check impedance angles before using simple magnitude sharing.
- Transformer T_2 has lower impedance, so it carries higher kVA.

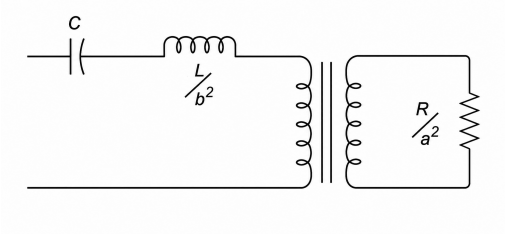
Q2.15.3 MCQ 2015 2M Ans: B ●●○

Using impedance reflection and resonance condition.
If the load resistance is referred to the second winding,



$$R' = \frac{R}{a^2}$$

Referring inductance to the primary side,



$$L' = \frac{L}{b^2}$$

$$X_L = X_C$$

$$\frac{\omega L}{b^2} = \frac{1}{\omega C}$$

$$\frac{5000 \times 10^{-3}}{b^2} = \frac{1}{5000 \times 10^{-5}}$$

$$\frac{5}{b^2} = 20$$

$$b = 0.5$$

$$Z_{in} = \frac{R}{a^2 b^2}$$

$$\frac{2.5}{a^2 b^2} = 2.5$$

$$a = 2$$

$$\boxed{B}$$

Key Points

- Resistance referred across transformer:
$$R' = \frac{R}{n^2}$$
- Inductance referred across transformer:
$$L' = \frac{L}{n^2}$$
- For purely resistive input:
$$X_L = X_C$$

Q2.15.4 MCQ | 2015 | 2M | Ans: A | ● ○ ○

$$\bar{I}_1 = \frac{N_2}{N_1}\bar{I}_2 + \frac{N_3}{N_1}\bar{I}_3$$

$$\bar{I}_1 = \frac{50}{100} \times 2\angle 30^\circ + \frac{50}{100} \times 2\angle 150^\circ$$

$$\bar{I}_1 = 1\angle 30^\circ + 1\angle 150^\circ$$

$$\bar{I}_1 = 1\angle 90^\circ \text{ A}$$

Therefore,

$$\bar{I}_1 = 1\angle 90^\circ \text{ A}$$

$$\boxed{A}$$

Q2.15.5 NAT | 2015 | 2M | Ans: 30 | ● ● ●

For the first connection, both HV and LV windings are connected in delta with identical polarity orientation.

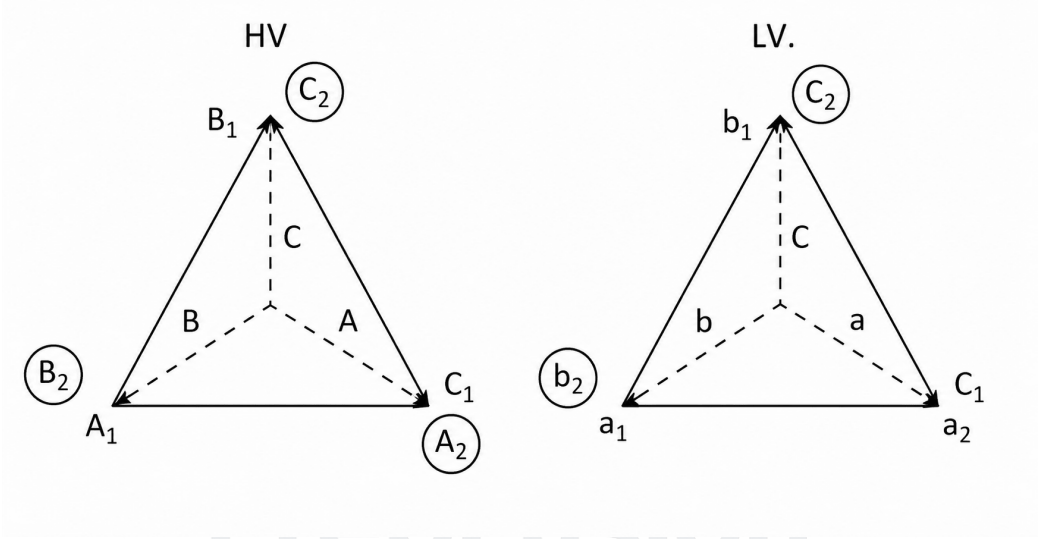


Fig. Solution Q2.15.5

The phase displacement between primary and secondary line voltages is

$$\theta = 0^\circ$$

Hence, the vector group is

Dd0

For the second connection, the LV side leads the corresponding primary line voltage phasor by 30°, which corresponds to clock number 11. Hence, the vector group is Dd11.

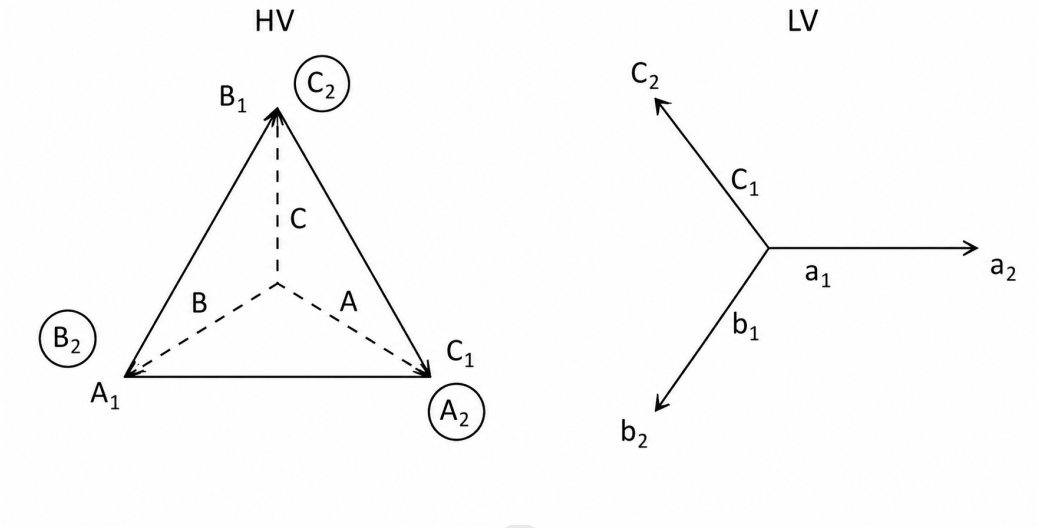


Fig. Solution Q2.15.5

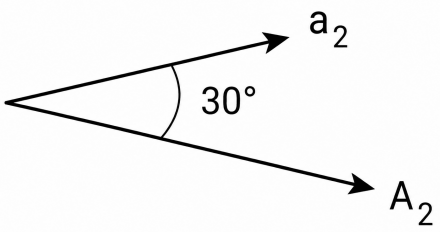


Fig. Solution Q2.15.5

From the phasor construction,

$$\angle V_2 - \angle V_1 = 30^\circ$$

Therefore,

V_2 leads V_1 by 30°

30

Key Points

- Vector group notation specifies winding connection and phase displacement between HV and LV line voltages.
- Clock notation is measured clockwise from the HV phasor to the LV phasor.
- Each clock number represents a phase shift of 30°.
- Dd0 corresponds to zero phase displacement.
- Dd11 corresponds to the LV side leading the HV side by 30°.

Mistakes to be avoided

- Do not confuse phase-voltage displacement with line-voltage displacement.
- A leading phase shift of 30° corresponds to clock number 11, not 1.

- Always verify winding polarities before constructing the phasor diagram.
- Ensure all phasors are compared using the same reference direction.

Q2.15.6 MCQ 2015 2M Ans: A ● ○ ○

The phase current in a delta connection leads the corresponding line current by:

$$30^\circ$$

Hence, per-phase current on the delta connected secondary side is

$$I_{\text{ph,s}} = \frac{I_{L,s}}{\sqrt{3}} \angle 30^\circ$$

$$I_{\text{ph,s}} = \frac{100}{\sqrt{3}} \angle 30^\circ \text{ A}$$

Turns ratio of transformer is

$$a = \frac{V_{\text{ph,p}}}{V_{\text{ph,s}}}$$

Primary is star connected and secondary is delta connected.

$$a = \frac{230/\sqrt{3}}{115} = \frac{2}{\sqrt{3}}$$

For an ideal transformer,

$$\frac{I_{\text{ph,s}}}{I_{\text{ph,p}}} = a$$

$$I_{\text{ph,p}} = \frac{I_{\text{ph,s}}}{a}$$

Substituting values,

$$I_{\text{ph,p}} = \frac{\frac{100}{\sqrt{3}} \angle 30^\circ}{\frac{2}{\sqrt{3}}} = 50 \angle 30^\circ \text{ A}$$

Since primary is star connected,

$$I_{L,p} = I_{\text{ph,p}} = 50 \angle 30^\circ \text{ A}$$

$$\boxed{A}$$

Q2.15.7 MCQ 2015 1M Ans: C ● ○ ○

The primary mmf is least affected by secondary terminal conditions in a current transformer (CT).

In a current transformer:

$$N_p I_p \approx N_s I_s$$

The primary current is dictated by the power circuit and remains nearly independent of the secondary burden.

Thus, variations in secondary load produce very small changes in the primary current and hence primary mmf remains almost unchanged.

Compare transformer types:

- Power transformer: Primary current changes with load.
- Distribution transformer: Primary current changes with load.
- Potential transformer (PT): Primary current depends on secondary burden.
- Current transformer (CT): Primary current is fixed by the line current.

Therefore,

$\boxed{C}$

Q2.15.8 MCQ 2015 1M Ans: A

Method 1

Using the T-equivalent model of coupled inductors.
The T-equivalent circuit of the transformer is shown below.

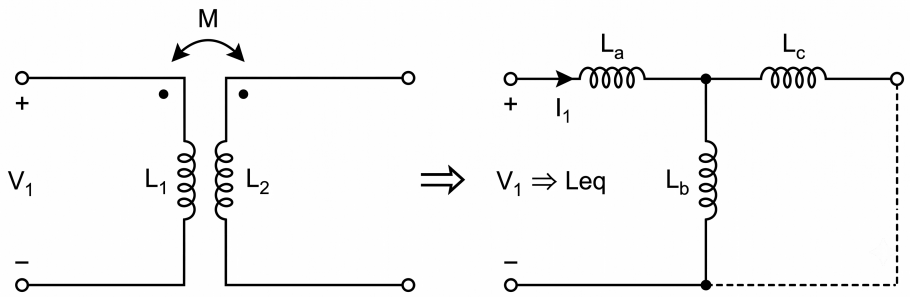


Fig. Solution Q2.15.8

The equivalent inductances are:

$$L_a = L_1 - M$$

$$L_c = L_2 - M$$

$$L_b = M$$

Substituting the given values,

$$L_a = 800 - 480 = 320\,\text{mH}$$

$$L_c = 600 - 480 = 120\,\text{mH}$$

$$L_b = 480\,\text{mH}$$

The equivalent inductance seen at the input is

$$L_{eq} = L_a + (L_b \parallel L_c)$$

$$L_{eq} = 320 + \frac{480 \times 120}{480 + 120}$$

$$L_{eq} = 320 + 96$$

$$L_{eq} = 416\,\text{mH}$$

Hence,

$$\boxed{A}$$

Method 2

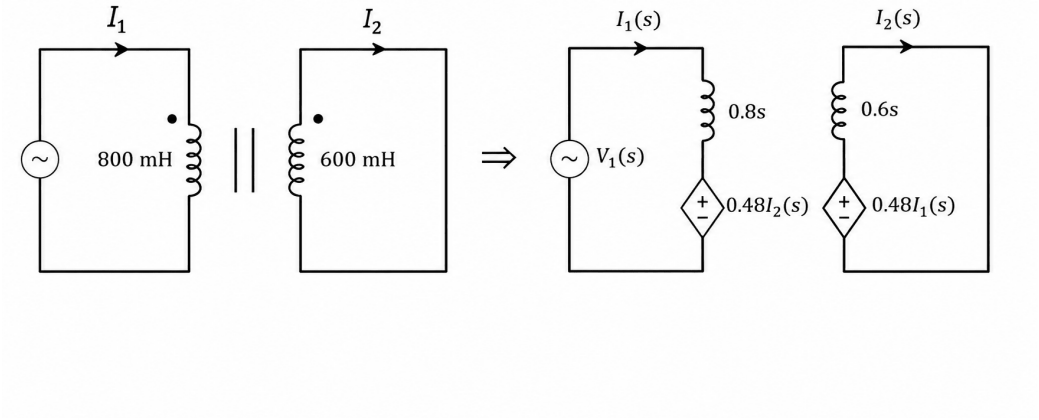


Fig. Solution Q2.15.8

PrepFusion

Since the secondary is short circuited, assuming dot polarities and current directions as shown

$$I_2(s) = \frac{0.48\,s\,I_1(s)}{0.6\,s} = 0.8\,I_1(s)$$

$$V_1(s) = 0.8\,s\,I_1(s) - 0.48\,s\,I_2(s)$$

$$= 0.8\,s\,I_1(s) - 0.48\,s\,(0.8\,I_1(s))$$

$$Z_{eq} = \frac{V_1(s)}{I_1(s)} = 0.8\,s - 0.48 \times 0.8\,s$$

$$= 0.52 \times 0.8\,s = 0.416\,s$$

$$L_{eq} = \frac{Z_{eq}}{s} = 0.416\,\text{H} = 416\,\text{mH}$$

Hence,

$$\boxed{A}$$

Key Points

- T-equivalent parameters of coupled coils:

$$L_a = L_1 - M$$

$$L_b = M$$

$$L_c = L_2 - M$$

- Input equivalent inductance:

$$L_{eq} = L_a + (L_b \parallel L_c)$$

- Mutual inductance contributes as the shunt arm of the T-network.

Mistakes to be avoided

- Do not directly add self-inductances.
- Remember that the mutual inductance branch is the middle branch in the T-equivalent representation.
- Use

$$L_b \parallel L_c = \frac{L_b L_c}{L_b + L_c}$$

instead of direct addition.

Q2.14.1 MCQ | 2014 | 2M | Ans: A | ● ○ ○

Core loss consists of hysteresis and eddy current losses.

$$P_h = K_h B_m^2 f$$

$$P_e = K_e B_m^2 f^2$$

Since,

$$B_m \propto \frac{V}{f}$$

For the two operating conditions,

$$\frac{V_1}{f_1} = \frac{230}{50} = \frac{138}{30} = \frac{V_2}{f_2}$$

Hence,

$$B_m = \text{constant}$$

Therefore,

$$P_h \propto f$$

and

$$P_e \propto f^2$$

At

$$V = 230\,\text{V}, \qquad f = 50\,\text{Hz}$$

Total core loss is

$$P_h + P_e = 1050 \qquad \cdots (i)$$

At

$$V = 138\,\text{V}, \qquad f = 30\,\text{Hz}$$

Total core loss is

$$P'_h + P'_e = 500 \qquad \cdots (ii)$$

Using proportional relations,

$$\frac{P'_h}{P_h} = \frac{30}{50} = 0.6$$

$$\frac{P'_e}{P_e} = \left(\frac{30}{50}\right)^2 = 0.36$$

Substituting in (ii),

$$0.6P_h + 0.36P_e = 500 \qquad \cdots (iii)$$

Solving (i) and (iii),

$$P_h = 508\,\text{W}$$

$$P_e = 542\,\text{W}$$

Hence,

$$\boxed{A}$$

Q2.14.2 NAT | 2014 | 1M | Ans: 550 | ●○○

For series additive polarity,

$$V_H = 1000 + 100 = 1100\,\text{V}$$

The voltage ratio is

$$K = \frac{\text{low voltage}}{\text{high voltage}} = \frac{1000}{1100}$$

Autotransformer kVA rating:

$$(\text{kVA})_{\text{auto}} = \frac{1}{1-K}(\text{kVA})_{\text{2-wdg}}$$

$$(\text{kVA})_{\text{auto}} = \frac{1}{1 - \frac{1000}{1100}} \times 50 = 550\,\text{kVA}$$

$$\boxed{550}$$

Q2.14.3 MCQ | 2014 | 2M | Ans: B ●●○

Core loss is the sum of hysteresis loss and eddy current loss.

$$P_c = P_h + P_e$$

Hysteresis and eddy current losses are given by

$$P_h = K_h (B_m)^2 f$$

$$P_e = K_e (B_m)^2 f^2$$

Since

$$B_m \propto \frac{V}{f}$$

Given that

$$\frac{V}{f}$$

is constant, therefore

$$B_m = \text{Constant}$$

Hence,

$$P_h = K_h' f$$

and

$$P_e = K_e' f^2$$

Therefore, core loss becomes

$$P_c = K_h' f + K_e' f^2$$

Dividing throughout by f,

$$\frac{P_c}{f} = K_h' + K_e' f$$

The intercept of the

$$\frac{P_c}{f} \text{ versus } f$$

curve gives K_h' and the slope gives K_e'.
From the graph,

$$K_h' = 10\,\text{W/Hz}$$

$$K_e' = \frac{15 - 10}{50} = \frac{1}{10}\ \text{W/Hz}^2$$

At

$$f = 25\ \text{Hz}$$

Hysteresis loss:

$$P_h = K_h' f$$

$$P_h = 10 \times 25$$

$$P_h = 250\ \text{W}$$

Eddy current loss:

$$P_e = K_e' f^2$$

$$P_e = \frac{1}{10}(25)^2$$

$$P_e = 62.5\ \text{W}$$

Hence,

$$P_h > P_e$$

Therefore, the correct option is

$$\boxed{B}$$

Q2.14.4 MCQ | 2014 | 2M | Ans: B

Given,

$$L_1 = 45\ \text{mH}, \qquad L_2 = 30\ \text{mH}, \qquad M = 20\ \text{mH}$$

Turns ratio:

$$a = \frac{N_1}{N_2} = \frac{220}{110} = 2$$

Magnetizing inductance referred to primary:

$$L_m = aM = 2 \times 20 = 40\ \text{mH}$$

Primary leakage inductance:

$$L_{l1} = L_1 - L_m = 45 - 40 = 5\ \text{mH}$$

Secondary leakage inductance referred to primary:

$$L_{l2} = a^2 L_2 - L_m = (2)^2(30) - 40 = 80\ \text{mH}$$

Therefore,

$$(L_{l1}, L_{l2}, L_m) = (5\,\text{mH}, 80\,\text{mH}, 40\,\text{mH})$$

Hence, the correct option is

$$\boxed{B}$$

Key Points

- Magnetizing inductance referred to primary:
$$L_m = aM$$
- Secondary self-inductance referred to primary:
$$L_2' = a^2 L_2$$
- Leakage inductance is obtained by subtracting magnetizing inductance from self-inductance.

Mistakes to be avoided

- Do not take M directly as L_m when referring to primary.
- Do not forget to refer secondary inductance to primary before finding L_{l2}.
- Use $a = \frac{N_1}{N_2} = \frac{220}{110} = 2$.

Q2.14.5 NAT 2014 2M Ans: 23.5 to 23.7 ● ● ○

The secondary load absorbs

$$Q_L = P\tan(\cos^{-1}0.89) = 4000\tan(\cos^{-1}0.89) \approx 2049.26\,\text{var}.$$

For the ideal 2 : 1 transformer, the primary winding voltage is 220 V and the reactive power is unchanged. The series 1 Ω resistance absorbs only real power, so unity input power factor requires the shunt branch to supply $Q_C = Q_L$.

$$|X_C| = \frac{220^2}{Q_C} \approx 23.618\,\Omega.$$

The branch must be capacitive, with impedance $-j23.618\,\Omega$; a positive inductive reactance would increase the lagging reactive power.

$$\boxed{|X_C| \approx 23.62\,\Omega}$$

Q2.14.6 MCQ 2014 1M Ans: A ● ○ ○

Hysteresis loss and eddy current loss are given by

$$P_h = K_h(B_m)^2 f$$

$$P_e = K_e(B_m)^2 f^2$$

Also,

$$B_m \propto \left(\frac{V}{f}\right)$$

Since voltage and frequency are increased by the same amount,

$$\frac{V}{f} = \text{Constant}$$

Hence,

$$B_m = \text{Constant}$$

Therefore,

$$P_h \propto f$$

and

$$P_e \propto f^2$$

Frequency increase:

$$f_2 = 1.1 f_1$$

For hysteresis loss,

$$\frac{P_{h2}}{P_{h1}} = \frac{f_2}{f_1} = 1.1$$

Therefore,

$$P_{h2} = 1.1 P_{h1}$$

Percentage change:

$$\% \text{ change} = \frac{P_{h2} - P_{h1}}{P_{h1}} \times 100$$

$$= \frac{1.1P_{h1} - P_{h1}}{P_{h1}} \times 100 = 10\%$$

For eddy current loss,

$$\frac{P_{e2}}{P_{e1}} = \left(\frac{f_2}{f_1}\right)^2$$

$$= (1.1)^2 = 1.21$$

Therefore,

$$P_{e2} = 1.21 P_{e1}$$

Percentage change:

$$\% \text{ change} = \frac{P_{e2} - P_{e1}}{P_{e1}} \times 100$$

$$= \frac{1.21P_{e1} - P_{e1}}{P_{e1}} \times 100 = 21\%$$

Hence hysteresis loss changes by 10% and eddy current loss changes by 21%.

$\boxed{A}$

Q2.14.7 MCQ | 2014 | 1M | Ans: A | ● ○ ○

Using transformer voltage and impedance transformation relations.

Voltage transformation relation:

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

Given,

$$\frac{N_s}{N_p} = \frac{2}{1}$$

Primary voltage:

$$V_p = \sin \omega t$$

Therefore,

$$V_s = \frac{2}{1} \sin \omega t$$

$$V_s = 2 \sin \omega t$$

Impedance transformation relation:

$$Z_s = \left(\frac{N_s}{N_p}\right)^2 Z_p$$

Substituting values,

$$Z_s = \left(\frac{2}{1}\right)^2 (1)$$

$$Z_s = 4 \times 1$$

$$Z_s = 4\,\Omega$$

Hence the correct option is

$$\boxed{A}$$

Key Points

- Voltage ratio equals turns ratio:

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

- Impedance transforms as square of turns ratio:

$$Z_s = \left(\frac{N_s}{N_p}\right)^2 Z_p$$

- Current transforms inversely with turns ratio:

$$\frac{I_s}{I_p} = \frac{N_p}{N_s}$$

Mistakes to be avoided

- Do not use linear turns ratio for impedance transformation.
- Voltage ratio and impedance ratio are different relations.
- Remember impedance varies as square of turns ratio.

Q2.14.8 MCQ 2014 1M Ans: C ● ● ●

The induced emf equation of a transformer is:

$$E = 4.44 f N \phi$$

When voltage and frequency are constant,

$$\phi \propto \frac{1}{N}$$

Flux in a magnetic circuit is:

$$\phi = \frac{N I_\mu}{\mathcal{R}}$$

Since,

$$\mathcal{R} = \frac{l}{\mu A}$$

therefore,

$$\phi = \frac{N I_\mu}{\frac{l}{\mu A}}$$

$$\phi = \frac{\mu N I_\mu A}{l}$$

Since,

$$\phi \propto \frac{1}{N}$$

then,

$$\frac{\mu N I_\mu A}{l} \propto \frac{1}{N}$$

Hence,

$$\frac{\mu N^2 I_\mu A}{l} = \text{constant}$$

For constant no-load (magnetizing) current,

$$N^2 A = \text{constant}$$

Therefore,

$$N^2 \propto \frac{1}{A}$$

For circular cross-section,

$$A = \pi r^2$$

Hence,

$$\left(\frac{N_2}{N_1}\right)^2 = \frac{\pi r_1^2}{\pi r_2^2}$$

$$= \left(\frac{r_1}{r_2}\right)^2$$

Given,

$$r_2 = \frac{r_1}{2}$$

Substituting,

$$\left(\frac{N_2}{N_1}\right)^2 = 4$$

$$\frac{N_2}{N_1} = 2$$

Hence, the number of turns becomes double.

$$\boxed{C}$$

Key Points

- Transformer emf equation:

$$E = 4.44 f N \phi$$

- Magnetic reluctance:

$$\mathcal{R} = \frac{l}{\mu A}$$

- For constant magnetizing current:

$$N^2 A = \text{constant}$$

Mistakes to be avoided

- Do not assume turns are directly proportional to area.
- Remember area for circular core:

$$A = \pi r^2$$

- Radius reduction affects area quadratically.

Q2.13.1 MCQ | 2013 | 1M | Ans: C | ● ○ ○

No-load loss corresponds to constant loss.

$$P_i = 64\,\text{W}$$

For short circuit test,

$$I = 0.9I_{fl}$$

Copper loss is

$$P_{cu} = I^2R = 81\,\text{W}$$

Substituting current value,

$$(0.9I_{fl})^2R = 81$$

$$0.81I_{fl}^2R = 81$$

$$I_{fl}^2R = 100$$

Hence, full-load copper loss is

$$P_{cu,FL} = 100\,\text{W}$$

For maximum efficiency,

$$x = \sqrt{\frac{P_i}{P_{cu,FL}}}$$

Substituting values,

$$x = \sqrt{\frac{64}{100}}$$

$$x = 0.8$$

Thus maximum efficiency occurs at

$$0.8 \times 100 = 80\%$$

of rated current.

$$\boxed{C}$$

Q2.13.2 MCQ | 2013 | 2M | Ans: B | ● ○ ○

When 100 V is applied at primary,

$$V_{wx1} = 100\,\text{V}$$

Secondary voltage is

$$V_{\text{sec}} = \frac{1.25}{1} \times 100$$

$$V_{sec} = 125\ \text{V}$$

Due to potential divider (attenuator),

$$V_{yz1} = 125 \times 0.8$$

$$V_{yz1} = 100\ \text{V}$$

Hence,

$$\frac{V_{wx1}}{V_{yz1}} = \frac{100}{100} = 1$$

When 100 V is applied at YZ,

$$V_{yz2} = 100\ \text{V}$$

Since primary is open-circuited, no current flows in primary and secondary. Therefore, the attenuator has no effect.

$$V_{sec} = 100\ \text{V}$$

Primary voltage becomes

$$V_{pri} = V_{wx2} = \frac{100}{1.25}$$

$$V_{wx2} = 80\ \text{V}$$

Hence,

$$\frac{V_{wx2}}{V_{yz2}} = \frac{80}{100} = 0.8$$

Thus, option (b) is correct.

B

Q2.12.1 MCQ | 2012 | 2M | Ans: B | ● ○ ○

For constant flux density,

$$B_m = \text{constant}$$

Voltage rating is proportional to core cross-sectional area:

$$V \propto A$$

Since,

$$A \propto (\text{dimension})^2$$

Therefore,

$$V \propto (\text{dimension})^2$$

Hence,

$$\frac{V_2}{V_1} = \frac{(\text{dimension})_2^2}{(\text{dimension})_1^2}$$

Substituting,

$$\frac{V_2}{V_1} = \left(\frac{\sqrt{2}}{1}\right)^2$$

$$V_2 = 2V_1$$

$$V_2 = 2 \times 1 = 2\,\text{kV}$$

No-load current varies directly with dimension:

$$I_\mu \propto \text{dimension}$$

Hence,

$$\frac{I_{\mu 2}}{I_{\mu 1}} = \frac{\text{dimension}_2}{\text{dimension}_1}$$

$$I_{\mu 2} = \sqrt{2} \times 0.5$$

$$I_{\mu 2} = 0.707\,\text{A}$$

Core loss is proportional to core volume:

$$P_c \propto (\text{dimension})^3$$

Thus,

$$\frac{P_{c2}}{P_{c1}} = \left(\frac{\text{dimension}_2}{\text{dimension}_1}\right)^3$$

$$= (\sqrt{2})^3 = 2\sqrt{2}$$

Hence,

$$P_{c2} = 55 \times 2\sqrt{2}$$

$$P_{c2} = 155.6\,\text{W}$$

$$\boxed{B}$$

Key Points

- For constant flux density:

$$V \propto A$$

- Core area:

$$A \propto (\text{dimension})^2$$

- No-load current:

$$I_\mu \propto \text{dimension}$$

- Core loss:

$$P_c \propto \text{Core Volume} \propto (\text{dimension})^3$$

Q2.11.1 MCQ | 2011 | 1M | Ans: C ● ○ ○

Since the supply voltage is sinusoidal, the core flux is also sinusoidal. The waveform of the magnetizing current then depends only on the B-H (flux-current) relation of the core.

$$\phi \propto \int v\, dt$$

For sinusoidal voltage,

$$v = V_m \sin \omega t$$

Therefore,

$$\phi \propto -\frac{V_m}{\omega} \cos \omega t$$

Thus, the flux is also sinusoidal.

For an air core, $\mu = \mu_0$ is constant: the ϕ-i relation is a straight line, with no saturation and no hysteresis. Hence

$$i_m \propto \phi$$

so the magnetizing current is also a pure sinusoid (in phase with the flux, lagging the voltage by 90°). The peaky or flat-topped waveforms in the other options occur only in iron cores because of saturation and hysteresis.

C

Q2.10.1 MCQ | 2010 | 2M | Ans: D ● ● ○

Method 1

Converting the star connected load to delta,

$$Z_\Delta = 3Z_Y$$

$$Z_\Delta = 3 \times 4$$

$$Z_\Delta = 12\,\Omega$$

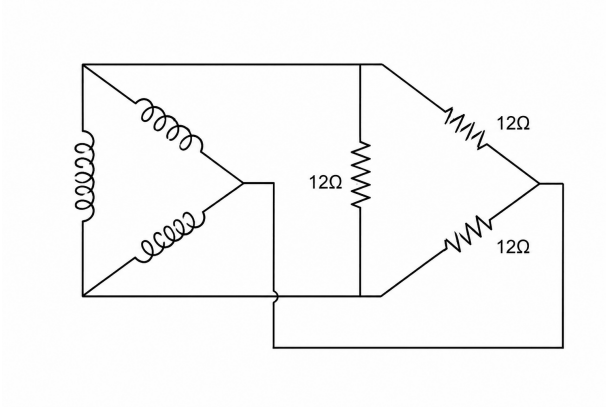


Fig. Solution Q2.10.1

Hence, the secondary side becomes a balanced delta load with impedance per phase

$$Z_s = 12\,\Omega$$

Referring the impedance to primary side,

$$Z_p = \frac{Z_s}{\left(\frac{N_s}{N_p}\right)^2}$$

Since turns ratio can be expressed in terms of phase voltages,

$$Z_p = \frac{Z_s}{\left(\frac{V_{ph,s}}{V_{ph,p}}\right)^2}$$

Substituting values,

$$Z_p = \frac{12}{\left(\frac{220}{110/\sqrt{3}}\right)^2}$$

$$Z_p = \frac{12}{\left(2\sqrt{3}\right)^2}$$

$$Z_p = \frac{12}{12} = 1\,\Omega$$

Therefore,

$$Z_p = (1 + j0)\,\Omega$$

Since primary is star connected, the equivalent load impedance is also star connected.

Method 2

Using power balance on both sides,

$$3\frac{(V_{ph,\Delta})^2}{4} = 3\frac{(V_{ph,Y})^2}{Z}$$

Substituting values,

$$3\frac{(220)^2}{3\times 4} = 3\frac{(110)^2}{3\times Z}$$

$$Z[22\times 22\times 100] = 4[11\times 11\times 100]$$

$$Z = 1\,\Omega$$

$$Z = (1 + j0)\,\Omega$$

Therefore, option D is correct

D

Q2.10.2 MCQ 2010 1M Ans: C ● ○ ○

Assume

$$\overline{V}_p = V_p\angle 0^\circ$$

For the ideal transformer,

$$\overline{V}_s = 2\overline{V}_p = 2V_p\angle 0^\circ$$

Since the load is purely resistive,

$$\overline{I}_L = 1\angle 0^\circ\ \text{A}$$

Using the transformer current ratio,

$$\overline{I}'_p = 2\overline{I}_L = 2\angle 0^\circ\ \text{A}$$

The magnetizing current lags the applied voltage by 90°,

$$\overline{I}_\mu = 1\angle(-90^\circ)\ \text{A}$$

Therefore, the total primary current is

$$\overline{I}_p = \overline{I}'_p + \overline{I}_\mu$$

$$\overline{I}_p = 2\angle 0^\circ + 1\angle(-90^\circ)$$

$$\overline{I}_p = (2 - j1)\ \text{A}$$

$$|\overline{I}_p| = \sqrt{2^2 + 1^2} = 2.236\ \text{A}$$

$$\angle\overline{I}_p = -\tan^{-1}\left(\frac{1}{2}\right) = -26.56^\circ$$

$$\overline{I}_p = 2.236\angle(-26.56^\circ)\ \text{A}$$

Hence, the correct option is

C

Key Points

- Magnetizing current always lags the applied voltage by approximately 90°.
- Total primary current is the phasor sum of reflected load current and magnetizing current.

Q2.09.1 MCQ 2009 1M Ans: B

The mutual inductance of a transformer is

$$M = \frac{N_1 N_2}{\mathcal{R}}$$

where $\mathcal{R}$ is the total reluctance of the magnetic circuit.

Initially, the windings are placed on the two vertical limbs. If they are shifted to the two opposite horizontal limbs, the magnetic flux still completes the same closed iron-core path.

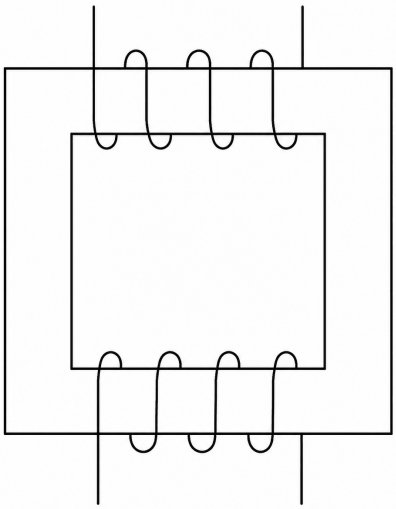


Fig. Solution Q2.09.1

The total reluctance of the magnetic circuit remains

$$\mathcal{R} = 2\mathcal{R}_v + 2\mathcal{R}_h$$

where

$$\mathcal{R}_v = \frac{l_v}{\mu A_v}, \qquad \mathcal{R}_h = \frac{l_h}{\mu A_h}.$$

Although the vertical limbs have a cross-sectional area of $20\,\text{cm}^2$ and the horizontal limbs have $10\,\text{cm}^2$, the magnetic flux still traverses all four limbs in series. Hence, the total reluctance of the magnetic circuit is unchanged.

Therefore,

$$M = \frac{N_1 N_2}{\mathcal{R}}$$

also remains unchanged.

$$\boxed{B}$$

Key Points

- Mutual inductance is inversely proportional to the total magnetic reluctance.
- Shifting the windings to another pair of opposite limbs does not change the magnetic path.
- The entire magnetic circuit determines the reluctance, not merely the limb on which the winding is placed.

Mistakes to be avoided

- Do not assume that mutual inductance is proportional only to the cross-sectional area of the limb carrying the winding.
- Do not calculate reluctance using a single limb; the complete magnetic path must be considered.
- Do not conclude that the mutual inductance halves because the horizontal limbs have half the cross-sectional area.

Q2.09.2 MCQ 2009 2M Ans: B ●●○

With both S_1 and S_2 open, the neutral paths are open. Hence, the third-harmonic magnetizing current cannot flow.

For a transformer core, if the third-harmonic component of magnetizing current is suppressed, the core flux becomes distorted and contains a third-harmonic component.

This distortion makes the flux waveform flat-topped.

$$\therefore \text{ Core flux waveform is flat-topped with third harmonic.}$$

B

Q2.09.3 MCQ 2009 2M Ans: C ●●○

When S_2 is closed and no load is connected, the secondary winding forms a closed delta.

Since no external load is connected, the fundamental component of secondary current is zero.

$$I_{2,\text{fundamental}} = 0$$

However, third-harmonic currents can circulate in the closed delta path.

$$I_2 = I_3 + I_9 + I_{15} + \cdots$$

Hence, the secondary current contains only third-harmonic components.

C

Key Points

- A closed delta provides a path for third-harmonic circulating currents.
- With no external load, the fundamental component of secondary current is zero.
- Triplen harmonics can circulate within the closed delta loop.

Mistakes to be avoided

- Do not assume load current flows when no load is connected.
- Do not ignore the closed delta path for third-harmonic currents.

Q2.09.4 MCQ | 2009 | 2M | Ans: D ●●○

The circuit referred to primary looks like as shown below.

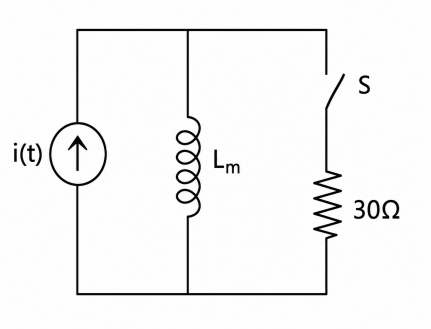


Fig. Solution Q2.09.4

If S is open, entire current flows into the magnetizing inductance.

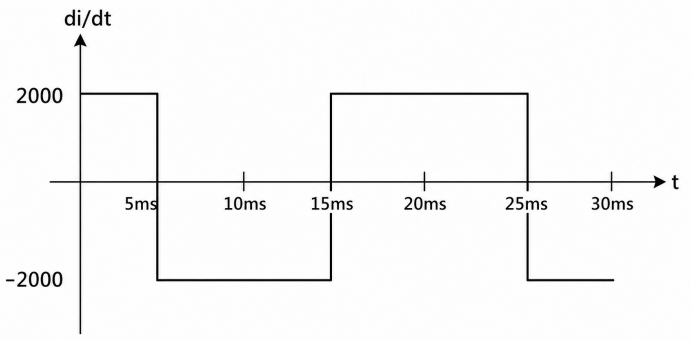


Fig. Solution Q2.09.4

From the waveform,

$$\left(\frac{di}{dt}\right)_{\max} = 2000$$

Voltage across magnetizing inductance is

$$V = L_m \frac{di}{dt}$$

Given,

$$L_m = \frac{400}{\pi}\text{ mH} = \frac{400}{\pi} \times 10^{-3}\text{ H}$$

Therefore,

$$V_p = \left(\frac{400}{\pi} \times 10^{-3}\right)(2000) = \frac{800}{\pi}\text{ V}$$

Hence, right option is

$$\boxed{D}$$

Key Points

- For an inductor, $v = L\dfrac{di}{dt}$.
- With S open, the full source current flows through L_m.
- Peak voltage occurs at maximum slope of $i(t)$.

Mistakes to be avoided

- Do not forget to convert mH into H.
- Do not use current magnitude; use slope $\frac{di}{dt}$.

Q2.09.5 MCQ | 2009 | 2M | Ans: B

Now when S is closed and sinusoidal current is applied, inductance is replaced by reactance.

$$X_L = \omega L$$

$$X_L = \frac{400}{\pi} \times 10^{-3} \times 100\pi = 40\,\Omega$$

Equivalent circuit is shown below.

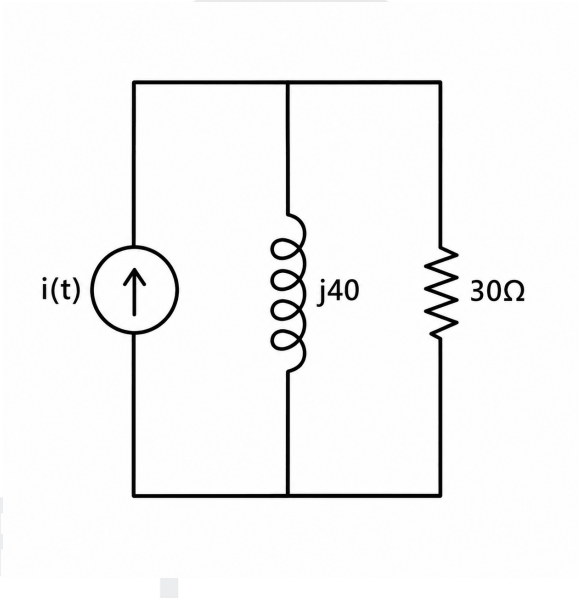


Fig. Solution Q2.09.5

Current through resistance branch is

$$I_R = I\left(\frac{jX_L}{R + jX_L}\right)$$

Substituting values,

$$I_R = 10\angle 0^\circ \left(\frac{j40}{30 + j40}\right)$$

$$I_R = \frac{10 \times 40}{50}\angle(90^\circ - 53.13^\circ)$$

$$I_R = 8\angle 36.87^\circ \text{ A}$$

The peak voltage across A and B is the voltage across the 30 Ω resistance (the current source is 10 A peak, so I_R is also a peak value):

$$V_{AB} = |I_R|\,R = 8 \times 30 = 240\,\text{V}$$

$\boxed{B}$

Voltage across resistance is

$$V_R = I_R R$$

$$V_R = 8\angle 36.87^\circ \times 30$$

$$V_R = 240\angle 36.87^\circ \text{ V}$$

Peak voltage is

$$240 \text{ V}$$

Hence, right option is

$\boxed{B}$

Q2.08.1 MCQ 2008 1M Ans: D ● ○ ○

In open circuit test, no-load current is drawn by the transformer which is 5%–8% of the full-load current.

Due to high percentage of magnetizing current in no-load current, the no-load power factor is low. So, W_2 will be used for no-load test.

In short circuit test, high current is drawn by the transformer and it has higher power factor as compared to no-load test. So, W_3 will be used for short-circuit test. Hence, right option is

$\boxed{D}$

Key Points

- Open circuit test has low current and low power factor.
- Short circuit test has high current and comparatively higher power factor.
- Wattmeter selection depends on current rating and power factor condition.

Mistakes to be avoided

- Do not use the same wattmeter for both tests without checking current and power factor conditions.
- Do not confuse no-load current with full-load current.

Q2.08.2 MCQ 2008 2M Ans: A ● ● ○

The polarity of induced emf is determined based on Lenz's law. Current must be induced in secondary in such a way that it opposes primary flux.

Hence secondary flux must be anti-clockwise and current must come out of terminal R.

Since secondary delivers power to the load, current must leave the positive terminal. Hence, R is a positive terminal and the polarities are correctly marked.

For $0 < t < 1$

$$\frac{d\phi}{dt} = \frac{0.12}{1} = 0.12 \text{ Wb/s}$$

$$e_{rs} = N\frac{d\phi}{dt}$$

$$e_{rs} = 200 \times 0.12 = 24 \text{ V}$$

For $1 < t < 2$

$$\frac{d\phi}{dt} = 0$$

$$e_{rs} = 0$$

For $2 < t < 2.5$

$$\frac{d\phi}{dt} = \frac{-0.12}{0.5} = -0.24\ \text{Wb/s}$$

$$e_{rs} = 200 \times (-0.24) = -48\ \text{V}$$

Hence, right option is

$$\boxed{\text{A}}$$

Q2.08.3 MCQ | 2008 | 1M | Ans: B

The phasor diagram for the transformer connection can be drawn as follows.

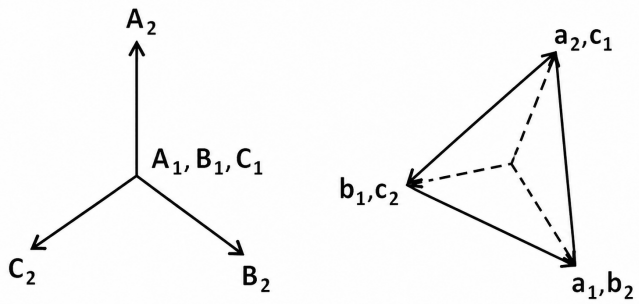


Fig. Solution Q2.08.3

From the phasor diagram,
Primary reference phasor:(High voltage)

$$A_1,\ B_1,\ C_1$$

Secondary phasor:(Low voltage)

$$a_2,\ b_2,\ c_2$$

The secondary phasor lags the primary phasor by

$$30^\circ$$

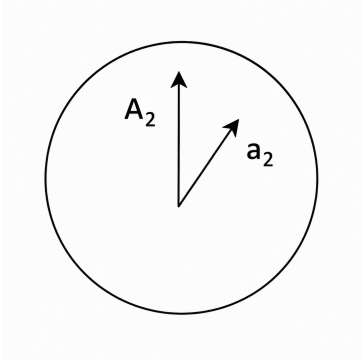


Fig. Solution Q2.08.3

Hence, the transformer clock notation becomes

$$Yd1$$

Hence, right option is

$$\boxed{B}$$

Q2.07.1 MCQ 2007 2M Ans: D

The negative terminal of capacitor is connected to dot.

So, if we refer both secondary quantities to primary side, we get the equivalent circuit shown below.

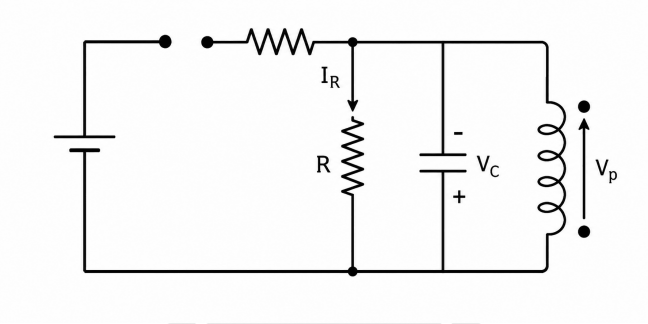


Fig. Solution Q2.07.1

At $t = 0^+$,

Since turns ratio is unity,

$$\frac{N_1}{N_2} = 1$$

Therefore, current and voltages remain same on both sides.

$$V_p = -V_c$$

$$V_p = -5\ \text{V}$$

Using Ohm's law,

$$I_R = \frac{V_p}{R}$$

$$I_R = \frac{-5}{10}$$

$$I_R = -0.5\ \text{A}$$

Hence, right option is

$$\boxed{D}$$

Q2.07.2 MCQ | 2007 | 2M | Ans: C

At full load and unity power factor,

$$P_{out} = 50\,\text{kW}$$

Efficiency is

$$\eta = \frac{P_{out}}{P_{out} + P_{loss}} \times 100$$

Substituting values,

$$95 = \frac{50}{50 + P_{loss}} \times 100$$

$$50 + P_{loss} = \frac{50}{0.95}$$

$$P_{loss} = 2.6315\,\text{kW}$$

When transformer is reconfigured as an autotransformer,

$$k = \frac{\text{LV}}{\text{HV}} = \frac{500}{750} = \frac{2}{3}$$

Rating of autotransformer:

$$\text{Rating} = \frac{\text{Rating of two winding transformer}}{1 - k}$$

$$= \frac{50}{1 - \frac{2}{3}}$$

$$= 150\,\text{kVA}$$

Efficiency of autotransformer:

$$\eta = \frac{(\text{kVA})\cos\phi}{(\text{kVA})\cos\phi + P_{loss}} \times 100$$

At full load and unity power factor,

$$\eta = \frac{150 \times 1}{150 \times 1 + 2.6315} \times 100$$

$$\eta = 98.275\%$$

Hence, right option is

$$\boxed{C}$$

Q2.07.3 MCQ | 2007 | 1M | Ans: C | ● ○ ○

The voltage regulation versus power factor angle curve for a transformer is shown below.

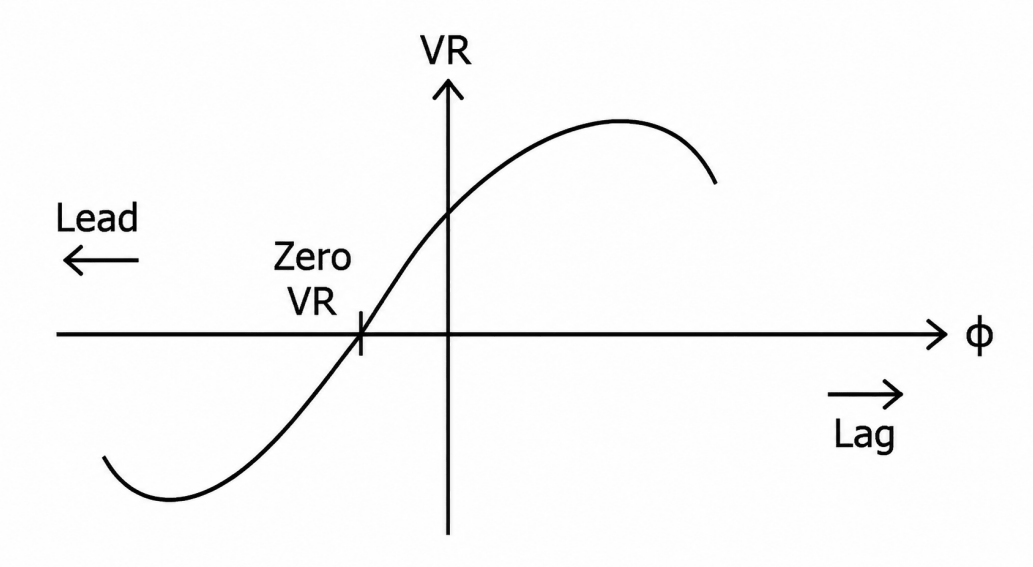


Fig. Solution Q2.07.3

For a transformer, approximate voltage regulation is

$$\%VR = \frac{I_2R_{eq}\cos\phi \pm I_2X_{eq}\sin\phi}{V_2} \times 100$$

For lagging power factor,

$$\%VR = \frac{I_2R_{eq}\cos\phi + I_2X_{eq}\sin\phi}{V_2} \times 100$$

For leading power factor,

$$\%VR = \frac{I_2R_{eq}\cos\phi - I_2X_{eq}\sin\phi}{V_2} \times 100$$

Zero voltage regulation is possible when

$$I_2R_{eq}\cos\phi = I_2X_{eq}\sin\phi$$

This condition occurs only at leading power factor of load. Hence, right option is

C

Key Points

- Voltage regulation can be positive, zero, or negative depending on load power factor.
- Lagging power factor generally gives positive voltage regulation.
- Leading power factor can give zero or negative voltage regulation.

Q2.06.1 MCQ 2006 1M Ans: D ● ○ ○

The transformer in open circuit (OC) test operates at no-load condition. Hence, only core (iron) losses are measured because copper losses are negligible.

No-load power factor is very low:

$$\cos \phi_0 = 0.2 \text{ to } 0.3$$

Therefore, power factor angle becomes:

$$\phi_0 = \cos^{-1}(0.2 \text{ to } 0.3)$$

$$\phi_0 \approx 70^\circ \text{ to } 80^\circ$$

Hence, current drawn in OC test is at a very low lagging power factor.

For short circuit (SC) test, applied voltage is small and core losses become negligible. Hence, copper losses are determined.

$$\text{OC test} \rightarrow \text{Core losses}$$

$$\text{SC test} \rightarrow \text{Copper losses}$$

Hence, option (d) is correct.

$$\boxed{D}$$

Q2.06.2 MCQ 2006 2M Ans: C ● ○ ○

The condition for proper load sharing between transformers in proportion to their kVA ratings is that the per-unit impedance on their own bases must be equal.

For proportional load sharing,

$$Z_{pu1} = Z_{pu2}$$

Given that the first transformer has

$$Z_{pu1} = 0.05\, pu$$

Hence the second transformer must also satisfy

$$Z_{pu2} = 0.05\, pu$$

$$\boxed{C}$$

Key Points

- Transformers connected in parallel share load according to their kVA ratings only when their per-unit impedances on their own bases are equal.
- Equal per-unit impedance prevents circulating currents and improper load division.
- Per-unit representation simplifies comparison of transformers having different ratings.

Mistakes to be avoided

- Do not compare actual impedances directly; compare per-unit impedances on respective bases.
- Do not confuse percentage impedance with actual impedance values.
- Ensure both transformers are referred to their own ratings before comparison.

Q2.06.3 MCQ | 2006 | 2M | Ans: C | ● ○ ○

The efficiency of a transformer is given by

$$\eta = \frac{x(\text{kVA})\cos\phi}{x(\text{kVA})\cos\phi + P_i + x^2 P_{Cu,fl}} \times 100\%$$

where x is the per-unit loading.
At full load,

$$x = 1$$

At half load,

$$x = 0.5$$

For full load condition,

$$0.95 = \frac{1 \times 300 \times 0.8}{1 \times 300 \times 0.8 + P_i + P_{Cu,fl}}$$

$$0.95 = \frac{240}{240 + P_i + P_{Cu,fl}}$$

$$P_i + P_{Cu,fl} = 12.631 \qquad (i)$$

For half load condition,

$$0.96 = \frac{0.5 \times 300 \times 1}{0.5 \times 300 \times 1 + P_i + 0.25 P_{Cu,fl}}$$

$$0.96 = \frac{150}{150 + P_i + 0.25 P_{Cu,fl}}$$

$$P_i + 0.25 P_{Cu,fl} = 6.25 \qquad (ii)$$

Solving equations (i) and (ii),

$$P_{Cu,fl} = 8.51\,\text{kW}$$

$$P_i = 4.123\,\text{kW}$$

$\boxed{C}$

Q2.06.4 MCQ 2006 2M Ans: B ● ○ ○

For maximum efficiency of a transformer,

$$x = \sqrt{\frac{P_i}{P_{Cu,fl}}}$$

Using $P_i = 4.123\,\text{kW}$ and $P_{Cu,fl} = 8.51\,\text{kW}$,

$$x = \sqrt{\frac{4.123}{8.51}} = 0.696$$

At maximum efficiency,

$$\eta_{\max} = \frac{x(\text{kVA})\cos\phi}{x(\text{kVA})\cos\phi + 2P_i} \times 100\%$$

$$\eta_{\max} = \frac{0.696 \times 300 \times 1}{0.696 \times 300 \times 1 + 2 \times 4.123} \times 100\%$$

$$\eta_{\max} = 96.2\%$$

$\boxed{B}$

Q2.05.1 MCQ 2005 1M Ans: B ● ○ ○

Out of the four types of transformer connections, the connections that fall into the 30° vector group are $Yd1$ and $Yd11$.

In $Yd1$,

HV leads LV by 30°

In $Yd11$,

HV lags LV by 30°

Other transformer connections do not belong to the 30° vector group and generally come under 0° and 180° vector groups.

$\boxed{B}$

Q2.05.2 MCQ 2005 1M Ans: D ● ○ ○

When we see the equivalent circuit of a transformer referred to the primary side, the primary leakage reactance and secondary leakage reactance are comparable in value.

But the magnetizing reactance has a very large value in comparison to leakage reactance because magnetizing reactance represents flux linkage in the core.

Primary leakage reactance = secondary leakage reactance

$$X_1 = X_2'$$

Also, the magnetizing reactance is much larger than leakage reactance,

$$X_m \gg X_1,\ X_2'$$

Hence,

$$X_1 = X_2' \ll X_m$$

$$\boxed{D}$$

Q2.04.1 MCQ | 2004 | 1M | Ans: B | ● ○ ○

For maximum efficiency, iron loss is equal to copper loss.

$$P_i = x^2 P_{Cu,FL}$$

where x is the fraction of full load.

$$300 = x^2 \times 600$$

$$x^2 = \frac{1}{2}$$

$$x = \frac{1}{\sqrt{2}} = 0.707$$

Hence, maximum efficiency occurs at 70.7% of full load.

$$x = 70.7\%$$

$$\boxed{B}$$

Key Points

- Maximum efficiency of a transformer occurs when copper loss equals iron loss.
- Copper loss varies as the square of load current.
- If x is the fraction of full load, copper loss at that load is $x^2 P_{Cu,FL}$.

Q2.04.2 MCQ | 2004 | 2M | Ans: D | ● ○ ○

In the given question, the transformer is connected in additive polarity as shown.

Input voltage is the same as the primary voltage of the two-winding transformer,

$$V_{in} = 3300\,\text{V}$$

Hence,

$$V_{out} = 3300 + 230$$

$$V_{out} = 3530\,\text{V}$$

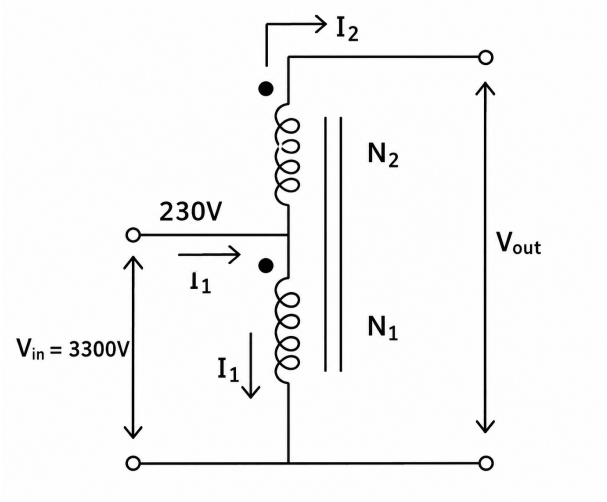


Fig. Solution Q2.04.2

Transformation ratio,

$$k = \frac{LV}{HV}$$

$$k = \frac{3300}{3530}$$

Rating of auto-transformer is,

$$S_{auto} = \frac{S_{2\text{-}wdg}}{1-k}$$

$$S_{auto} = \frac{50}{1-\frac{3300}{3530}}$$

$$S_{auto} = 767.4\,\text{kVA}$$

D

Q2.04.3 MCQ 2004 2M Ans: D ● ○ ○

Per-unit impedance is the same on both primary and secondary sides of the transformer.

$$Z_e(pu) = 0.02 + j0.07$$

The actual impedance is given by

$$Z_e(\Omega) = Z_e(pu) \times Z_e(base)$$

Base impedance on the primary side of transformer is

$$Z_e(base)_\Delta = \frac{3V_L^2}{S_{base}}$$

$$Z_e(base)_\Delta = \frac{3 \times (11 \times 10^3)^2}{10^5}$$

$$Z_e(base)_\Delta = 3630\,\Omega$$

Hence,

$$Z_e(\Omega) = (0.02 + j0.07) \times 3630$$

$$Z_e(\Omega) = (72.6 + j254.1)\,\Omega$$

D

Key Points

- Per-unit impedance remains the same on both primary and secondary sides of a transformer.
- Actual impedance can be obtained using:

$$Z(\Omega) = Z(pu) \times Z(base)$$

- Base impedance for a three-phase system:

$$Z_{base} = \frac{3V_L^2}{S_{base}}$$

Q2.03.1 MCQ | 2003 | 1M | Ans: D | ● ○ ○

Efficiency is,

$$\eta = \frac{\text{Output}}{\text{Input}}$$

$$\eta = \frac{\text{Output}}{\text{Output + Losses}}$$

At x fraction of full load,

$$\eta = \frac{xS\cos\phi}{xS\cos\phi + P_i + x^2P_{Cu,FL}}$$

For maximum efficiency,

$$P_i = x^2P_{Cu,FL}$$

Hence,

$$\eta_{\max} = \frac{xS\cos\phi}{xS\cos\phi + 2P_i}$$

It is given that maximum efficiency occurs at full load.

$$x = 1$$

$$0.9 = \frac{1 \times S \times 1}{1 \times S \times 1 + 2P_i}$$

$$P_i = 0.055S$$

Since,

$$P_i = (1)^2 P_{Cu,FL}$$

$$P_{Cu,FL} = 0.055S$$

At half full load and UPF,

$$x = \frac{1}{2}$$

$$\eta = \frac{\frac{1}{2} \times S \times 1}{\frac{1}{2} \times S \times 1 + 0.055S + \left(\frac{1}{2}\right)^2 (0.055S)} \times 100$$

$$\eta = 87.8\%$$

D

Q2.03.2 MCQ 2003 2M Ans: A ●●○

The induced EMF E_2 lags flux ϕ by 90°.

The terminal voltage E_2' is opposite to the induced EMF E_2 and hence leads flux ϕ by 90°.

Since the load is resistive,

$$I_2 \text{ is in phase with } E_2'$$

The supply voltage E_1 opposes the induced EMF E_1' and therefore,

$$I_1 \text{ is in phase with } E_1$$

Also, since

$$N_1 > N_2$$

the secondary current is greater than the primary current,

$$I_2 > I_1$$

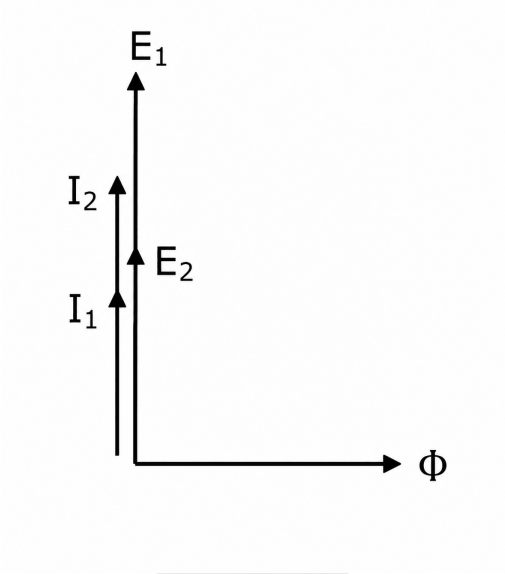


Fig. Solution Q2.03.2

A

Key Points

- Induced EMF in a transformer lags the mutual flux by 90°.
- For a resistive load, current and voltage remain in phase.

Q2.03.3 MCQ | 2003 | 2M | Ans: B

Turns ratio is,

$$a = \frac{N_1}{N_2} = \frac{V_{1,ph}}{V_{2,ph}}$$

$$a = \frac{11\,\text{kV}}{415/\sqrt{3}}$$

$$a = 240$$

As S_2 is closed, points B and C are at the same potential.
Hence,

$$V_{BC} = V_B - V_C = 0$$

For Δ connection,

$$V_{AB} + V_{BC} + V_{CA} = 0$$

$$V_{AB} = -V_{CA}$$

$$V_{AB} = 11 \times 10^3 \angle 0°\ \text{V}$$

$$V_{CA} = -11 \times 10^3 \angle 0°$$

$$V_{CA} = 11 \times 10^3 \angle 180^\circ \text{ V}$$

Since winding AB on primary is parallel to an of secondary,

$$V_{an} = 240 \text{ V}$$

Since winding CA on primary is parallel to cn of secondary,

$$V_{cn} = -240 \text{ V}$$

Therefore,

$$V_{ac} = V_{an} - V_{cn}$$

$$V_{ac} = 240 - (-240)$$

$$V_{ac} = 480 \text{ V}$$

B

Key Points

- Transformer turns ratio:
$$a = \frac{N_1}{N_2} = \frac{V_1}{V_2}$$
- In a closed Δ connection:
$$V_{AB} + V_{BC} + V_{CA} = 0$$
- Voltage polarity must be considered carefully while referring voltages between windings.

Mistakes to be avoided

- Do not neglect phase reversal when changing from V_{AB} to V_{CA}.
- Do not use line voltage instead of phase voltage incorrectly.
- Carefully track voltage polarity while combining voltages.

Q2.03.4 MCQ 2003 2M Ans: C ●○○

As we know,

$$\frac{E_1}{E_2} = \frac{N_1}{N_2}$$

$$E_2 = \frac{N_2}{N_1} E_1 = \left(\frac{2}{4}\right) \times 400 \angle 0^\circ = 200 \angle 0^\circ \text{ V}$$

Also,

$$\frac{E_1}{E_3} = \frac{N_1}{N_3}$$

$$E_3 = \frac{N_3}{N_1} E_1 = \frac{1}{4} \times 400 \angle 0^\circ = 100 \angle 0^\circ \text{ V}$$

Secondary current is,

$$I_2 = \frac{E_2}{R} = \frac{200\angle 0^\circ}{10} = 20\angle 0^\circ \text{ A}$$

Tertiary current is,

$$I_3 = \frac{E_3}{-jX_c} = \frac{100\angle 0^\circ}{-j \times 2.5} = 40\angle 90^\circ \text{ A}$$

I_2 referred to primary is,

$$I_2' = \left(\frac{N_2}{N_1}\right) I_2 = \left(\frac{2}{4}\right) \times 20\angle 0^\circ = 10\angle 0^\circ \text{ A}$$

I_3 referred to primary is,

$$I_3' = \left(\frac{N_3}{N_1}\right) I_3 = \frac{1}{4}(40\angle 90^\circ) = 10\angle 90^\circ \text{ A}$$

Therefore, primary current is,

$$I_1 = I_2' + I_3'$$

$$I_1 = 10\angle 0^\circ + 10\angle 90^\circ$$

$$I_1 = (10 + j10) \text{ A}$$

C

Q2.02.1 MCQ 2002 1M Ans: D ● ○ ○

When the supply is sinusoidal, the induced EMF would also be sinusoidal. Due to sinusoidal EMF, the flux would also be sinusoidal.

To produce sinusoidal flux in the non-linear core, the current drawn from supply should be peaky in nature.

If the current were sinusoidal, then the flux waveform would be flat topped.

D

Q2.02.2 MCQ 2002 1M Ans: A ● ○ ○

Since the supply is connected across the primary winding,

$$V_{AB} = 400 \text{ V}$$

Based on dot polarity, the corresponding dotted terminals in secondary and tertiary winding have positive polarity.

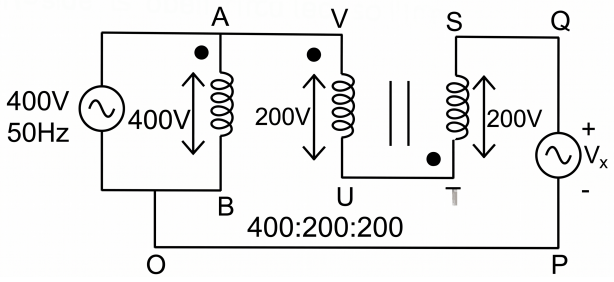


Fig. Solution Q2.02.2

From turns ratio,

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{200}{400}$$

Hence,

$$V_s = V_{VU} = 200\,\text{V}$$

Similarly for tertiary winding,

$$\frac{V_T}{V_s} = \frac{N_T}{N_s} = \frac{200}{400}$$

Hence,

$$V_T = V_{ST} = -200\,\text{V}$$

Applying KVL to determine voltmeter reading,

$$V_{AB} - V_{VU} - V_{ST} - V_x = 0$$

$$400 - 200 - 200 + V_x = 0$$

$$V_x = 0$$

$\boxed{A}$

Q2.02.3 NAT 2002 2M Ans: 652.83 ● ● ●

If the transformer is replaced by its equivalent circuit and all quantities are referred to the HV side,

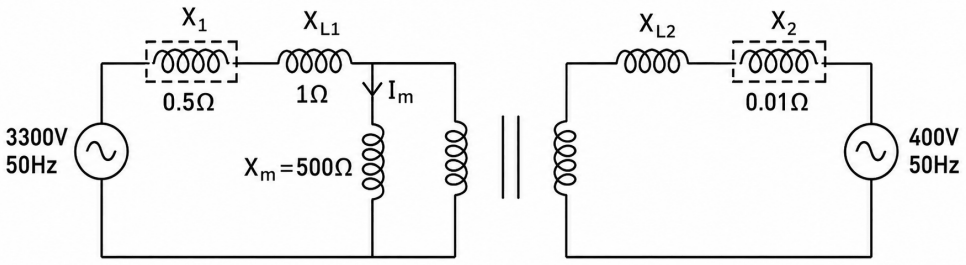


Fig. Solution Q2.02.3

$$k = \frac{N_2}{N_1} = \frac{12}{99} = 0.1212$$

Secondary reactance referred to HV side:

$$X_2' = \frac{X_2}{k^2}$$

$$X_2' = \frac{0.01}{(0.1212)^2} = 0.68\,\Omega$$

Leakage reactance referred to HV side:

$$X'_{L2} = \frac{X_{L2}}{k^2}$$

$$X'_{L2} = \frac{0.012}{(0.1212)^2} = 0.8167\,\Omega$$

Secondary voltage referred to HV side:

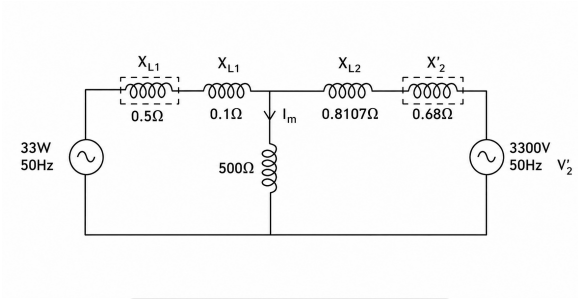


Fig. Solution Q2.02.3

$$V'_2 = \frac{V_2}{k}$$

$$V'_2 = \frac{400}{0.1212} = 3300\,\text{V}$$

Using superposition theorem,

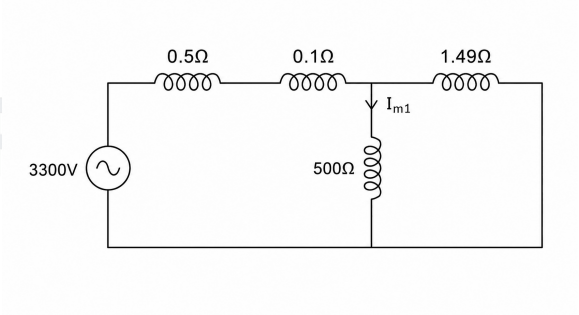


Fig. Solution Q2.02.3

$$I_m = I_{m1} + I_{m2}$$

For source on left side,

$$I_{m1} = \frac{3300}{(500 \parallel 1.49) + 0.6} \times \frac{1.49}{501.49}$$

$$I_{m1} = 4.7012\,\text{A}$$

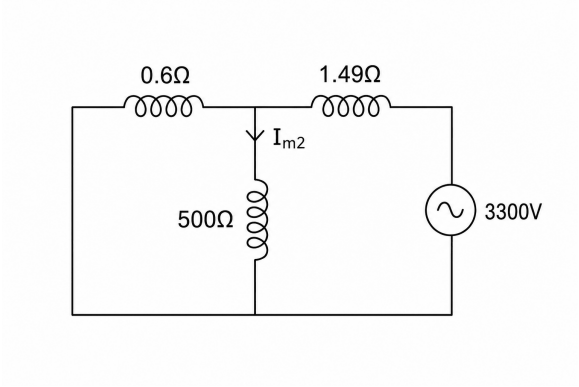


Fig. Solution Q2.02.3

For source on right side,

$$I_{m2} = \frac{3300}{(500 \parallel 0.6) + 1.49} \times \frac{0.6}{500.6}$$

$$I_{m2} = 1.8931 \text{ A}$$

Hence,

$$I_m = 4.7012 + 1.8931$$

$$I_m = 6.5943 \text{ A}$$

Magnetizing ampere-turns:

$$F_m = N_1 I_m$$

$$F_m = 99 \times 6.5943$$

$$F_m = 652.83 \text{ A-T}$$

652.83

Q2.01.1 MCQ 2001 1M Ans: A

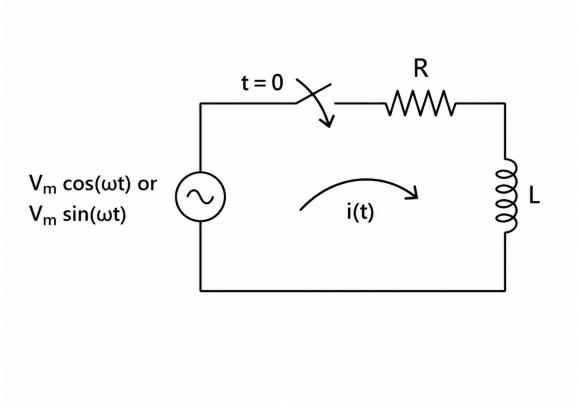


Fig. Solution Q2.01.1

Whenever switching takes place in an inductive circuit, transient current is produced and the total current becomes

$$i(t) = i_t(t) + i_{ss}(t)$$

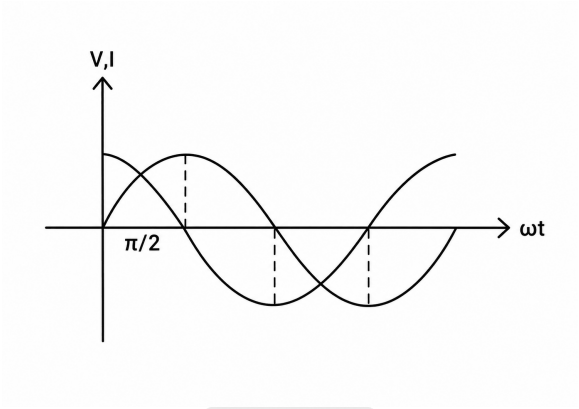


Fig. Solution Q2.01.1

where,

$i_t(t) \rightarrow$ Transient component

$i_{ss}(t) \rightarrow$ Steady-state component

Assume an AC input of the form,

$$v(t) = V_m \sin(\omega t + \alpha)$$

The total current at any instant is

$$i(t) = \left[-\frac{V_m}{\sqrt{R^2 + X_L^2}} \sin\left(\alpha - \tan^{-1}\left(\frac{\omega L}{R}\right)\right) e^{-Rt/L}\right]$$
$$+\frac{V_m}{\sqrt{R^2 + X_L^2}} \sin\left[\omega t + \alpha - \tan^{-1}\left(\frac{\omega L}{R}\right)\right]$$

For transient current to become zero,

$$\alpha = \tan^{-1}\left(\frac{\omega L}{R}\right)$$

Generally in transformers,

$$X \gg R$$

Hence,

$$\phi = 90^\circ$$

Thus voltage-current phase difference is approximately

$$90^\circ$$

From the waveform, if switching occurs at voltage maximum, current at the switching instant becomes zero and heavy inrush current can be avoided.

Switching at voltage minimum (0°) results in maximum current and high inrush current.

$\boxed{A}$

Key Points

- Total current consists of transient and steady-state components.
- In transformers,

$$X_L \gg R$$

therefore current lags voltage by nearly

$$90^\circ$$

- Transformer inrush current strongly depends on switching instant.
- Switching at voltage maximum minimizes transient current.

Mistakes to be avoided

- Do not assume switching at zero voltage minimizes inrush current.

Q2.01.2 MCQ 2001 1M Ans: A ● ○ ○

The core flux of a transformer is determined by the applied voltage-to-frequency ratio:

$$\phi \propto \frac{\text{Applied Voltage}}{\text{Applied Frequency}}$$

Hence, once the applied voltage and frequency are fixed, the core flux becomes independent of load variations due to the compensation effect in a transformer.

During loading, the secondary current produces a flux that opposes the main flux. To maintain the flux constant, the primary draws an additional current component.

$$I_1 = \frac{V_1 - E_1}{Z}$$

This extra current compensates for the demagnetizing effect of secondary current.

Therefore, the core flux remains approximately constant.

$\boxed{A}$

Q2.01.3 MCQ 2001 2M Ans: B ● ○ ○

Base impedance for Δ connection is,

$$(Z_b)_\Delta = \frac{3(V_{base})^2}{S_{base}}$$

$$(Z_b)_\Delta = \frac{3(33\,\text{kV})^2}{20\,\text{MVA}}$$

Since per-unit impedance is,

$$(Z)_{pu} = 0.12$$

Actual impedance is,

$$(Z)_{actual} = (Z)_{base} \times (Z)_{pu}$$

$$(Z)_{actual} = \frac{3 \times (33)^2}{20} \times \frac{12}{100}$$

$$(Z)_{actual} = 19.6\,\Omega$$

B

Key Points

- Base impedance for Δ connection:

$$(Z_b)_{\Delta} = \frac{3V_{base}^2}{S_{base}}$$

- Actual impedance is obtained from per-unit impedance as:

$$Z_{actual} = Z_{base} Z_{pu}$$

- Percentage impedance 12% is equal to $0.12\,pu$.

Q2.01.4 NAT 2001 5M Ans: *

For a linear B-H curve,

$$\phi \propto I_{exciting}$$

Hence both flux and exciting current have the same waveform.

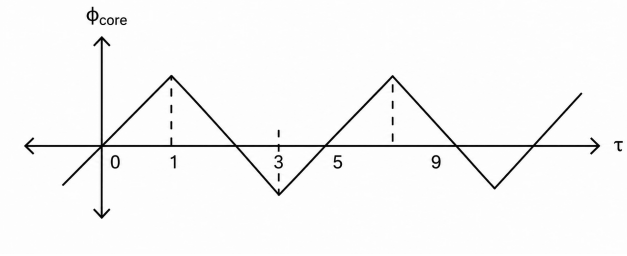


Fig. Solution Q2.01.4

Induced emf is given by,

$$V_2(t) = -N\frac{d\phi}{dt}$$

Thus the induced emf waveform is obtained from the derivative of the triangular flux waveform.

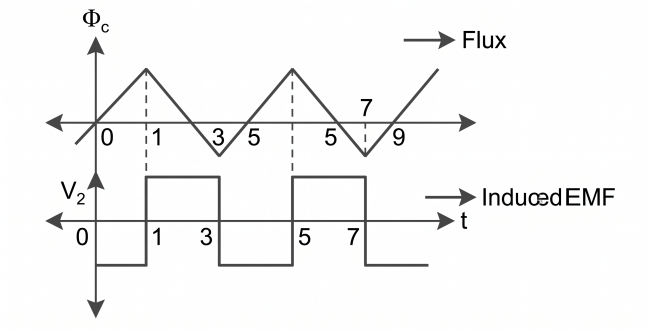


Fig. Solution Q2.01.4

For short-circuit condition,

$$R_2 \rightarrow 0$$

Hence the transformer behaves as a purely inductive circuit and short-circuit current lags induced emf approximately by

$$90^\circ$$

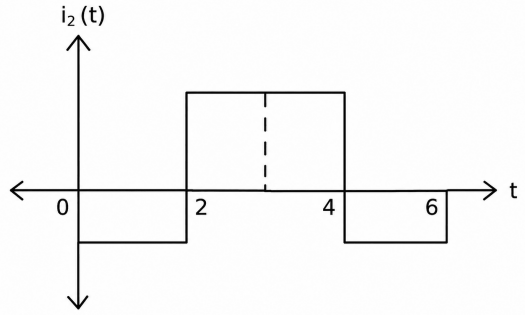


Fig. Solution Q2.01.4

Due to transformer action, secondary flux is compensated by additional primary current, therefore net core flux remains unchanged.

*

Key Points

- Core flux in a transformer is determined mainly by applied voltage and frequency.
- Induced emf is proportional to the rate of change of flux:

$$e = -N\frac{d\phi}{dt}$$

- Under short-circuit condition, additional primary current compensates secondary flux.
- Therefore transformer core flux waveform remains nearly constant.

Mistakes to be avoided

- Do not assume short-circuit condition changes the core flux significantly.
- Do not confuse flux waveform with induced emf waveform; emf is proportional to the derivative of flux.
- Current waveform in a highly inductive circuit lags voltage approximately by 90°, not in phase.

Q2.01.5 MCQ 2001 2M Ans: C ●●○

The area under the hysteresis curve represents the hysteresis energy loss per cycle.

Method 1

Area under curve:

$$= 5\,\text{cm}^2$$

Given scale:

$$1\,\text{cm}^2 = (2\,\text{AT})(50\,\text{mWb})$$

Hence hysteresis energy loss per cycle:

$$= 5 \times (2\,\text{AT} \times 50\,\text{mWb}) = 500\,\text{mJ}$$

For a frequency of:

$$f = 50\,\text{Hz}$$

Power loss:

$$P_h = (500\,\text{mJ}) \times 50$$

$$= 500 \times 10^{-3} \times 50 = 25\,\text{W}$$

C

Key Points

- Area of the hysteresis loop gives hysteresis energy loss per cycle.
- Hysteresis power loss is:

$$P_h = (\text{Energy loss per cycle}) \times f$$

- Frequency directly affects hysteresis power loss.

Q2.00.1 NAT 2000 5M Ans: 2.5,0.6 ●○○

Given turns ratio:

$$N_1 : N_2 : N_3 = 20 : 4 : 1$$

Power factor at tertiary side:

$$\cos\phi = 0.6$$

Hence,

$$\phi = \cos^{-1}(0.6) = 53.13^\circ$$

Tertiary current:

$$I_3 = 50\angle -53.13^\circ$$

Using MMF balance principle of transformer:

$$N_1 I_3' = N_3 I_3$$

where I_3' is the tertiary current reflected to primary.

$$I_3' = \left(\frac{N_3}{N_1}\right) I_3$$

$$I_3' = \frac{1}{20} \times 50\angle -53.13^\circ$$

$$I_3' = 2.5\angle -53.13^\circ$$

Hence primary current magnitude:

$$I_1 = 2.5\ \text{A}$$

Primary power factor:

$$\text{p.f.} = \cos(53.13^\circ)$$

$$= 0.6$$

No-load current is neglected and secondary is assumed open-circuited.

$$\boxed{2.5, 0.6}$$

Q2.00.2 MCQ 2000 2M Ans: B ●○○

Voltage regulation of a transformer is given by

$$\text{V.R.} = R\cos\phi + X\sin\phi$$

where R and X are in p.u. or percentage.
Given:

$$R = 1\% = 0.01\ \text{p.u.}$$

$$X = 5\% = 0.05\ \text{p.u.}$$

Hence,

$$\text{V.R.} = (1\cos\phi + 5\sin\phi)\%$$

Since,

$$\cos\phi = 0.8$$

$$\sin\phi = 0.6$$

Therefore,

$$\text{V.R.} = (1 \times 0.8 + 5 \times 0.6)\% = 3.8\%$$

Voltage regulation relation:

$$\frac{E - V}{E} = 3.8\%$$

Given line-to-line voltage:

$$V = 415\,\text{V}$$

Thus,

$$E_{(L-L)} = \frac{3.8}{100}E_{(L-L)} + 415$$

$$E_{(L-L)} = \frac{415 \times 100}{96.2} = 431.33\,\text{V}$$

Induced EMF per phase:

$$E_2 = \frac{431.33}{1.732} = 249.06\,\text{V}$$

For delta connection:

$$E_1 = 6000\,\text{V}$$

Turns ratio:

$$\frac{E_1}{E_2} = \frac{6000}{249.06} = 24.08$$

$$\approx 24$$

$\boxed{B}$

Q2.00.3 MCQ | 2000 | 1M | Ans: A | ● ○ ○

In a constant voltage transformer, a capacitor is connected across the secondary winding. This increases the current in the secondary winding and drives the magnetic circuit into saturation.

The saturation effect causes the secondary terminal voltage to remain nearly constant over a large range of input voltage variations.

However, the output waveform is not perfectly sinusoidal. It becomes approximately sinusoidal with flattened peaks due to magnetic saturation.

$\boxed{A}$

Q2.99.1 MCQ | 1999 | 2M | Ans: *

Let the per-phase winding impedance be Z.

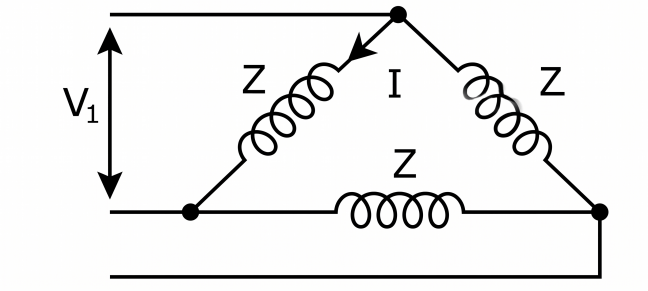


Fig. Solution Q2.99.1

When all windings are reconnected as a single-phase equivalent circuit, all windings are connected in additive polarity for maximum voltage.

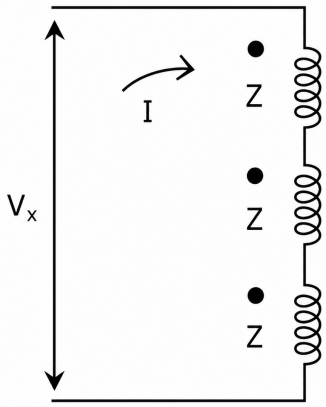


Fig. Solution Q2.99.1

Current rating of a winding remains unchanged as long as the cross-sectional area of the conductor remains constant.
For single-phase connection,

$$V_x = 3V_1$$

KVA rating in delta connection:

$$Q_\Delta = 3V_1 I$$

KVA rating in equivalent single-phase connection:

$$Q_{\text{single}} = (3V_1)I$$

$$Q_{\text{single}} = Q_\Delta$$

Hence KVA rating remains unchanged.
Therefore voltage ratio becomes

$$\frac{3V_1}{3V_2}$$

which is equal to the original turns ratio.

If windings are connected in parallel, voltage rating remains unchanged but current rating becomes three times, therefore KVA rating still remains the same.

Hence none of the given options satisfy the required condition.

*

Key Points

- KVA rating depends on both voltage and current ratings.
- In additive polarity connection, voltages are added directly.
- Changing winding configuration does not change total transformer KVA if conductor ratings remain unchanged.
- Current rating of individual winding remains fixed by conductor size.

Q2.99.2 MCQ 1999 2M Ans: C ● ○ ○

Given transformer rating:

$$10\,\text{kVA}, \quad 400\,\text{V}/200\,\text{V}, \quad Z = 10\%$$

Rated current on 400 V side:

$$I_{\text{rated}} = \frac{10 \times 10^3}{400} = 25\,\text{A}$$

Short-circuit current:

$$I_{sc} = \frac{I_{\text{rated}}}{Z_{\text{p.u.}}} = \frac{25}{0.1} = 250\,\text{A}$$

C

Q2.99.3 MCQ 1999 2M Ans: B ● ○ ○

Fraction of full-load at maximum efficiency:

$$x = \sqrt{\frac{\text{Iron loss}}{\text{F.L. copper loss}}} = \sqrt{\frac{0.01}{0.02}} = 0.707$$

At maximum efficiency:

$$\cos\phi = 1$$

$$\eta_{\max} = \frac{x}{x + 2P_i} \times 100$$

$$= \frac{0.707}{0.707 + 0.02} \times 100$$

$$= 97.25\%$$

$\boxed{B}$

Key Points

- Maximum efficiency occurs when:

$$P_i = P_{cu}$$

- Fractional load for maximum efficiency:

$$x = \sqrt{\frac{P_i}{P_{cu,\text{F.L.}}}}$$

- Maximum efficiency is generally evaluated at unity power factor unless specified.

Q2.99.4 MCQ | 1999 | 2M | Ans: A | ● ○ ○

Given:

$$400\,\text{V}/100\,\text{V}, \quad 10\,\text{kVA}$$

For maximum rating, use additive polarity.
Rated currents:

$$I_p = \frac{10 \times 10^3}{400} = 25\,\text{A}$$

$$I_s = \frac{10 \times 10^3}{100} = 100\,\text{A}$$

New voltage:

$$V = 400 + 100 = 500\,\text{V}$$

Hence,

$$\text{Rating} = 500 \times 100 = 50\,\text{kVA}$$

$\boxed{A}$

Q2.99.5 MCQ | 1999 | 2M | Ans: A | ● ○ ○

Percentage voltage regulation is given by

$$\%\ \text{Regulation} = \left(\frac{IR\cos\phi + IX\sin\phi}{V}\right) \times 100$$

For a resistive load,

$$\phi = 0^\circ$$

Using per-unit quantities,

$$\%\ \text{Regulation} = \left[R_{\text{p.u.}}\cos\phi + X_{\text{p.u.}}\sin\phi\right] \times 100$$

Substituting values,

$$\% \text{ Regulation} = 0.03\cos 0^\circ + 0.06\sin 0^\circ$$

$$= 0.03(1) + 0.06(0) = 3\%$$

Voltage regulation relation:

$$3\% = \left(\frac{E_f - V_t}{E_f}\right)$$

Given,

$$E_f = 200\,\text{V}$$

Hence,

$$\frac{3}{100} \times 200 = E_f - V_t$$

$$= 6\,\text{V}$$

Therefore,

$$V_t = 200 - 6 = 194\,\text{V}$$

$$\boxed{A}$$

Q2.99.6 NAT | 1999 | 5M | Ans: 7.466,6.787 | ● ○ ○

Load kVA:

$$S_L = \frac{12}{0.8}\angle 36.87^\circ = 15\angle 36.87^\circ\,\text{kVA}$$

Taking 10 kVA as base,

$$Z_B = (0.04 + j0.03) \times \frac{10}{5} = 0.08 + j0.06\,\text{p.u.}$$

Load shared by transformer A:

$$S_A^* = \frac{Z_B}{Z_A + Z_B} S_L^*$$

$$S_A^* = \frac{0.08 + j0.06}{(0.03 + j0.04) + (0.08 + j0.06)} (15\angle -36.87^\circ)$$

$$S_A^* = (7.466 - j6.787)\,\text{kVA}$$

Therefore,

$$P_A = 7.466\,\text{kW}, \qquad Q_A = 6.787\,\text{kVAR}$$

$$\boxed{7.466, 6.787}$$

Key Points

- Load sharing is inversely proportional to transformer impedance.
- For parallel transformers:

$$S_A^* = \frac{Z_B}{Z_A + Z_B} S_L^*$$

- Positive kVA angle indicates lagging power factor.

Mistakes to be avoided

- Convert impedances to the same kVA base before load sharing.
- Use S_L^* while applying the complex power sharing relation.

Q2.98.1 MCQ | 1998 | 1M | Ans: C | ● ○ ○

Assume unity power factor.

At full load:

$$0.98 = \frac{\text{kVA}}{\text{kVA} + P_i + P_c} \quad (1)$$

At half load:

$$0.98 = \frac{0.5\,\text{kVA}}{0.5\,\text{kVA} + P_i + 0.25P_c}$$

$$= \frac{\text{kVA}}{\text{kVA} + 2P_i + 0.5P_c} \quad (2)$$

From (1) and (2),

$$P_i + P_c = 2P_i + 0.5P_c$$

$$P_c = 2P_i$$

Hence,

$$P_c > P_i$$

$\boxed{C}$

Q2.98.2 MCQ | 1998 | 1M | Ans: A | ● ○ ○

If flux in a transformer were purely sinusoidal, due to the nonlinear B-H characteristic, magnetizing current would contain harmonics, predominantly the 3rd harmonic.

The induced EMF becomes:

$$E = E_1 + E_3$$

where E_1 is the fundamental component and E_3 is the third harmonic component.

A large E_3 component creates excessive electric stress on winding insulation, which may damage the insulation.

Hence, magnetizing current is rich in third harmonic.

$\boxed{A}$

Key Points

- Transformer core operates in the nonlinear region of the B-H curve.
- Magnetizing current contains harmonics even when flux is sinusoidal.
- The dominant harmonic in magnetizing current is the 3rd harmonic.

Q2.98.3 MCQ | 1998 | 2M | Ans: C ● ● ○

Hysteresis loss and eddy current loss relations are:

$$P_h \propto f(B_m)^2, \qquad P_e \propto V^2$$

Given,

$$V_2 = 0.9V_1, \qquad f_2 = 45\,\text{Hz} = 0.9f_1$$

Since,

$$\frac{V_2}{f_2} = \frac{V_1}{f_1}$$

flux density remains constant.
Hence,

$$P_h \propto f$$

$$P_{h2} = 0.9P_{h1}$$

Also,

$$P_{e2} = (0.9)^2 P_{e1} = 0.81P_{e1}$$

Initially,

$$P_{h1} = P_{e1} = \frac{P_{\text{core1}}}{2}$$

Therefore,

$$P_{\text{core2}} = 0.9\left(\frac{P_{\text{core1}}}{2}\right) + 0.81\left(\frac{P_{\text{core1}}}{2}\right)$$

$$P_{\text{core2}} = 0.855P_{\text{core1}}$$

$$\%\text{ change} = \frac{0.855P_{\text{core1}} - P_{\text{core1}}}{P_{\text{core1}}} \times 100 = -14.5\%$$

Therefore, core loss reduces by 14.5%.

$\boxed{C}$

Key Points

- Hysteresis loss: $P_h \propto f(B_m)^2$
- Eddy current loss: $P_e \propto V^2$

- Constant $\frac{V}{f}$ implies constant flux density.

Mistakes to be avoided

- Do not directly scale hysteresis loss with voltage.
- Check whether $\frac{V}{f}$ remains constant before changing B_m.

Q2.98.4 NAT | 1998 | 5M | Ans: 97.84,500,45.65 | ● ● ○

Given,

$$(P_i)_{fl} = 500\,\text{W}, \qquad (P_{cu})_{fl} = 600\,\text{W}$$

Full-load efficiency:

$$\eta = \frac{50 \times 10^3}{50 \times 10^3 + 500 + 600} \times 100$$

$$\eta = 97.84\%$$

For maximum efficiency,

$$x = \sqrt{\frac{(P_i)_{fl}}{(P_{cu})_{fl}}}$$

$$x = \sqrt{\frac{500}{600}} = 0.913$$

At maximum efficiency,

$$(P_{cu})_x = x^2 (P_{cu})_{fl}$$

$$(P_{cu})_x = (0.913)^2 (600) = 500\,\text{W}$$

Loading at maximum efficiency:

$$= 0.913 \times 50$$

$$= 45.65\,\text{kVA}$$

$$\boxed{97.84, 500, 45.65}$$

Q2.97.1 MCQ | 1997 | 1M | Ans: C | ● ○ ○

For a transformer,

$$\frac{V}{f} \propto \phi_m$$

For constant core flux,

$$\frac{V_1}{f_1} = \frac{V_2}{f_2}$$

Substituting values,

$$\frac{230}{50} = \frac{V_2}{25}$$

$$V_2 = 115\,\text{V}$$

$$\boxed{C}$$

Q2.97.2 MCQ | 1997 | 1M | Ans: C | ● ○ ○

Given,

$$v = 400\sin(314.16t)$$

$$i_0 = 2\sin(314.16t - 85°)$$

Hence phasors are:

$$V = 400\angle 0°, \qquad I_0 = 2\angle -85°$$

Magnetizing branch impedance:

$$Z = \frac{V}{I_0} = \frac{400\angle 0°}{2\angle -85°} = 200\angle 85°\,\Omega$$

$$\boxed{C}$$

Q2.97.3 NAT | 1997 | 2M | Ans: 10 | ● ● ○

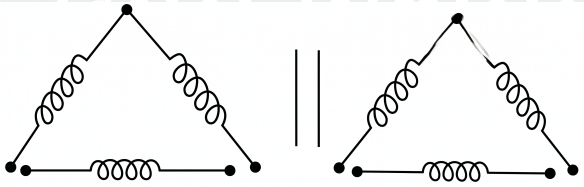


Fig. Solution Q2.97.3

For open-Δ connection, the maximum kVA rating of the bank is:

$$\text{Bank Rating} = \frac{1}{\sqrt{3}}(3 \times 15)$$

$$= 25.98\,\text{kVA} \approx 26\,\text{kVA}$$

Since two transformers share the load equally,

$$\text{Load per transformer} = \frac{20}{2} = 10\,\text{kVA}$$

$$\boxed{10}$$

Key Points

- Open-Δ (V-V) connection has capacity:

$$S_{\text{open-}\Delta} = 0.577\, S_{\text{closed-}\Delta}$$

- Open-Δ bank capacity is not twice the individual transformer rating because of unequal phase-wise apparent power sharing.

Mistakes to be avoided

- Do not assume bank rating = $2 \times 15 = 30\,\text{kVA}$.
- Use $\frac{1}{\sqrt{3}}$ factor for open-Δ rating.

Q2.96.1 MCQ 1996 1M Ans: D ● ○ ○

The advantages of an autotransformer are more when the transformation ratio is nearer to unity.

That is, when the ratio is smaller, the saving of copper and kVA rating advantage are higher as compared to a two-winding transformer.

Hence, the advantages of an autotransformer are maximum when:

$$\frac{V_2}{V_1} \approx 1$$

$\boxed{D}$

Q2.96.2 MCQ 1996 1M Ans: D ● ○ ○

For parallel operation of transformers, the essential requirement is that both transformers must belong to the same vector group.

Hence, the phase displacement between primary and secondary line voltages of both transformers should be identical.

$$\theta_1 = \theta_2$$

$\boxed{D}$

Q2.96.3 MCQ 1996 1M Ans: A ● ○ ○

Transformer oil mainly serves two purposes:

(i) Provides electrical insulation

(ii) Dissipates transformer heat (acts as a coolant)

Hence, transformer oil acts both as an insulating and cooling medium.

$\boxed{A}$

Q2.96.4 MCQ | 1996 | 1M | Ans: B | ● ○ ○

When one supply line fails in a star-delta transformer, due to magnetic circuit symmetry the voltage across the failed phase is shared equally by the other two phases. Given line voltage:

$$V = 230\,\text{V}$$

Hence voltage across each of the other two phases:

$$\frac{230}{2} = 115\,\text{V}$$

Therefore, secondary side voltages are:

$$230\,\text{V}, \quad 115\,\text{V}, \quad 115\,\text{V}$$

B

Key Points

- Magnetic circuit symmetry makes phase reluctances equal.
- When one line opens, the remaining two phases share the voltage equally.
- For a 1 : 1 transformer, secondary voltages follow primary phase voltages.

Q2.95.1 MCQ | 1995 | 1M | Ans: B | ● ● ○

Let the supply to one terminal (assume terminal A) of the delta-connected primary fail.

The winding connected between the active terminals B and C will still receive the full line-to-line supply voltage, V.

The other two primary windings (A-B and C-A) are now connected in series across the active supply terminals B and C.

Assuming magnetic circuit symmetry, the voltage will divide equally across these two series-connected windings.

Voltage across winding B-C:

$$V_{BC} = V$$

Voltage across windings A-B and C-A:

$$V_{AB} = V_{CA} = \frac{V}{2}$$

In a delta-wye transformer, the induced phase voltages on the secondary side are directly proportional to the voltages across the corresponding primary windings.

Therefore, the secondary phase voltages will be in the ratio of:

$$V : \frac{V}{2} : \frac{V}{2}$$

$$\text{Ratio} = 2 : 1 : 1$$

Looking at the given options, the voltages 230, 115, 115 satisfy this 2 : 1 : 1 ratio.

Q2.95.2 NAT | 1995 | 1M | Ans: 0.75 | ● ○ ○

For autotransformer starter with tapping K:

$$I_{\text{AT}} = K^2 I_{\text{DOL}}$$

For 50% tapping,

$$I_{\text{AT}} = (0.5)^2 I_{\text{DOL}} = 0.25 I_{\text{DOL}}$$

For star-delta starter:

$$I_{SD} = \frac{1}{3} I_{DOL}$$

Hence,

$$\frac{I_{AT}}{I_{SD}} = \frac{0.25}{1/3} = \frac{3}{4}$$

$$\boxed{0.75}$$

Key Points

- Autotransformer starter line current is K^2 times DOL current.
- Star-delta starter line current is $\frac{1}{3}$ times DOL current.

Mistakes to be avoided

- Do not use K instead of K^2 for supply current in autotransformer starting.
- Convert 50% tapping into $K = 0.5$.

Q2.95.3 NAT | 1995 | 1M | Ans: 247.5 ● ● ○

Given,

$$S_{rated} = 100\,\text{kVA}$$

$$V_{HV} = 11\,\text{kV}, \qquad V_{LV} = 400\,\text{V}$$

$$Z_{pu} = 0.045$$

Under short-circuit condition,

$$I_{rated} = 1\,\text{pu}$$

Actual current:

$$I = 0.5\, I_{rated}$$

Hence,

$$I = 0.5\,\text{pu}$$

By KVL,

$$E = IZ$$

$$E = (0.5)(0.045)$$

$$E = 0.0225\,\text{pu}$$

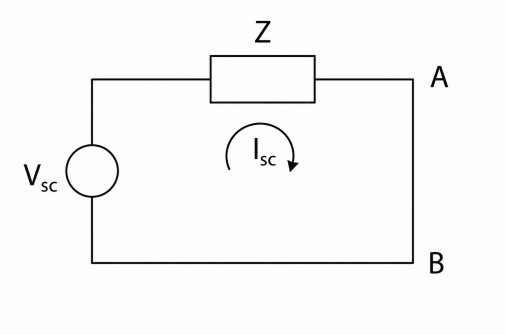


Fig. Solution Q2.95.3

Since per-unit voltage remains same on either side of the transformer,

$$V_{HV} = 0.0225 \times 11 \times 10^3$$

$$V_{HV} = 247.5\,\text{V}$$

$$\boxed{247.5}$$

Key Points

- Under short-circuit condition, induced voltage is calculated as $E = IZ$ in per-unit.
- Per-unit voltage values are identical on both sides of a transformer.
- Base quantities simplify transformer calculations significantly.

Mistakes to be avoided

- Do not directly use actual current instead of converting into per-unit.
- Do not multiply by LV voltage when HV side voltage is required.

Q2.95.4 NAT 1995 5M Ans: 0.389 ●●○

Under no-load condition:

$$P_{\text{input}} = P_c = 80\,\text{W}$$

Core loss remains the same irrespective of the side on which it is measured.
Mutual inductance referred to HV side:

$$M_{HV} = 1.91 \times \left(\frac{400}{200}\right)$$

$$M_{HV} = 3.82\,\text{H}$$

Magnetizing reactance on HV side:

$$X_m = 2\pi f M$$

$$X_m = 2\pi \times 50 \times 3.82$$

$$X_m = 1200.088\,\Omega$$

Magnetizing current:

$$I_\mu = \frac{V}{X_m}$$

$$I_\mu = \frac{400}{1200.088}$$

$$I_\mu = 0.334\,\text{A}$$

Core loss current:

$$I_w = \frac{P_c}{V}$$

$$I_w = \frac{80}{400}$$

$$I_w = 0.2\,\text{A}$$

Since both currents are orthogonal,

$$I_0 = \sqrt{I_\mu^2 + I_w^2}$$

$$I_0 = \sqrt{(0.334)^2 + (0.2)^2}$$

$$I_0 = 0.389\,\text{A}$$

$$\boxed{0.389}$$

Q2.94.1 MCQ | 1994 | 1M | Ans: C | ● ○ ○

For satisfactory load sharing during parallel operation of transformers, the per-unit impedances should be the same.

This ensures proportional load sharing among transformers.

Also, the transformers should have the same X/R ratio so that both transformers operate at the same power factor.

$$Z_{\text{pu},1} = Z_{\text{pu},2}$$

$$\frac{X_1}{R_1} = \frac{X_2}{R_2}$$

These conditions ensure equal power-factor loading and proper current sharing.

$$\boxed{C}$$

Q2.93.1 MCQ 1993 2M Ans: C ● ○ ○

Hysteresis loss is given by:

$$P_h \propto B_m^n f$$

Taking Steinmetz coefficient,

$$n = 2$$

Since,

$$V \propto B_m f$$

Therefore,

$$B_m \propto \frac{V}{f}$$

Substituting in hysteresis loss expression:

$$P_h \propto \left(\frac{V}{f}\right)^2 f$$

$$P_h \propto \frac{V^2}{f}$$

For eddy current loss,

$$P_e \propto B_m^2 f^2$$

Substituting,

$$P_e \propto \left(\frac{V}{f}\right)^2 f^2$$

$$P_e \propto V^2$$

Since applied voltage remains constant, eddy current loss remains constant.
However, as frequency decreases,

$$P_h \propto \frac{1}{f}$$

Hence hysteresis loss increases.

$$\boxed{C}$$

Q2.92.1 MCQ 1992 1M Ans: A ● ● ○

Transformer rating is proportional to:

$$(\text{kVA})_{\text{rating}} \propto EI$$

where induced EMF is

$$E = 4.44 f N \phi$$

and,

$$\phi = B_m A$$

When linear dimensions are doubled, both length and breadth become twice the original values.
Hence core area becomes:

$$A' = (2L)(2B) = 4A$$

Since,

$$E \propto A$$

$$E' = 4E$$

Current carrying capability of winding also increases in proportion to conductor cross-sectional area:

$$I' = 4I$$

Therefore,

$$(\text{kVA})' = V'I'$$

$$(\text{kVA})' = (4V)(4I)$$

$$(\text{kVA})' = 16(VI)$$

Thus transformer rating becomes:

$$(\text{kVA})' = 16(\text{kVA})$$

$\boxed{A}$

Key Points

- If linear dimensions increase by factor k, area becomes k^2 times.
- Current carrying capability also varies as k^2.
- Transformer kVA rating varies as k^4.

Q2.92.2 MCQ 1992 1M Ans: B ● ○ ○

For transformers of different kVA ratings operating in parallel, load sharing is proportional to their ratings when their per-unit leakage impedances on their own respective kVA ratings are equal.

$$Z_{\text{pu},1} = Z_{\text{pu},2}$$

Hence, each transformer carries current proportional to its kVA rating.

$\boxed{B}$

Q2.92.3 MCQ 1992 1M Ans: A ●○○

A Buchholz relay is used for protection of an oil-immersed transformer against internal faults.
It operates due to gas accumulation or sudden oil movement caused by internal faults inside the transformer tank.

$\boxed{A}$

Key Points

- Buchholz relay is used in oil-immersed transformers.
- It detects internal faults such as winding faults, core faults and insulation failure.
- It is placed between the transformer tank and conservator.

Mistakes to be avoided

- Buchholz relay does not protect against external faults.
- It is not used for induction motor protection.

Q2.91.1 MCQ 1991 1M Ans: B ●●●

Let,

$$V_1 = 200\,\text{V}, \qquad V_2 = 160\,\text{V}$$

$$f_1 = 50\,\text{Hz}, \qquad f_2 = 40\,\text{Hz}$$

For a single-phase transformer:
Eddy current loss is,

$$P_e \propto f^2 B_m^2$$

Since,

$$B_m \propto \frac{V}{f}$$

Substituting:

$$\frac{P_{e1}}{P_{e2}} = \frac{f_1^2 B_{m1}^2}{f_2^2 B_{m2}^2} = \frac{V_1^2}{V_2^2}$$

$$\frac{P_{e2}}{P_{e1}} = \left(\frac{V_2}{V_1}\right)^2$$

$$= \left(\frac{160}{200}\right)^2 = 0.64$$

Hence,

$$\text{Percentage reduction in eddy current loss} = (1 - 0.64) \times 100 = 36\%$$

Hysteresis loss:

$$P_h \propto B_m^{1.6} f$$

Substituting,

$$P_h \propto \left(\frac{V}{f}\right)^{1.6} f$$

$$P_h \propto V^{1.6} f^{-0.6}$$

Thus,

$$\frac{P_{h2}}{P_{h1}} = \left(\frac{V_2}{V_1}\right)^{1.6} \left(\frac{f_2}{f_1}\right)^{-0.6}$$

$$= \left(\frac{160}{200}\right)^{1.6} \left(\frac{40}{50}\right)^{-0.6} = 0.8$$

Hence,

$$\text{Percentage reduction in hysteresis loss} = (1 - 0.8) \times 100$$

$$= 20\%$$

$$\boxed{B}$$

Key Points

- Eddy current loss: $P_e \propto f^2 B_m^2$
- Hysteresis loss: $P_h \propto B_m^{1.6} f$
- Maximum flux density varies as $B_m \propto V/f$

Mistakes to be avoided

- Do not directly vary losses only with frequency; include variation of B_m.
- Use the correct Steinmetz exponent for hysteresis loss.

Q2.91.2 MSQ 1991 1M Ans: A,B ● ● ○

In a transformer, the flux produced by secondary current must oppose the primary flux according to Lenz's law.

Hence, using right-hand grip rule, the correct connections are those in which:

$$\phi_1 \text{ and } \phi_2$$

oppose each other.

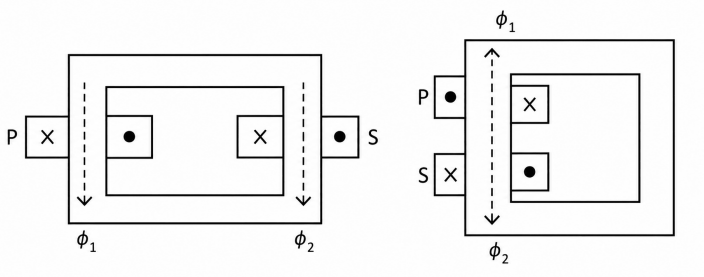


Fig. Solution Q2.91.2

From the given figures, this condition is satisfied in options (a) and (b).

A, B

Key Points

- Secondary current produces flux opposing the primary flux.
- This is a direct consequence of Lenz's law.
- Right-hand grip rule is used to determine flux direction.

Mistakes to be avoided

- Do not assume primary and secondary fluxes aid each other.
- Always check dot polarity/current direction before deciding flux direction.

PrepFusion

SOLUTIONS

III

DC Machines

Topics

1. Types of DC Machines - Separately Excited, Series and Shunt
2. Construction of DC Machines
3. Armature Winding, Equivalent Resistance and Voltage across Brushes
4. Armature Reaction: Concept and Method to Improve
5. Commutation
6. Losses and Testing in DC Machines
7. Starter of DC Motor
8. DC Generator Characteristics - Series and Compound
9. DC Motor Characteristics/Mode of Operation
10. Speed Control and Braking of DC Motors

Unit Snapshot*

Stream: EE	**Questions:** 78	**MCQ:** 46	**MSQ:** 0	**NAT:** 32
1M: 25	**2M:** 45	**Descriptive:** 8	**Avg Weightage** ~3M	**Range:** 2M(2025)–5M(2015)

**See preface for how Avg Weightage and Range are calculated.*

Q3.25.1 MCQ 2025 2M Ans: A ●●○

Given,

$$V_t = 400\,\text{V}, \qquad I_a = 40\,\text{A}$$

$$(r_a + r_{se}) = 0\,\Omega$$

For a DC series motor,

$$\phi_{se} \propto I_a$$

$$E_a \propto \phi_{se} N \propto I_a N$$

Since armature and series field resistances are neglected,

$$V_t = E_a$$

$$V_t \propto I_a N \tag{i}$$

Given,

$$N' = \frac{N}{2}$$

With external resistance added,

$$E'_a = V_t - I'_a R_{ext}$$

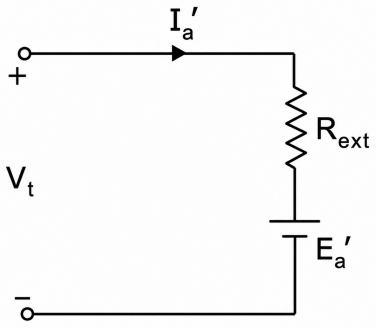


Fig. Solution Q3.25.1

Using (i),

$$\frac{V_t - I'_a R_{ext}}{V_t} = \frac{I'_a}{I_a} \cdot \frac{N'}{N}$$

$$\frac{400 - I'_a R_{ext}}{400} = \frac{I'_a}{40} \cdot \frac{1}{2} \tag{ii}$$

Also,

$$T_e \propto \phi_{se} I_a \propto I_a^2$$

Given,

$$T_L \propto N^3$$

At equilibrium,

$$T_e = T_L$$

$$\frac{I_a'^2}{40^2} = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

$$I'_a = 14.142\,\text{A}$$

Substituting in (ii),

$$\frac{400 - 14.142R_{ext}}{400} = \frac{14.142}{40} \times \frac{1}{2}$$

$$R_{ext} = 23.28\,\Omega$$

$$\boxed{A}$$

Key Points

- In a DC series motor, $\phi_{se} \propto I_a$ before saturation.
- Electromagnetic torque is proportional to I_a^2.
- Back emf relation is $E_a \propto \phi N$.

Mistakes to be avoided

- Do not use constant flux relation as in shunt motors.
- Use $T_L \propto N^3$ carefully while comparing torque.
- Use the new speed ratio as $N'/N = 1/2$.

Q3.24.1 NAT | 2024 | 2M | Ans: 27.19 to 27.39 | ● ○ ○

Given,

$$I_f = 1\,\text{A}$$

$$I_{a1} = 3 - 1 = 2\,\text{A}$$

At no-load,

$$E_1 = V - I_{a1}r_a$$

$$E_1 = 220 - (2)(0.5) = 219\,\text{V}$$

Constant loss,

$$P_c = E_1 I_{a1} = 219 \times 2 = 438\,\text{W}$$

At rated load,

$$P_{sh} = 5000\,\text{W}$$

Developed power,

$$P_{em} = P_{sh} + P_c = 5000 + 438 = 5438\,\text{W}$$

$$E_2 I_{a2} = 5438$$

$$(220 - 0.5I_{a2})I_{a2} = 5438$$

$$220I_{a2} - 0.5I_{a2}^2 = 5438$$

Solving,

$$I_{a2} = 26.29\,\text{A}$$

The lower root is chosen, since the higher root ($I_{a2} \approx 413.7$ A) gives a back emf of only about 13 V and is not a practical operating point.

$$I_{L2} = I_{a2} + I_f = 26.29 + 1 = 27.29 \text{ A}$$

$$\boxed{27.29}$$

Key Points

- Constant loss is approximately equal to no-load developed power.
- Developed power is $P_{em} = EI_a$.
- For DC shunt motor, $I_L = I_a + I_f$.

Q3.23.1 MCQ | 2023 | 1M | Ans: D ●●○

Magnetic field produced by stator is as shown in the figure.

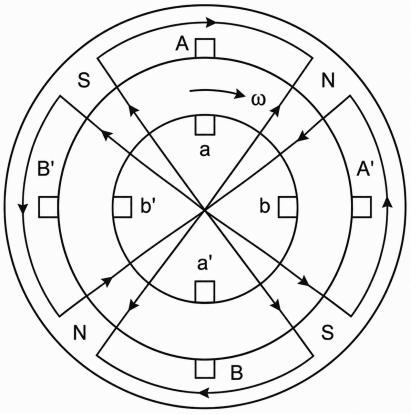


Fig. Solution Q3.23.1

Number of poles,

$$P = 4$$

Relation between electrical and mechanical angle is

$$\theta_e = \frac{P}{2}\theta_m$$

$$\theta_m = \frac{2}{P}\theta_e$$

Given,

$$\theta_e = 90^\circ$$

$$\theta_m = \frac{2}{4} \times 90^\circ = 45^\circ$$

Thus, geometrical neutral axis lies at 90° electrical from pole axis, i.e. 45° mechanical from pole axis.

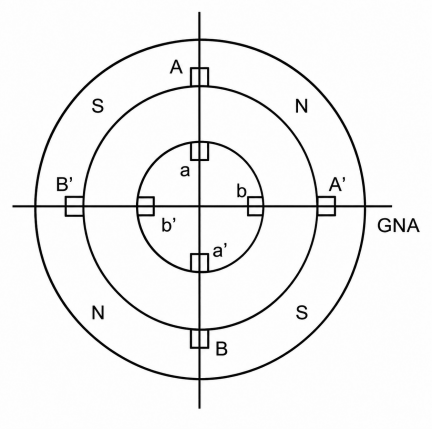


Fig. Solution Q3.23.1

Since coils $a - a'$ and $b - b'$ lie along GNA, emf induced in each of them is zero.

$\boxed{D}$

Key Points

- GNA is perpendicular to the magnetic axis.
- For a 4-pole machine, 90° electrical equals 45° mechanical.
- Coils lying along GNA have zero induced emf.

Mistakes to be avoided

- Do not confuse electrical angle with mechanical angle.
- Do not use 90° mechanical directly for a 4-pole machine.

Q3.23.2 NAT 2023 1M Ans: 1340 to 1345 ● ○ ○

Load torque is constant,

$$T = \text{constant}$$

Under rated condition,

$$E_1 = V_t - I_a r_a$$

$$E_1 = 400 - (15)(1.2) = 382\,\text{V}$$

Since the motor is separately excited,

$$\phi = \text{constant}$$

When supply voltage is reduced by 10%,

$$V_{t2} = 400 - 400 \times \frac{10}{100} = 360\,\text{V}$$

For constant torque,

$$T \propto \phi I_a$$

Since ϕ is constant,

$$I_a = \text{constant} = 15\,\text{A}$$

$$E_2 = V_{t2} - I_a r_a$$

$$E_2 = 360 - (15)(1.2) = 342\,\text{V}$$

As ϕ = constant,

$$E \propto N$$

$$\frac{E_2}{E_1} = \frac{N_2}{N_1}$$

$$\frac{342}{382} = \frac{N_2}{1500}$$

$$N_2 = 1343\,\text{rpm}$$

1343

Key Points

- For a separately excited DC motor, flux remains constant if field excitation is unchanged.
- For constant torque, armature current remains constant.
- Speed is proportional to back emf when flux is constant.

Q3.22.1 NAT | 2022 | 2M | Ans: 483 | ● ○ ○

Given:

$$V = 280\,\text{V}, \quad R_a = 1\,\Omega, \quad R_{ext} = 10\,\Omega$$

$$I_{a1} = 30\,\text{A}, \quad N_1 = 1000\,\text{rpm}$$

For a separately excited DC motor with constant flux,

$$T \propto \phi I_a$$

Since flux is constant,

$$T \propto I_a$$

Also, given $T \propto N$,

$$\frac{T_2}{T_1} = \frac{N_2}{N_1}$$

Hence,

$$\frac{I_{a2}}{I_{a1}} = \frac{N_2}{N_1}$$

Substituting values,

$$I_{a2} = \frac{30}{1000} N_2$$

$$I_{a2} = 0.03 N_2$$

For constant flux,

$$N \propto E_b$$

Therefore,

$$\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}}$$

Using back emf equation,

$$\frac{N_2}{1000} = \frac{280 - I_{a2}(R_a + R_{ext})}{280 - I_{a1}R_a}$$

Substituting values,

$$\frac{N_2}{1000} = \frac{280 - 0.03N_2(1 + 10)}{280 - 30(1)}$$

$$\frac{N_2}{1000} = \frac{280 - 0.33N_2}{250}$$

$$250N_2 = 280000 - 330N_2$$

$$580N_2 = 280000$$

$$N_2 = 482.76\,\text{rpm}$$

$$\boxed{483}$$

Key Points

- For separately excited DC motors, flux remains constant.
- Torque is directly proportional to armature current.
- Speed is proportional to back emf when flux is constant.

Q3.21.1 NAT | 2021 | 2M | Ans: 275.10 to 275.30 | ● ● ○

Given:

$$P = 100\,\text{kW}, \quad V = 200\,\text{V}, \quad R_a = 0.025\,\Omega$$

$$R_{sh} = 50\,\Omega, \quad N_g = 300\,\text{rpm}$$

Current drawn from grid,

$$I_L = \frac{100 \times 10^3}{200} = 500\,\text{A}$$

Shunt field current,

$$I_{sh} = \frac{200}{50} = 4\,\text{A}$$

Generator armature current,

$$I_a = I_L + I_{sh} = 504\,\text{A}$$

Generated emf,

$$E_g = V + I_a R_a + \text{Brush drop}$$

$$E_g = 200 + 504(0.025) + 2$$

$$E_g = 214.6\,\text{V}$$

In motoring condition,

$$VI = 10\,\text{kW}$$

$$I_L = \frac{10000}{200} = 50\,\text{A}$$

Field current,

$$I_f = 4\,\text{A}$$

Motor armature current,

$$I_a = I_L - I_f = 46\,\text{A}$$

Back emf,

$$E_b = V - I_a R_a - \text{Brush drop}$$

$$E_b = 200 - 46(0.025) - 2$$

$$E_b = 196.85\,\text{V}$$

Since flux is constant,

$$\frac{N_m}{N_g} = \frac{E_b}{E_g}$$

$$N_m = \frac{E_b}{E_g} \times N_g$$

$$N_m = \frac{196.85}{214.6} \times 300$$

$$N_m = 275.19\,\text{rpm}$$

$$\boxed{275.19}$$

Key Points

- For a DC machine with constant flux, speed is proportional to back emf.
- In generator mode: $E_g = V + I_a R_a +$ brush drop.
- In motor mode: $E_b = V - I_a R_a -$ brush drop.

Q3.20.1 MCQ | 2020 | 2M | Ans: C | ● ○ ○

Given: No load current is 5 A

$$V = 250\,\text{V}, \quad R_a = 0.2\,\Omega, \quad R_{sh} = 100\,\Omega$$

$$I_L = 50\,\text{A}, \quad N_1 = 1200\,\text{rpm}$$

Brush resistance drop:

$$1\,\text{V per brush} \Rightarrow \text{Total B.R.D} = 2\,\text{V}$$

Shunt field current,

$$I_{sh} = \frac{250}{100} = 2.5\,\text{A}$$

At no-load,

$$I_{a0} = 2.5\,\text{A}$$

At loaded condition,

$$I_{aL} = I_L - I_{sh}$$

$$I_{aL} = 50 - 2.5 = 47.5\,\text{A}$$

Back emf at no-load,

$$E_{b1} = V - I_{a0}R_a - \text{B.R.D}$$

$$E_{b1} = 250 - 2.5(0.2) - 2$$

$$E_{b1} = 247.5\,\text{V}$$

Back emf at load,

$$E_{b2} = 250 - 47.5(0.2) - 2$$

$$E_{b2} = 238.5\,\text{V}$$

Using speed relation,

$$\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}} \times \frac{\phi_1}{\phi_2}$$

Given,

$$\phi_2 = 0.95\phi_1$$

$$\frac{N_2}{1200} = \frac{238.5}{247.5} \times \frac{\phi_1}{0.95\phi_1}$$

$$N_2 = 1217.22\,\text{rpm}$$

$\boxed{C}$

Key Points

- For DC motors, speed is proportional to $\frac{E_b}{\phi}$.
- Shunt field current remains nearly constant as it is connected across supply voltage.
- Brush drop must be included while calculating back emf.

Q3.19.1 NAT | 2019 | 2M | Ans: 1577 to 1583 | ● ● ○

Given:

$$V_t = 220\,\text{V}, \quad R_f = 220\,\Omega, \quad r_a = 0.5\,\Omega$$

$$N_1 = 1500\,\text{rpm}$$

Field current,

$$I_f = \frac{220}{220} = 1\,\text{A}$$

At full load,

$$I_L = 25\,\text{A}$$

Armature current,

$$I_{a1} = I_L - I_f$$

$$I_{a1} = 25 - 1 = 24\,\text{A}$$

Induced emf at full load,

$$E_1 = V_t - I_{a1} r_a$$

$$E_1 = 220 - 24(0.5)$$

$$E_1 = 208\,\text{V}$$

At no-load,

$$I_L = 3\,\text{A}$$

Armature current,

$$I_{a2} = I_L - I_f$$

$$I_{a2} = 3 - 1 = 2\,\text{A}$$

Induced emf at no-load,

$$E_2 = V_t - I_{a2} r_a$$

$$E_2 = 220 - 2(0.5)$$

$$E_2 = 219\,\text{V}$$

Since I_f is constant,

$$\phi = \text{constant}$$

Hence,

$$E \propto N$$

$$\frac{E_2}{E_1} = \frac{N_2}{N_1}$$

$$N_2 = \frac{219}{208} \times 1500$$

$$N_2 = 1579.32\,\text{rpm}$$

$$\boxed{1579.32}$$

Key Points

- In a DC shunt motor, field current is approximately constant.
- For constant flux, speed is directly proportional to induced emf.
- Induced emf of motor is given by $E = V_t - I_a r_a$.

Q3.18.1 NAT 2018 2M Ans: 825 ● ○ ○

At original condition,

$$I_{a1} = 10\,\text{A}$$

$$E_1 = V - I_{a1}R_a = 200 - 10(1) = 190\,\text{V}$$

When torque becomes 1.44 times,

$$T_2 = 1.44T_1$$

For a DC series motor with linear magnetization,

$$T \propto I_a^2$$

$$I_{a2}^2 = 1.44I_{a1}^2$$

$$I_{a2} = 1.2I_{a1} = 1.2(10) = 12\,\text{A}$$

$$E_2 = V - I_{a2}R_a = 200 - 12(1) = 188\,\text{V}$$

For a DC series motor,

$$E \propto \phi N$$

Since $\phi \propto I_a$,

$$\frac{E_2}{E_1} = \frac{I_{a2}}{I_{a1}} \times \frac{N_2}{N_1}$$

$$N_2 = \frac{188}{190} \times \frac{10}{12} \times 1000$$

$$N_2 = 824.56\,\text{rpm}$$

Therefore, nearest integer speed is

$$\boxed{825}$$

Key Points

- For a DC series motor with linear magnetization, $\phi \propto I_a$.
- Torque relation is $T \propto \phi I_a$, hence $T \propto I_a^2$.
- Back emf relation is $E \propto \phi N$.

Mistakes to be avoided

- Do not use $T \propto I_a$ for a series motor in the linear region.
- Do not forget that flux changes with armature current.
- Use $E = V - I_aR_a$ for motor operation.

Q3.18.2 NAT 2018 1M Ans: 600 ● ● ○

At zero speed,

$$\omega_m = 0$$

Hence,

$$E = k\phi\omega_m = 0$$

Armature current at starting,

$$I_a = \frac{V_t - E}{R_a}$$

$$I_a = \frac{100 - 0}{0.05} = 2000\,\text{A}$$

Given torque,

$$T = k\phi I_a$$

$$k\phi = \frac{500}{2000} = \frac{1}{4}$$

Since the motor is separately excited,

$$k\phi = \text{constant}$$

At no-load, neglecting mechanical losses,

$$I_a = 0$$

Therefore,

$$V_t = E = 150\,\text{V}$$

Using emf equation,

$$E = k\phi\omega_m$$

$$150 = \frac{1}{4}\omega_m$$

$$\omega_m = 600\,\text{rad/s}$$

$$\boxed{600}$$

Key Points

- At starting, back emf of a DC motor is zero.
- In a separately excited motor, flux remains constant.
- Back emf equation is $E = k\phi\omega_m$.

Mistakes to be avoided

- Do not assume non-zero back emf at zero speed.
- Use $I_a = \frac{V_t - E}{R_a}$, not $\frac{V_t}{R_a + E}$.
- At no-load with ignored losses, armature current is approximately zero.

Q3.17.1 NAT | 2017 | 2M | Ans: 10.75 | ● ○ ○

Initial back emf,

$$E_1 = V_t - I_a(R_a + R_{se})$$

$$E_1 = 220 - 30(0.5)$$

$$E_1 = 205\,\text{V}$$

If speed is reduced by 50%,

$$\frac{N_2}{N_1} = 0.5$$

For fan-type load,

$$T \propto N^2$$

$$\frac{T_2}{T_1} = \left(\frac{N_2}{N_1}\right)^2 = (0.5)^2 = 0.25$$

For a DC series motor with linear magnetization,

$$T \propto I_a^2$$

$$\left(\frac{I_{a2}}{I_{a1}}\right)^2 = 0.25$$

$$\frac{I_{a2}}{I_{a1}} = 0.5$$

$$I_{a2} = 0.5(30) = 15\,\text{A}$$

For series motor,

$$E \propto \phi N \propto I_a N$$

$$\frac{E_2}{E_1} = \frac{I_{a2}}{I_{a1}} \times \frac{N_2}{N_1}$$

$$E_2 = 205 \times 0.5 \times 0.5$$

$$E_2 = 51.25\,\text{V}$$

Total resistance required,

$$R_2 = \frac{V - E_2}{I_{a2}}$$

$$R_2 = \frac{220 - 51.25}{15}$$

$$R_2 = 11.25\,\Omega$$

External resistance,

$$R_{ext} = R_2 - (R_a + R_{se})$$

$$R_{ext} = 11.25 - 0.5$$

$$R_{ext} = 10.75\,\Omega$$

$$\boxed{10.75}$$

Key Points

- For fan-type loads, torque varies as square of speed.
- In a DC series motor with linear magnetization, $T \propto I_a^2$.

- Back emf of a series motor is proportional to I_aN.

Q3.17.2 NAT | 2017 | 2M | Ans: 550 | ●●○

By KVL,

$$E_1 = V_t + I_aR_a$$

$$E_1 = 145 + 150(0.1)$$

$$E_1 = 160\,\text{V}$$

For a separately excited DC machine,

$$\phi = \text{constant}$$

Hence,

$$E \propto N$$

$$\frac{E_2}{E_1} = \frac{N_2}{N_1}$$

$$E_2 = 160 \times \frac{1000}{800}$$

$$E_2 = 200\,\text{V}$$

Armature current,

$$I_a = \frac{E_2 - V}{R_a}$$

$$I_a = \frac{200 - 145}{0.1}$$

$$I_a = 550\,\text{A}$$

$$\boxed{550}$$

Key Points

- In a separately excited DC machine, flux remains constant.
- Back emf is directly proportional to speed when flux is constant.
- Generator equation is $E = V + I_aR_a$.

Mistakes to be avoided

- Do not use motor emf equation $E = V - I_aR_a$ for generator operation.
- Keep speed ratio in correct order while using $\frac{E_2}{E_1} = \frac{N_2}{N_1}$.
- Do not forget unit conversion while calculating current.

Q3.17.3 MCQ 2017 2M Ans: B ● ○ ○

In a separately excited DC machine,

$$\phi = \text{constant}$$

Hence,

$$T \propto I_a$$

$$T = k_m I_a$$

Given,

$$70 = k_m I_a$$

$$I_a = \frac{70}{k_m} \quad (1)$$

Back emf equation,

$$E = k_m \omega$$

At 900 rpm,

$$\omega = 900 \times \frac{2\pi}{60}$$

$$\omega = 30\pi \text{ rad/s}$$

$$E = k_m (30\pi)$$

By KVL,

$$E = V - I_a R_a$$

$$k_m (30\pi) = 220 - \frac{70}{k_m}(0.02)$$

$$30\pi k_m^2 = 220 k_m - 1.4$$

$$30\pi k_m^2 - 220 k_m + 1.4 = 0$$

$$k_m = 2.3278$$

Using Eq. (1),

$$I_a = \frac{70}{2.3278}$$

$$I_a = 30.07 \text{ A}$$

$$\boxed{B}$$

Q3.17.4 NAT | 2017 | 2M | Ans: 1241.5 to 1242.0 | ● ○ ○

Shunt field current,

$$I_f = \frac{V_t}{R_f}$$

$$I_f = \frac{120}{240} = 0.5\,\text{A}$$

No-load armature current,

$$I_{a(NL)} = 2 - 0.5$$

$$I_{a(NL)} = 1.5\,\text{A}$$

Full-load armature current,

$$I_{a(FL)} = 7 - 0.5$$

$$I_{a(FL)} = 6.5\,\text{A}$$

Induced emf at no-load,

$$E_{NL} = 120 - 1.5(0.8)$$

$$E_{NL} = 118.8\,\text{V}$$

Induced emf at full-load,

$$E_{FL} = 120 - 6.5(0.8)$$

$$E_{FL} = 114.8\,\text{V}$$

Since field current is constant,

$$\phi = \text{constant}$$

Hence,

$$E \propto N$$

$$\frac{E_{NL}}{E_{FL}} = \frac{N_{NL}}{N_{FL}}$$

$$N_{NL} = \frac{118.8}{114.8} \times 1200$$

$$N_{NL} = 1241.81\,\text{rpm}$$

$$\boxed{1241.81}$$

Q3.16.1 MCQ 2016 1M Ans: C ● ○ ○

For lap winding,

$$A = P = 4$$

Current in each parallel path,

$$I = \frac{I_a}{A}$$

$$I = \frac{40}{4} = 10\,\text{A}$$

Armature speed,

$$600\,\text{rpm} = 10\,\text{rps}$$

Time for one revolution,

$$T = \frac{1}{10} = 0.1\,\text{s} = 100\,\text{ms}$$

Since there are 4 poles, time to cover one pole pitch is

$$\frac{100}{4} = 25\,\text{ms}$$

Assuming linear commutation, current reverses from 10 A to −10 A.
Hence, half cycle duration is

$$25\,\text{ms}$$

$$\boxed{C}$$

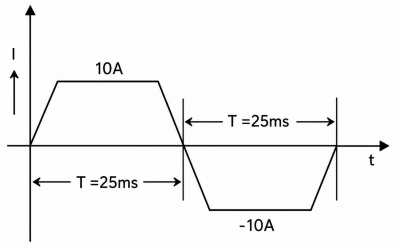


Fig. Solution Q3.16.1

Key Points

- In lap winding, number of parallel paths equals number of poles.
- Pole pitch time is obtained by dividing revolution time by number of poles.
- Current reversal occurs during commutation.

Mistakes to be avoided

- Do not confuse rpm with rps.
- For lap winding, use $A = P$, not $A = 2$.
- Half cycle corresponds to one pole pitch, not full revolution.

Q3.16.2 NAT | 2016 | 2M | Ans: 86.84 | ● ○ ○

The circuit diagram of DC shunt generator is shown below.

$$I_f = \frac{V_t}{R_f} = \frac{220}{44} = 5\,\text{A}$$

By KCL,

$$I_a = I_L + I_f = 45 + 5 = 50\,\text{A}$$

By KVL,

$$E_g = V_t + I_a R_a$$

$$E_g = 220 + 50 \times 0.01 = 220.5\,\text{V}$$

Developed power,

$$P_{\text{dev}} = E_g I_a = 220.5 \times 50 = 11025\,\text{W}$$

Mechanical input power,

$$P_m = P_{\text{dev}} + P_{\text{rot}}$$

$$P_m = 11025 + 375 = 11400\,\text{W}$$

Electrical output power,

$$P_e = 220 \times 45 = 9900\,\text{W}$$

Efficiency,

$$\eta = \frac{P_e}{P_m} \times 100$$

$$\eta = \frac{9900}{11400} \times 100 = 86.84\%$$

86.84

Key Points

- In a DC shunt generator, armature current is given by $I_a = I_L + I_f$.
- Generated emf equation is $E_g = V_t + I_a R_a$.
- Efficiency is the ratio of electrical output power to mechanical input power.
- Developed electrical power is calculated using $P_{\text{dev}} = E_g I_a$.

Mistakes to be avoided

- Do not use $I_a = I_L$ for a shunt generator; field current must also be included.
- Do not subtract armature drop in generator mode; use $E_g = V_t + I_a R_a$.
- Do not forget to include rotational losses while calculating mechanical input power.
- Ensure efficiency is multiplied by 100 to express it in percentage.

Q3.15.1 MCQ | 2015 | 1M | Ans: B | ● ○ ○

Induced emf in a DC generator is given by

$$E_g = \frac{\phi NZ}{60}\left(\frac{P}{A}\right)$$

Hence,

$$E_g \propto \phi N$$

Therefore,

$$\frac{E_{g2}}{E_{g1}} = \frac{\phi_2 N_2}{\phi_1 N_1}$$

Since field current and speed are halved,

$$\frac{\phi_2}{\phi_1} = 0.5 \quad \text{and} \quad \frac{N_2}{N_1} = 0.5$$

$$\frac{E_{g2}}{E_{g1}} = 0.5 \times 0.5 = 0.25$$

$$E_{g2} = 200 \times 0.25 = 50\,\text{V}$$

For charging RC circuit,

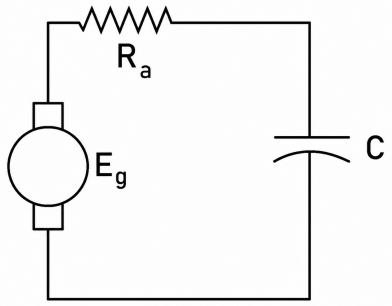


Fig. Solution Q3.15.1

$$V_c = E_g\left(1 - e^{-t/RC}\right)$$

$$25 = 50\left(1 - e^{-t/RC}\right)$$

$$e^{-t/RC} = 0.5$$

$$t = -RC\ln(0.5)$$

$$t = 69.3\,\mu\text{sec}$$

$$\boxed{B}$$

Key Points

- Generated emf of a DC generator is proportional to flux and speed, i.e., $E_g \propto \phi N$.
- In an RC charging circuit, capacitor voltage is given by $V_c = E_g\left(1 - e^{-t/RC}\right)$.
- Halving both flux and speed reduces generated emf to one-fourth of its original value.
- Time constant of RC circuit is $\tau = RC$.

Mistakes to be avoided

- Do not forget that both flux and speed are halved; emf becomes 0.25 times, not 0.5 times.
- Do not use discharging capacitor equation in place of charging equation.
- Ensure natural logarithm ln is used while solving exponential equations.
- Be careful with unit conversion; final answer is in μsec.

Q3.15.2 NAT | 2015 | 2M | Ans: 8 | ● ○ ○

Under no load,

$$I_a = 0$$

Therefore,

$$V_t = E_b$$

$$E_{b1} = 200\,\text{V} \quad \text{at} \quad N_1 = 1000\,\text{rpm}$$

Since the motor is separately excited, flux can be assumed constant.

$$E_b \propto N$$

$$\frac{E_{b2}}{E_{b1}} = \frac{N_2}{N_1} = \frac{500}{1000}$$

$$E_{b2} = 100\,\text{V}$$

For rated torque, let armature current be I_a.
Since ϕ is constant,

$$T \propto I_a$$

At half rated torque,

$$I_{a2} = \frac{I_a}{2}$$

At $N_3 = 520\,\text{rpm}$,

$$\frac{E_{b3}}{E_{b1}} = \frac{N_3}{N_1} = \frac{520}{1000}$$

$$E_{b3} = 104\,\text{V}$$

Using voltage equation,

$$V_t = E_{b2} + I_a R_a$$

$$V_t = E_{b3} + \frac{I_a}{2} R_a$$

Substituting values,

$$100 + I_a R_a = 104 + \frac{I_a}{2} R_a$$

$$4 = \frac{I_a}{2} R_a$$

$$I_a = \frac{8}{R_a}$$

Given $R_a = 1\,\Omega$,

$$I_a = 8\,\text{A}$$

Therefore, full load armature current is

$$\boxed{8}$$

Key Points

- For separately excited DC motors, flux ϕ remains constant.
- Back emf is directly proportional to speed, i.e., $E_b \propto N$.
- Torque in a separately excited motor is proportional to armature current, i.e., $T \propto I_a$.
- Motor voltage equation is $V_t = E_b + I_a R_a$.

Mistakes to be avoided

- Do not assume varying flux in separately excited motors unless specified.
- Do not forget that half torque implies half armature current when flux is constant.
- Be careful while applying proportionality between back emf and speed.
- Ensure correct substitution in voltage equations before simplifying.

Q3.15.3 NAT | 2015 | 2M | Ans: 57.789 | ● ○ ○

Given,

$$\phi = 0.01\,\text{Wb}, \quad N = 1000\,\text{rpm}, \quad Z = 666, \quad A = 2, \quad P = 4$$

Induced emf of DC motor is

$$E_b = \frac{\phi N Z}{60}\left(\frac{P}{A}\right)$$

Substituting the values,

$$E_b = \frac{0.01 \times 1000 \times 666}{60}\left(\frac{4}{2}\right)$$

$$E_b = \frac{6660}{60} \times 2 = 222\,\text{V}$$

Armature current,

$$I_a = \frac{V_t - E_b}{R_a}$$

$$I_a = \frac{230 - 222}{0.267} = 29.9625\,\text{A}$$

Developed power,

$$P_{\text{dev}} = E_b I_a$$

$$P_{\text{dev}} = 222 \times 29.9625 = 6651.685\,\text{W}$$

Shaft power,

$$P_{\text{sh}} = P_{\text{dev}} - \text{Rotational loss}$$

$$P_{\text{sh}} = 6651.685 - 600 = 6051.685\,\text{W}$$

Angular speed,

$$\omega = \frac{2\pi N}{60}$$

$$\omega = \frac{2\pi}{60} \times 1000$$

Shaft torque,

$$T_{\text{sh}} = \frac{P_{\text{sh}}}{\omega}$$

$$T_{\text{sh}} = \frac{6051.685}{\left(\frac{2\pi}{60} \times 1000\right)} = 57.789\,\text{N m}$$

$$\boxed{57.789}$$

Mistakes to be avoided

- Do not use terminal voltage directly in developed power calculation; use back emf.

Q3.15.4 NAT | 2015 | 1M | Ans: 179.5 to 180.5 | ● ● ○

The speed-torque characteristic of the DC shunt motor is represented by a straight line.

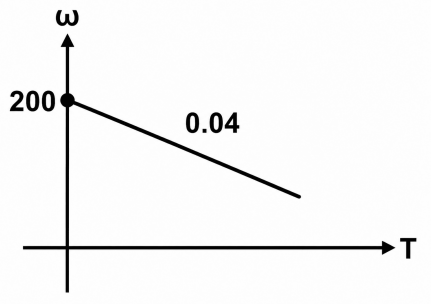


Fig. Solution Q3.15.4

Hence, the equation of the straight line is

$$\omega = 200 - \frac{(200 - 180)}{500}T$$

$$\omega = 200 - \frac{20}{500}T$$

$$\omega = 200 - 0.04T$$

The load torque is given by

$$T_L = 2.78\omega$$

Under steady-state operation, neglecting rotational losses, the motor torque equals the load torque.

$$T = T_L = 2.78\omega$$

Substituting in the speed–torque equation,

$$\omega = 200 - 0.04(2.78\omega)$$

$$\omega = 200 - 0.1112\omega$$

$$1.1112\omega = 200$$

$$\omega = \frac{200}{1.1112}$$

$$\omega = 179.985 \text{ rad/s}$$

$$\boxed{179.985}$$

Q3.15.5 MCQ 2015 1M Ans: B ● ○ ○

Induced emf of a DC machine is

$$E = \frac{\phi NZ}{60}\left(\frac{P}{A}\right)$$

For lap connected armature,

$$P = A = 4$$

For wave connected armature,

$$A = 2$$

Since all other quantities remain same,

$$E \propto \frac{P}{A}$$

Therefore,

$$\frac{E_{\text{lap}}}{E_{\text{wave}}} = \frac{\left(\frac{P}{A}\right)_{\text{lap}}}{\left(\frac{P}{A}\right)_{\text{wave}}}$$

$$\frac{E_{\text{lap}}}{E_{\text{wave}}} = \frac{\frac{4}{4}}{\frac{4}{2}} = \frac{1}{2}$$

$$E_{\text{lap}} = \frac{1}{2}E_{\text{wave}}$$

$$E_{\text{lap}} = 115\ \text{V}$$

In lap connection, the number of parallel paths is doubled, so current rating is doubled.

$$I_{\text{lap}} = 2I_{\text{wave}}$$

Power rating,

$$P = EI = \text{constant}$$

Therefore, power rating remains

$$5\ \text{kW}$$

$$\boxed{B}$$

Key Points

- DC machine induced emf depends on $\frac{P}{A}$.
- For lap winding, $A = P$.
- For wave winding, $A = 2$.
- Converting from wave to lap halves the voltage but doubles the current rating.

Mistakes to be avoided

- Do not assume voltage rating remains same after changing winding connection.
- Do not forget that lap winding increases the number of parallel paths.
- Power rating remains unchanged because voltage halves and current doubles.

Q3.15.6 NAT 2015 2M Ans: 47.5 ● ○ ○

With an armature voltage of 100 V and rated field winding voltage, the speed of a separately excited DC motor driving a fan is 1000 rpm, and its armature current is 10 A. The armature resistance is 1 Ω. The load torque of the fan load is proportional to the square of the rotor speed. Neglecting rotational losses, the value of the armature voltage in Volt which will reduce the rotor speed to 500 rpm is _____.

Since the field voltage is rated,

$$\phi = \text{constant}$$

Initial back emf,

$$E_{b1} = V_1 - I_{a1}R_a = 100 - 10(1) = 90\,\text{V}$$

For fan load,

$$T \propto N^2$$

$$\frac{T_2}{T_1} = \left(\frac{500}{1000}\right)^2 = \frac{1}{4}$$

Since ϕ is constant,

$$T \propto I_a$$

$$I_{a2} = \frac{10}{4} = 2.5\,\text{A}$$

Also,

$$E_b \propto N$$

$$E_{b2} = 90 \times \frac{500}{1000} = 45\,\text{V}$$

Required armature voltage,

$$V_2 = E_{b2} + I_{a2}R_a$$

$$V_2 = 45 + 2.5(1) = 47.5\,\text{V}$$

47.5

Q3.14.1 NAT 2014 2M Ans: 1240.63 ● ○ ○

From the circuit diagram shown,

$$I_L = I_f + I_a$$

At no load,

$$I_L = 5\,\text{A}$$

$$I_f = \frac{V}{R_f} = \frac{230}{230} = 1\,\text{A}$$

$$I_a = I_L - I_f = 5 - 1 = 4\,\text{A}$$

Back emf at no load,

$$(E_b)_{NL} = V - I_aR_a$$

$$(E_b)_{NL} = 230 - 4 \times 0.4 = 228.4\,\text{V}$$

Under full load,

$$I_L = 70\,\text{A}$$

Since terminal voltage is constant, field current remains constant.

$$I_a = 70 - 1 = 69\,\text{A}$$

Back emf at full load,

$$(E_b)_{FL} = V - I_a R_a$$

$$(E_b)_{FL} = 230 - 69 \times 0.4 = 202.4\,\text{V}$$

We know,

$$E_b = K\phi\omega_m$$

Since flux is constant,

$$\frac{(E_b)_{FL}}{(E_b)_{NL}} = \frac{N_{FL}}{N_{NL}}$$

$$\frac{N_{FL}}{1400} = \frac{202.4}{228.4}$$

$$N_{FL} = 1400 \times \frac{202.4}{228.4} = 1240.63\,\text{rpm}$$

1240.63

Q3.14.2 NAT 2014 2M Ans: 1.27

When the machine operates as a generator,

$$I_a = I_L + I_f$$

$$I_f = \frac{V_t}{R_f} = \frac{250}{125} = 2\,\text{A}$$

$$I_a = 50 + 2 = 52\,\text{A}$$

Generated emf,

$$E_g = V_t + I_a R_a$$

$$E_g = 250 + 52(0.6) = 281.2\,\text{V}$$

When the machine operates as a motor,

$$I_a = I_L - I_f = 50 - 2 = 48\,\text{A}$$

Back emf,

$$E_b = V_t - I_a R_a$$

$$E_b = 250 - 48(0.6) = 221.2\,\text{V}$$

Since I_f is constant,

$$\phi = \text{constant}$$

Hence,

$$N \propto E$$

$$\frac{N_m}{N_g} = \frac{E_b}{E_g}$$

$$\frac{N_g}{N_m} = \frac{E_g}{E_b} = \frac{281.2}{221.2}$$

$$\frac{N_g}{N_m} = 1.27$$

$$\boxed{1.27}$$

Key Points

- In generator operation, $I_a = I_L + I_f$.
- In motor operation, $I_a = I_L - I_f$.
- Generator emf equation is $E_g = V_t + I_a R_a$.
- Motor back emf equation is $E_b = V_t - I_a R_a$.

Q3.14.3 NAT | 2014 | 1M | Ans: 12.55 | ● ○ ○

Since the motor is separately excited, the flux is constant.

$$\omega = \frac{E_b}{K\phi} = \frac{E_b}{K'}$$

At no load,

$$V_t = E_b$$

$$K' = \frac{E_b}{\omega} = \frac{V_t}{\omega}$$

$$K' = \frac{230}{1400\left(\frac{2\pi}{60}\right)} = 1.568$$

Torque developed is,

$$T = K\phi I_a = K' I_a$$

$$T = 1.568 \times 8 = 12.55 \text{ N-m}$$

$$\boxed{12.55}$$

Q3.14.4 NAT | 2014 | 2M | Ans: 880.3 | ● ○ ○

Terminal voltage equation,

$$V_t = E_b + I_a R_a$$

Under no-load condition,

$$300 = E_{b1} + 2(0.5)$$

$$E_{b1} = 299 \text{ V}$$

Under loaded condition,

$$300 = E_{b2} + 15(0.5)$$

$$E_{b2} = 292.5 \text{ V}$$

Since the machine is separately excited, flux remains constant.

$$N \propto E_b$$

$$\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}}$$

$$N_2 = \frac{292.5}{299} \times 900$$

$$N_2 = 880.43 \text{ rpm}$$

$$\boxed{880.43}$$

Q3.12.1 MCQ | 2012 | 2M | Ans: D | ● ○ ○

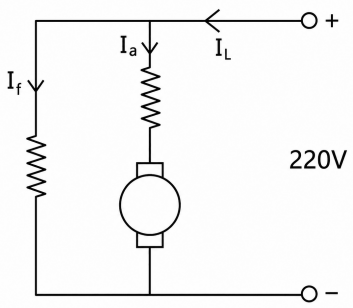


Fig. Solution Q3.12.1

At rated condition,

$$I_L = I_a + I_f$$

$$68 = I_a + 2.2$$

$$I_{a1} = 65.8\,\text{A}$$

By KVL,

$$V_t = E_a + I_a R_a$$

$$220 = E_{a1} + 65.8 \times 0.25$$

$$E_{a1} = 220 - \frac{65.8}{4} = 203.55\,\text{V}$$

Under loaded condition,

$$I_L = 52.8\,\text{A}$$

$$I_f = 1.8\,\text{A}$$

$$I_{a2} = I_L - I_f = 52.8 - 1.8 = 51\,\text{A}$$

Again, by KVL,

$$220 = E_{a2} + 51 \times 0.25$$

$$E_{a2} = 207.25\,\text{V}$$

Since,

$$E_a \propto \phi N$$

$$\frac{E_{a1}}{E_{a2}} = \frac{\phi_1}{\phi_2} \times \frac{N_1}{N_2}$$

$$\frac{203.55}{207.25} = \frac{\phi_1}{\phi_2} \times \frac{1000}{1600}$$

$$\frac{\phi_2}{\phi_1} = \frac{1000}{1600} \times \frac{207.25}{203.55} = 0.6363$$

$$\phi_2 = 63.63\% \text{ of } \phi_1$$

Hence, percentage decrease in flux is,

$$(100 - 63.63)\% = 36.36\%$$

$$\boxed{D}$$

Q3.11.1 MCQ 2011 1M Ans: B ● ○ ○

A point starter in combined armature voltage control and flux control can control the speed of DC shunt motor above and below base speed.

Flux weakening control is used for speeds above base speed and armature voltage control for speeds below base speed.

B

Key Points

- Armature voltage control is used for speeds below base speed.
- Flux weakening control is used for speeds above base speed.
- Combined control provides a wide range of speed control in DC shunt motors.
- Base speed corresponds to rated voltage and rated flux condition.

Q3.11.2 MCQ 2011 2M Ans: A ● ● ○

Since excitation of the machine is reduced by 10%, the flux is also reduced by 10%.

$$E_a \propto \phi N$$

Since speed is constant,

$$E_a \propto \phi$$

Also,

$$T \propto \phi I_a$$

Since torque is constant,

$$I_a \propto \frac{1}{\phi}$$

Initially,

$$R_{a1} = 1\,\Omega, \qquad I_{a1} = 10\,\text{A}$$

$$E_{a1} = V_t - I_{a1}R_{a1}$$

$$E_{a1} = 220 - 10 \times 1 = 210\,\text{V}$$

Since flux is reduced by 10%,

$$\frac{E_{a2}}{E_{a1}} = \frac{\phi_2}{\phi_1} = 0.9$$

$$E_{a2} = 0.9 \times 210 = 189\,\text{V}$$

Also,

$$\frac{I_{a2}}{I_{a1}} = \frac{\phi_1}{\phi_2} = \frac{1}{0.9}$$

$$I_{a2} = \frac{10}{0.9} = 11.11\,\text{A}$$

Now,

$$V_t = E_{a2} + I_{a2}R_2$$

$$220 = 189 + (11.11)R_2$$

$$R_2 = 2.79\,\Omega$$

Hence, external resistance added is

$$R_{\text{ext}} = 2.79 - 1 = 1.79\,\Omega$$

$$\boxed{A}$$

Q3.10.1 MCQ | 2010 | 2M | Ans: B | ● ○ ○

For separately excited machine, at no-load,

$$P = 0, \qquad I_a = 0$$

Hence,

$$V_t = E_a + I_aR_a = E_a$$

Given induced emf is 200 V at 1500 rpm.

$$E = K_m\phi\omega$$

$$K_m\phi = \frac{E}{\omega}$$

$$K_m\phi = \frac{200}{1500\left(\frac{2\pi}{60}\right)} = 1.273$$

Since field voltage is unchanged,

$$K_m\phi = \text{constant}$$

Torque equation,

$$T = (K_m\phi)I_a$$

$$5 = 1.273I_a$$

$$I_a = 3.927\,\text{A}$$

Under loaded condition,

$$N = 1400\,\text{rpm}$$

Back emf,

$$E_a = 1.273 \times 1400 \times \frac{2\pi}{60}$$

$$E_a = 186.63\,\text{V}$$

Armature resistance,

$$R_a = \frac{V_t - E_a}{I_a}$$

$$R_a = \frac{200 - 186.63}{3.927}$$

$$R_a = 3.4\,\Omega$$

$\boxed{B}$

Q3.10.2 MCQ | 2010 | 2M | Ans: B ● ○ ○

Since,

$$K_m\phi = \text{constant}$$

Therefore,

$$T \propto I_a$$

When torque is halved, armature current is also halved.

$$I_a = \frac{3.927}{2} = 1.9635\,\text{A}$$

Since speed is constant, back emf remains constant.

$$E_a = 186.63\,\text{V}$$

Using motor voltage equation,

$$V_t = E_a + I_a R_a$$

$$V_t = 186.63 + 1.9635 \times 3.4$$

$$V_t = 193.3\,\text{V}$$

$\boxed{B}$

Q3.09.1 MCQ 2009 2M Ans: A ●●○

The given diagram is for motor operation. If we analyse the induced currents in the armature, the developed torque and direction of rotation is same.so, we can say that the given diagram is about a motor and the brushes are shifted in the direction of rotation.

In motor operation, when brushes are shifted in the direction of rotation, the leading pole tips will be strengthened and the trailing pole tips will be weakened.

Due to this unequal flux distribution, the voltage waveform $V_{A'B'}$ becomes distorted and non-symmetrical. Hence, the waveform resembles option *A*.

$\boxed{A}$

Key Points

- In motor operation, brush shift in the direction of rotation affects armature reaction.
- Leading pole tips are strengthened.
- Trailing pole tips are weakened.
- Unequal flux distribution causes distorted voltage waveform.

Mistakes to be avoided

- Do not confuse motor brush shift effect with generator brush shift effect.

Q3.08.1 MCQ 2008 2M Ans: D ●○○

Input current,

$$I_{in} = 15\,\text{A}$$

Field current,

$$I_f = \frac{240}{80} = 3\,\text{A}$$

Armature current,

$$I_a = I_{in} - I_f = 15 - 3 = 12\,\text{A}$$

Induced emf,

$$E_a = V_t - I_a R_a$$

$$E_a = 240 - 12(0.5) = 234\,\text{V}$$

During plugging, armature terminals are reversed.

Therefore, voltage across armature resistance is

$$V_{R_a} = V_t + E_a$$

$$V_{R_a} = 240 + 234 = 474\,\text{V}$$

$\boxed{D}$

Key Points

- During plugging, supply voltage and back emf act in the same direction across armature resistance.

Q3.08.2 MCQ 2008 2M Ans: A ● ○ ○

If an external resistance is added to the armature circuit, the net armature resistance becomes:

$$R_{\text{net}} = R_a + R_{\text{ex}}$$

During plugging (or reverse current braking), the supply voltage polarity is reversed to oppose the motion. The total voltage causing the plugging current is the sum of the supply voltage and the back EMF ($V_t + E_a$).

The plugging armature current I'_a is given by:

$$I'_a = \frac{V_t + E_a}{R_a + R_{\text{ex}}} = 1.25 \times I_{a\,(\text{rated})}$$

Substituting the given values from the problem statement:

$$\frac{474}{R_a + R_{\text{ex}}} = 1.25 \times 12$$

$$\frac{474}{R_a + R_{\text{ex}}} = 15$$

Solving for the total circuit resistance:

$$R_a + R_{\text{ex}} = \frac{474}{15} = 31.6\,\Omega$$

Given that the internal armature resistance $R_a = 0.5\,\Omega$, the external resistance to be added is:

$$R_{\text{ex}} = 31.6 - 0.5 = 31.1\,\Omega$$

A

Key Points

- During plugging of a DC motor, the armature connections are reversed so that the back EMF E_a and the supply voltage V_t act in the same direction.
- Total loop voltage during plugging is equal to $V_t + E_a$.
- External series resistance R_{ex} is always inserted during braking to limit the high initial plugging current and protect the winding.

Q3.07.1 MCQ 2007 1M Ans: D ● ○ ○

For zero speed regulation, the motor speed must remain constant from no-load to full-load.

For a DC motor,

$$N \propto \frac{E_b}{\phi}$$

$$E_b = V - I_a R_a$$

At full-load, I_a increases, so the armature voltage drop $I_a R_a$ increases and E_b decreases.

For speed to remain constant, the flux ϕ must also decrease in the same proportion.

This condition is achieved in a differential compound motor, because the series field opposes the shunt field. Hence, as load increases, the net flux decreases and compensates for the fall in back emf.

$\boxed{D}$

Key Points

- Zero speed regulation means no change in speed from no-load to full-load.
- In a differential compound motor, net flux decreases with increase in load.
- Decrease in flux compensates the decrease in back emf, keeping speed nearly constant.

Mistakes to be avoided

- Do not choose DC shunt motor; it gives good but not zero speed regulation.
- Do not choose DC series motor; it has very poor speed regulation.
- Remember that differential compounding reduces flux with load.

Q3.06.1 MCQ 2006 1M Ans: B ● ○ ○

The compensating winding and interpoles perform different functions in a DC machine.

Compensating windings are embedded in the pole faces and neutralize the cross-magnetizing effect of armature reaction under the main poles.

Interpoles (commutating poles) are placed along the interpolar axis and produce a commutating emf that neutralizes the reactance voltage induced in the coil undergoing commutation.

Hence, the correct statement corresponds to option B.

$\boxed{B}$

Key Points

- Compensating windings are located under the main pole faces.
- Their primary function is to neutralize armature reaction under the pole arc.
- Interpoles are placed between the main poles on the interpolar axis.
- Interpoles improve commutation by counteracting the reactance voltage in the commutating coil.

Q3.06.2 MCQ 2006 2M Ans: A ● ○ ○

When the machine is operated as a generator,

$$E_1 = V_t + I_a r_a$$

$$E_1 = 200 + (20)(0.2) = 204\,\text{V}$$

When the machine is operated as a motor,

$$E_2 = V_t - I_a r_a$$

$$E_2 = 200 - (20)(0.2) = 196\,\text{V}$$

The flux in motor operation is 10% more than that in generator operation.

$$\phi_2 = 1.1\,\phi_1$$

Since

$$E \propto \phi N$$

$$\frac{E_2}{E_1} = \frac{\phi_2}{\phi_1} \cdot \frac{N_2}{N_1}$$

Therefore,

$$\frac{N_2}{N_1} = \frac{E_2}{E_1} \cdot \frac{\phi_1}{\phi_2}$$

$$\frac{N_2}{N_1} = \frac{196}{204} \cdot \frac{1}{1.1}$$

$$\frac{N_2}{N_1} = 0.873$$

Therefore, the correct option is

$$\boxed{A}$$

Q3.05.1 MCQ | 2005 | 2M | Ans: A | ● ○ ○

Assume the poles are stationary and the rotor rotates in the anti-clockwise direction. This causes the same relative motion between conductors and magnetic field.

The induced emf in a conductor is given by

$$e = |\vec{V} \times \vec{B}|$$

The magnetic field is directed from North pole to South pole, i.e., from right to left.

For conductors belonging to coil C_2, the velocity of conductors is parallel or anti-parallel to the magnetic field. Hence,

$$\vec{V} \times \vec{B} = 0$$

and no emf is induced in coil C_2.

For conductors belonging to coil C_1, the conductor velocity is perpendicular to the magnetic field. Therefore,

$$|\vec{V} \times \vec{B}|$$

is maximum and the induced emf in coil C_1 is maximum.

Using Fleming's right-hand rule, the direction of induced emf is outward for conductor C_1 and inward for the opposite conductor of the same coil.

Hence, the correct option is

$$\boxed{A}$$

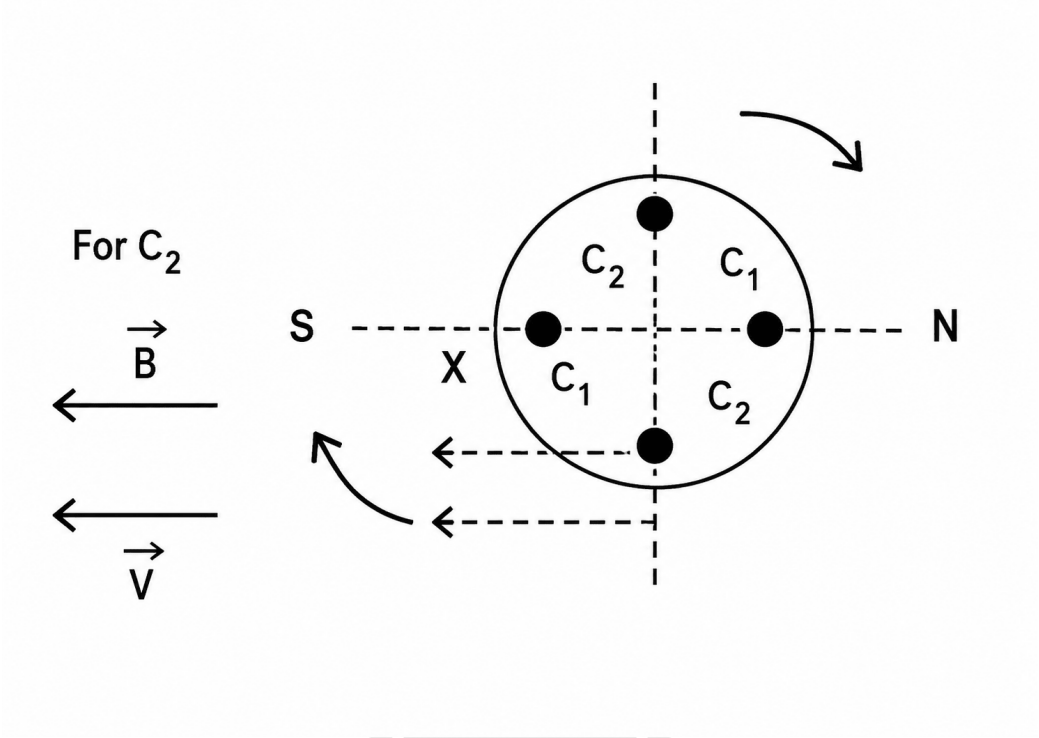


Fig. Solution Q3.05.1

Key Points

- Magnitude of induced emf is proportional to the cross product of velocity and magnetic field:

$$e \propto |\vec{V} \times \vec{B}|$$

- Maximum emf is induced when the conductor motion is perpendicular to the magnetic field.
- Zero emf is induced when the conductor motion is parallel to the magnetic field.
- Fleming's right-hand rule determines the direction of induced emf in a generator.

Mistakes to be avoided

- Do not confuse the direction of rotor rotation with the direction of magnetic field.
- Remember that conductors moving parallel to the magnetic field do not cut flux lines.
- Apply Fleming's right-hand rule only after identifying the directions of $\vec{V}$ and $\vec{B}$ correctly.
- Do not assume all conductors induce equal emf; conductor position is important.

Q3.05.2 MCQ 2005 2M Ans: B ●○○

In armature voltage control, the DC motor operates as a constant torque drive.

$$P = T\omega$$

Since torque remains constant,

$$P \propto \omega$$

At half the rated speed,

$$P_2 = \frac{1}{2}P_1 = \frac{1}{2} \times 50 = 25\,\text{kW}$$

In field control, the motor operates as a constant power drive. Therefore,

$$P = 50\,\text{kW}$$

$\boxed{B}$

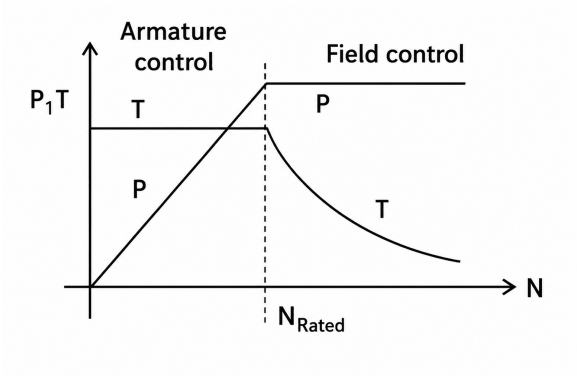


Fig. Solution Q3.05.2

Key Points

- Armature voltage control is used below rated speed and provides constant torque operation.
- For constant torque drives, $P = T\omega \Rightarrow P \propto \omega$.
- Field control is used above rated speed and provides constant power operation.

Mistakes to be avoided

- Do not confuse field control with constant torque operation.
- At half speed under constant torque operation, power becomes half.

Q3.05.3 MCQ | 2005 | 2M | Ans: D | ● ○ ○

For a DC machine,
(P) Induced emf is

$$E = k\phi\omega$$

Hence,

$$E \propto \phi\omega$$

Therefore,

$$P \rightarrow 2$$

(Q) Torque developed is

$$T = k\phi I_a$$

Hence,

$$T \propto \phi I_a$$

Therefore,

$$Q \rightarrow 3$$

(R) Power developed is

$$P = T\omega = E_b I_a$$

Substituting

$$E_b = k\phi\omega$$

gives

$$P = (k\phi\omega) I_a$$

$$P \propto \phi\omega I_a$$

Therefore,

$$R \rightarrow 1$$

Thus, the correct matching is

$$P \rightarrow 2, \quad Q \rightarrow 3, \quad R \rightarrow 1$$

$$\boxed{D}$$

Q3.04.1 MCQ | 2004 | 1M | Ans: C

A permanent magnet stepper motor has a rotor made of ferrite or rare-earth material that is permanently magnetized. The motor rotates through a fixed angle, called the *step angle*, for each electrical pulse received by its control unit.

Since the rotor position is directly related to the number of input pulses, position and speed control can be achieved without feedback sensors.

$$\boxed{C}$$

Key Points

- The rotor of a permanent magnet stepper motor is permanently magnetized.
- Each input pulse produces a discrete angular displacement called a step.
- Stepper motors convert digital pulse inputs into mechanical motion.
- Open-loop operation eliminates the need for position or speed sensors in many applications.

Q3.04.2 MCQ | 2004 | 2M | Ans: A

For a PMDC motor, there is no field winding; hence field losses are absent.
At no-load,

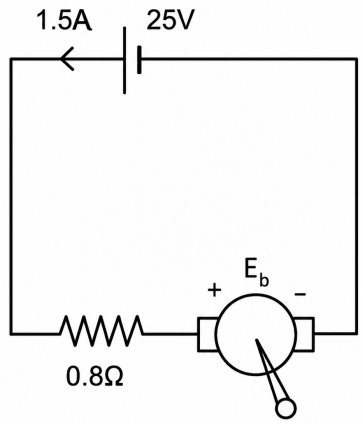


Fig. Solution Q3.04.2

$$E_{b,\text{NL}} = V_t - I_a R_a = 25 - (1.5)(0.8) = 23.8\,\text{V}$$

Rotational losses are

$$P_{\text{rot}} = E_{b,\text{NL}} I_a = 23.8 \times 1.5 = 35.7\,\text{W}$$

At loaded condition,

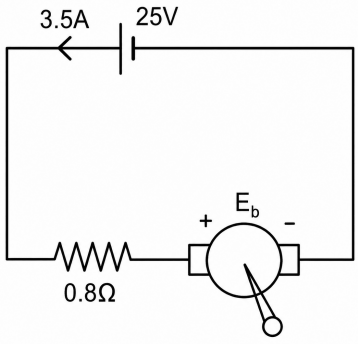


Fig. Solution Q3.04.2

$$P_{\text{in}} = VI = 25 \times 3.5 = 87.5\,\text{W}$$

Armature copper loss is

$$P_{cu} = I_a^2 R_a = (3.5)^2 (0.8) = 9.8\,\text{W}$$

Output power is

$$P_{out} = 87.5 - 9.8 - 35.7 = 42\,W$$

Hence,

$$\eta = \frac{P_{out}}{P_{in}} \times 100 = \frac{42}{87.5} \times 100 = 48\%$$

$$\boxed{A}$$

Key Points

- PMDC motors have no field winding; therefore field copper losses are absent.
- Rotational losses are assumed constant from no-load to load.
- Back emf is given by $E_b = V_t - I_a R_a$.

Q3.04.3 MCQ | 2004 | 2M | Ans: C | ● ○ ○

Induced emf in a DC machine is

$$E = \frac{NP\phi Z}{60A}$$

Given,

$$N = 250\,\text{rpm}, \quad P = 8, \quad \phi = 0.06\,\text{Wb}$$

Number of conductors,

$$Z = 32 \times 6 \times 2 = 384$$

For wave winding,

$$A = 2$$

Therefore,

$$E = \frac{250 \times 8 \times 0.06 \times 384}{60 \times 2} = 384\,V$$

$$\boxed{C}$$

Q3.03.1 MCQ | 2003 | 2M | Ans: B | ● ● ○

For a DC series motor,

$$E = V - I(R_a + R_{se})$$

Let,

$$R_a + R_{se} = R$$

Then,

$$E = V - IR$$

Also,

$$E = k_m \phi N$$

Hence,

$$N = \frac{V - IR}{k_m \phi}$$

For a series motor,

$$\phi \propto I$$

At constant power load,

$$T\omega = \text{constant}$$

If speed is reduced to one-fourth of its rated value, torque must increase four times to maintain constant power. Since,

$$T \propto \phi I$$

and for a series motor $\phi \propto I$,

$$T \propto I^2$$

Therefore,

$$I_2 = 2I_1$$

Developed power is

$$P = EI = \text{constant}$$

Hence, when current doubles,

$$E_2 = \frac{E_1}{2}$$

Neglecting armature and series field resistance,

$$E \simeq V$$

Therefore,

$$V_2 = \frac{V_1}{2}$$

Thus the supply voltage becomes

0.5 p.u.

$\boxed{B}$

Key Points

- In a DC series motor, $\phi \propto I$ (before saturation).
- Torque developed is $T \propto \phi I \Rightarrow T \propto I^2$.
- For constant power load, $T\omega$ = constant.

Mistakes to be avoided

- Do not use $T \propto I$ for a series motor in the unsaturated region.
- Remember that constant power implies torque varies inversely with speed.
- Do not forget the assumption $E \simeq V$ when armature and series field resistances are neglected.

Q3.03.2 MCQ 2003 2M Ans: C ●●○

In the figure, the stator mmf axis F_s points towards D (to the right), and the rotor mmf axis F_r points towards d, lagging F_s by the angle δ in the clockwise direction.

Stator poles. The stator mmf drives flux across the bore in the direction of F_s. On the left, this flux leaves the stator surface ABC and crosses the air gap; on the right, it crosses the air gap and enters the stator surface CDA. In the air gap, flux leaves an N-pole and enters an S-pole. Hence,

$$ABC \rightarrow N\text{-pole}, \qquad CDA \rightarrow S\text{-pole}$$

This rules out options B and D.

Torque direction. The electromagnetic torque tends to align the rotor mmf axis with the stator mmf axis,

$$T \propto F_s F_r \sin\delta$$

Since F_r lies at an angle δ clockwise from F_s, the torque on the rotor acts to reduce δ, that is, the torque is **counter-clockwise**. This rules out option A, which has clockwise torque.

Only row C has ABC as N, CDA as S and a counter-clockwise torque.

$$\boxed{C}$$

Note: by the same air-gap rule, the rotor half cda (towards which F_r points) acts as an N-pole and abc as an S-pole; row C lists these two the other way round, but it is the only row with the correct stator poles and torque direction, and the accepted key is C.

Key Points

- Magnetic flux flows from N to S in the air-gap and from S to N inside the magnetic material.
- Electromagnetic torque tends to align the rotor and stator magnetic axes, $T \propto F_s F_r \sin\delta$.
- Torque direction can be determined from the relative positions of the rotor and stator MMF axes.

Mistakes to be avoided

- Do not confuse flux direction in the air-gap with flux direction inside the magnetic material.
- The rotor always tends to move so as to reduce the angular displacement between the rotor and stator MMF axes.
- Do not interchange clockwise and anti-clockwise torque directions.

Q3.03.3 MCQ 2003 2M Ans: C ●○○

The schematic of motor and generator shown in below.

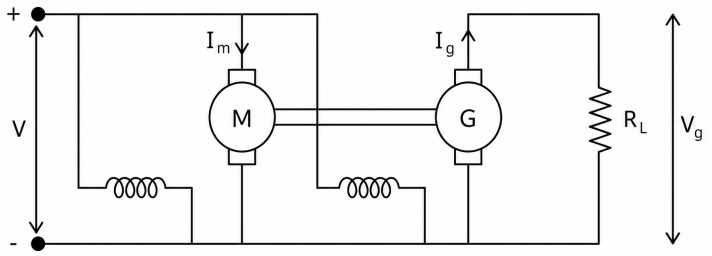


Fig. Solution Q3.03.3

All calculations are in per unit.

$$V = 1, \quad I_g = I_m = 1, \quad R_m = R_g = 0.02$$

For the motor,

$$E_b = V - I_m R_m = 1 - (1)(0.02) = 0.98 \text{ p.u.}$$

Motor output power is fed to the generator. Hence,

$$E_g I_g = E_b I_m$$

Since $I_g = I_m = 1$,

$$E_g = E_b = 0.98 \text{ p.u.}$$

Generator terminal voltage is

$$V_g = E_g - I_g R_g = 0.98 - (1)(0.02) = 0.96 \text{ p.u.}$$

Therefore, the load resistance is

$$R_L = \frac{V_g}{I_g} = \frac{0.96}{1} = 0.96 \text{ p.u.}$$

$\boxed{B}$

Q3.03.4 MCQ | 2003 | 1M | Ans: B | ● ○ ○

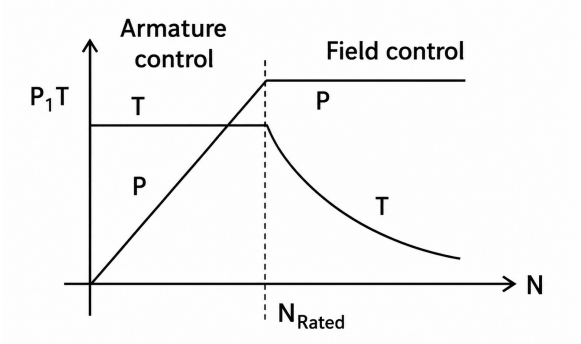


Fig. Solution Q3.03.4

Field control is used for speed control above base speed.
Since,

$$N \propto \frac{1}{\phi}$$

the field flux can only be weakened (reduced) and cannot be increased beyond its rated value.
For a given armature current,

$$T_{\text{developed}} \propto \phi$$

Hence, reducing flux reduces torque capability and the motor operates in the constant power region.
Armature voltage control is used below base speed. By maintaining rated field flux and controlling armature voltage,

$$T \propto \phi I_a$$

and since ϕ is constant, rated torque can be obtained up to full-load armature current.
Thus, armature control provides constant torque operation below base speed, while field control provides constant power operation above base speed.

$\boxed{B}$

Key Points

- Field control provides speed above base speed and operates in the constant power region.
- Armature voltage control provides speed below base speed and operates in the constant torque region.
- Speed of a DC motor varies inversely with field flux:

$$N \propto \frac{1}{\phi}$$

- For constant armature current:

$$T \propto \phi$$

Mistakes to be avoided

- Field control weakens flux; it does not increase flux beyond rated value.
- Constant torque operation is associated with armature control, not field control.
- Constant power operation is associated with field weakening above base speed.

Q3.02.1 MCQ 2002 1M Ans: B ●●○

When a DC series motor is overloaded, its magnetic circuit becomes saturated. Hence, flux becomes practically constant and is no longer proportional to field current.
Since,

$$T = k\phi I_a$$

For constant flux,

$$T = k_1 I_a$$

Also,

$$\omega = \frac{E}{k\phi} = \frac{V - I_a(R_a + R_f)}{k_1}$$

Using $I_a = \frac{T}{k_1}$,

$$\omega = -\frac{(R_a + R_f)}{k_1^2}T + \frac{V}{k_1}$$

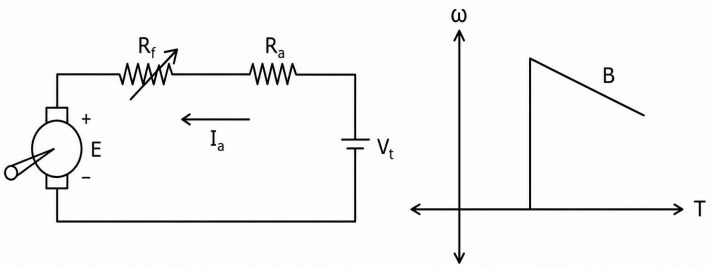


Fig. Solution Q3.02.1

Thus, the speed-torque characteristic is a straight line with negative slope.

$\boxed{B}$

Key Points

- In saturation, flux becomes nearly constant.
- For a DC series motor under saturation, $T \propto I_a$.
- Speed varies linearly and decreases with increase in torque.
- The speed-torque characteristic has negative slope.

Mistakes to be avoided

- Do not use $\phi \propto I_a$ after saturation.
- Do not assume $T \propto I_a^2$ in the saturated region.
- Remember that overloading causes flux saturation in a DC series motor.

Q3.02.2 MCQ 2002 2M Ans: A ●●○

Due to separate excitation, the flux is constant.
By KVL at rated current,

$$E = V_t - I_a R_a = 200 - (10 \times 2) = 180\,\text{V}$$

Since,

$$E = k\phi\omega$$

At rated speed of 2000 rpm,

$$k\phi = \frac{E}{\omega} = \frac{180}{\frac{2\pi}{60} \times 2000} = 0.8598\ \text{V} \cdot \text{s/rad}$$

Developed torque,

$$T = k\phi I_a$$

When the motor draws 5 A from the source,

$$T = 0.8598 \times 5 = 4.299 \approx 4.3\ \text{N} \cdot \text{m}$$

$\boxed{A}$

Key Points

- Developed torque is proportional to armature current:

$$T = k\phi I_a$$

- Torque varies linearly with armature current when flux is constant.

Mistakes to be avoided

- Do not use terminal voltage directly in the torque equation; first calculate back emf.
- Convert speed from rpm to rad/s before evaluating $k\phi$.
- Do not neglect armature resistance drop while calculating E.
- Use the new armature current (5 A) while computing torque.

Q3.02.3 NAT | 2002 | 2M | Ans: 300.8,27.15 | ● ○ ○

Since it is a separately excited motor, flux is constant.

$$E = k'\omega$$

At rated condition,

$$E_1 = V_t - I_a R_a = 230 - (100)(0.5) = 180\ \text{V}$$

$$k' = \frac{180}{\left(\frac{2\pi}{60}\right)250} = 6.875\ \text{V} \cdot \text{s/rad}$$

Since flux is constant,

$$T_2 = k' I_{a2}$$

Also,

$$I_{a2} = \frac{230 - E_2}{0.5} = \frac{230 - k'\omega_r}{0.5}$$

Therefore,

$$T_2 = k'\left(\frac{230 - k'\omega_r}{0.5}\right)$$

At steady state,

$$T_2 = T_L$$

$$2k'(230 - k'\omega_r) = 500 - 10\omega_r$$

$$\omega_r = 31.48\,\text{rad/s}$$

$$N_r = \frac{60\omega_r}{2\pi} = 300.8\,\text{rpm}$$

$$I_{a2} = \frac{230 - (6.875)(31.48)}{0.5} = 27.15\,\text{A}$$

$$\boxed{300.8, 27.15}$$

Q3.01.1 NAT | 2001 | 5M | Ans: 1140 | ● ○ ○

Given that the flux in the machine is constant, the air-gap flux density B remains constant.

For rotating machines,

$$P_h \propto B^{1.6} f$$

Since B is constant,

$$P_h \propto f \propto N$$

Also,

$$P_e \propto B^2 f^2$$

Hence,

$$P_e \propto f^2 \propto N^2$$

At $N = 2000\,\text{rpm}$,

$$P_h = 500\,\text{W}$$

Therefore,

$$P_h = k_h N$$

$$500 = k_h \times 2000$$

$$k_h = \frac{500}{2000} = 0.25$$

Similarly,

$$P_e = 200\,\text{W}$$

$$P_e = k_e N^2$$

$$200 = k_e (2000)^2$$

$$k_e = \frac{200}{(2000)^2} = 5 \times 10^{-5}$$

Initially, total iron loss is

$$P_i = P_h + P_e = 500 + 200 = 700\,\text{W}$$

When the iron losses are halved,

$$P_i = 350\,\text{W}$$

Let the new speed be N'.

$$k_h N' + k_e N'^2 = 350$$

$$0.25N' + 5 \times 10^{-5} N'^2 = 350$$

Multiplying throughout by 20000,

$$5000N' + N'^2 = 700 \times 10^4$$

Solving,

$$N' = 1140\,\text{rpm}$$

1140

Key Points

- Hysteresis loss varies directly with frequency and hence speed when flux density is constant.
- Eddy current loss varies as the square of frequency and hence as the square of speed.
- Total iron loss is the sum of hysteresis and eddy current losses.

Q3.01.2 NAT | 2001 | 5M | Ans: 2678.57 | ● ○ ○

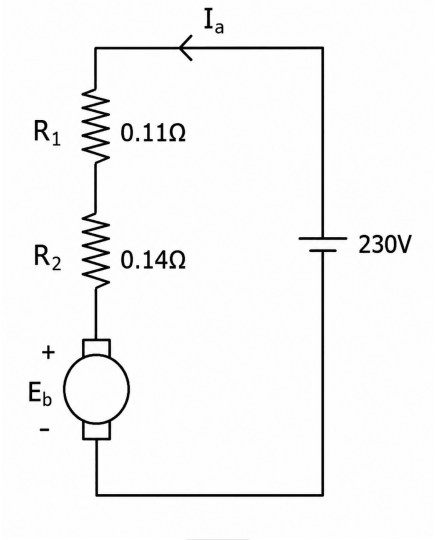


Fig. Solution Q3.01.2

Total resistance of field and armature winding,

$$R = 0.11 + 0.14 = 0.25\,\Omega$$

Given a 230 V, 1000 rpm, 80 A series DC motor.
At the initial condition,

$$E_{b1} = 230 - (80)(0.25) = 210\,\text{V}$$

$$E_{b1} = k\phi_1 N_1$$

Hence,

$$k = \frac{210}{\phi_1 N_1}$$

At the new operating condition,

$$E_{b2} = 230 - (20)(0.25) = 225\,\text{V}$$

$$E_{b2} = k\phi_2 N_2$$

Given,

$$\phi_2 = 0.4\phi_1$$

Therefore,

$$N_2 = \frac{225}{k\phi_2} = \frac{225(\phi_1 N_1)}{210(0.4\phi_1)}$$

$$N_2 = \frac{225 \times 1000}{210 \times 0.4} = 2678.57\,\text{rpm}$$

$$\boxed{2678.57}$$

Q3.01.3 MCQ 2001 2M Ans: D ● ○ ○

For a constant power drive,

$$P = \frac{2\pi N}{60} T = T\omega = \text{Constant}$$

Hence,

$$T\omega = \text{Constant}$$

$$T \propto \frac{1}{\omega}$$

Since,

$$\omega \propto N$$

Therefore,

$$T \propto \frac{1}{N}$$

The torque-speed characteristic is therefore a rectangular hyperbola.

$$T\,N = \text{Constant}$$

$\boxed{D}$

Q3.00.1 MCQ 2000 1M Ans: C ● ○ ○

Compensating windings are placed in slots provided in pole faces parallel to the rotor conductors.

They are used to overcome the cross-magnetizing effect of armature reaction, which causes distortion of the main flux distribution.

During severe overloads or rapidly changing loads, the voltage between adjacent commutator segments may become very high. This ionizes the surrounding air and can establish an arc from brush to brush. This phenomenon is known as flashover.

Compensating windings are connected in series with the armature winding so that they carry the same armature current and produce an opposing M.M.F. to neutralize the armature M.M.F.

As a result, the flux per pole remains substantially undistorted.

$\boxed{C}$

Q3.00.2 MCQ 2000 2M Ans: D ● ○ ○

In a DC series motor,

$$T = kI_a^2$$

Since the developed torque remains at the rated value,

$$I_a = \text{constant} = 40\,\text{A}$$

Under rated conditions,

$$E_1 = V - I_a R$$

$$E_1 = 240 - 40 \times 0.3 = 228\ \text{V}$$

For a series motor,

$$\phi \propto I_f = I_a$$

Since I_a is constant,

$$\phi = \text{constant}$$

Hence,

$$E \propto \phi N \propto N$$

Therefore,

$$\frac{E_1}{E_2} = \frac{N_1}{N_2}$$

$$E_2 = \frac{1000}{1500} \times 228 = 152\ \text{V}$$

Since $I_a = 40\ \text{A}$ remains constant,

$$240 - 40\,(0.3 + R_x) = 152$$

$$40\,(0.3 + R_x) = 88$$

$$0.3 + R_x = 2.2$$

$$R_x = 1.9\,\Omega$$

$\boxed{D}$

Key Points

- In an unsaturated DC series motor, $T \propto I_a^2$.
- For constant torque, armature current remains constant.
- With constant flux, back EMF is directly proportional to speed.
- Additional series resistance reduces back EMF and hence motor speed.

Q3.00.3 MCQ | 2000 | 2M | Ans: B | ● ● ○

As the motor is a permanent magnet DC motor, the field flux remains constant for any load current.

$$T = kI_a$$

$$E = k\omega$$

At no-load, friction and windage losses are neglected. Therefore,

$$I_a = 0$$

and

$$E_1 = V_{dc} = 120\,\text{V}$$

Given no-load speed,

$$N_1 = 6000\,\text{rpm}$$

$$\omega_1 = \frac{2\pi}{60} \times 6000$$

Hence,

$$120 = k\left(\frac{2\pi}{60} \times 6000\right)$$

$$k = 0.191\,\frac{\text{V} \cdot \text{s}}{\text{rad}}$$

For developed torque,

$$T = 0.5\,\text{N} \cdot \text{m}$$

$$I_a = \frac{T}{k} = \frac{0.5}{0.191} = 2.617\,\text{A}$$

When terminal voltage is reduced to 60 V,

$$E_2 = 60 - (2.617)(2.5)$$

$$E_2 = 53.45\,\text{V}$$

Therefore,

$$\omega_2 = \frac{E_2}{k} = \frac{53.45}{0.191} = 279.87\,\text{rad/s}$$

Converting to rpm,

$$N_2 = \frac{60}{2\pi}\omega_2$$

$$N_2 = \frac{279.87 \times 60}{6.28} = 2673\,\text{rpm}$$

$$\boxed{B}$$

Q3.00.4 NAT | 2000 | 5M | Ans: 1387 | ● ● ○

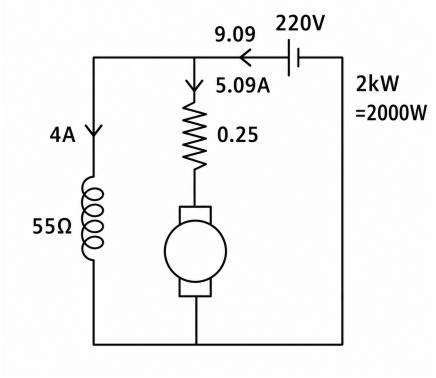


Fig. Solution Q3.00.4

When the machine operates as a generator, it delivers a load of 10 kW.

$$I_L = \frac{10000}{220} = 45.45\ \text{A}$$

Field current,

$$I_f = \frac{220}{55} = 4\ \text{A}$$

Since the machine is shunt connected,

$$I_a = I_L + I_f = 45.45 + 4 = 49.45\ \text{A}$$

Generated EMF,

$$E_g = 220 + (49.45)(0.25) + 2 = 234.36\ \text{V}$$

When the machine operates as a motor, it draws 2000 W power.

$$I_L = \frac{2000}{220} = 9.09\ \text{A}$$

Armature current,

$$I_a = 9.09 - 4 = 5.09\ \text{A}$$

Back EMF,

$$E_m = 220 - (5.09)(0.25) - 2 = 216.7275\ \text{V}$$

Since the field current remains unchanged, flux can be assumed constant.

$$\frac{E_g}{E_m} = \frac{N_g}{N_m} = \frac{234.36}{216.7275} = 1.081$$

Given,

$$N_g = 1500\ \text{rpm}$$

Therefore,

$$N_m = \frac{1500}{1.081} = 1387.11\ \text{rpm}$$

1387

Q3.99.1 MCQ | 1999 | 1M | Ans: C ●●○

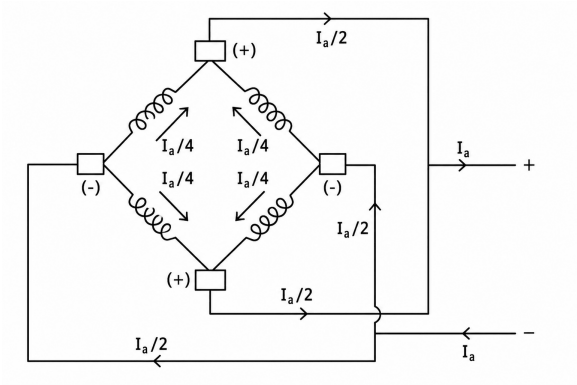


Fig. Solution Q3.99.1

For a 4-pole lap wound generator, the number of parallel paths is $A = 4$ with alternately placed brushes. One brush is connected to the outgoing current path and the opposite brush to the incoming current path as shown. Developed power is

$$P = EI_a$$

If two adjacent brushes are removed, the induced EMF across the remaining brushes remains unchanged because the parallel paths are still connected across the same generated voltage.

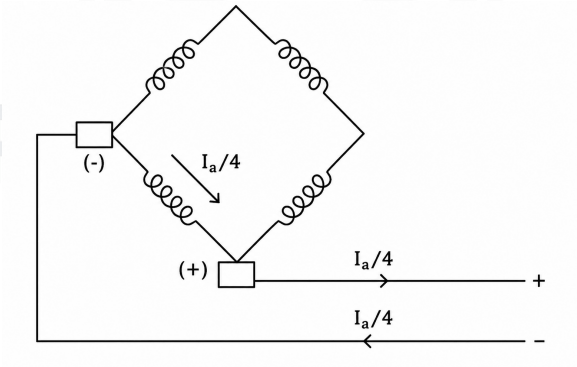


Fig. Solution Q3.99.1

However, the armature current gets reduced to one-fourth of its original value:

$$I_a' = \frac{I_a}{4}$$

Therefore, the developed power becomes

$$P' = E\left(\frac{I_a}{4}\right)$$

$$P' = \frac{P}{4}$$

C

Key Points

- In a lap-wound generator, the number of parallel paths equals the number of poles.
- Developed power is proportional to EI_a.
- Removal of alternate brushes opens the circuit and no power is developed.
- Removal of adjacent brushes reduces armature current to one-fourth while generated EMF remains unchanged.

Mistakes to be avoided

- Do not confuse alternate-brush removal with adjacent-brush removal.
- Remember that power depends on both generated EMF and armature current.

Q3.99.2 MCQ 1999 2M Ans: D ●○○

As the losses in the DC shunt motor are negligible, armature resistance can be neglected.

$$V_t = E_b$$

where E_b is back EMF and V_t is terminal voltage.
Field current is

$$I_f = \frac{V_t}{R_f}$$

Hence,

$$\phi \propto I_f$$

$$\phi \propto V_t$$

Speed of a DC motor is

$$\omega = \frac{E_b}{k\phi}$$

Since both E_b and ϕ are proportional to terminal voltage,

$$\omega = \text{constant}$$

Therefore, the speed remains constant for a lossless DC shunt motor.

D

Q3.99.3 NAT 1999 5M Ans: 156 ●●○

Method 1

Nature of load torque:

$$T \propto \omega^2$$

or

$$T = k'\omega^2$$

Since it is a separately excited motor, field flux is constant.

$$E_b = k\omega$$

At the initial operating condition,

$$E_{b1} = k_1\left(\frac{2\pi}{60}N_1\right) = k_1 \times 52.36$$

$$E_{b1} = V_{t1} - I_{a1}R_a = 100 - 8 \times 1 = 92\,\text{V}$$

$$k_1 = \frac{92}{52.36} = 1.757\,\text{V-sec/rad}$$

Since mechanical losses are negligible, developed torque equals load torque.

$$T_1 = k_1 I_{a1}$$

$$T_1 = 1.757 \times 8 = 14.056\,\text{N-m}$$

$$k' = \frac{T_1}{\omega_1^2}$$

$$k' = \frac{14.056}{(52.36)^2} = 5.13 \times 10^{-3}\,\text{N-m-sec}^2/\text{rad}^2$$

For the new speed,

$$\omega_2 = 750\,\text{rpm} \times \frac{2\pi}{60} = 78.54\,\text{rad/sec}$$

$$T_2 = k'\omega_2^2$$

$$T_2 = 5.13 \times 10^{-3} \times (78.54)^2 = 31.64\,\text{N-m}$$

$$T_2 = k_1 I_{a2}$$

$$I_{a2} = \frac{31.64}{1.757} = 18\,\text{A}$$

$$E_{b2} = k_1\omega_2 = 1.757 \times 78.54 \approx 138\,\text{V}$$

$$V_{t2} = E_{b2} + I_{a2}R_a$$

$$V_{t2} = 138 + 18 \times 1 = 156\,\text{V}$$

156

Key Points

- Fan-type load torque varies as the square of speed: $T \propto \omega^2$.

Q3.98.1 MCQ | 1998 | 2M | Ans: A | ● ○ ○

At starting condition, speed is zero. Hence, back EMF becomes zero and the DC motor draws maximum current.

$$V = I_{\text{stall}}(R_a + R_{se})$$

$$240 = I_{\text{stall}}(0.5 + 1)$$

$$I_{\text{stall}} = \frac{240}{1.5} = 160\,\text{A}$$

Since flux in a DC shunt motor remains unchanged,

$$T \propto I_a$$

Therefore,

$$\frac{T_{\text{stall}}}{T_{\text{full-load}}} = \frac{I_{\text{stall}}}{I_a} = \frac{160}{40} = 4$$

$\boxed{A}$

Key Points

- At stall, speed is zero and back EMF is zero.
- Stall current is limited only by armature and series resistance.
- In a DC shunt motor, flux remains approximately constant.

Q3.98.2 NAT | 1998 | 5M | Ans: 343.15 | ● ○ ○

Given:

$$P = 60\,\text{kW}, \qquad V = 240\,\text{V}, \qquad N_g = 360\,\text{rpm}$$

$$R_a = 0.015\,\Omega, \qquad R_f = 60\,\Omega$$

Brush drop = 2 V.
When operating as a generator,

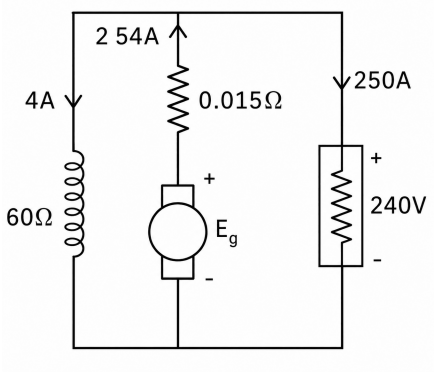


Fig. Solution Q3.98.2

$$I_L = \frac{60 \times 10^3}{240} = 250\,\text{A}$$

$$I_f = \frac{240}{60} = 4\,\text{A}$$

$$(I_a)_g = I_L + I_f = 250 + 4 = 254\,\text{A}$$

Generated EMF:

$$E_g = 240 + 254 \times 0.015 + 2 = 245.81\,\text{V}$$

Since,

$$E_g = K\phi\omega_g$$

and

$$E_g = K'\phi N_g$$

$$245.81 = K'\phi(360) \tag{1}$$

Now when the machine operates as a motor, the field voltage remains the same.

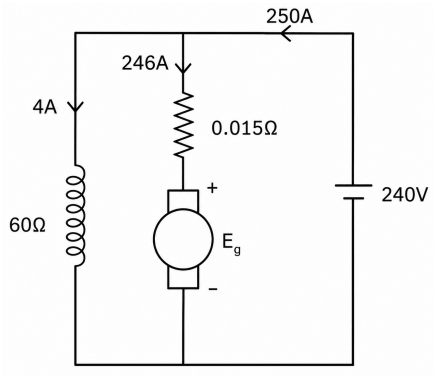


Fig. Solution Q3.98.2

Hence,

$$I_f = 4\,\text{A}$$

and the flux remains unchanged.

$$(I_a)_m = 250 - 4 = 246\,\text{A}$$

Back EMF:

$$E_m = 240 - 246 \times 0.015 - 2 = 234.31\,\text{V}$$

$$E_m = K'\phi N_m$$

$$234.31 = K'\phi N_m \tag{2}$$

From (1) and (2),

$$\frac{245.81}{234.31} = \frac{360}{N_m}$$

$$N_m = \frac{360 \times 234.31}{245.81}$$

$$N_m = 343.15\,\text{rpm}$$

$$\boxed{343.15}$$

Q3.97.1 NAT | 1997 | 2M | Ans: 784,7.846 | ● ○ ○

Given: 5 kW, 200 V D.C. shunt motor,

$$R_a = 1\,\Omega, \qquad R_f = 100\,\Omega$$

$$(\omega)_{nl} = 1000 \times \frac{2\pi}{60} = 104.72\,\text{rad/s}$$

$$I_f = \frac{V}{R_f} = \frac{200}{100} = 2\,\text{A}$$

$$I_a = 6 - 2 = 4\,\text{A}$$

$$E_g = 200 - 4 \times 1 = 196\,\text{V}$$

$$P_{\text{rotational loss}} = E_g I_a = 196 \times 4 = 784\,\text{W}$$

$$\tau_{nl} = \frac{P_{\text{rotational loss}}}{(\omega)_{nl}} = \frac{784}{104.72} = 7.486\,\text{N m}$$

$$\boxed{784, 7.486}$$

Q3.97.2 NAT 1997 2M Ans: 4 ● ○ ○

Given: 200 V, 10 kW, lap-wound DC generator,

$$P = 10, \qquad Z = 500$$

Since,

$$\frac{\text{Pole arc}}{\text{Pole pitch}} = 0.8$$

For a lap-wound generator,

$$A = P$$

Compensating conductors required per pole are

$$Z_c = \frac{Z}{AP} \times 0.8$$

$$Z_c = \frac{500}{10 \times 10} \times 0.8$$

$$Z_c = 4$$

$$\boxed{4}$$

Key Points

- Compensating conductors are placed in pole faces to neutralize armature reaction.
- Pole arc to pole pitch ratio decides the fraction of armature conductors under the pole face.

Q3.97.3 MCQ 1997 2M Ans: 2 and <1 ● ● ○

From the equivalent circuit of a DC series motor, armature current is constant and armature resistance varies with the field connection.

For the linear region of the B-H curve,

$$\phi \propto NI_a$$

$$T \propto \phi I_a$$

The motor takes the rated current in both cases, hence

$$I_{a1} = I_{a2} = I$$

When both field coils are connected in series,

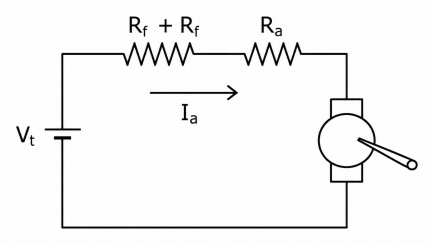


Fig. Solution Q3.97.3

$$\phi_1 \propto (NI + NI) = 2NI$$

When the two field sections are connected in parallel,

$$\phi_2 \propto \left(\frac{NI}{2} + \frac{NI}{2}\right) = NI$$

Therefore,

$$\frac{T_1}{T_2} = \frac{\phi_1}{\phi_2} = \frac{2NI}{NI} = \frac{2}{1} = 2$$

Now,

$$E = k'\phi\omega$$

Since I_a is constant,

$$E_1 = V_t - I_a(R_f + R_f + R_a)$$

When the two field sections are connected in parallel,

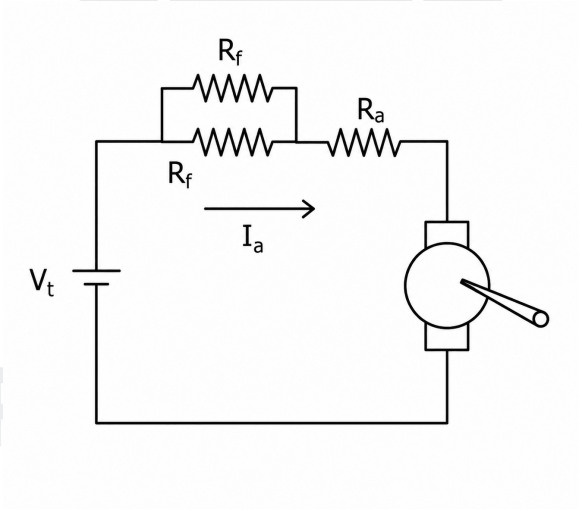


Fig. Solution Q3.97.3

$$E_2 = V_t - I_a\left(R_a + \frac{R_f}{2}\right)$$

Since the net resistance decreases in the parallel connection,

$$E_2 > E_1$$

Also,

$$\frac{E_1}{E_2} = \frac{\phi_1\omega_1}{\phi_2\omega_2}$$

$$\frac{\omega_1}{\omega_2} = \frac{E_1}{E_2} \cdot \frac{\phi_2}{\phi_1} = \frac{E_1}{E_2} \cdot \frac{NI}{2NI} = \frac{E_1}{2E_2}$$

Since

$$E_2 > E_1$$

$$\frac{\omega_1}{\omega_2} < 1$$

Hence

$$\boxed{2 \text{ and } <1}$$

Key Points

- For a DC series motor in the linear magnetic region, $\phi \propto NI_a$.
- Torque is proportional to ϕI_a.
- Back EMF is given by $E = k'\phi\omega$.
- Connecting field sections in parallel reduces field resistance and increases speed.

Q3.97.4 NAT | 1997 | 5M | Ans: 28 | ● ● ○

Given,

$$I_{a1} = 40\,\text{A}$$

Initial induced EMF is

$$E_1 = 200 - 40 \times 1 = 160\,\text{V}$$

Let the reduced flux be

$$\frac{\phi_2}{\phi_1} = x$$

Torque increases by 10%, hence

$$T_2 = 1.1T_1$$

Since,

$$T \propto \phi I_a$$

$$\frac{T_2}{T_1} = \frac{\phi_2 I_{a2}}{\phi_1 I_{a1}} = \frac{xI_{a2}}{40}$$

$$\frac{xI_{a2}}{40} = 1.1$$

$$I_{a2} = \frac{40 \times 1.1}{x}$$

New induced EMF is

$$E_2 = 200 - I_{a2}R_a = 200 - \frac{44}{x}$$

Speed increases by 20%, therefore

$$\omega_2 = 1.2\omega_1$$

Also,

$$E \propto \phi\omega$$

$$E_2 = x(1.2)E_1$$

$$200 - \frac{44}{x} = 1.2x(160)$$

$$192x^2 - 200x + 44 = 0$$

Solving,

$$x = 0.315,\ 0.72$$

For $x = 0.315$, current becomes very large. Hence permissible value is

$$x = 0.72$$

Percentage reduction in flux is

$$\frac{\phi_1 - \phi_2}{\phi_1} \times 100 = (1 - 0.72) \times 100 = 28\%$$

$$\boxed{28}$$

Key Points

- Reduction in flux increases speed but also changes armature current for the required torque.

Q3.96.1 MCQ | 1996 | 1M | Ans: D | ● ○ ○

In a double-layer winding, each slot contains two coil sides, i.e., two conductors per slot.

Given number of coils:

$$16 \text{ coils}$$

Assuming single-turn coils,

$$16 \text{ coils} = 32 \text{ conductors}$$

Since,

$$2 \text{ conductors} = 1 \text{ slot}$$

$$32 \text{ conductors} = 16 \text{ slots}$$

Total number of slots is

$$S = 16$$

For a 4-pole machine,

$$\text{Pole pitch} = \frac{\text{slots}}{\text{poles}} = \frac{16}{4} = 4$$

$$\boxed{D}$$

Key Points

- In double-layer winding, each slot contains two coil sides.
- For single-turn coils, one coil has two conductors.
- Pole pitch in slots is calculated as total slots divided by number of poles.

Mistakes to be avoided

- Do not forget that double-layer winding has two coil sides per slot.
- Use total slots, not total conductors, for calculating pole pitch.

Q3.96.2 MCQ 1996 2M Ans: C ● ○ ○

For a DC shunt generator,

$$V_t = E_g - I_a R_a$$

Thus, terminal voltage decreases either due to a reduction in generated EMF E_g or due to an increase in the internal voltage drop $I_a R_a$.

Examining each factor:

(i) **Interpole**: Interpoles improve commutation and do not directly affect terminal voltage.

(ii) **Armature resistance**: Causes a voltage drop $I_a R_a$, thereby reducing terminal voltage.

(iii) **Armature leakage reactance**: Reactance is relevant in AC machines. In a DC generator, it does not contribute to steady-state terminal voltage reduction.

(iv) **Armature reaction**: Weakens the main field flux, reducing generated EMF.

(v) **Reduction in field current**: Since

$$E_g \propto \phi\omega$$

and

$$\phi \propto I_f,$$

a reduction in field current decreases flux and hence generated EMF.

Therefore, the factors responsible for a decrease in terminal voltage are

$$(ii),\ (iv) \text{ and } (v)$$

Hence,

$$\boxed{C}$$

Q3.96.3 MCQ 1996 1M Ans: B ● ○ ○

Given a 4-pole wave-wound DC generator,

$$P = 4, \qquad A = 2$$

Total number of conductors is

$$Z = 51 \times 20 = 1020$$

Given,

$$E = 357\,\text{V}, \qquad N = 8500\,\text{rpm}$$

For a DC generator,

$$E = \frac{NP\phi Z}{60A}$$

Therefore,

$$\phi = \frac{60AE}{NPZ}$$

$$\phi = \frac{60 \times 2 \times 357}{8500 \times 4 \times 51 \times 20}$$

$$\phi = 1.23 \times 10^{-3}\,\text{Wb} = 1.23\,\text{mWb}$$

$\boxed{B}$

Q3.95.1 MCQ | 1995 | 1M | Ans: B | ● ○ ○

In a compound machine, both series and shunt windings are present.

When the terminal voltage connection is reversed while running as a generator, the field pole polarity remains unchanged because the shunt field current direction remains the same.

The direction of rotation also remains the same, hence the generated EMF retains the same polarity. Therefore, the armature current must reverse.

Since interpoles are connected in series with the armature, their polarity automatically reverses with armature current. Hence, there is no need to interchange the interpole connections.

In a differentially compounded machine, the shunt-field flux and series-field flux oppose each other. When the armature current reverses, the series-field flux reverses while the shunt-field flux remains unchanged.

Thus, the two fluxes now act in the same direction and add together, causing the machine to behave as a cumulatively compounded machine without reversing the interpole connections.

$\boxed{B}$

Q3.95.2 NAT | 1995 | 5M | Ans: 940 | ● ○ ○

Given:

$$V_t = 220\,\text{V}, \qquad R_a = 2.5\,\Omega$$

Armature voltage, field current and losses are assumed to remain constant from no-load to full-load.

Rated armature current:

$$(I_a)_{\text{rated}} = 8\,\text{A}$$

Full-load copper loss:

$$P_{Cu} = I_{a,\text{rated}}^2 R_a = 8^2 \times 2.5 = 160\,\text{W}$$

Under full-load condition,

$$E_1 = 220 - 8 \times 2.5 = 200\,\text{V}$$

Developed power:

$$P_{\text{dev}} = E_1 I_a = 200 \times 8 = 1600\,\text{W}$$

Rated shaft power:

$$P_{sh} = 1500\,\text{W}$$

Therefore, constant losses:

$$P_{\text{const}} = 1600 - 1500 = 100\,\text{W}$$

At no-load, developed power equals constant losses:

$$E_b I_a = (V_t - I_a R_a) I_a = 100$$

$$(220 - 2.5 I_a) I_a = 100$$

$$220 I_a - 2.5 I_a^2 = 100$$

$$2.5 I_a^2 - 220 I_a + 100 = 0$$

Solving,

$$I_a = 0.4569\,\text{A}$$

Hence, induced EMF at no-load:

$$E_2 = 220 - 0.4569 \times 2.5 = 218.86\,\text{V}$$

Since field current is constant, flux remains constant and

$$E \propto N$$

Therefore,

$$\frac{E_2}{E_1} = \frac{N_2}{N_1}$$

$$N_2 = \frac{218.86}{200} \times 859 = 939.99\,\text{rpm}$$

$$\boxed{940}$$

PrepFusion

Key Points

- For constant field current, flux remains constant and induced EMF is directly proportional to speed.
- At no-load, the developed power supplies only the constant losses.

Q3.93.1 MCQ | 1993 | 1M | Ans: A | ● ○ ○

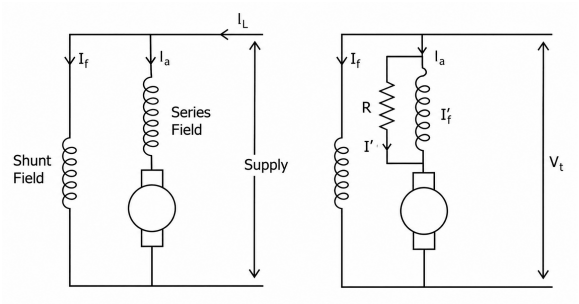


Fig. Solution Q3.93.1

If the series field is shunted by a resistance, then the current through the series field decreases. Hence, the series-field flux ϕ_{se} decreases and the resultant flux becomes

$$\phi_{\text{net}} = \phi_{sh} + \phi_{se}$$

Since ϕ_{se} is reduced,

$$\phi_{\text{net,new}} < \phi_{\text{net,old}}$$

For a DC motor,

$$T \propto \phi I_a$$

Assuming the load torque remains constant,

$$T = \text{constant}$$

Therefore,

$$\phi_{\text{net}} I_a = \text{constant}$$

When ϕ_{net} decreases, the armature current must increase to maintain the same torque.

$$I_{a,\text{new}} > I_{a,\text{old}}$$

Hence, shunting the series field causes the armature current to increase.

$\boxed{A}$

Key Points

- In a cumulative compound motor, resultant flux is $\phi_{\text{net}} = \phi_{sh} + \phi_{se}$.
- A diverter (shunt resistance) across the series field reduces series-field current.
- Reduction in series-field current reduces series-field flux.
- For constant load torque, $T \propto \phi I_a$; therefore, a reduction in flux causes an increase in armature current.

Q3.92.1 NAT 1992 5M Ans: 15.28 ● ○ ○

Given,

$$R_a = 0.5\,\Omega, \qquad V_t = 250\,\text{V}, \qquad I_{a1} = 20\,\text{A}$$

$$N_1 = 1500\,\text{rpm}$$

Since the field current remains unchanged,

$$\phi \propto I_f$$

Hence, flux is constant.

Back EMF at 1500 rpm:

$$(E_b)_{1500\text{rpm}} = V_t - I_{a1}R_a = 250 - 20 \times 0.5 = 240\,\text{V}$$

Torque at 1500 rpm:

$$T_1 = \frac{E_b I_{a1}}{\omega} = \frac{240 \times 20}{\left(\frac{2\pi}{60}\right) \times 1500} = 30.56\,\text{N-m}$$

Since flux is constant,

$$T \propto \phi I_a$$

$$\frac{T_1}{T_2} = \frac{I_{a1}}{I_{a2}}$$

For $I_{a2} = 10\,\text{A}$,

$$T_2 = \left(\frac{I_{a2}}{I_{a1}}\right) T_1 = \left(\frac{10}{20}\right) \times 30.56 = 15.28\,\text{N-m}$$

$$\boxed{15.28}$$

Q3.92.2 MCQ 1992 1M Ans: B ● ○ ○

If supply voltage in a separately excited DC motor is constant, then input power is

$$P = V_t I_a$$

Since V_t is constant,

$$P \propto I_a \qquad (1)$$

Torque developed is given by

$$T = K\phi I_a$$

For constant field voltage, field current remains constant and hence flux is constant.

$$\phi = \text{constant}$$

Therefore,

$$T \propto I_a \qquad (2)$$

From (1) and (2),

$$T \propto P$$

Hence, the correct option is

$$\boxed{B}$$

Q3.91.1 NAT 1991 1M Ans: 0.0235,340,63.69 ● ● ○

Given, a 10 kW, 6-pole lap wound DC generator,

$$E = 200\,\text{V}, \qquad N = 1500\,\text{rpm}, \qquad B = 0.9\,\text{T}$$

$$L = 0.25\,\text{m}, \qquad D = 0.2\,\text{m}$$

For a lap wound DC machine,

$$A = P = 6$$

Method 1

(a) Flux per pole

Total flux,

$$P\phi = BA$$

where

$$A = \pi DL$$

Hence,

$$\phi = \frac{B\pi DL}{P}$$

$$\phi = \frac{0.9 \times 3.14 \times 0.2 \times 0.25}{6} = 0.02355\ \text{Wb}$$

(b) Number of armature conductors

Using the EMF equation,

$$E = \frac{NP\phi Z}{60A}$$

$$Z = \frac{E \times 60A}{NP\phi}$$

$$Z = \frac{200 \times 60 \times 6}{1500 \times 6 \times 0.02355} \approx 340$$

(c) Developed torque

Armature current,

$$I_a = \frac{10 \times 10^3}{200} = 50\ \text{A}$$

Angular speed,

$$\omega = \frac{2\pi N}{60} = \frac{2\pi}{60} \times 1500 \approx 157\ \text{rad/s}$$

Torque developed,

$$\tau = \frac{EI_a}{\omega}$$

$$\tau = \frac{200 \times 50}{157} = 63.69\ \text{N m}$$

$$\boxed{0.02355,\ 340,\ 63.69}$$

Q3.91.2 MCQ 1991 1M Ans: A ●○○

Matching the windings with their functions:

- Series field winding generates the working flux $\Rightarrow A \rightarrow R$.
- Shunt field winding prevents runaway speed under no-load conditions $\Rightarrow B \rightarrow S$.
- Commutating pole (interpole) winding improves commutation and avoids sparking by neutralizing the reactance voltage during commutation $\Rightarrow C \rightarrow Q$.
- Compensating winding is connected in series with the armature winding and compensates armature reaction under the pole arc, thereby avoiding field distortion $\Rightarrow D \rightarrow P$.

Hence,

$$A \to R, \qquad B \to S, \qquad C \to Q, \qquad D \to P$$

In a DC series motor, the series field is responsible for producing the working flux.

For no-load operation of a series motor, the armature current becomes very small, reducing the series flux and causing dangerously high speed. Therefore, a shunt field is added in a compound machine to provide adequate flux under no-load conditions.

The commutating pole winding improves commutation by counteracting the reactance voltage in the coil undergoing commutation, thereby preventing sparking.

The compensating winding neutralizes armature reaction under the pole faces and prevents field distortion.

$$\boxed{A}$$

Q3.91.3 NAT 1991 1M Ans: decreses by 20% ● ○ ○

Given,

$$\omega_2 = 1.25\,\omega_1, \qquad T_2 = 0.5\,T_1$$

Assuming negligible armature voltage drop,

$$I_a R_a \approx 0$$

Hence, back emf remains constant.

$$E = K\phi\omega_m$$

Therefore,

$$\phi_1\omega_1 = \phi_2\omega_2$$

$$\phi_1\omega_1 = \phi_2(1.25\,\omega_1)$$

$$\phi_2 = 0.8\,\phi_1$$

Thus, excitation (flux) decreases by

$$(1 - 0.8) \times 100 = 20\%$$

$$\boxed{\text{Decreases by } 20\%}$$

Q3.91.4 MCQ 1991 1M Ans: D ● ○ ○

A level compounded DC generator is designed such that the series field compensates exactly for the voltage drops due to armature resistance and armature reaction.

Hence, the terminal voltage remains nearly constant from no-load to full-load.

Therefore, level compounded generators are suitable for applications requiring constant voltage supply such as lighting systems.

$$V_t \approx \text{constant}$$

D

PrepFusion

SOLUTIONS

IV

Three Phase Induction Machines

Topics

1. Principle of Operation of Rotating Machines
2. Construction of 3-Phase Induction Motor
3. Equivalent Circuit of Induction Motor
4. Power Flow
5. Torque-Slip, Torque-Speed Characteristics and Performance of IM
6. Stability and Testing – No-Load (OC) and Blocked-Rotor (SC) Tests
7. Starting of Induction Motors
8. Speed Control and Double Cage Rotor
9. Dynamic Braking
10. Plugging, Cogging and Crawling
11. Single-Phase Induction Motors – Operating Principle

Unit Snapshot*

Stream: EE	**Questions:** 96	**MCQ:** 67	**MSQ:** 0	**NAT:** 29
1M: 45	**2M:** 42	**Descriptive:** 9	**Avg Weightage** ~3M	**Range:** 1M(2020)–4M(2022)

**See preface for how Avg Weightage and Range are calculated.*

Q4.26.1 NAT | 2026 | 2M | Ans: 0.09 | ● ● ○

Synchronous speed is

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{4} = 1500\,\text{rpm}$$

With shorted slip rings,

$$s_1 = \frac{1500 - 1476}{1500} = 0.016$$

With external rotor resistance,

$$s_2 = \frac{1500 - 1404}{1500} = 0.064$$

For constant torque load,

$$\frac{s_1}{R_r} = \frac{s_2}{R_r + 0.27}$$

$$\frac{0.016}{R_r} = \frac{0.064}{R_r + 0.27}$$

$$0.016(R_r + 0.27) = 0.064R_r$$

$$R_r = 0.09\,\Omega$$

0.09

Key Points

- For small slip region, torque is proportional to $\frac{s}{R_r}$.
- For constant torque load, the ratio $\frac{s}{R_r}$ remains constant.
- Adding rotor resistance increases slip for the same load torque.

Q4.26.2 MCQ | 2026 | 1M | Ans: A | ● ○ ○

In a double-squirrel-cage induction motor, the outer cage has high resistance and low reactance, whereas the inner cage has low resistance and high reactance.

At starting, the rotor frequency is equal to the supply frequency, so the rotor reactance is high. The current therefore prefers the path having lower impedance, i.e., the outer cage.

$$Z_{\text{outer}} = \sqrt{R_{\text{outer}}^2 + X_{\text{outer}}^2}$$

$$Z_{\text{inner}} = \sqrt{R_{\text{inner}}^2 + X_{\text{inner}}^2}$$

Since

$$X_{\text{inner}} \gg X_{\text{outer}}$$

at starting,

$$Z_{\text{inner}} > Z_{\text{outer}}$$

Hence, a larger portion of the rotor current flows in the outer cage.

At normal operating speed, the rotor frequency becomes very small, making reactances negligible. Due to its lower resistance, the inner cage carries most of the current.

A

Q4.25.1 MCQ 2025 2M Ans: C ● ○ ○

For maximum starting torque in an induction motor, the starting slip ($s = 1$) must be equal to the slip corresponding to maximum torque.

$$s_{max} = \frac{R_r'}{X}$$

At maximum starting torque,

$$1 = s_{max} = \frac{R_r'}{X}$$

$$R_r' = X$$

$$R_r' = X_s + X_r'$$

At 50 Hz,

$$X_s + X_r' = 2 + 2 = 4\,\Omega$$

With constant V/f control, reactance is proportional to frequency.

$$1 = 4\left(\frac{f}{50}\right)$$

$$f = 12.5\,\text{Hz}$$

Since V/f is constant,

$$\frac{V}{400} = \frac{12.5}{50}$$

$$V = 100\,\text{V}$$

$\boxed{C}$

Key Points

- For maximum starting torque, $R_r' = X_s + X_r'$.
- Under constant V/f control, voltage and frequency reduce in the same ratio.

Q4.25.2 MCQ 2025 1M Ans: A ● ○ ○

For minimum transition time, the accelerating torque must be maximum.

$$T_{acc} = T_{em} - T_L$$

To get maximum acceleration, T_{em} should be kept at its maximum permissible value within the shaded operating region. Hence, the drive should move from O to Q, then operate along the maximum torque boundary $Q - R$, and finally reach S.

$$O \rightarrow Q \rightarrow R \rightarrow S$$

$\boxed{A}$

Key Points

- Shortest time trajectory requires maximum available accelerating torque.
- Accelerating torque is $T_{em} - T_L$.
- The upper boundary $Q - R$ gives maximum permissible T_{em}.

Mistakes to be avoided

- Do not choose the geometrically shortest path.
- The fastest path is decided by maximum acceleration, not by minimum distance on the plot.

Q4.24.1 NAT 2024 2M Ans: 0.26-0.29 ● ○ ○

For maximum starting torque,

$$\frac{R'_{r,\text{total}}}{s} = \sqrt{R_s^2 + (X_s + X'_r)^2}$$

At starting, $s = 1$.

$$R'_{r,\text{total}} = \sqrt{3^2 + (2+2)^2} = 5\,\Omega$$

Existing rotor resistance referred to stator is

$$R'_r = 2.5\,\Omega$$

So, extra resistance referred to stator is

$$R'_{\text{ext}} = 5 - 2.5 = 2.5\,\Omega$$

Given stator to rotor turns ratio is 3 : 1,

$$R'_{\text{ext}} = 3^2 R_{\text{ext}}$$

$$R_{\text{ext}} = \frac{2.5}{9} = 0.2777\,\Omega$$

$$\boxed{0.28}$$

Key Points

- For maximum starting torque, total rotor resistance referred to stator must satisfy $R'_{r,\text{total}} = \sqrt{R_s^2 + (X_s + X'_r)^2}$.
- Resistance referred from rotor to stator is multiplied by the square of turns ratio.

Q4.24.2 MCQ 2024 1M Ans: C ● ○ ○

For a 6-pole, 50 Hz induction motor,

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{6} = 1000\,\text{rpm}$$

Slip is

$$s = \frac{N_s - N}{N_s} = \frac{1000 - 960}{1000} = 0.04$$

Losses such as stator copper loss, core loss, and rotational losses are neglected. However, rotor copper loss cannot be neglected.

Mechanical power developed is

$$P_m = (1 - s)P_g$$

Since stator losses are neglected,

$$P_{in} = P_g$$

Therefore,

$$\eta = \frac{P_{out}}{P_{in}} = \frac{P_m}{P_g} = 1 - s$$

$$\eta = 1 - 0.04 = 0.96$$

$$\eta = 96\%$$

$$\boxed{C}$$

Key Points

- Rotor copper loss = sP_g.
- Mechanical power developed = $(1 - s)P_g$.
- When stator and rotational losses are neglected, $P_{in} = P_g$.
- Efficiency of an induction motor under these assumptions is $\eta = 1 - s$.

Q4.23.1 NAT | 2023 | 2M | Ans: 760 | ● ○ ○

Given that both voltage and frequency are reduced by 20%,

$$\frac{V}{f} = \text{constant}$$

The motor drives a constant torque load. Hence,

$$N_s - N_r = \text{constant}$$

Initially,

$$f_1 = 50\,\text{Hz}$$

After a 20% reduction,

$$f_2 = 50 - \frac{20}{100} \times 50 = 40\,\text{Hz}$$

For a 6-pole motor,

$$N_{s1} = \frac{120 \times 50}{6} = 1000\,\text{rpm}$$

$$N_{s2} = \frac{120 \times 40}{6} = 800\,\text{rpm}$$

Using the constant torque condition,

$$N_{s1} - N_{r1} = N_{s2} - N_{r2}$$

$$1000 - 960 = 800 - N_{r2}$$

$$N_{r2} = 760\,\text{rpm}$$

$$\boxed{760}$$

Key Points

- For constant flux operation in V/f control, $\frac{V}{f}$ is maintained constant.
- For a constant torque load, slip speed $(N_s - N_r)$ remains approximately constant.

Mistakes to be avoided

- Do not assume slip remains constant; slip *speed* remains constant for constant torque operation.
- Use the reduced frequency to calculate the new synchronous speed.

Q4.23.2 MCQ | 2023 | 1M | Ans: A | ● ○ ○

Torque-slip characteristics of an induction machine are shown below.

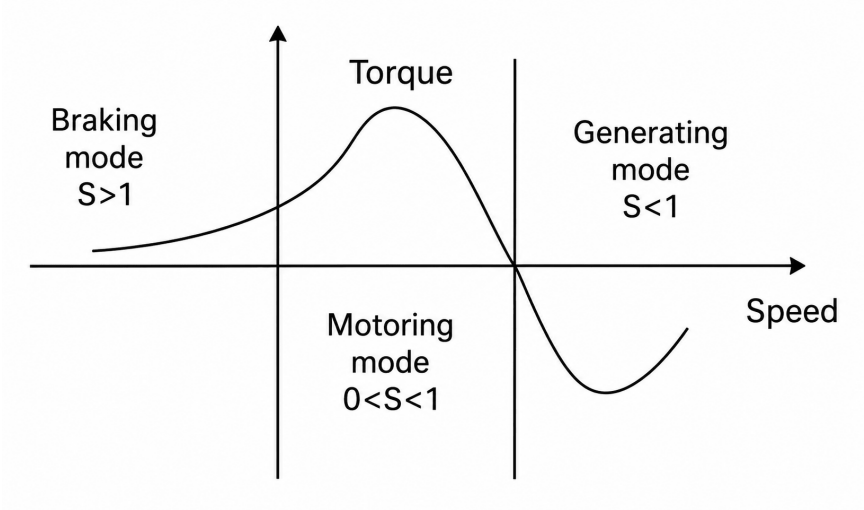


Fig. Solution Q4.23.2

For an induction machine:

$$0 < s < 1 \Rightarrow \text{Motoring mode}$$

$$s < 0 \Rightarrow \text{Generating mode}$$

$$s > 1 \Rightarrow \text{Braking mode}$$

From the torque-slip characteristic shown, the correct option is

$$\boxed{A}$$

Q4.22.1 NAT 2022 1M Ans: 585 ● ○ ○

Given,

$$f = 40\,\text{Hz}, \qquad f_r = 1\,\text{Hz}, \qquad P = 8$$

Slip is

$$s = \frac{f_r}{f} = \frac{1}{40} = 0.025$$

Synchronous speed is

$$N_s = \frac{120f}{P} = \frac{120 \times 40}{8} = 600\,\text{rpm}$$

Rotor speed is

$$N = N_s(1 - s)$$

$$N = 600(1 - 0.025) = 585\,\text{rpm}$$

$$\boxed{585}$$

Q4.22.2 NAT 2022 1M Ans: 1733 ● ● ●

From the speed-time characteristic between 4 s and 8 s,

$$N = 125t + 500$$

for

$$4\,\text{s} \le t \le 8\,\text{s}$$

The equation of motion is

$$J\frac{d\omega}{dt} = T_e - T_L$$

or

$$T_e = J\frac{d\omega}{dt} + T_L$$

Multiplying by ω,

$$P_e = J\omega\frac{d\omega}{dt} + \omega T_L$$

Since

$$P_e = \frac{dE}{dt}$$

$$\frac{dE}{dt} = J\omega\frac{d\omega}{dt} + \omega T_L$$

$$dE = J\omega\, d\omega + T_L\omega\, dt$$

Hence,

$$E = J\int \omega\, d\omega + T_L \int \omega\, dt$$

Using

$$\omega = \frac{2\pi N}{60}$$

$$E = J\left(\frac{2\pi}{60}\right)^2 \int_{1000}^{1500} N\, dN + T_L \frac{2\pi}{60}\int_4^8 N\, dt$$

Substituting

$$J = 0.1, \qquad T_L = 2, \qquad N = 125t + 500$$

$$E = 0.1\left(\frac{2\pi}{60}\right)^2 \left[\frac{N^2}{2}\right]_{1000}^{1500} + 2\frac{2\pi}{60}\left[\frac{125t^2}{2} + 500t\right]_4^8$$

$$E = 0.1\left(\frac{2\pi}{60}\right)^2 \left(\frac{1500^2 - 1000^2}{2}\right) + 2\frac{2\pi}{60}\left[\frac{125t^2}{2} + 500t\right]_4^8$$

$$E = 685.389 + 1047.197$$

$$E = 1732.5865\,\text{J}$$

$$\boxed{1733}$$

Key Points

- Mechanical dynamics: $J\frac{d\omega}{dt} = T_e - T_L$.
- Mechanical power: $P = T\omega$.
- Angular speed conversion: $\omega = \frac{2\pi N}{60}$.
- Energy is obtained by integrating power with respect to time.

Mistakes to be avoided

- Do not use speed in rpm directly; convert it to rad/s.
- Include both kinetic energy change and load energy components.
- Apply the correct integration limits for speed and time.

Q4.22.3 NAT 2022 2M Ans: 0.63 ●○○

Let the transformation ratio of the autotransformer be k.

At rated voltage, starting current drawn by the motor is

$$I_{sc} = 5I_r$$

When voltage applied to the motor is reduced to kV, the motor current becomes

$$I_m = kI_{sc} = 5kI_r$$

For an autotransformer starter, supply current is

$$I_s = kI_m$$

$$I_s = k(5kI_r) = 5k^2I_r$$

Given that supply current is limited to $2I_r$,

$$5k^2I_r = 2I_r$$

$$k^2 = \frac{2}{5} = 0.4$$

$$k = \sqrt{0.4} = 0.632$$

$$\boxed{0.63}$$

Key Points

- In an autotransformer starter, motor current reduces in proportion to voltage.
- Supply current reduces as the square of the transformation ratio.
- Hence, $I_s = k^2 I_{sc}$.

Q4.21.1 MCQ | 2021 | 1M | Ans: B | ● ○ ○

Match each constructional feature with the corresponding machine and the phenomenon it mitigates:

- Damper bars are provided in synchronous machines to reduce hunting.

$$P \rightarrow U, \quad X$$

- Skewed rotor slots are used in induction motors to reduce magnetic locking (cogging).

$$Q \rightarrow S, \quad Y$$

- Compensating winding is used in DC machines to neutralize armature reaction.

$$R \rightarrow V, \quad Z$$

Hence,

$$P - U - X, \qquad Q - S - Y, \qquad R - V - Z$$

which corresponds to

$$\boxed{B}$$

Q4.21.2 NAT | 2021 | 2M | Ans: 0.52 | ● ● ○

Given,

$$N_{\max} = 650\,\text{rpm}, \qquad P = 8, \qquad f = 50\,\text{Hz}$$

Synchronous speed is

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{8} = 750\,\text{rpm}$$

Slip corresponding to maximum torque is

$$s_m = \frac{N_s - N_{\max}}{N_s} = \frac{750 - 650}{750} = 0.1333$$

For maximum torque,

$$s_m = \frac{R_2}{X_2}$$

Hence,

$$X_2 = \frac{R_2}{s_m} = \frac{0.08}{0.1333} \approx 0.60\,\Omega$$

$$R_2 = 0.08\,\Omega, \qquad X_2 = 0.60\,\Omega$$

For maximum starting torque,

$$R_2 = X_2$$

Therefore,

$$R_2 + R_{ext} = X_2$$

$$R_{ext} = X_2 - R_2$$

$$R_{ext} = 0.60 - 0.08 = 0.52\,\Omega$$

$$\boxed{0.52}$$

Key Points

- Slip at maximum torque is given by $s_m = \frac{R_2}{X_2}$.
- Maximum starting torque occurs when total rotor resistance equals rotor standstill reactance.
- External rotor resistance is added in slip-ring induction motors to improve starting torque.

Q4.21.3 NAT 2021 1M Ans: 90 ● ○ ○

Given,

$$P_{in} = 40\,\text{kW}$$

Stator losses

$$P_{stator} = 1\,\text{kW}$$

Friction and windage losses

$$P_{fw} = 2.025\,\text{kW}$$

Stator output (air-gap power):

$$P_g = P_{in} - P_{stator} = 40 - 1 = 39\,\text{kW}$$

Slip:

$$s = \frac{N_s - N_r}{N_s} = \frac{1000 - 975}{1000} = 0.025$$

Mechanical power developed:

$$P_m = P_g(1 - s) = 39(1 - 0.025) = 38.025\,\text{kW}$$

Motor shaft output:

$$P_{out} = P_m - P_{fw} = 38.025 - 2.025 = 36\,\text{kW}$$

Efficiency:

$$\eta = \frac{P_{out}}{P_{in}} \times 100$$

$$\eta = \frac{36}{40} \times 100 = 90\%$$

90

Key Points

- Air-gap power $P_g = P_{in}$ − stator losses.
- Rotor copper loss equals sP_g.
- Mechanical power developed is $P_m = P_g(1 - s)$.
- Shaft output power is obtained after subtracting friction and windage losses from P_m.

Q4.20.1 NAT 2020 1M Ans: 4.21 ● ○ ○

Synchronous speed is

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{4} = 1500\,\text{rpm}$$

No-load speed:

$$N_{nl} = N_s(1 - s_{nl}) = 1500(1 - 0.01) = 1485\,\text{rpm}$$

Full-load speed:

$$N_{fl} = N_s(1 - s_{fl}) = 1500(1 - 0.05) = 1425\,\text{rpm}$$

Percentage speed regulation is

$$\%\,\text{regulation} = \frac{N_{nl} - N_{fl}}{N_{fl}} \times 100$$

$$\%\,\text{regulation} = \frac{1485 - 1425}{1425} \times 100 = 4.21$$

$$\boxed{4.21}$$

Q4.19.1 NAT | 2019 | 2M | Ans: 14.95 | ● ○ ○

At starting,

$$s = 1, \qquad \frac{R_2}{s} = R_2$$

At 10 Hz, the reactance becomes

$$X' = X\frac{10}{50} = \frac{X}{5}$$

Since the motor is delta connected,

$$V_{ph} = 100\,\text{V}$$

Starting current per phase is

$$I_{st} = \frac{100}{(5.39 + 5.72) + j\left(\frac{16.44}{5}\right)}$$

$$|I_{st}| = \frac{100}{\sqrt{(11.11)^2 + \left(\frac{16.44}{5}\right)^2}}$$

$$|I_{st}| = \frac{100}{\sqrt{123.43 + \frac{270.2736}{25}}} = \frac{100}{11.586}$$

Line current is

$$I_L = \sqrt{3} \times \frac{100}{11.586}$$

$$I_L = 14.95\,\text{A}$$

$$\boxed{14.95}$$

Key Points

- At starting, slip is unity.
- Inductive reactance varies directly with frequency.
- For delta connection, line current is $\sqrt{3}$ times phase current.

Mistakes to be avoided

- Do not use the original reactance at 50 Hz for 10 Hz operation.
- Do not confuse phase current with line current in delta connection.

Q4.19.2 MCQ 2019 1M Ans: C ● ○ ○

In the equivalent circuit of a three-phase induction motor:

$$R_1,\ R_2',\ X_1,\ X_2'$$

are determined by the physical dimensions and winding parameters of the machine and are practically independent of the supply voltage.

The magnetizing reactance branch is represented by

$$X_m = \omega L_m$$

When the rms supply voltage is reduced at rated frequency, the air-gap flux decreases due to magnetic saturation effects. This changes the magnetizing inductance L_m and hence the magnetizing reactance X_m.

Therefore, the equivalent circuit parameter affected by reducing the rms value of the supply voltage at rated frequency is the magnetizing reactance.

$$\boxed{C}$$

Q4.17.1 MCQ 2017 1M Ans: C ● ○ ○

For a 4-pole induction machine operating as a generator,

$$N_s = \frac{120f}{P} = \frac{120 \times 60}{4} = 1800\,\text{rpm}$$

Given rotor current frequency,

$$f_r = 5\,\text{Hz}$$

For an induction machine,

$$f_r = |s| f$$

$$|s| = \frac{5}{60} = \frac{1}{12} = 0.0833$$

Since the machine is operating as an induction generator,

$$s < 0$$

Hence,

$$s = -0.0833$$

Using

$$s = \frac{N_s - N_r}{N_s}$$

$$N_r = N_s(1 - s)$$

$$N_r = 1800\,(1 + 0.0833) = 1950\,\text{rpm}$$

$$\boxed{C}$$

Q4.17.2 NAT 2017 2M Ans: 70.0533 ● ○ ○

For star connected motor,

$$I_{st} = \frac{V_{ph}}{\sqrt{(R_1 + R_2')^2 + (X_1 + X_2')^2}}$$

At 20 Hz frequency, the reactance changes as

$$X \propto f$$

$$X' = \frac{f'}{f}X = \frac{20}{60}X = \frac{X}{3}$$

$$X_1' = X_2' = \frac{0.41}{3}$$

$$I_{st} = \frac{\frac{80}{\sqrt{3}}}{\sqrt{0.6^2 + \left(\frac{0.82}{3}\right)^2}} = 70.0533\,\text{A}$$

$$\boxed{70.0533}$$

Mistakes to be avoided

- Do not use line voltage directly for a star-connected motor; convert to phase voltage first.
- Scale only reactances with frequency; resistances remain unchanged.
- Ensure all reactances are referred to the same side before addition.
- Do not forget to use standstill condition ($s = 1$) while calculating starting current.

Q4.17.3 MCQ 2017 1M Ans: C ● ○ ○

Synchronous speed is

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{4} = 1500\,\text{rpm}$$

The rotor flux rotates at synchronous speed with respect to the stationary observer. Therefore, its mechanical angular speed is

$$\omega_s = \frac{2\pi N_s}{60}$$

$$\omega_s = \frac{2\pi \times 1500}{60} = 157.08\,\text{rad/s}$$

$$\boxed{C}$$

Q4.16.1 MCQ | 2016 | 1M | Ans: C | ● ○ ○

For maximum torque,

$$s_{m,T} = \frac{R_2}{X_2}$$

Since

$$X_2 = 2\pi f L_2$$

$$s_{m,T} = \frac{R_2}{2\pi f L_2}$$

Hence,

$$s_{m,T} \propto \frac{1}{f}$$

The synchronous speed is

$$N_s = \frac{120f}{P}$$

Therefore,

$$N_s \propto f$$

Combining the above relations,

$$s_{m,T} \propto \frac{1}{N_s}$$

So, slip for maximum torque is inversely proportional to synchronous speed.

$$\boxed{C}$$

Q4.16.2 NAT | 2016 | 2M | Ans: 31.80 | ● ○ ○

Starting line current is

$$I_{st} = \frac{V}{Z_{sc}}$$

where Z_{sc} is the short-circuit impedance or standstill impedance.

$$I_1 = \frac{V_1}{Z_{sc}}$$

$$I_2 = \frac{V_2}{Z_{sc}}$$

Therefore,

$$\frac{I_2}{I_1} = \frac{V_2}{V_1}$$

$$I_2 = \frac{V_2}{V_1} I_1 = \frac{110}{415} \times 120 = 31.80\,\text{A}$$

$$\boxed{31.80}$$

Q4.15.1 NAT | 2015 | 2M | Ans: 1.192 ● ● ○

Given slip,

$$s = \frac{N_s - N_r}{N_s} = 0.04$$

$$1 - \frac{N_r}{N_s} = 0.04$$

$$N_r = 0.96N_s$$

For the 5th harmonic, the synchronous speed is

$$N_{s5} = 5N_s$$

The magnetic field produced by the 5th harmonic rotates in the direction opposite to the fundamental field. Hence, the slip with respect to the 5th harmonic field is

$$s_5 = \frac{N_{s5} + N_r}{N_{s5}} = \frac{5N_s + 0.96N_s}{5N_s}$$

$$s_5 = \frac{5 + 0.96}{5} = 1.192$$

$$\boxed{1.192}$$

Key Points

- Slip is defined as $s = \frac{N_s - N_r}{N_s}$.
- The synchronous speed of the n^{th} harmonic field is nN_s.
- Negative-sequence harmonics (such as the 5th harmonic) rotate opposite to the fundamental field.
- For an oppositely rotating harmonic field, relative speed becomes the sum of rotor speed and harmonic synchronous speed.

Q4.15.2 NAT | 2015 | 1M | Ans: 401.66 ● ● ○

At slip,

$$s = 0.05$$

The effective rotor resistance becomes

$$\frac{R_2'}{s} = \frac{0.05}{0.05} = 1\,\Omega$$

Given,

$$V_g = 210\,\text{V}$$

Current in the rotor branch is

$$I = \frac{V_g}{1 + j0.22}$$

$$I = \frac{210}{1 + j0.22} = 205.09\angle -12.40^\circ \text{ A}$$

Air-gap power is

$$P_g = 3I^2R$$

$$P_g = 3 \times (205.09)^2 \times 1$$

$$P_g = 126185.72 \text{ W}$$

Electromagnetic torque is

$$T_e = \frac{P_g}{\omega_s}$$

$$T_e = \frac{126185.72}{2\pi \times 50}$$

$$T_e = 401.66 \text{ N m}$$

$$\boxed{401.66}$$

Key Points

- Air-gap power is given by $P_g = 3I^2\left(\frac{R_2'}{s}\right)$.
- Electromagnetic torque is obtained using $T_e = \frac{P_g}{\omega_s}$.
- At a given slip, rotor resistance in the equivalent circuit becomes $\frac{R_2'}{s}$.

Q4.15.3 NAT | 2015 | 2M | Ans: 0.17 | ●○○

Given,

$$\frac{T_{\max}}{T_{FL}} = 2.25$$

Slip corresponding to maximum torque is

$$s_{m,T} = 0.15$$

For an induction motor,

$$s_{m,T} = \frac{R_2}{X_2}$$

$$0.15 = \frac{0.03}{X_2}$$

$$X_2 = 0.2\,\Omega$$

For maximum torque to occur at starting,

$$s_{m,T} = 1$$

Hence,

$$\frac{R_2 + R_{ext}}{X_2} = 1$$

$$R_2 + R_{ext} = X_2$$

$$R_{ext} = X_2 - R_2$$

$$R_{ext} = 0.2 - 0.03$$

$$R_{ext} = 0.17\,\Omega$$

$$\boxed{0.17}$$

Q4.14.1 NAT | 2014 | 1M | Ans: 3.33 | ● ○ ○

Given,

$$P = 8, \qquad f = 50\,\text{Hz}, \qquad N_r = 700\,\text{rpm}$$

Synchronous speed is

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{8} = 750\,\text{rpm}$$

Slip is

$$s = \frac{N_s - N_r}{N_s} = \frac{750 - 700}{750} = 0.0667$$

Rotor current frequency is

$$f_r = sf$$

$$f_r = 0.0667 \times 50 = 3.33\,\text{Hz}$$

$$\boxed{3.33}$$

Q4.14.2 MCQ | 2014 | 2M | Ans: C | ● ● ○

The ratio of torque at slip s to maximum torque is

$$\frac{T}{T_{\max}} = \frac{2}{\frac{s}{s_{m,T}} + \frac{s_{m,T}}{s}}$$

Slip corresponding to maximum torque is

$$s_{m,T} = \frac{R_2}{X_2} = \frac{0.1}{0.92} = 0.1087$$

Hence,

$$\frac{T_{FL}}{T_{\max}} = \frac{2}{\frac{s_{FL}}{s_{m,T}} + \frac{s_{m,T}}{s_{FL}}}$$

$$\frac{T_{FL}}{T_{\max}} = \frac{2}{\frac{0.03}{0.1087} + \frac{0.1087}{0.03}} = 0.5129$$

Therefore,

$$\frac{T_{\max}}{T_{FL}} = \frac{1}{0.5129} = 1.949$$

$\boxed{C}$

Q4.14.3 MCQ | 2014 | 1M | Ans: C | ● ○ ○

For a 4-pole induction generator running at 1500 rpm,

$$N_s = \frac{120f}{P}$$

At 50 Hz,

$$N_s = \frac{120 \times 50}{4} = 1500 \text{ rpm}$$

For induction generator operation,

$$N_r > N_s$$

Since,

$$N_r = 1500 \text{ rpm}$$

the generated frequency must satisfy

$$N_s < 1500 \text{ rpm}$$

Hence,

$$f < 50 \text{ Hz}$$

When load is partially removed, slip magnitude decreases and frequency increases.

$$f_1 < 50 \text{ Hz}, \qquad f_2 < 50 \text{ Hz}, \qquad f_2 > f_1$$

$\boxed{C}$

Key Points

- In induction generator operation, $N_r > N_s$.
- For fixed speed and fixed poles, frequency is proportional to synchronous speed.
- Reducing load reduces slip magnitude, so frequency increases.

Mistakes to be avoided

- Do not assume frequency remains 50 Hz because speed is 1500 rpm.
- For generator operation, synchronous speed must be less than rotor speed.

Q4.14.4 MCQ 2014 1M Ans: B ●●○

In constant V/f control, if the voltage is reduced linearly with frequency, then at low frequencies the stator voltage drop cannot be neglected.

The air-gap flux is proportional to:

$$\phi \propto \frac{E}{f}$$

where

$$E = V - I_a R_a$$

Therefore,

$$\phi \propto \frac{V - I_a R_a}{f}$$

At low frequencies, the stator resistance drop $I_a R_a$ becomes significant. Hence, if the voltage is reduced strictly in proportion to frequency, the effective induced emf E decreases, causing the air-gap flux to reduce below its rated value.

Therefore, to maintain constant flux at low frequencies, a higher voltage than the ideal V/f value must be applied (voltage boost).

$\boxed{B}$

Q4.14.5 MCQ 2014 2M Ans: A ●●●

The stator is supplied with a 3-phase ac supply of frequency f.

When the rotor is driven at mechanical frequency f_r in the anti-clockwise direction, the slip-ring frequency is the relative frequency between stator field and rotor speed.

Since the motor normally rotates clockwise when rotor windings are closed, anti-clockwise rotation is opposite to the rotating magnetic field. Hence,

$$f_1 = f + f_r$$

The commutator winding gives a frequency corresponding to the stator supply frequency, independent of rotor motion.

$$f_2 = f$$

Therefore,

$$f_1 = f + f_r, \qquad f_2 = f$$

$\boxed{A}$

Key Points

- Slip-ring voltage frequency depends on relative speed between rotating field and rotor.
- If rotor is driven opposite to the rotating field, relative speed increases.
- Hence, slip-ring frequency becomes $f + f_r$.

- Frequency across commutator brushes remains equal to supply frequency f.

Mistakes to be avoided

- Do not use $f - f_r$ when rotor is driven opposite to the rotating field.
- Do not confuse slip-ring frequency with commutator brush frequency.
- Pay attention to the given direction of rotation.

Q4.14.6 MCQ | 2014 | 1M | Ans: B

For an induction motor, the stability criterion is

$$\frac{dT_L}{dn} > \frac{dT_e}{dn}$$

where T_L is the load torque and T_e is the electromagnetic torque developed by the motor.

In option (a), the load torque is constant, therefore

$$\frac{dT_L}{dn} = 0$$

At point P,

$$\frac{dT_e}{dn} < 0$$

Hence,

$$\frac{dT_L}{dn} > \frac{dT_e}{dn}$$

So, P is a stable operating point.

At point Q,

$$\frac{dT_e}{dn} > 0$$

Hence,

$$\frac{dT_L}{dn} < \frac{dT_e}{dn}$$

So, Q is an unstable operating point.

In figure (b), the motor characteristic T_M is a straight line through the origin, and the load curve T_L cuts it at R and S. Between R and S the load curve lies to the left of the motor line, i.e. at the same speed

$$T_e > T_L$$

At point R, the load curve is steeper than the motor line on the speed-torque plane, so

$$\frac{dT_L}{dn} < \frac{dT_e}{dn}$$

A small rise in speed above R gives $T_e > T_L$, and the motor accelerates further away from R. So, R is an unstable operating point.

At point S, the load curve is flatter than the motor line, so

$$\frac{dT_L}{dn} > \frac{dT_e}{dn}$$

Below S the motor accelerates ($T_e > T_L$) and above S it decelerates ($T_L > T_e$), so the speed returns to S. So, S is a stable operating point.

Therefore, the stable operating points are P and S.

$\boxed{B}$

Key Points

- Stability of an operating point requires $\frac{dT_L}{dn} > \frac{dT_e}{dn}$.
- For constant torque load, $\frac{dT_L}{dn} = 0$.
- With a constant-torque load, points on the negative-slope region of the motor characteristic are stable.
- When the load torque varies with speed, compare the two slopes at the crossing point; a crossing on a rising motor characteristic (like S) can still be stable.

Mistakes to be avoided

- Using the stability criterion for torque-speed curves without checking the axis orientation.
- Confusing speed-torque curves with torque-speed curves.
- Ignoring the sign of $\frac{dT_e}{dn}$ at the operating point.
- Assuming every intersection of load and motor characteristics is stable.

Q4.13.1 MCQ 2013 1M Ans: D ●○○

Leakage flux in an induction motor is the flux that links only one winding (either stator or rotor) and does not link the other winding.

Mutual flux is responsible for energy transfer and links both stator and rotor windings, whereas leakage flux does not contribute to mutual coupling.

Therefore, leakage flux is defined as the flux that links the stator winding or the rotor winding but not both.

$\boxed{D}$

Q4.13.2 MCQ 2013 2M Ans: B ●○○

For a 4-pole induction motor supplied by a 50 Hz source,

$$N_s = \frac{120f}{P}$$

$$N_s = \frac{120 \times 50}{4} = 1500\,\text{rpm}$$

Rotor speed is

$$N_r = 1440\,\text{rpm}$$

For negative sequence current, the rotating magnetic field rotates opposite to the rotor. Hence, relative speed is

$$N_{\text{rel}} = N_s + N_r$$

$$N_{\text{rel}} = 1500 + 1440 = 2940\,\text{rpm}$$

Therefore, induced negative sequence rotor frequency is

$$f_{2n} = \frac{N_{\text{rel}}}{N_s} f$$

$$f_{2n} = \frac{2940}{1500} \times 50 = 98\,\text{Hz}$$

$\boxed{B}$

Q4.12.1 MCQ | 2012 | 1M | Ans: D | ● ○ ○

Slip of an induction motor is defined as

$$s = \frac{N_s - N_r}{N_s}$$

where N_s is the synchronous speed and N_r is the rotor speed.
The rotor speed varies with load torque. Therefore, slip depends on the shaft torque.
Also, synchronous speed appears explicitly in the expression for slip, so slip depends on synchronous speed.
However, the core-loss component is associated with iron losses in the machine and does not directly affect the slip.
Hence, slip normally does not depend on the core-loss component.

$\boxed{D}$

Q4.12.2 MCQ | 2012 | 2M | Ans: B | ● ○ ○

Neglecting rotor resistance, the blocked rotor current is

$$I_2 = \frac{V_{\text{ph}}}{X_2}$$

Since

$$X_2 = 2\pi f L_2$$

At rated condition,

$$50 = \frac{230/\sqrt{3}}{2\pi(50)L_2} \quad (1)$$

Under loaded condition,

$$I_2 = \frac{236/\sqrt{3}}{2\pi(57)L_2} \quad (2)$$

Dividing equation (2) by equation (1),

$$\frac{I_2}{50} = \frac{236}{230} \times \frac{50}{57}$$

$$I_2 = 45\,\text{A}$$

$\boxed{B}$

Q4.11.1 MCQ | 2011 | 2M | Ans: * | ● ○ ○

Synchronous speed of the stator magnetic field is

$$N_s = \frac{120f}{P}$$

$$N_s = \frac{120 \times 50}{6} = 1000\,\text{rpm}$$

Slip is given as

$$s = 5\% = 0.05$$

Rotor speed is

$$N_r = N_s(1 - s)$$

$$N_r = 1000(1 - 0.05) = 950\,\text{rpm}$$

The rotor magnetic field rotates with respect to rotor at slip speed,

$$sN_s = 0.05 \times 1000 = 50\,\text{rpm}$$

Hence, speed of rotor magnetic field with respect to stator magnetic field is

$$0$$

because both rotate at synchronous speed with respect to stator.
Speed of rotor with respect to stator magnetic field is

$$N_r - N_s = 950 - 1000 = -50\,\text{rpm}$$

So, no option is matching the final answer

*

Key Points

- Stator magnetic field rotates at synchronous speed N_s.
- Rotor speed is $N_r = N_s(1 - s)$.
- Rotor magnetic field also rotates at synchronous speed with respect to stator.
- Therefore, relative speed between stator magnetic field and rotor magnetic field is zero.

Q4.10.1 MCQ | 2010 | 1M | Ans: A | ● ○ ○

The equivalent circuit of an induction motor referred to the stator is shown below.

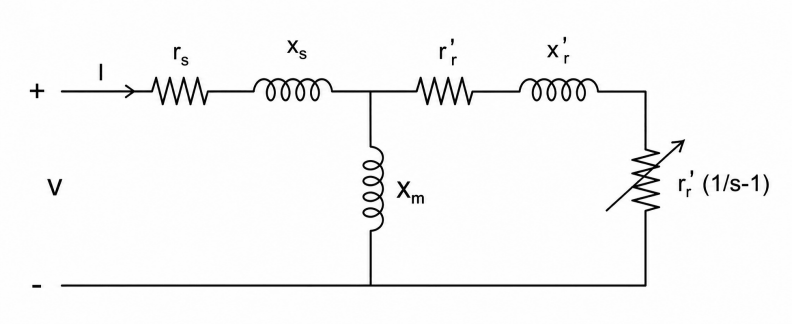


Fig. Solution Q4.10.1

At standstill,

$$s = 1$$

Hence,

$$I = I_{st}$$

The starting current is

$$I_{st} = \frac{V}{\sqrt{(r_s + r'_r)^2 + (X_s + X'_r)^2}}$$

If the magnetizing reactance X_m is neglected, the current is determined only by the series impedance.
Therefore,

$$I_{st} \propto \frac{1}{\sqrt{(r_s + r'_r)^2 + (X_s + X'_r)^2}}$$

Using the standstill equivalent circuit, the required result corresponds to option (a).

$\boxed{A}$

Q4.10.2 MCQ 2010 2M Ans: C ●●○

Synchronous speed of the induction motor is

$$N_s = \frac{120f}{P}$$

$$N_s = \frac{120 \times 50}{4} = 1500\,\text{rpm}$$

Since both machines are mechanically coupled, both rotate at the same speed of

$$N = 1600\,\text{rpm}$$

For the induction machine,

$$N_r > N_s$$

Hence, it operates in generating mode.
The DC machine, being mechanically coupled to the induction machine, acts as a motor.
Therefore, the correct answer is option (c).

$\boxed{C}$

Q4.09.1 MCQ | 2009 | 1M | Ans: A | ● ○ ○

For an induction motor, the condition for stable operation is

$$\frac{dT_e}{dn} < \frac{dT_L}{dn}$$

where T_e is the electromagnetic torque and T_L is the load torque.
The tangents at the operating points are shown in the figure.
At point A,

$$\frac{dT_L}{dn} > \frac{dT_e}{dn}$$

Hence, point A satisfies the stability criterion and is a stable operating point.
At point B, the slope of the load torque characteristic is more negative than that of the motor torque characteristic.
Therefore,

$$\frac{dT_L}{dn} < \frac{dT_e}{dn}$$

Hence, point B does not satisfy the stability criterion and is an unstable operating point.
Therefore, the correct answer is

$\boxed{A}$

Key Points

- Stability of an induction motor operating point is determined by comparing the slopes of motor and load torque characteristics.
- Stable operation requires $\frac{dT_e}{dn} < \frac{dT_L}{dn}$.
- At a stable operating point, a small speed disturbance produces a restoring torque.
- The sign and magnitude of the slopes at the intersection point are more important than the torque values themselves.

Q4.08.1 MCQ | 2008 | 2M | Ans: C | ● ○ ○

Input power to the three-phase induction motor is

$$P_{\text{in}} = \sqrt{3} V_L I_L \cos\phi$$

$$P_{\text{in}} = \sqrt{3} \times 400 \times 50 \times 0.8$$

$$P_{\text{in}} = 27.71\,\text{kW}$$

Stator losses are

$$P_{\text{stator loss}} = P_{\text{stator Cu}} + P_{\text{core}}$$

$$P_{\text{stator loss}} = 1.5 + 1.2 = 2.7\,\text{kW}$$

Air-gap power is

$$P_g = P_{\text{in}} - P_{\text{stator loss}}$$

$$P_g = 27.71 - 2.7 = 25.01\,\text{kW}$$

$\boxed{C}$

Q4.08.2 MCQ | 2008 | 2M | Ans: B | ● ● ○

Slip for maximum torque is

$$s_m = \frac{R_r'}{X_s + X_r'}$$

$$s_m = \frac{1}{1.5 + 1.5} = \frac{1}{3}$$

When frequency is changed, maximum torque occurs at starting. Hence,

$$s_m = 1$$

Therefore,

$$\frac{R_r'}{2\pi f\,(L_s + L_r')} = 1 \quad (1)$$

Initially, at $f = 50\,\text{Hz}$,

$$\frac{R_r'}{2\pi(50)\,(L_s + L_r')} = \frac{1}{3} \quad (2)$$

Dividing equation (2) by equation (1),

$$\frac{f}{50} = \frac{1}{3}$$

$$f = 16.67\,\text{Hz}$$

Due to constant $\frac{V}{f}$ control,

$$\frac{400}{50} = \frac{V}{16.67}$$

$$V = \frac{400}{3} = 133.33\,\text{V}$$

$\boxed{B}$

Key Points

- Slip at maximum torque is $s_m = \frac{R_r'}{X_s + X_r'}$.
- Reactance is directly proportional to frequency.
- For maximum torque at starting, $s_m = 1$.
- In constant V/f control, voltage is reduced in the same ratio as frequency.

Q4.08.3 MCQ 2008 2M Ans: D ● ● ○

Since the motor is fed from rotor side, it is a case of inverted induction motor.

Synchronous speed is

$$N_s = \frac{120f}{P}$$

$$N_s = \frac{120 \times 50}{4} = 1500\,\text{rpm}$$

Speed of stator magnetic field with respect to rotor is equal to the speed of rotor magnetic field.

Hence, speed of stator magnetic field with respect to rotor is

$$1500\,\text{rpm}$$

Speed of rotor magnetic field with respect to rotor is also

$$1500\,\text{rpm}$$

Given rotor speed is

$$N_r = 1410\,\text{rpm}$$

Slip is

$$s = \frac{N_s - N_r}{N_s}$$

$$s = \frac{1500 - 1410}{1500} = \frac{90}{1500} = \frac{3}{50}$$

For an inverted induction motor, the rotor moves opposite to the magnetic field.

Therefore, the correct answer is

$$\boxed{D}$$

Q4.08.4 MCQ 2008 2M Ans: C ● ○ ○

Power input to the induction motor is

$$P_{\text{in}} = W_1 + W_2$$

$$P_{\text{in}} = 1600\,\text{W}$$

Neglecting all losses,

$$P_{\text{in}} = P_g = 1600\,\text{W}$$

From the previous result,

$$s = \frac{3}{50}$$

Mechanical power developed by the induction motor is

$$P_m = P_g(1 - s)$$

$$P_m = 1600\left(1 - \frac{3}{50}\right)$$

$$P_m = 1504\,\text{W}$$

This mechanical power is supplied to the generator. Since all losses are neglected,

$$P_{\text{out}} = 1504\,\text{W}$$

The generator feeds a resistive load of $10\,\Omega$.

$$P_{\text{out}} = I^2 R$$

$$1504 = I^2 \times 10$$

$$I^2 = 150.4$$

$$I = 12.26\,\text{A}$$

$$\boxed{C}$$

Q4.07.1 MCQ | 2007 | 1M | Ans: A ●●○

For stability of an induction motor operating point,

$$\frac{dT_e}{dn} < \frac{dT_L}{dn}$$

where T_e is the electromagnetic torque and T_L is the load torque.
At point A,

$$\frac{dT_e}{dn} < 0, \qquad \frac{dT_L}{dn} > 0$$

Hence,

$$\frac{dT_e}{dn} < \frac{dT_L}{dn}$$

So, A is a stable operating point.
At point B,

$$\frac{dT_e}{dn} > \frac{dT_L}{dn}$$

So, B is an unstable operating point.
At point C,

$$\frac{dT_e}{dn} < \frac{dT_L}{dn}$$

So, C is a stable operating point.
At point D, both slopes are negative, but

$$\frac{dT_e}{dn}$$

is more negative as compared to

$$\frac{dT_L}{dn}$$

Therefore,

$$\frac{dT_L}{dn} > \frac{dT_e}{dn}$$

Hence, D is also a stable operating point.
Therefore, the stable operating points are A, C and D.

$$\boxed{A}$$

Q4.07.2 MCQ | 2007 | 2M | Ans: D | ● ○ ○

Given,

$$T_{st} = 150\% T_{rated}$$

$$T_{max} = 300\% T_{rated}$$

Therefore,

$$\frac{T_{st}}{T_{max}} = \frac{150}{300} = \frac{1}{2}$$

For an induction motor neglecting stator resistance,

$$\frac{T}{T_{max}} = \frac{2\frac{s}{s_m}}{1 + \left(\frac{s}{s_m}\right)^2}$$

At starting,

$$s = 1$$

Hence,

$$\frac{1}{2} = \frac{2\frac{1}{s_m}}{1 + \left(\frac{1}{s_m}\right)^2}$$

$$\frac{1}{2} = \frac{2s_m}{1 + s_m^2}$$

$$s_m^2 - 4s_m + 1 = 0$$

$$s_m = 2 - \sqrt{3} = 0.2679$$

$$s_m = 26.79\%$$

$$\boxed{D}$$

Key Points

- For maximum torque, slip is denoted by s_m.
- Torque ratio is $\frac{T}{T_{\max}} = \frac{2\frac{s}{s_m}}{1+\left(\frac{s}{s_m}\right)^2}$.

Q4.07.3 MCQ | 2007 | 2M | Ans: C | ● ○ ○

Given,

$$s_{FL} = 0.05$$

$$\frac{I_{SC}}{I_{FL}} = 7$$

For autotransformer starter:

$$\frac{T_{st}}{T_{FL}} = x^2\left(\frac{I_{SC}}{I_{FL}}\right)^2 s_{FL}$$

$$1.5 = x^2(7)^2(0.05)$$

$$x^2 = \frac{1.5}{49\times 0.05} = \frac{30}{49}$$

$$x = \sqrt{\frac{30}{49}} = 0.7825$$

$$x = 78.25\%$$

$\boxed{C}$

Key Points

- Starting torque is proportional to the square of the applied voltage.
- For an autotransformer starter,

$$\frac{T_{st}}{T_{FL}} = x^2\left(\frac{I_{SC}}{I_{FL}}\right)^2 s_{FL}$$

where x is the tapping ratio.

Q4.07.4 MCQ | 2007 | 2M | Ans: B | ● ○ ○

For a star-delta starter, the starting torque is

$$\frac{T_{st}}{T_{FL}} = \frac{1}{3}\left(\frac{I_{SC}}{I_{FL}}\right)^2 s_{FL}$$

Given,

$$\frac{I_{SC}}{I_{FL}} = 7 \quad \text{and} \quad s_{FL} = 0.05$$

Therefore,

$$\frac{T_{st}}{T_{FL}} = \frac{1}{3}(7)^2(0.05)$$

$$\frac{T_{st}}{T_{FL}} = \frac{1}{3} \times 49 \times 0.05$$

$$\frac{T_{st}}{T_{FL}} = 0.816$$

Hence, the correct answer is

$$\boxed{B}$$

Key Points

- In star-delta starting, starting torque becomes one-third of direct-on-line starting torque.

Q.07.5 MCQ | 2007 | 2M | Ans: C | ● ○ ○

Given starting torque,

$$T_{st} = 0.5 \text{ p.u.}$$

Hence,

$$\frac{T_{st}}{T_{FL}} = 0.5$$

For an induction motor,

$$\frac{T_{st}}{T_{FL}} = \left(\frac{I_{st}}{I_{FL}}\right)^2 s_{FL}$$

Substituting $s_{FL} = 0.05$,

$$0.5 = \left(\frac{I_{st}}{I_{FL}}\right)^2 (0.05)$$

$$\left(\frac{I_{st}}{I_{FL}}\right)^2 = \frac{0.5}{0.05} = 10$$

$$\frac{I_{st}}{I_{FL}} = \sqrt{10} = 3.16$$

Therefore,

$$\boxed{C}$$

Q4.06.1 MCQ | 2006 | 1M | Ans: D | ● ○ ○

To keep the air-gap flux constant in the constant torque region,

$$\phi \propto \frac{V}{f}$$

Hence,

$$\frac{V}{f} = \text{constant}$$

Therefore, the inverter output voltage must increase linearly with frequency up to rated frequency.

For operation above rated frequency, the inverter voltage cannot be increased beyond rated voltage. Hence, the voltage remains constant and flux weakens.

Thus, the voltage-frequency characteristic is linear up to rated frequency and constant beyond rated frequency.

$\boxed{D}$

Q4.06.2 MCQ 2006 2M Ans: C ●○○

For constant load torque operation under constant $\frac{V}{f}$ control of an induction motor,

$$N_s - N_r = \text{constant}$$

Initially, at

$$f_1 = 50\,\text{Hz}$$

$$N_{s1} = \frac{120 f_1}{P} = \frac{120 \times 50}{4} = 1500\,\text{rpm}$$

Given,

$$N_{r1} = 1440\,\text{rpm}$$

Hence,

$$N_{s1} - N_{r1} = 1500 - 1440 = 60\,\text{rpm}$$

At

$$f_2 = 30\,\text{Hz}$$

$$N_{s2} = \frac{120 f_2}{P} = \frac{120 \times 30}{4} = 900\,\text{rpm}$$

Since rotor speed deficit remains constant,

$$N_{s1} - N_{r1} = N_{s2} - N_{r2}$$

$$60 = 900 - N_{r2}$$

$$N_{r2} = 900 - 60 = 840\,\text{rpm}$$

Therefore,

$\boxed{C}$

Key Points

- Under constant $\frac{V}{f}$ control, air-gap flux remains approximately constant.

- For constant load torque, slip frequency remains nearly constant.
- Hence, the speed difference $(N_s - N_r)$ remains constant.
- Synchronous speed is given by $N_s = \dfrac{120f}{P}$.

Q4.06.3 MCQ | 2006 | 2M | Ans: A | ●●○

In the approximate equivalent circuit model, X_m is neglected.

At starting,

$$s = 1$$

The starting torque is given by

$$T_{st} = \frac{3}{\omega_s}\left[\frac{V_{ph}^2\left(\frac{R_2'}{s}\right)}{\left(R_1 + \frac{R_2'}{s}\right)^2 + \left(X_1 + X_2'\right)^2}\right]$$

For $s = 1$,

$$T_{st} = \frac{3}{\omega_s}\left[\frac{V_{ph}^2 R_2'}{\left(R_1 + R_2'\right)^2 + \left(X_1 + X_2'\right)^2}\right]$$

Synchronous speed:

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{4} = 1500\,\text{rpm}$$

$$\omega_s = \frac{2\pi N_s}{60} = \frac{2\pi \times 1500}{60} = 50\pi\,\text{rad/s}$$

Phase voltage:

$$V_{ph} = \frac{400}{\sqrt{3}}\,\text{V}$$

Substituting

$$R_1 = 1\,\Omega, \quad R_2' = 0.5\,\Omega, \quad X_1 = 1.2\,\Omega, \quad X_2' = 1.2\,\Omega$$

$$T_{st} = \frac{3}{50\pi}\left[\frac{\left(\frac{400}{\sqrt{3}}\right)^2 (0.5)}{(1 + 0.5)^2 + (1.2 + 1.2)^2}\right]$$

$$T_{st} = 63.6\,\text{N m}$$

Hence,

$$\boxed{A}$$

Q4.06.4 MCQ | 2006 | 2M | Ans: B | ● ● ○

Given,

$$I_{FL} = 20\,\text{A}$$

$$I_{BR} = 15\,\text{A}$$

Copper loss obtained from blocked rotor test:

$$P_{cu,BR} = 762\,\text{W}$$

Since copper loss varies as the square of current,

$$P_{cu,FL} = \left(\frac{I_{FL}}{I_{BR}}\right)^2 P_{cu,BR}$$

$$P_{cu,FL} = \left(\frac{20}{15}\right)^2 \times 762 = 1354.67\,\text{W}$$

Core loss at rated voltage:

$$P_{core} = 1002\,\text{W}$$

Hence, total losses are

$$P_{loss} = P_{cu,FL} + P_{core}$$

$$P_{loss} = 1354.67 + 1002 = 2356.67\,\text{W}$$

Given output power,

$$P_{out} = 10000\,\text{W}$$

Efficiency,

$$\eta = \frac{P_{out}}{P_{out} + P_{loss}} \times 100$$

$$\eta = \frac{10000}{10000 + 2356.67} \times 100 = 80.93\%$$

$$\eta \approx 81\%$$

$\boxed{B}$

Key Points

- Copper loss varies as the square of current, i.e., $P_{cu} \propto I^2$.
- Blocked rotor test is used to determine the full-load copper loss.
- Core loss is assumed constant at rated voltage.
- Efficiency is given by $\eta = \frac{P_{out}}{P_{out} + P_{loss}} \times 100$.

Mistakes to be avoided

- Using a linear current ratio instead of a squared current ratio for copper loss calculation.
- Omitting core loss while calculating total losses.

Q4.05.1 MCQ 2005 1M Ans: A ● ○ ○

Slip greater than unity indicates that the induction machine is operating in the braking (plugging) region.

In the motoring region,

$$s = \frac{N_s - N_r}{N_s} < 1$$

In the braking (plugging) region, the rotor rotates opposite to the rotating magnetic field. Therefore,

$$s = \frac{N_s + N_r}{N_s} > 1$$

Since the rotor and rotating magnetic field move in opposite directions, the developed torque opposes the rotor motion, producing braking action.

Hence, the correct option is

$\boxed{A}$

Key Points

- Motoring operation: $0 < s < 1$.
- Generating operation: $s < 0$.
- Plugging (braking) operation: $s > 1$.
- During plugging, torque and rotor speed act in opposite directions.

Mistakes to be avoided

- Assuming slip cannot exceed unity.
- Confusing plugging operation with generating operation.
- Using $s = \frac{N_s - N_r}{N_s}$ in the braking region where rotor rotates opposite to the field.
- Forgetting that braking torque opposes the direction of rotor motion.

Q4.05.2 MCQ 2005 1M Ans: B ● ○ ○

For an induction motor,

$$P_{ag} = \text{Air-gap power}$$

Rotor copper loss is

$$P_{cu} = sP_{ag}$$

Hence, gross mechanical power developed (gross power output) is

$$P_g = P_{ag} - P_{cu}$$

$$P_g = P_{ag} - sP_{ag}$$

$$P_g = (1 - s)P_{ag}$$

Therefore,

$$\frac{P_g}{P_{ag}} = 1 - s$$

Hence,

$$\boxed{B}$$

Q4.05.3 MCQ | 2005 | 2M | Ans: D | ● ● ○

At no-load, the slip of an induction motor is nearly zero. Therefore, the rotor branch of the equivalent circuit can be neglected and the circuit reduces to the shunt magnetizing branch.

$$Z = (R_c) \parallel (jX_m)$$

The no-load current is

$$I_0 = \frac{V}{|Z|}$$

If the supply voltage V is reduced, the no-load current I_0 also reduces approximately in the same proportion. However, since the motor is operating at no-load,

$$s \approx 0$$

and hence the rotor speed is nearly equal to synchronous speed,

$$N_r \approx N_s$$

$$N_s = \frac{120f}{P}$$

Since frequency remains unchanged, synchronous speed remains unchanged. Therefore, the no-load speed of the induction motor remains practically constant even though the supply voltage is reduced.

$$N_r \approx N_s = \text{constant}$$

Hence, the correct option is

$$\boxed{D}$$

Q4.05.4 MCQ | 2005 | 2M | Ans: B | ● ○ ○

Developed torque in an induction motor is

$$T = \frac{3}{\omega_s} I_2'^2 \frac{R_2'}{s}$$

Hence, the ratio of starting torque to full-load torque is

$$\frac{(T)_{\text{starting}}}{(T)_{\text{full-load}}} = \left(\frac{I_{\text{st}}}{I_{\text{fl}}}\right)^2 s_{\text{fl}}$$

In the above expression, rotor current is replaced by stator current since the ratio of rotor and stator currents remains the same.

Given,

$$\frac{I_{st}}{I_{fl}} = 6$$

and

$$s_{fl} = 0.04$$

Therefore,

$$\frac{(T)_{starting}}{(T)_{full\text{-}load}} = 6^2 \times 0.04$$

$$\frac{(T)_{starting}}{(T)_{full\text{-}load}} = 36 \times 0.04 = 1.44$$

Hence,

$$(T)_{starting} = 1.44\,(T)_{full\text{-}load}$$

$\boxed{B}$

Q4.05.5 MCQ | 2005 | 2M | Ans: A | ●●○

The resultant air-gap flux per pole is proportional to the ratio of voltage to frequency.

$$\phi_m \propto \frac{V}{f}$$

To avoid saturation of the stator and rotor cores, the flux must be kept constant. Therefore,

$$\frac{V}{f} = \text{constant}$$

The maximum torque of an induction motor is

$$T_{max} = \frac{3}{2\omega_s}\frac{V^2}{X_2'}$$

Since

$$\omega_s \propto f$$

and

$$X_2' \propto f$$

therefore,

$$T_{max} \propto \frac{V^2}{f^2}$$

or

$$T_{max} \propto \left(\frac{V}{f}\right)^2$$

If both V and f are reduced while maintaining a constant $\frac{V}{f}$ ratio, then

$$T_{max} = \text{constant}$$

Hence, both the assertion and reason are true, and the reason correctly explains the assertion.

$$\boxed{A}$$

Q4.04.1 MCQ | 2004 | 1M | Ans: C | ● ○ ○

Reversing the phase sequence of currents in the stator winding reverses the direction of the rotating magnetic field.

Since the rotor of an induction motor follows the direction of the rotating magnetic field, the rotor direction also reverses.

If clockwise rotation is produced by the positive phase sequence

$$A - B - C$$

then anticlockwise rotation is produced by the negative phase sequence

$$A - C - B$$

Hence, the required phase sequence is

$$A - C - B$$

$$\boxed{C}$$

Key Points

- Direction of rotation of a 3-phase induction motor depends on the phase sequence of the supply.
- Interchanging any two supply lines reverses the phase sequence.
- Reversing the phase sequence reverses the rotating magnetic field direction.
- The rotor rotates in the same direction as the rotating magnetic field.

Q4.04.2 MCQ | 2004 | 2M | Ans: C | ● ○ ○

For a 4-pole, 50 Hz induction motor,

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{4} = 1500\,\text{rpm}$$

Rotor speed at full load is

$$N_r = N_s(1 - s) = 1500(1 - 0.04) = 1440\,\text{rpm}$$

Given output power,

$$P_{out} = 15\,\text{kW} = 15000\,\text{W}$$

Torque developed is

$$T = \frac{P_{out}}{\omega_r}$$

where

$$\omega_r = \frac{2\pi N_r}{60} = \frac{2\pi \times 1440}{60}$$

Hence,

$$T = \frac{15000}{\frac{2\pi \times 1440}{60}} = 99.47\,\text{N} \cdot \text{m}$$

$\boxed{C}$

Q4.04.3 MCQ | 2004 | 2M | Ans: A ● ● ○

For an 8-pole, 50 Hz induction machine, the synchronous speed of the fundamental rotating magnetic field is

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{8} = 750\,\text{rpm}$$

The synchronous speed of the n^{th} order space harmonic is

$$N_{s,n} = \frac{N_s}{n}$$

For the 7^{th} space harmonic,

$$N_{s,7} = \frac{750}{7} = 107.14\,\text{rpm}$$

Also, harmonics of order $(3k+1)$ are positive-sequence harmonics and rotate in the forward direction, whereas harmonics of order $(3k-1)$ are negative-sequence harmonics and rotate in the reverse direction.

Since

$$7 = 3(2) + 1$$

the 7^{th} harmonic is a positive-sequence harmonic and rotates in the forward direction.

$\boxed{A}$

Key Points

- Fundamental synchronous speed: $N_s = \frac{120f}{P}$.
- Synchronous speed of n^{th} space harmonic: $N_{s,n} = \frac{N_s}{n}$.
- Harmonics of order $(3k + 1)$ are positive-sequence (forward rotating).
- Harmonics of order $(3k - 1)$ are negative-sequence (backward rotating).

Q4.03.1 MCQ | 2003 | 1M | Ans: D ● ● ○

At no-load, shaft output is zero. Hence, input power supplies only the losses:

$$P_0 = P_i + P_{wf}$$

where P_i is the iron loss and P_{wf} is the friction and windage loss.

Iron loss varies approximately as the square of voltage:

$$P_i \propto V^2$$

Since slip changes very little at no-load, friction and windage loss remain almost constant.

Therefore, by plotting P_0 versus V and extrapolating the curve to $V = 0$, the iron loss becomes zero:

$$P_i = 0$$

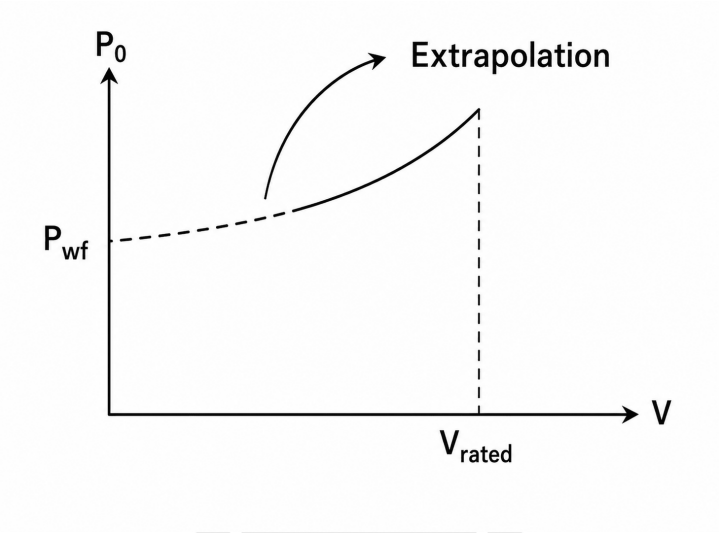


Fig. Solution Q4.03.1

Hence,

$$P_0 = P_{wf}$$

$$\boxed{D}$$

Q4.03.2 MCQ | 2003 | 2M | Ans: A | ● ○ ○

For a D.C. machine, the stator field winding carries D.C. current, and the rotor armature current is alternating in nature but is commutated by the commutator. Also, the stator has salient poles and the rotor has a commutator.

$$\text{D.C. machine} \rightarrow P, S$$

For a synchronous machine, the stator winding carries A.C. current and the rotor field winding carries D.C. current. The rotor generally has salient poles with slip rings.

$$\text{Synchronous machine} \rightarrow Q, T$$

For an induction machine, both stator and rotor currents are A.C. Also, both stator and rotor may have poly-phase windings.

$$\text{Induction machine} \rightarrow R, U$$

Hence, the correct matching is

$$\boxed{A}$$

Q4.03.3 MCQ | 2003 | 2M | Ans: B | ● ● ○

For an induction motor,

$$\phi \propto \frac{V}{f}$$

Since both voltage and frequency are halved,

$$\frac{V_2}{f_2} = \frac{V_1/2}{f_1/2} = \frac{V_1}{f_1}$$

Hence, the air-gap flux remains constant.
Under normal running conditions, the electromagnetic torque can be approximated as

$$T = \frac{3}{\omega_s}\frac{sV^2}{R_2'}$$

Since the load torque remains constant,

$$\frac{3}{\omega_{s1}}\frac{s_1V_1^2}{R_2'} = \frac{3}{\omega_{s2}}\frac{s_2V_2^2}{R_2'}$$

or

$$\frac{s_2}{s_1} = \frac{\omega_{s2}}{\omega_{s1}}\frac{V_1^2}{V_2^2}$$

Since frequency is halved,

$$\frac{\omega_{s2}}{\omega_{s1}} = \frac{1}{2}$$

and

$$\frac{V_2}{V_1} = \frac{1}{2}$$

Therefore,

$$\frac{s_2}{s_1} = \frac{1}{2} \times 4 = 2$$

$$s_2 = 2s_1$$

Using the slip relation,

$$s = \frac{N_s - N_r}{N_s}$$

$$\frac{N_{s2} - N_{r2}}{N_{s2}} = 2\frac{N_{s1} - N_{r1}}{N_{s1}}$$

Since

$$\frac{N_{s2}}{N_{s1}} = \frac{1}{2}$$

we get

$$N_{s2} - N_{r2} = 2\left(N_{s1} - N_{r1}\right)\frac{N_{s2}}{N_{s1}} = N_{s1} - N_{r1}$$

Hence, the difference between synchronous speed and rotor speed remains unchanged.

Also,

$$T \propto \phi I_2$$

Since load torque is constant and flux is constant,

$$I_2 = \text{constant}$$

Therefore, the rotor current and its reflected stator component remain unchanged.

$$\boxed{B}$$

Q4.03.4 MCQ | 2003 | 2M | Ans: C | ● ○ ○

The rated synchronous speed nearest to the rated speed is

$$N_{s1} = 3000\,\text{rpm}$$

Hence, the rated slip is

$$s_1 = \frac{N_{s1} - N_{r1}}{N_{s1}} = \frac{3000 - 2850}{3000} = 0.05$$

At half the rated slip,

$$s_2 = \frac{s_1}{2} = 0.025$$

At 40 Hz, the synchronous speed becomes

$$N_{s2} = N_{s1}\left(\frac{40}{50}\right) = 3000 \times \frac{40}{50} = 2400\,\text{rpm}$$

The rotor speed is

$$N_{r2} = N_{s2}(1 - s_2)$$

$$N_{r2} = 2400(1 - 0.025) = 2400 \times 0.975 = 2340\,\text{rpm}$$

The correct option is

$$\boxed{C}$$

Q4.02.1 MCQ | 2002 | 1M | Ans: A | ● ○ ○

The torque developed in an electrical machine is proportional to the product of air-gap flux and rotor (or armature) current.

$$T \propto \phi I_2'$$

Since the motor draws rated current in both cases,

$$I_2' = \text{constant}$$

Also,

$$\phi \propto \frac{V}{f}$$

For a constant supply voltage of 400 V,

$$\phi \propto \frac{1}{f}$$

Therefore,

$$T \propto \phi \propto \frac{1}{f}$$

Comparing the two operating frequencies,

$$\frac{T_{75}}{T_{50}} = \frac{\phi_{75}}{\phi_{50}} = \frac{50}{75} = \frac{2}{3}$$

Hence, the torque developed at 75 Hz, 400 V is less than the torque developed at 50 Hz, 400 V.

$$\boxed{A}$$

Key Points

- Electromagnetic torque is proportional to the product of air-gap flux and rotor current:

$$T \propto \phi I_2'$$

- Air-gap flux is approximately proportional to the V/f ratio:

$$\phi \propto \frac{V}{f}$$

- For constant voltage operation, increasing frequency reduces flux and hence reduces torque capability.

Q4.02.2 MCQ 2002 2M Ans: C ● ○ ○

Since the stator and rotor are wound for different numbers of poles, the magnetic fields produced by them do not remain locked with each other.

Hence, a unidirectional average torque cannot be developed. The developed torque is oscillatory in nature, resulting in zero average torque.

Therefore, the motor cannot sustain rotation and the steady-state speed becomes

$$N_r = 0$$

Hence,

$$\boxed{C}$$

Key Points

- In a conventional induction motor, stator and rotor must effectively produce magnetic fields of the same pole number.
- Electromagnetic torque is produced by the interaction of synchronously rotating stator and rotor magnetic fields.
- Different pole numbers lead to an oscillating torque with zero average value.
- Zero average torque implies zero steady-state speed.

Q4.02.3 NAT | 2002 | 2M | Ans: 1040,268.666 | ● ● ○

The equivalent circuit of the induction motor referred to the stator is shown below.

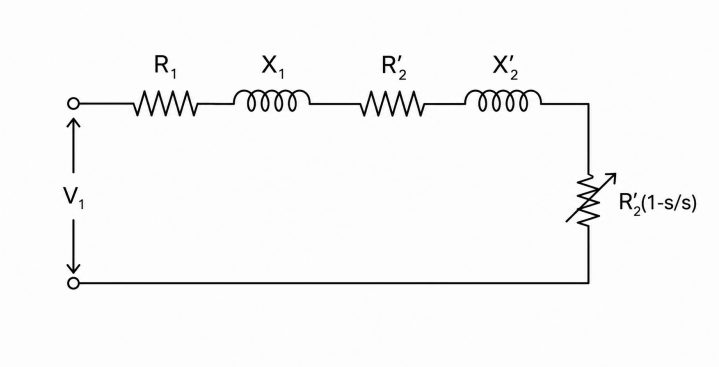


Fig. Solution Q4.02.3

The synchronous speed is

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{6} = 1000\,\text{rpm}$$

The slip at rated speed is

$$s = \frac{N_s - N_r}{N_s} = \frac{1000 - 960}{1000} = 0.04$$

If the machine is run above rated speed with the same stator current, it operates as an induction generator with negative slip of the same magnitude.

$$s = -0.04$$

Hence,

$$N = 1000(1 + 0.04) = 1040\,\text{rpm}$$

The stator current becomes

$$I_1' = \frac{440}{\sqrt{3}\sqrt{\left(0.6 - \frac{0.3}{0.04}\right)^2 + 2^2}}$$

$$I_1' = 35.361\,\text{A}$$

The developed torque is

$$T = \frac{180}{2\pi \times 1000}(35.361)^2 \frac{0.3}{0.04}$$

$$T = 268.66\,\text{N} \cdot \text{m}$$

Therefore,

$$\boxed{1040, 268.66}$$

Key Points

- Negative slip indicates induction generator operation.
- Developed torque is proportional to rotor power divided by synchronous angular speed.

Q4.00.1 MCQ 2000 1M Ans: C ● ○ ○

For the production of steady torque in an induction machine, the rotor must have the same number of poles as the stator.

Hence,

$$\text{Number of rotor poles} = \text{Number of stator poles} = 4$$

The number of rotor phases is given by

$$\text{Number of rotor phases} = \frac{\text{Number of rotor slots}}{\text{Number of rotor poles}}$$

$$= \frac{28}{4} = 7$$

Therefore,

$$\boxed{C}$$

Key Points

- Stator and rotor must have the same number of poles to develop steady electromagnetic torque.
- Rotor phase number can be determined from the ratio of rotor slots to rotor poles.
- In a polyphase wound rotor, adjacent slots belonging to one phase form a repeating pattern over the pole pitch.

Q4.00.2 MCQ 2000 2M Ans: B ● ○ ○

The synchronous speed is

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{6} = 1000\,\text{rpm}$$

Slip is

$$s = \frac{N_s - N_r}{N_s} = \frac{1000 - 975}{1000} = 0.025$$

Air-gap power is

$$P_g = P_{\text{in}} - \text{stator losses} = 40 - 1 = 39\,\text{kW}$$

Mechanical power developed is

$$P_m = P_g(1 - s) = 39(1 - 0.025) = 38.025\,\text{kW}$$

Output power is

$$P_{\text{out}} = P_m - \text{friction and windage losses} = 38.025 - 2 = 36.025\,\text{kW}$$

Efficiency is

$$\eta = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100 = \frac{36.025}{40} \times 100 = 90.06\%$$

Therefore,

$$\boxed{B}$$

Q4.00.3 MCQ 2000 2M Ans: D ●○○

It is given that the output (developed) power is P_o.

The air-gap power is related to the developed power by

$$P_o = P_g(1 - s)$$

Hence,

$$P_g = \frac{P_o}{1 - s}$$

Rotor copper loss is

$$P_{cu} = sP_g$$

Substituting for P_g,

$$P_{cu} = s\left(\frac{P_o}{1 - s}\right) = \frac{sP_o}{1 - s}$$

Rotor copper losses can be recovered at the slip-ring terminals and utilized to drive another induction motor in a cascade arrangement.

Therefore,

$$\boxed{D}$$

Q4.00.4 NAT 2000 5M Ans: 1100.7 ●○○

Rated synchronous speed is

$$N_{s1} = \frac{120 f_1}{P} = \frac{120 \times 60}{6} = 1200\,\text{rpm}$$

Rated slip is

$$s_1 = \frac{N_{s1} - N_{r1}}{N_{s1}} = \frac{1200 - 1175}{1200} = \frac{1}{48}$$

At high speed, the developed torque is approximated as

$$T_{\text{dev}} = \frac{3}{\omega_s}\frac{sV_{\text{ph}}^2}{R_2}$$

Since synchronous speed is proportional to frequency,

$$\omega_s \propto f$$

A 6% drop in frequency gives

$$\frac{\omega_{s2}}{\omega_{s1}} = \frac{f_2}{f_1} = 0.94$$

Due to 10% drop in voltage,

$$\frac{V_2}{V_1} = 0.90$$

For the same developed torque,

$$T_2 = T_1$$

$$\frac{3}{\omega_{s2}} \frac{s_2 V_2^2}{R_2} = \frac{3}{\omega_{s1}} \frac{s_1 V_1^2}{R_2}$$

Hence,

$$s_2 = s_1 \frac{\omega_{s2}}{\omega_{s1}} \left(\frac{V_1}{V_2}\right)^2$$

$$s_2 = \frac{1}{48} \times 0.94 \times \left(\frac{1}{0.9}\right)^2 = 0.0242$$

New synchronous speed is

$$N_{s2} = 0.94 N_{s1} = 0.94 \times 1200 = 1128 \text{ rpm}$$

Rotor speed is

$$N_2 = N_{s2}(1 - s_2) = 1128(1 - 0.0242) = 1100.7 \text{ rpm}$$

Therefore,

$$\boxed{1100.7}$$

Key Points

- At low slip, induction motor torque is approximately

$$T_{\text{dev}} = \frac{3}{\omega_s} \frac{s V_{\text{ph}}^2}{R_2}$$

- For constant torque, slip changes according to both voltage and frequency.

Q4.99.1 MCQ | 1999 | 1M | Ans: C | ● ○ ○

The effectiveness of a starting method is often judged by the ratio.

$$\frac{\text{Starting Torque}}{\text{Line Current}}$$

For different induction motor starters:

- Direct-on-line (DOL) starter provides full voltage and hence a comparatively high torque per ampere.
- Autotransformer starter provides better torque per line current due to voltage transformation.
- Star-delta starter gives reduced current with reasonable starting torque.
- Series reactor (series inductor) starter reduces voltage by inserting inductive reactance in series with the motor. This causes a poor power factor and results in the lowest starting torque per ampere of line current drawn.

Hence, the series inductor method is inferior from the viewpoint of starting torque per ampere of line current drawn.

$\boxed{C}$

Q4.99.2 NAT | 1999 | 5M | Ans: 6.89,0.348 | ● ● ●

Given:

$$\text{Rating: } 3\,\text{kW},\ 400\,\text{V}/200\,\text{V},\ \Delta - Y,\ 50\,\text{Hz}$$

Under blocked rotor conditions,

$$s = 1$$

Synchronous angular speed is

$$\omega_s = \frac{120f}{P} \times \frac{2\pi}{60} = 104.72\,\text{rad/s}$$

Turns ratio is

$$a = \frac{400}{200/\sqrt{3}} = 2\sqrt{3} : 1$$

Rotor parameters referred to the stator are

$$R_2' = a^2 R_2 = (2\sqrt{3})^2 R_2 = 12R_2$$

$$X_2' = a^2 X_2 = 12X_2$$

For the equivalent circuit referred to the stator,

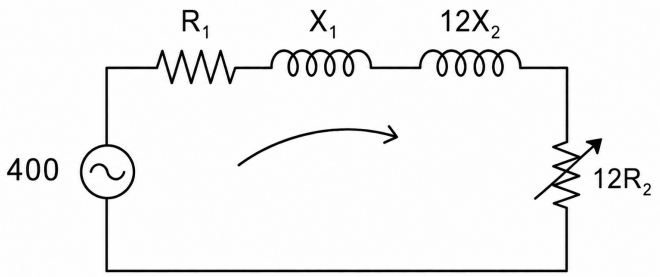


Fig. Solution Q4.99.2

$$R_{eq} = R_1 + 12R_2$$

$$X_{eq} = X_1 + 12X_2$$

Since the current drawn has power factor 0.4,

$$\tan\left[\cos^{-1}(0.4)\right] = \frac{X_{eq}}{R_{eq}}$$

$$X_{eq}^2 = 5.25\, R_{eq}^2 \tag{1}$$

Input impedance is

$$Z = \frac{400}{25/\sqrt{3}} = 27.71\,\Omega$$

Hence,

$$(27.71)^2 = R_{eq}^2 + X_{eq}^2$$

Using (1),

$$(27.71)^2 = 6.25\, R_{eq}^2$$

$$R_{eq}^2 = \frac{(27.71)^2}{6.25} = 122.85$$

$$R_{eq} = 11.084\,\Omega$$

Therefore,

$$R_1 + 12R_2 = 11.084 \tag{2}$$

The developed torque is

$$T = \frac{P_g}{\omega_s} = \frac{3}{\omega_s}(I_s)^2 R_2'$$

Given

$$T = 25\,\text{N}\cdot\text{m}$$

$$\frac{3}{104.72}\left(\frac{25}{\sqrt{3}}\right)^2 (12R_2) = 25$$

$$R_2 = 0.3489\,\Omega$$

Substituting in (2),

$$R_1 = 11.084 - 12(0.3489) = 6.8972\,\Omega$$

Hence,

$$\boxed{6.8972, 0.3489}$$

Key Points

- Rotor parameters referred to stator:
$$R_2' = a^2 R_2, \qquad X_2' = a^2 X_2$$
- Power factor relation:
$$\tan\phi = \frac{X_{eq}}{R_{eq}}$$
- Blocked rotor condition implies:
$$s = 1$$

- Developed torque:

$$T = \frac{P_g}{\omega_s}$$

Mistakes to be avoided

- Do not forget to refer rotor parameters to the stator side using a^2.
- Use phase current $\left(\frac{25}{\sqrt{3}}\right)$ A for the Δ-connected stator.
- Under blocked rotor condition, always take $s = 1$.
- Do not confuse impedance magnitude with resistance component.

Q4.98.1 MCQ | 1998 | 1M | Ans: D | ● ○ ○

Given,

$$(\eta)_{fl} = 0.8$$

$$(\eta)_{\max} = 0.9$$

and

$$s = 0.6$$

The efficiency of an induction motor depends on stator copper loss, rotor copper loss, core loss, friction and windage loss, and load condition.

Although the full-load efficiency and maximum efficiency are specified, the fixed losses (core loss and mechanical losses) are not given.

Without the fixed-loss information, the operating point corresponding to $s = 0.6$ cannot be uniquely determined, and hence the efficiency at the given slip cannot be calculated.

Therefore, the data provided is insufficient to determine the efficiency.

$\boxed{D}$

Q4.98.2 MCQ | 1998 | 2M | Ans: A_Q, B_P, C_S | ● ○ ○

The correct matching is obtained from the applications and purposes of the respective tests.

(A) No-load and blocked rotor tests are conducted on an induction machine to determine its equivalent circuit parameters.

$$(A) \rightarrow (Q)$$

The no-load test is used to obtain shunt branch parameters, whereas the blocked rotor test is used to obtain series parameters.

(B) Sumpner's test is performed on transformers to determine the maximum temperature rise and efficiency under full-load conditions.

$$(B) \rightarrow (P)$$

(C) Swinburne's test is performed on DC machines and is used to estimate no-load losses and efficiency.

$$(C) \rightarrow (S)$$

Hence,

$$\boxed{(A) \rightarrow (Q),\ (B) \rightarrow (P),\ (C) \rightarrow (S)}$$

Q4.98.3 NAT | 1998 | 5M | Ans: 0.27 | ● ● ○

Given:

$$50\,\text{hp},\quad 6\text{-pole},\quad 50\,\text{Hz}$$

$$(N_r)_{fl} = 960\,\text{rpm}$$

$$(I_2)_{fl} = 40\,\text{A}$$

$$(P_{cu})_{\text{rotor, ext}} = 300\,\text{W}$$

$$(P)_{\text{mech}} = 1200\,\text{W}$$

The synchronous speed is

$$N_s = \frac{120 \times 50}{6} = 1000\,\text{rpm}$$

The full-load slip is

$$s = \frac{N_s - N_r}{N_s} = \frac{1000 - 960}{1000} = 0.04$$

Mechanical power developed is

$$P_d = P_g - (P_{cu})_{\text{rotor}} = P_g - sP_g = (1 - s)P_g$$

where P_g is the air-gap power.
The shaft output power is

$$P_{\text{shaft}} = 50 \times 746 = 37.3\,\text{kW}$$

Hence,

$$P_d = P_{\text{shaft}} + P_{\text{mech}} = 37.3 + 1.2 = 38.5\,\text{kW}$$

Therefore,

$$P_g = \frac{P_d}{1 - s} = \frac{38500}{0.96}$$

Rotor copper loss is

$$P_{rcu} = sP_g = 3I_2^2R_2 + 300$$

Substituting the values,

$$0.04\left(\frac{38500}{0.96}\right) = 3(40)^2R_2 + 300$$

$$1604.167 = 4800R_2 + 300$$

$$4800R_2 = 1304.167$$

$$R_2 = \frac{1304.167}{3(40)^2} = 0.2717\,\Omega$$

Hence,

$$\boxed{0.27}$$

Q4.97.1 NAT | 1997 | 2M | Ans: 240,800 | ●●○

Given:

$$3-\phi, \quad 20\,\text{kW}, \quad 400\,\text{V}, \quad 1470\,\text{rpm}, \quad 50\,\text{Hz}$$

For constant air-gap flux,

$$\frac{V_1}{f_1} = \frac{V_2}{f_2}$$

Hence,

$$V_2 = \frac{f_2}{f_1}V_1 = \frac{30}{50} \times 400 = 240\,\text{V}$$

The nearest synchronous speed is

$$N_{s1} = 1500\,\text{rpm}$$

Slip is

$$s_1 = \frac{N_{s1} - N_{r1}}{N_{s1}} = \frac{1500 - 1470}{1500} = \frac{1}{50}$$

For constant air-gap flux, torque is proportional to rotor current. Since torque is constant, rotor current remains the same.

$$S_1\omega_{s1} = S_2\omega_{s2}$$

$$S_2 = \frac{\omega_{s1}}{\omega_{s2}}S_1 = \frac{f_1}{f_2}S_1$$

$$S_2 = \frac{50}{30} \times \frac{1}{15} = \frac{1}{9}$$

At 30 Hz,

$$N_{s2} = \frac{120 \times 30}{4} = 900\,\text{rpm}$$

Therefore,

$$N_2 = N_{s2}(1 - S_2) = 900\left(1 - \frac{1}{9}\right) = 800\,\text{rpm}$$

Hence,

$$\boxed{240,\ 800}$$

Q4.97.2 NAT | 1997 | 5M | Ans: * | ● ● ●

Given rated output power

$$P = 50 \times 10^3 \text{ W}$$

Assuming the induction motor is delta connected,

$$I_{fl} = \frac{P}{\sqrt{3}V} = \frac{50 \times 10^3}{\sqrt{3} \times 400} = 72.17 \text{ A}$$

Let the tapping ratio of the autotransformer be x.
The short-circuit impedance per phase is

$$Z_{ph} = 0.866\,\Omega$$

Hence, short circuit current can be calculated as

$$I_{sc} = \frac{400}{Z_{ph}} = \frac{400}{0.866} = 461.89 \text{ A}$$

Here, the calculated short-circuit current is lower than the specified maximum allowable starting supply current of 1000, A, which is physically inconsistent and does not represent a practical starting condition. Therefore, for a meaningful analysis, the maximum allowable starting supply current is assumed to be 100, A instead of 1000, A.

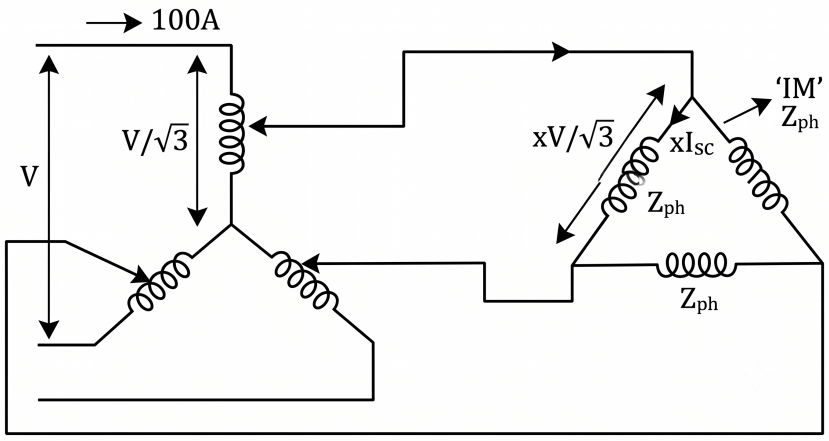


Fig. Solution Q4.97.2

Since input kVA and output kVA of an autotransformer are equal,

$$\left(\frac{V}{\sqrt{3}}\right)(100) = \left(x\frac{V}{\sqrt{3}}\right)(xI_{sc})$$

$$x^2 = \frac{100}{I_{sc}} = \frac{100}{461.89} = 0.2165$$

$$x = \sqrt{0.2165} = 0.4652$$

Starting torque is proportional to the square of applied voltage:

$$\frac{T_{st}}{T_{fl}} = x^2 \left(\frac{I_{sc}}{I_{fl}}\right)^2 s_{fl}$$

Given

$$s_{fl} = 0.05$$

Therefore,

$$\frac{T_{st}}{T_{fl}} = (0.4652)^2 \left(\frac{461.89}{72.17}\right)^2 (0.05)$$

$$\frac{T_{st}}{T_{fl}} = 0.4432$$

Hence,

$$\boxed{*}$$

Key Points

- For an autotransformer starter, motor voltage equals xV.
- Starting current varies directly with applied voltage.
- Starting torque varies as the square of applied voltage.
- Input kVA and output kVA of an ideal autotransformer are equal.

Mistakes to be avoided

- Using line current instead of phase current inconsistently.
- Forgetting that starting torque varies as V^2.
- Using I_{sc} instead of xI_{sc} for motor current at reduced voltage.

Q4.97.3 MCQ | 1997 | 1M | Ans: B | ● ○ ○

The slip of an induction machine is

$$s = \frac{N_s - N_r}{N_s}$$

where N_s is the synchronous speed and N_r is the rotor speed.

When the machine is driven above synchronous speed,

$$N_r > N_s$$

Hence,

$$s = \frac{N_s - N_r}{N_s} < 0$$

A negative slip reverses the direction of rotor current and electromagnetic torque. The machine now delivers electrical power to the supply instead of absorbing it.

Therefore, the induction machine operates as an Induction Generator.

Hence, the correct option is

$$\boxed{B}$$

Q4.96.1 MCQ | 1996 | 1M | Ans: C | ● ○ ○

An unbalanced three-phase supply can be resolved into symmetrical components as

$$V = V_1 + V_2 + V_0$$

where

- V_1 = positive-sequence component,
- V_2 = negative-sequence component,
- V_0 = zero-sequence component.

Any unbalanced set of three-phase voltages must contain positive- and negative-sequence components. The presence of a zero-sequence component depends on the availability of a return path and is not mandatory.

In a three-phase induction motor, the negative-sequence component produces a backward rotating magnetic field, resulting in negative-sequence currents and additional heating.

Therefore, an unbalanced supply voltage will cause Negative-sequence component current.

Hence, the correct option is

$$\boxed{C}$$

Q4.96.2 MCQ | 1996 | 1M | Ans: C | ● ○ ○

For an induction motor, the maximum torque is given by

$$T_{max} \propto V^2$$

where V is the supply voltage.

If the supply voltage is reduced by 10%,

$$V' = 0.9V$$

Hence,

$$\frac{T'_{max}}{T_{max}} = \left(\frac{V'}{V}\right)^2 = (0.9)^2 = 0.81$$

Therefore,

$$T'_{max} = 0.81\, T_{max}$$

Percentage decrease in maximum torque:

$$(1 - 0.81) \times 100 = 19\%$$

Hence, the correct option is

$$\boxed{C}$$

Q4.96.3 MCQ | 1996 | 1M | Ans: A | ● ○ ○

In delta connection, the full line voltage appears across each phase:

$$V_{p\Delta} = V_L$$

Hence, the starting current in delta is

$$(I_{st})_\Delta = \frac{V_L}{Z_{sc}}$$

In star connection, the phase voltage is

$$V_{pY} = \frac{V_L}{\sqrt{3}}$$

Therefore, the starting current in star is

$$(I_{st})_Y = \frac{V_L}{\sqrt{3}\, Z_{sc}}$$

Since

$$T_{st} \propto (I_{st})^2,$$

we get

$$\frac{(T_{st})_Y}{(T_{st})_\Delta} = \left(\frac{(I_{st})_Y}{(I_{st})_\Delta}\right)^2 = \left(\frac{1}{\sqrt{3}}\right)^2 = \frac{1}{3}$$

Given that the starting torque in delta connection is

$$(T_{st})_\Delta = 120\,\text{N-m},$$

the starting torque in star connection is

$$(T_{st})_Y = \frac{1}{3} \times 120 = 40\,\text{N-m}$$

$\boxed{A}$

Q4.96.4 NAT | 1996 | 3M | Ans: a_T,b_P,C_Q | ● ○ ○

(a) Load vs Efficiency

For any electrical machine, efficiency increases with load, attains a maximum value at a particular load, and then decreases.

At zero load, output power is zero; hence efficiency is also zero.

Therefore, the curve must start from the origin and rise to a maximum value.

Thus,

$$(a) \rightarrow T$$

(b) Speed vs Current

As the speed of an induction motor decreases, slip increases.

The rotor impedance is

$$Z_r = \sqrt{\left(\frac{R_2}{s}\right)^2 + X_2^2}$$

An increase in slip causes a decrease in effective rotor impedance.
Hence, rotor current increases.
At high speed (low slip),

$$I_r = \frac{V}{Z_r} = \frac{V}{\sqrt{\left(\frac{R_2}{s}\right)^2 + X_2^2}} \approx \frac{sV}{R_2}$$

Thus, current increases as speed decreases.
Therefore,

$$(b) \rightarrow P$$

(c) Speed vs Power Factor
The rotor power-factor angle is

$$\theta = \tan^{-1}\left(\frac{sX_2}{R_2}\right)$$

Hence,

$$\cos\theta = \frac{1}{\sqrt{1 + \left(\frac{sX_2}{R_2}\right)^2}}$$

At high speed (low slip), the power factor is high.
As speed decreases, slip increases, causing the power factor to decrease.
Therefore,

$$(c) \rightarrow Q$$

Hence, the correct matching is

$$\boxed{a - T,\ b - P,\ c - Q}$$

PrepFusion

Q4.96.5 MCQ | 1996 | 1M | Ans: C,D | ● ○ ○

The air-gap flux is given by

$$\phi = \frac{\text{MMF}}{\mathcal{R}}$$

where $\mathcal{R}$ is the magnetic reluctance.

If the air-gap length increases, the reluctance increases. To maintain the same flux, the magnetizing MMF must increase. Hence, the magnetizing current drawn from the supply increases.

An increase in magnetizing current causes the input power factor of the induction motor to decrease.

Therefore, statement (c) is correct.

Further, an increase in air-gap length increases the leakage flux. Leakage flux is represented by the leakage reactance X_2 of the induction motor.

Since

$$T_{\max} \propto \frac{1}{X_2}$$

an increase in leakage reactance causes the maximum torque to decrease.

Therefore, statement (d) is also correct.

Hence, the correct options are

$$\boxed{C, D}$$

Q4.95.1 MCQ | 1995 | 1M | Ans: C | ● ● ○

When one transformer of a delta-delta transformer bank is removed, the remaining two transformers operate in open-delta or V-V connection.

In open-delta connection, three-phase supply is still available at the motor terminals. Hence, single phasing will not occur.

The capacity of an open-delta bank is

$$\text{Open-delta capacity} = 57.7\% \text{ of closed-delta capacity}$$

Therefore, the induction motor can be loaded only up to

$$57.7\%$$

of its rating, with balanced three-phase voltage at its terminals.

Hence, the correct option is

$$\boxed{C}$$

Q4.95.2 NAT | 1995 | 1M | Ans: 4 | ● ○ ○

Given:

$$N_r = 1250\,\text{rpm}$$

$$f = 50\,\text{Hz}$$

The synchronous speed of an induction motor is

$$N_s = \frac{120f}{P}$$

Substituting the given frequency,

$$N_s = \frac{120 \times 50}{P} = \frac{6000}{P}$$

Since the rotor speed of an induction motor is slightly less than the synchronous speed, the nearest synchronous speed greater than 1250 rpm must be selected.

Checking standard synchronous speeds:

$$P = 2 \Rightarrow N_s = 3000\,\text{rpm}$$

$$P = 4 \Rightarrow N_s = 1500\,\text{rpm}$$

$$P = 6 \Rightarrow N_s = 1000\,\text{rpm}$$

The nearest synchronous speed above 1250 rpm is

$$N_s = 1500\,\text{rpm}$$

which corresponds to

$$P = \frac{6000}{1500} = 4$$

Hence,

$$\boxed{4}$$

Q4.95.3 MCQ | 1995 | 5M | Ans: B

The correct matching is

$$(a) \rightarrow R, \quad (b) \rightarrow U, \quad (c) \rightarrow P, \quad (d) \rightarrow S, \quad (e) \rightarrow Q$$

(a) Speed-torque characteristic of induction motor

The speed-torque characteristic of an induction motor is the inverse of the torque-speed characteristic.

Hence,

$$(a) \rightarrow R$$

(b) DC series motor

In a DC series motor, at low values of armature current, the machine is unsaturated and flux is proportional to current.

$$\phi \propto I_a$$

Since

$$T \propto \phi I_a$$

we get

$$T \propto I_a^2$$

At high values of armature current, the machine saturates, so flux becomes almost constant.

$$T \propto I_a$$

Thus, curve U is the closest representation.

Hence,

$$(b) \rightarrow U$$

(c) No-load induction motor

Under no-load, the current drawn by the motor is mainly magnetizing and core-loss current. With increase in voltage and constant frequency, flux increases linearly with voltage and hence magnetizing as well as core-loss current increases.

Therefore,

$$(c) \rightarrow P$$

(d) DC braking of induction motor

When DC voltage is applied, the stator field is stationary and cuts the rotor conductors. This induces emf and produces braking torque.

At zero relative speed, induced emf is zero and hence braking torque is zero.

Therefore,

$$(d) \rightarrow S$$

(e) Speed variation of DC series motor

In a DC series motor,

$$N \propto \frac{1}{\phi}$$

At light load, current and flux are small, so speed becomes very high. At higher load, flux increases and speed decreases. Hence,

$$(e) \rightarrow Q$$

Therefore, the final answer is

$$\boxed{B}$$

Q4.95.4 NAT | 1995 | 5M | Ans: 53.72 | ●●○

Given,

$$f_1 = 50\,\text{Hz}, \qquad P = 4$$

The original synchronous speed is

$$N_{s1} = \frac{120 f_1}{P} = \frac{120 \times 50}{4} = 1500\,\text{rpm}$$

The motor speed is

$$N_1 = 1490\,\text{rpm}$$

Hence, the original slip is

$$s_1 = \frac{N_{s1} - N_1}{N_{s1}} = \frac{1500 - 1490}{1500} = 0.00667 = 0.67\%$$

The motor is required to run at

$$N_2 = 1600\,\text{rpm}$$

For small-slip operation,

$$T \propto sV^2$$

Since the load torque and applied voltage remain unchanged, the slip remains approximately constant. Therefore,

$$s_2 = s_1 = 0.00667$$

Using

$$N_2 = N_{s2}(1 - s_2)$$

$$N_{s2} = \frac{N_2}{1 - s_2} = \frac{1600}{1 - 0.00667} = 1610.7\,\text{rpm}$$

Now,

$$f_2 = \frac{N_{s2} P}{120} = \frac{1610.7 \times 4}{120} = 53.72\,\text{Hz}$$

Hence,

$$\boxed{53.72}$$

Q4.94.1 MCQ | 1994 | 1M | Ans: C | ● ○ ○

In an induction motor, the stator and rotor slots produce periodic variations in the air-gap permeance. This gives rise to slot harmonics, which cause crawling, cogging, noise, vibration, and torque pulsations.

To reduce the effect of slot harmonics, the rotor slots are skewed with respect to the shaft axis.

Hence, skewing smoothens the torque and reduces harmonic torque pulsations caused by slot harmonics.

Therefore, the correct option is

$$\boxed{C}$$

Q4.94.2 NAT | 1994 | 1M | Ans: False | ● ● ○

For a three-phase induction motor operating on a fan-type or pump-type load, if one supply line gets disconnected, the motor enters the single-phasing condition.

Under single phasing, the motor continues to run if it was already running, but the available torque reduces significantly.

Since torque is proportional to the square of the applied voltage,

$$T \propto V^2$$

Under single-phasing operation, the effective torque becomes approximately

$$T_{sp} = \frac{1}{3}V^2$$

instead of

$$T_{3\phi} = V^2$$

Hence, the developed torque decreases.

For fan and pump loads, the load torque decreases rapidly with speed. Therefore, the motor settles at a lower speed where the reduced motor torque balances the load torque.

Thus, the motor does not stop immediately; it continues to run at a reduced speed.

Therefore, the statement that the motor will stop is

$$\boxed{\text{False}}$$

Q4.94.3 NAT | 1994 | 1M | Ans: Generating | ● ○ ○

For a 6-pole, 50 Hz induction motor, the synchronous speed is

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{6} = 1000\,\text{rpm}$$

Given rotor speed,

$$N_r = 1400\,\text{rpm}$$

Since

$$N_r > N_s$$

the slip is

$$s = \frac{N_s - N_r}{N_s} = \frac{1000 - 1400}{1000} = -0.4$$

The slip is negative.

An induction machine operates

- as a motor for $0 < s < 1$,
- as a generator for $s < 0$,
- in plugging mode for $s > 1$.

Since $s = -0.4$,

Generating

Q4.94.4 NAT 1994 1M Ans: Directly proportional ● ○ ○

In a variable-frequency induction motor drive, the air-gap flux is given by

$$\phi \propto \frac{V}{f}$$

To maintain a constant flux and avoid saturation at low frequencies or weakening of flux at high frequencies, the ratio

$$\frac{V}{f}$$

must be kept constant.

Therefore, the applied voltage must be varied directly proportional to the frequency.

$$V \propto f$$

Hence, the required answer is

Directly proportional

Q4.93.1 MCQ 1993 1M Ans: A ● ○ ○

This is the case of an **inverted induction motor**, where the three-phase supply is applied to the rotor terminals and the stator winding is short-circuited.

A rotating magnetic field is produced, which rotates at synchronous speed N_s with respect to the rotor.

This rotating field cuts the stator conductors and induces an emf in the short-circuited stator winding, causing current to flow.

To reduce the relative speed between the magnetic field and the stator conductors (Lenz's law), the rotor rotates in the direction **opposite** to that of the rotating field set up by the rotor currents. If the rotor runs at speed N_r, the field moves relative to the stator at $N_s - N_r$, which is the slip speed.

Hence, the frequency of the emf induced in the stator is equal to the slip frequency.

$$f_s = sf$$

Therefore, the correct option is

A

Q4.91.1 MCQ 1991 1M Ans: B,D ● ○ ○

In a deep-bar rotor, during starting, the frequency of rotor current is high. Due to high rotor current frequency, leakage reactance is high and current is forced to flow near the outer portion of the rotor bar.

Hence, the effective cross-sectional area decreases and the effective rotor resistance increases. This gives high starting torque.

As the rotor speeds up, slip decreases and the rotor current frequency decreases. The current spreads uniformly through the rotor conductor, reducing the effective rotor resistance.

Thus, deep-bar rotor gives high starting resistance and low running resistance.

Therefore, option (B) is correct.

In a double-cage rotor, there are two cages:

- outer cage: high resistance and low reactance,
- inner cage: low resistance and high reactance.

At starting, rotor frequency is high, so the inner cage reactance is large and most of the current flows through the outer high-resistance cage. This gives high starting torque.

During running, rotor frequency becomes small, so the inner cage reactance decreases and current flows mainly through the low-resistance inner cage, giving better efficiency.

Therefore, option (D) is correct.

Hence, the final answer is

$$\boxed{B,\ D}$$

Q4.91.2 MCQ 1991 5M Ans: B ● ● ●

Given:

$$P = 6, \qquad f_s = 60\,\text{Hz}$$

The synchronous speed is

$$N_s = \frac{120 f_s}{P} = \frac{120 \times 60}{6} = 1200\,\text{rpm}$$

The rotor speed is

$$N_r = 3000\,\text{rpm}$$

For torque production, the rotor and stator magnetic fields must be stationary with respect to each other.

Case 1: Assume that both rotor field and stator field rotate in the same direction as the rotor.

The stator field speed with respect to space is

$$N_s = 1200\,\text{rpm}$$

Hence,

$$3000 - N_{rf} = 1200$$

$$N_{rf} = 1800\,\text{rpm}$$

where N_{rf} is the rotor field speed with respect to the rotor structure.

Using

$$N_{rf} = \frac{120 f_r}{P}$$

$$1800 = \frac{120 f_r}{6}$$

$$f_r = \frac{1800 \times 6}{120} = 90\,\text{Hz}$$

Case 2: Assume that both rotor field and stator field rotate opposite to the rotor direction.
Then,

$$N_{rf} - 3000 = 1200$$

$$N_{rf} = 4200\,\text{rpm}$$

Using

$$N_{rf} = \frac{120 f_r}{P}$$

$$4200 = \frac{120 f_r}{6}$$

$$f_r = \frac{4200 \times 6}{120} = 210\,\text{Hz}$$

This corresponds to plugging (braking) operation with

$$s > 1$$

Thus, the possible rotor frequencies are

$$f_r = 90\,\text{Hz}$$

and

$$f_r = 210\,\text{Hz}$$

The correct option is

$$\boxed{B}$$

Q4.91.3 NAT | 1991 | 5M | Ans: 5.377 | ● ● ●

When the motor is connected in star, the starting torque is less than that in delta because the phase voltage in star is lower.
For equal starting torque in star and delta,

$$T_Y = T_\Delta$$

Since starting torque is proportional to square of phase current,

$$T \propto I_2^2 R_2$$

Therefore,

$$I_2 = I_2'$$

Here, both I_2 and I_2' are phase currents.
The equivalent per-phase rotor parameters referred to stator are

$$R_{eq} = 1 + 1.4 = 2.4\,\Omega$$

$$X_{eq} = 4.5 + 4.5 = 9\,\Omega$$

For star connection,

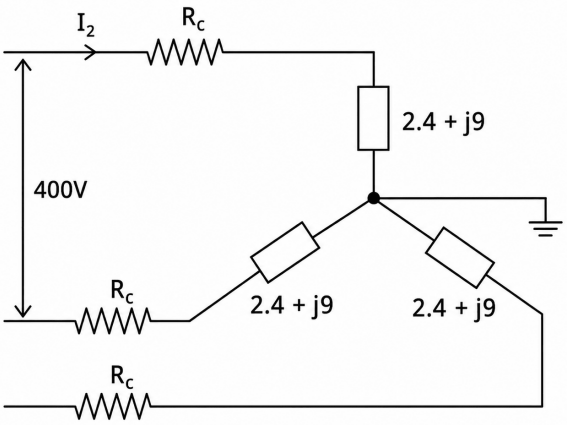


Fig. Solution Q4.91.3

$$I_2 = \frac{400/\sqrt{3}}{\sqrt{(R_c + 2.4)^2 + 9^2}}$$

$$I_2 = \frac{400/\sqrt{3}}{\sqrt{(R_c + 2.4)^2 + 81}}$$

For delta connection, converting delta impedance to equivalent star,

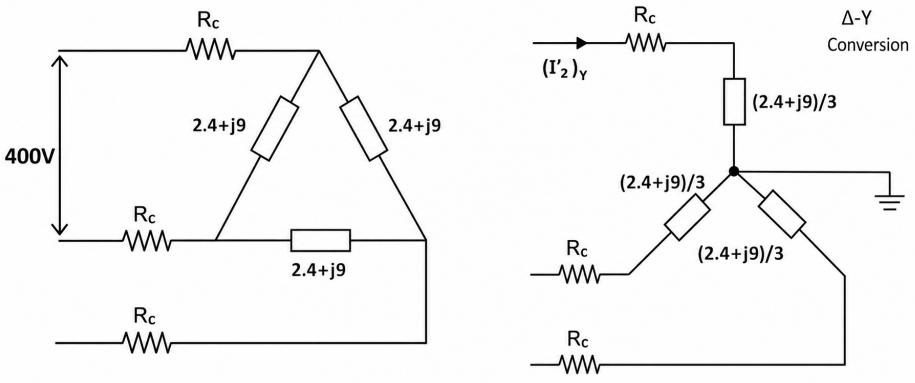


Fig. Solution Q4.91.3

$$Z_Y = \frac{2.4 + j9}{3}$$

The line current in the equivalent star circuit is

$$(I_2')_Y = \frac{400/\sqrt{3}}{\sqrt{\left(R_c + \frac{2.4}{3}\right)^2 + 3^2}}$$

The phase current in delta is

$$I_2' = \frac{(I_2')_Y}{\sqrt{3}}$$

Therefore,

$$I_2' = \frac{400/3}{\sqrt{\left(R_c + \frac{2.4}{3}\right)^2 + 9}}$$

Equating currents,

$$\frac{400/3}{\sqrt{\left(R_c + \frac{2.4}{3}\right)^2 + 9}} = \frac{400/\sqrt{3}}{\sqrt{(R_c + 2.4)^2 + 81}}$$

Squaring and simplifying,

$$3\left(R_c + \frac{2.4}{3}\right)^2 - (R_c + 2.4)^2 = 54$$

$$2R_c^2 = 57.84$$

$$R_c = \sqrt{28.92}$$

$$R_c = 5.377\,\Omega$$

Hence,

$$\boxed{5.377}$$

Key Points

- For equal starting torque, rotor phase currents must be equal.
- In delta connection, phase current is line current divided by $\sqrt{3}$.
- Delta impedance can be converted to equivalent star by dividing by 3.
- Starting torque is proportional to $I_2^2 R_2$.

Mistakes to be avoided

- Do not compare line currents directly; compare phase currents.
- Do not forget delta-to-star conversion of impedance.
- Do not omit the feeder resistance R_c while calculating voltage drop.

SOLUTIONS

V

Synchronous Machines

Topics

1. Working Principle, Equivalent Circuits,Torques of Cylindrical and Salient Pole Machines
2. Electrical Power in Alternators
3. Behaviour at Over,Under and Nominal Excitation
4. Voltage Regulation of Synchronous Generator
5. Parallel Operation of Generators/ Droop Characteristics
6. Starting of Synchronous Motors
7. Stability of Synchronous Machines
8. Hunting Phenomenon
9. Open Circuit Characteristics and Short Circuit Characteristics(OCC/SCC)
10. Armature Winding and Slot Harmonics

Unit Snapshot*

Stream: EE	**Questions:** 80	**MCQ:** 55	**MSQ:** 2	**NAT:** 23
1M: 33	**2M:** 40	**Descriptive:** 7	**Avg Weightage** ~3M	**Range:** 1M(2018)–5M(2020)

**See preface for how Avg Weightage and Range are calculated.*

Q5.26.1 MCQ | 2026 | 1M | Ans: B | ● ○ ○

Method 1

For a synchronous machine,

$$N_s = \frac{120f}{P}$$

The motor is connected to a 50 Hz supply. Hence,

$$N_m = \frac{120 \times 50}{P_m}$$

Since the motor drives the generator mechanically,

$$N_g = N_m$$

For the generator,

$$f_g = \frac{P_g N_g}{120}$$

Testing option (B),

$$N_m = \frac{120 \times 50}{6} = 1000\,\text{rpm}$$

$$f_g = \frac{2 \times 1000}{120} = 16.67\,\text{Hz}$$

Hence option (B) satisfies the requirement.

Method 2

Both the motor and generator are connected to the same shaft. Therefore, they rotate at the same speed.

$$N_m = N_g$$

Using the synchronous speed relation,

$$\frac{120f_m}{P_m} = \frac{120f_g}{P_g}$$

Substituting the given frequencies,

$$\frac{120 \times 50}{P_m} = \frac{120 \times 16.67}{P_g}$$

$$\frac{P_m}{P_g} = \frac{50}{16.67} \approx 3$$

$$\frac{P_m}{P_g} = \frac{6}{2}$$

Thus, the motor and generator pole combination is

$$P_m = 6, \qquad P_g = 2$$

which corresponds to option (B).

$$\boxed{B}$$

Q5.26.2 MCQ | 2026 | 1M | Ans: C ● ● ○

The V-curve of a synchronous machine is shown below.

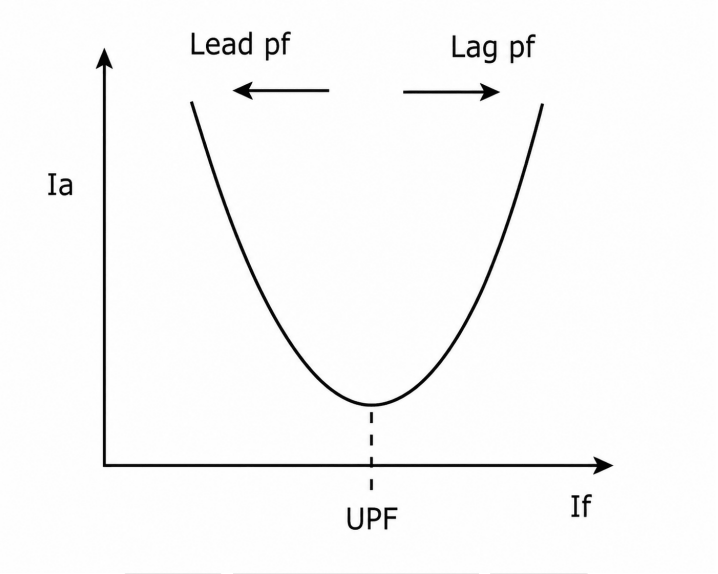


Fig. Solution Q5.26.2

At the minimum point of the V-curve, the machine operates at unity power factor.

If the field current I_f is increased beyond the unity power factor point, the armature current I_a increases and the machine operates at lagging power factor (for generator operation). From the V-curve,

$$I_a \text{ is minimum at UPF}$$

As excitation increases beyond the UPF point,

$$I_f \uparrow \quad \Rightarrow \quad I_a \uparrow$$

and the generator operates at lagging power factor.

Further, increased rotor field strength requires a smaller load angle δ to transfer the same real power.

$$P = \frac{EV}{X_s} \sin\delta$$

Since E increases,

$$\delta \downarrow$$

$$\boxed{C}$$

Key Points

- The V-curve represents variation of armature current I_a with field current I_f at constant load.
- Minimum armature current corresponds to unity power factor operation.
- For a synchronous generator, over-excitation results in lagging power factor operation.
- Real power transfer is governed by

$$P = \frac{EV}{X_s} \sin\delta.$$

Q5.25.1 NAT | 2025 | 1M | Ans: 66.25 to 66.75 | ● ● ○

Given,

$$V_L = 3.3\,\text{kV}$$

$$V = \frac{V_L}{\sqrt{3}} = \frac{3300}{\sqrt{3}} \approx 1905.26\,\text{V}$$

The induced emf E is in phase with V at no-load and makes a mechanical angle

$$\delta = 2^\circ$$

with V when loaded.
The electrical angle is

$$\delta_{elec} = 2 \times \frac{\text{number of poles}}{2} = 4^\circ$$

Also,

$$X_s = 2\,\Omega$$

and stator resistance is negligible.

Method 1

The armature current is

$$I = \frac{V - E}{jX_s}$$

Hence,

$$|I| = \frac{|V - E|}{X_s}$$

Since $|V| = |E|$ and the angle between them is 4°,

$$|V - E| = \sqrt{V^2 + E^2 - 2VE\cos(4^\circ)}$$

$$= \sqrt{2V^2\,(1 - \cos(4^\circ))}$$

$$= 1905.26\sqrt{2\,(1 - \cos(4^\circ))}$$

Using

$$\cos(4^\circ) \approx 0.99756$$

$$1 - \cos(4^\circ) \approx 0.00244$$

$$|V - E| = 1905.26\sqrt{2(0.00244)}$$

$$= 1905.26\sqrt{0.00488}$$

$$\approx 1905.26(0.06984) \approx 133.5\,\text{V}$$

Therefore,

$$|I| = \frac{133.5}{2} \approx 66.75\ \text{A}$$

$$\boxed{66.75}$$

Key Points

- For negligible armature resistance,
$$I = \frac{V - E}{jX_s}.$$
- The magnitude of current depends on the phasor difference between V and E.
- Mechanical angle and electrical angle are related by
$$\delta_{elec} = \delta_{mech} \times \frac{P}{2}.$$
- For equal magnitudes of V and E, the cosine law provides the quickest evaluation of $|V - E|$.

Q5.25.2 MSQ | 2025 | 1M | Ans: A,C | ● ○ ○

Before an alternator is connected (synchronized) to the grid, the following conditions must be satisfied at the instant of closing the breaker:

- The terminal voltage (magnitude) of the incoming alternator must be equal to the grid voltage.
- The frequency of the incoming alternator must be equal to the grid frequency.
- The phase sequence must be the same as that of the grid.
- The incoming alternator voltage must be in phase with the grid voltage.

Instruments used to check these conditions:

- **(A) Voltmeter:** required, to match the alternator terminal voltage with the grid voltage (by adjusting the field excitation).
- **(B) Wattmeter:** not required. It measures power, and no power is exchanged before the breaker is closed.
- **(C) Synchroscope:** required. It indicates whether the incoming machine is faster or slower than the grid (frequency difference) and the phase difference, and shows the correct instant to close the breaker.
- **(D) Stroboscope:** not required. It is used for measuring or observing speed of rotating parts, and it is not part of the synchronizing equipment.

$$\boxed{\text{(A) and (C)}}$$

Key Points

- Voltage magnitude, frequency, phase sequence and phase angle must match the grid before synchronizing.
- A voltmeter checks the voltage and a synchroscope checks frequency and phase angle; synchronizing lamps can be used in place of a synchroscope.
- Real power is exchanged only after the breaker is closed and the prime mover input is increased.

Q5.24.1 MCQ | 2024 | 2M | Ans: A | ● ● ○

The reactance of lossless line is,

$$X = 5\,\Omega$$

Load power factor angle is,

$$\phi = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = 30^\circ$$

Generator terminal phase voltage is,

$$V_t = \frac{11000}{\sqrt{3}} = 6350.85\,\text{V}$$

Load current is,

$$I_a = \frac{10 \times 10^6}{\sqrt{3} \times 11 \times 10^3} = 524.87\,\text{A}$$

Assume receiving end voltage as reference,

$$V_r = V\angle 0$$

By KVL,

$$6350.85\angle\delta = V\angle 0 + j5 \times 524.87\angle -30^\circ$$

$$6350.85\angle\delta = V\angle 0 + 2624.35\angle 60^\circ$$

$$6350.85\angle\delta = (V + 1312.175) + j2272.735$$

Taking magnitude on both sides,

$$6350.85^2 = (V + 1312.175)^2 + 2272.735^2$$

$$V = 4618.077\,\text{V}$$

Since the line is lossless, power at sending end i.e. generator terminals must be same as load power.

$$3 \times 6350.85 \times I \times \cos\theta = 3 \times 4618.077 \times I \times \cos 30^\circ$$

$$\cos\theta = \frac{4618.077 \cos 30^\circ}{6350.85}$$

$$\cos\theta \approx 0.63$$

Hence, sending end power factor is

A

Key Points

- For a lossless line, sending end real power is equal to receiving end real power.

- For a three-phase system,

$$P = \sqrt{3} V_L I_L \cos\phi.$$

Q5.23.1 NAT | 2023 | 2M | Ans: 1.58 | ● ● ●

Given,

$$Z_s = (0.1 + j0.3)$$

$$|Z_s| = \sqrt{0.1^2 + 0.3^2} = 0.3162$$

$$\theta = \tan^{-1}\left(\frac{0.3}{0.1}\right) = 71.56^\circ$$

Hence,

$$Z_s = 0.3162\angle 71.56^\circ \text{ p.u.}$$

The developed power is maximum when

$$\delta = \theta$$

For a synchronous machine having impedance angle θ,

$$P_m = \frac{E_f V}{|Z_s|} - \frac{E_f^2}{|Z_s|}\cos\theta$$

Substituting

$$P_m = 2.5, \qquad V = 1, \qquad |Z_s| = 0.3162, \qquad \theta = 71.56^\circ$$

$$2.5 = \frac{E_f(1)}{0.3162} - \frac{E_f^2}{0.3162}\cos(71.56^\circ)$$

Since

$$\cos(71.56^\circ) \approx 0.3162$$

$$2.5 = 3.162E_f - E_f^2$$

$$E_f^2 - 3.162E_f + 2.5 = 0$$

Solving,

$$E_f = \frac{3.162 \pm \sqrt{(3.162)^2 - 4(2.5)}}{2}$$

$$E_f = 1.58$$

Taking the required value,

$$\boxed{1.58}$$

Key Points

- Convert the synchronous impedance into polar form before applying the power equation.
- Maximum developed power occurs when the load angle equals the impedance angle, i.e.,

$$\delta = \theta.$$

- For a synchronous machine with non-zero resistance,

$$P_m = \frac{E_f V}{|Z_s|}\cos(\theta - \delta) - \frac{E_f^2}{|Z_s|}\cos\theta.$$

- At maximum power condition,

$$P_{m,\max} = \frac{E_f V}{|Z_s|} - \frac{E_f^2}{|Z_s|}\cos\theta.$$

Mistakes to be avoided

- Do not use only the reactance value; use the magnitude of Z_s.
- Do not forget to convert Z_s into polar form.
- For maximum power transfer, use $\delta = \theta$, not $\delta = 90°$.
- Check the quadratic roots carefully before selecting the final answer.

Q5.22.1 NAT | 2022 | 2M | Ans: 245 | ●○○

Given,

$$V_L = 400\,\text{V}$$

$$X_s = 1\,\Omega/\text{phase}$$

$$P_{out} = 10\,\text{kW}$$

$$\cos\phi = 0.8 \text{ (leading)}, \qquad P_{loss} = 2\,\text{kW}$$

Neglect armature resistance.

$$P_{in} = P_{out} + P_{loss} = 10 + 2 = 12\,\text{kW}$$

$$I_a = \frac{P_{in}}{\sqrt{3}V_L\cos\phi} = \frac{12000}{\sqrt{3}\times 400\times 0.8} = 21.65\,\text{A}$$

$$V = \frac{400}{\sqrt{3}}$$

For a synchronous motor,

$$E_b = \sqrt{(V\cos\phi)^2 + (V\sin\phi + I_a X_s)^2}$$

$$E_b = \sqrt{\left(\frac{400}{\sqrt{3}} \times 0.8\right)^2 + \left(\frac{400}{\sqrt{3}} \times 0.6 + 21.65 \times 1\right)^2}$$

$$E_b = \sqrt{(184.75)^2 + (138.56 + 21.65)^2}$$

$$E_b = 244.55 \text{ V per phase}$$

$$\boxed{245}$$

Q5.20.1 MCQ | 2020 | 1M | Ans: A

For a synchronous generator with negligible armature resistance,

$$\underline{E_A} = \underline{V_A} + jX_s\underline{I_a}$$

Take terminal voltage as reference,

$$\underline{V_A} = V_A\angle 0^\circ$$

For a lagging load,

$$\underline{I_a} = I_a\angle(-\phi)$$

Therefore,

$$jX_s\underline{I_a} = X_sI_a\angle(90^\circ - \phi)$$

This voltage drop has a positive component along $\underline{V_A}$, hence the resultant induced emf is always greater than terminal voltage.

$$E_A > V_A$$

So, statement Q is true.

For a leading load,

$$\underline{I_a} = I_a\angle(+\phi)$$

$$jX_s\underline{I_a} = X_sI_a\angle(90^\circ + \phi)$$

Here, the component of $jX_s\underline{I_a}$ along $\underline{V_A}$ is negative. Hence, depending on the magnitude of the current and the power factor angle, E_A may be less than or greater than V_A.

Therefore, it is not always true that

$$V_A > E_A$$

So, statement P is false.

$$\boxed{A}$$

Key Points

- For a synchronous generator,
$$\underline{E_A} = \underline{V_A} + jX_s\underline{I_a}.$$

- For lagging load, synchronous reactance drop increases the magnitude of induced emf.
- For leading load, the magnitude relation between V_A and E_A is not fixed for all cases.

Q5.20.2 MCQ 2020 2M Ans: A ●●●

At the rated open-circuit voltage, the graph gives $I_f = 8\,\text{A}$ and $V_{oc} = 400\,\text{V}$ per phase. The short-circuit characteristic is $I_{sc} = 5I_f$, so

$$Z_{s(\text{sat})} = \frac{400}{5(8)} = 10\,\Omega.$$

At the marked end of the unsaturated region, $(I_f, V_{oc}) = (2, 210)$, the corresponding short-circuit current is 10 A. Using that point gives

$$Z_{s(\text{unsat})} = \frac{210}{10} = 21\,\Omega, \qquad \frac{Z_{s(\text{unsat})}}{Z_{s(\text{sat})}} = 2.100.$$

This gives option A using the indicated unsaturated-region point.

Note: The drawn air-gap line has a nonzero intercept, $V = 100I_f + 10$. Extrapolating this line to $I_f = 8\,\text{A}$ instead gives $810/400 = 2.025$, option B. These constructions differ because of the intercept. The official GATE 2020 EE answer key selects A, corresponding to the marked unsaturated-region point. It is not a general rule that unsaturated impedance must always be evaluated at the knee.

Key Points

- Unsaturated synchronous impedance is obtained using the air-gap line voltage.
- Saturated synchronous impedance is obtained using the actual OCC voltage.
- $Z_s = \dfrac{V_{oc}}{I_{sc}}$ for the same field current.
- Saturation reduces the effective synchronous impedance.

Q5.20.3 NAT 2020 2M Ans: 125.29 ●○○

At constant power,

$$P = \sqrt{3}VI_a\cos\phi$$

Therefore,

$$I_{a1}\cos\phi_1 = I_{a2}\cos\phi_2$$

Given,

$$I_{a1} = 100\,\text{A}$$

$$\cos\phi_1 = 0.9$$

Hence,

$$\tan\phi_1 = \frac{\sqrt{1-\cos^2\phi_1}}{\cos\phi_1} = \frac{\sqrt{1-0.9^2}}{0.9} = 0.484$$

Since reactive power is doubled,

$$\tan\phi_2 = \frac{2Q}{P} = 2\tan\phi_1 = 2(0.484) = 0.968$$

Therefore,

$$\cos\phi_2 = \frac{1}{\sqrt{1+\tan^2\phi_2}} = \frac{1}{\sqrt{1+0.968^2}} = 0.7182$$

Using

$$I_{a1}\cos\phi_1 = I_{a2}\cos\phi_2$$

$$100 \times 0.9 = I_{a2} \times 0.7182$$

$$I_{a2} = \frac{90}{0.7182} = 125.29\,\text{A}$$

$$\boxed{125.29}$$

Q5.19.1 MCQ | 2019 | 1M | Ans: D

For a three-phase synchronous motor, real power input is

$$P = \sqrt{3}V_L I_L \cos\phi$$

For the same line voltage and same load, real power remains constant. Hence,

$$I_{L1}\cos\phi_1 = I_{L2}\cos\phi_2$$

Given,

$$I_{L1} = 200\,\text{A}$$

$$\cos\phi_1 = 1$$

$$\cos\phi_2 = 0.5$$

Therefore,

$$200 \times 1 = I_{L2} \times 0.5$$

$$I_{L2} = 400\,\text{A}$$

$$\boxed{D}$$

Q5.19.2 NAT | 2019 | 2M | Ans: 245 to 246

For a Y-connected synchronous motor,

$$V = \frac{V_L}{\sqrt{3}} = \frac{220}{\sqrt{3}} = 127.02\,\text{V}$$

Given,

$$Z_s = 0.25 + j2.5\,\Omega$$

$$I_a = 10\,\text{A}$$

At 0.8 power factor leading,

$$\cos\phi = 0.8, \qquad \sin\phi = 0.6$$

$$I_a = 10(0.8 + j0.6) = 8 + j6\,\text{A}$$

For a synchronous motor,

$$E_A = V - I_a Z_s$$

$$I_a Z_s = (8 + j6)(0.25 + j2.5)$$

$$I_a Z_s = 2 + j20 + j1.5 - 15$$

$$I_a Z_s = -13 + j21.5$$

Therefore,

$$E_A = 127.02 - (-13 + j21.5)$$

$$E_A = 140.02 - j21.5$$

$$|E_A| = \sqrt{(140.02)^2 + (21.5)^2}$$

$$|E_A| = 141.66\,\text{V per phase}$$

Hence, line-to-line internal voltage is

$$E_{A,L} = \sqrt{3} \times 141.66$$

$$E_{A,L} = 245.36\,\text{V}$$

$$\boxed{245.36}$$

Q5.18.1 MCQ | 2018 | 1M | Ans: B | ● ○ ○

For a salient pole synchronous motor, the reluctance power is given by

$$P_r = \frac{V^2}{2}\left(\frac{X_d - X_q}{X_d X_q}\right)\sin 2\delta$$

The corresponding reluctance torque is

$$T_r = \frac{P_r}{\omega_s}$$

Hence,

$$T_r \propto \sin 2\delta$$

The reluctance torque attains its maximum value when

$$\sin 2\delta = 1$$

$$2\delta = 90^\circ$$

$$\delta = 45^\circ$$

B

Key Points

- Reluctance torque exists only in salient pole machines where $X_d \neq X_q$.
- The reluctance torque component is proportional to $\sin 2\delta$.
- Maximum value of $\sin 2\delta$ occurs when $2\delta = 90^\circ$.
- Cylindrical rotor machines do not develop reluctance torque because $X_d = X_q$.

Q5.17.1 NAT | 2017 | 1M | Ans: 838.31 | ● ○ ○

For unity power factor operation, the synchronous motor must be normally excited.
Hence,

$$E_f \cos\delta = V_t$$

Given,

$$V_t = 6.6\,\text{kV}$$

$$\delta = 30^\circ$$

Therefore,

$$E_f = \frac{V_t}{\cos\delta} = \frac{6.6}{\cos 30^\circ} = 7.621\,\text{kV}$$

Since armature resistance is negligible, the power developed by the synchronous motor is

$$P = \frac{E_f V_t}{X_s}\sin\delta$$

Substituting,

$$P = \frac{\left(\frac{6.6}{\cos 30^\circ}\right)(6.6)}{30}\sin 30^\circ$$

$$P = \frac{(6.6)^2}{30}\tan 30^\circ$$

$$P = 0.8383125\,\text{MW}$$

$$P = 838.31\,\text{kW}$$

$$\boxed{838.31}$$

Q5.17.2 NAT 2017 2M Ans: 0.74 to 0.75 ● ● ○

A synchronous condenser is an over-excited synchronous machine running without load. It supplies only reactive power, so the current drawn by it is purely reactive and leads the terminal voltage by 90°. Equivalently, the current supplied to the grid lags the voltage by 90°.

Both machines supply equal currents, so each supplies half of the reactive power:

$$Q_A = Q_B = \frac{50}{2} = 25\,\text{MVAR}$$

Current supplied by each machine,

$$I = \frac{Q}{\sqrt{3}V_L} = \frac{25 \times 10^6}{\sqrt{3} \times 11 \times 10^3} = 1312.16\,\text{A}$$

Per-phase terminal voltage,

$$V = \frac{11 \times 10^3}{\sqrt{3}} = 6350.85\,\text{V}$$

Because the current is exactly 90° away from V, the reactance drop IX_s (which is 90° away from I) lies along V. So E and V are in phase and the magnitudes add arithmetically, not as a vector sum:

$$E = V + IX_s$$

For machine A ($X_s = 1\,\Omega$),

$$E_A = 6350.85 + 1312.16 \times 1 = 7663.01\,\text{V}$$

For machine B ($X_s = 3\,\Omega$),

$$E_B = 6350.85 + 1312.16 \times 3 = 10287.33\,\text{V}$$

Since the magnetic circuit is linear, $E \propto I_f$:

$$\frac{I_{fA}}{I_{fB}} = \frac{E_A}{E_B} = \frac{7663.01}{10287.33} = 0.745$$

$$\boxed{0.745}$$

Q5.17.3 MCQ 2017 1M Ans: B ● ○ ○

For a synchronous motor,

$$V_t = E_f + jX_sI_a$$

At leading power factor operation, the motor is **overexcited**. Therefore, the excitation emf E_f is greater than the terminal voltage V_t.

Using the phasor relation,

$$E_f = V_t - jX_sI_a$$

For a leading power factor current, the term jX_sI_a is oriented such that the magnitude of E_f becomes greater than V_t. Hence,

$$|E_f| > |V_t|$$

Therefore, the correct option is

$$\boxed{\text{B}}$$

Key Points

- A synchronous motor operates at leading power factor when it is over-excited.
- For normal excitation:
$$E_f \approx V_t$$
and the motor operates at unity power factor.
- Over-excitation $\Rightarrow$ leading power factor.
- Under-excitation $\Rightarrow$ lagging power factor.
- Increasing field current increases the excitation emf E_f.

Mistakes to be avoided

- Do not confuse motor and generator phasor diagrams.
- Unity power factor does not imply $E_f = V_t$ under all operating conditions.
- Leading power factor corresponds to over-excitation, not under-excitation.
- Terminal voltage is generally fixed by the supply; power factor is controlled by excitation.

Q5.17.4 NAT 2017 2M Ans: -14.55 ●●○

Given,

$$V_{oc} = 360\,\text{V}$$

Base voltage,

$$V_{base} = 400\,\text{V}$$

Therefore,

$$V_{oc(\text{pu})} = \frac{360}{400} = 0.9\,\text{pu}$$

Short-circuit current is

$$I_{sc} = 1\,\text{pu}$$

Hence, synchronous reactance is

$$X_s = \frac{V_{oc}}{I_{sc}} = \frac{0.9}{1} = 0.9\,\text{pu}$$

Since the machine delivers rated load at 0.8 leading power factor and rated terminal voltage,

$$I_a = 1\,\text{pu}$$

$$V_t = 1\,\text{pu}$$

Power factor angle,

$$\phi = \cos^{-1}(0.8) = 36.86^\circ$$

For a synchronous generator,

$$E_f = V_t + jX_sI_a$$

$$E_f = 1 + j(0.9)\,(1\angle 36.86^\circ)$$

$$E_f = 1 + 0.9\angle 126.86^\circ$$

Converting to rectangular form,

$$E_f = 1 + (-0.54 + j0.72)$$

$$E_f = 0.46 + j0.72$$

$$|E_f| = \sqrt{(0.46)^2 + (0.72)^2} = 0.854\,\text{pu}$$

Voltage regulation,

$$\%VR = \frac{E_f - V_t}{V_t} \times 100$$

$$\%VR = \frac{0.854 - 1}{1} \times 100 = -14.55\%$$

$$\boxed{-14.55}$$

Q5.16.1 NAT | 2016 | 2M | Ans: 1.608 | ● ● ○

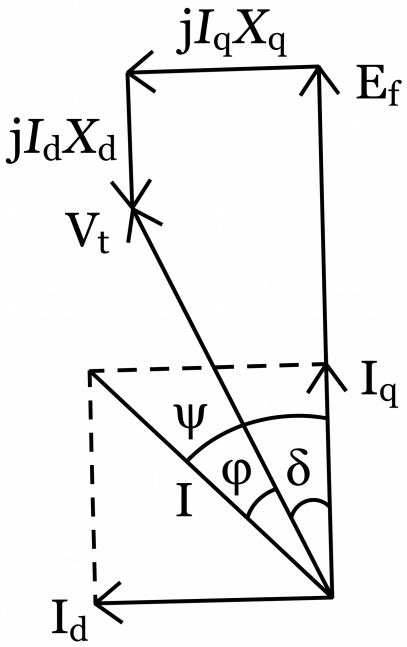


Fig. Solution Q5.16.1

For a synchronous motor operating at leading power factor, the internal power factor angle is

$$\psi = \delta + \phi$$

From the phasor diagram,

$$\tan\psi = \frac{V\sin\phi + I_a X_d}{V\cos\phi - I_a R_a}$$

At full-load,

$$I_a = 1\text{ pu}$$

Given,

$$\cos\phi = 0.8$$

$$\sin\phi = 0.6$$

Substituting the given values,

$$\psi = \tan^{-1}\left(\frac{1\times 0.6 + 1\times 0.6}{1\times 0.8 - 1\times 0}\right)$$

$$\psi = \tan^{-1}\left(\frac{1.2}{0.8}\right) = 56.31^\circ$$

Hence,

$$\delta = \psi - \phi = 56.31^\circ - \cos^{-1}(0.8)$$

$$\delta = 56.31^\circ - 36.87^\circ = 19.44^\circ$$

From the phasor diagram,

$$E_f = V_t \cos\delta + I_d X_d$$

where

$$I_d = I_a \sin(\delta + \phi)$$

Therefore,

$$I_d = 1 \times \sin(56.31^\circ) = 0.832$$

Substituting,

$$E_f = 1 \times \cos(19.44^\circ) + 0.832 \times 0.8$$

$$E_f = 0.943 + 0.666 = 1.608 \text{ pu}$$

$$\boxed{1.608}$$

Key Points

- For a salient-pole synchronous motor,
$$\psi = \delta + \phi$$
where δ is the load angle and ϕ is the power factor angle.
- Direct-axis current is
$$I_d = I_a \sin(\delta + \phi)$$
- Leading power factor operation corresponds to over-excitation.

PrepFusion

Q5.16.2 MCQ 2016 1M Ans: A

For reactive power exchange:

- A synchronous motor operating in an overexcited condition draws leading current and hence supplies reactive power to the system. Therefore, it acts as a source of reactive power.
- A synchronous generator operating at lagging power factor supplies lagging current to the load. Hence, it delivers reactive power and acts as a source of reactive power.
- A directly grid-connected induction generator requires magnetizing current from the grid for excitation. Therefore, it absorbs reactive power and acts as a sink of reactive power.

Thus,

Synchronous motor	→ Source of reactive power
Synchronous generator	→ Source of reactive power
Induction generator	→ Sink of reactive power

Hence, the correct statement is:

$$\boxed{\text{A}}$$

Q5.15.1 NAT | 2015 | 2M | Ans: -47.26 | ● ● ●

Given,

$$V_t = \frac{11000}{\sqrt{3}}\,\text{V}$$

$$I_a = 100\,\text{A}$$

$$X_s = 50\,\Omega$$

Initially,

$$\overline{E}_f = V_t\angle 0^\circ - jI_aX_s$$

$$\overline{E}_f = \frac{11000}{\sqrt{3}} - j(100)(50) = 8082.9\angle -38.21^\circ$$

When the load torque is increased, the power factor becomes lagging.
For a synchronous motor,

$$\overline{E}_f = V_t\angle 0^\circ - j(I_a\angle -\phi)X_s$$

$$\overline{E}_f = V_t - I_aX_s\sin\phi - jI_aX_s\cos\phi$$

Taking magnitude,

$$|E_f| = \sqrt{(V_t - I_aX_s\sin\phi)^2 + (I_aX_s\cos\phi)^2}$$

$$|E_f|^2 = V_t^2 + I_a^2X_s^2 - 2I_aX_sV_t\sin\phi$$

Since excitation is unchanged,

$$|E_f| = 8082.9\,\text{V}$$

Substituting values,

$$(8082.9)^2 = \left(\frac{11000}{\sqrt{3}}\right)^2 + (100\times 50)^2 - 2\left(\frac{11000}{\sqrt{3}}\right)(100)(50)\sin\phi$$

$$\sin\phi = 0.1443$$

$$\phi = 8.298^\circ$$

Now,

$$\overline{E}_f = \frac{11000}{\sqrt{3}} - j\,(100\angle -8.298^\circ)\,(50)$$

$$\overline{E}_f = 8082.97\angle -47.26^\circ\,\text{V}$$

Hence,

$$\boxed{-47.26}$$

Q5.14.1 NAT | 2014 | 1M | Ans: 15.06 | ● ○ ○

Given open-circuit test data,

$$V_{oc} = 400\,\text{V (line)} \quad \text{at} \quad I_f = 2.3\,\text{A}$$

Short-circuit test data,

$$I_{sc} = 10\,\text{A} \quad \text{at} \quad I_f = 1.5\,\text{A}$$

The synchronous impedance must be found at the same field current. "At rated voltage" means at the field current that gives 400 V on open circuit, i.e. $I_f = 2.3$ A.

The open-circuit characteristic may saturate, so it should not be interpolated linearly. The short-circuit characteristic, however, is a straight line through the origin (the machine is unsaturated on short circuit). Hence the short-circuit current at $I_f = 2.3$ A is

$$I'_{sc} = 10 \times \frac{2.3}{1.5}$$

$$I'_{sc} = 15.33\,\text{A}$$

For a star-connected machine, the per-phase open-circuit voltage is

$$V_{oc,ph} = \frac{400}{\sqrt{3}}$$

$$V_{oc,ph} = 230.94\,\text{V}$$

Therefore, per phase synchronous impedance is

$$Z_s = \frac{V_{oc,ph}}{I'_{sc}}$$

$$Z_s = \frac{230.94}{15.33}$$

$$Z_s = 15.06\,\Omega$$

15.06

Q5.14.2 MCQ | 2014 | 2M | Ans: A | ● ● ●

For the infinite bus, the phase sequence is

$$R - Y - B$$

The incoming generator phase sequence is given as

$$R' - Y' - B'$$

From the lamp connections shown, if the lamps become dark successively instead of becoming dark simultaneously, the phase sequence of the incoming generator is opposite to that of the infinite bus.

The observed dark sequence is

$$L_a - L_b - L_c$$

This direction of darkness indicates that the incoming generator frequency is higher than the infinite bus frequency.

Therefore,

A

Key Points

- If all lamps become dark simultaneously, phase sequence is same.
- If lamps become dark successively, phase sequence is opposite.
- The direction of lamp darkening indicates whether the incoming generator frequency is higher or lower.
- Sequence $L_a - L_b - L_c$ indicates that the incoming generator frequency is more than the infinite bus frequency.

Q5.14.3 MCQ | 2014 | 2M | Ans: D | ● ● ○

Given,

$$\text{Number of slots} = 180$$

$$\text{Conductors per slot} = 6$$

For both cases,

$$\beta = \frac{180^\circ}{\text{slots per pole}} = \frac{180^\circ}{180/20} = 20^\circ$$

For single-phase connection,

$$m = \frac{\text{Number of slots}}{\text{Number of poles} \times \text{Number of phases}} = \frac{180}{20 \times 1} = 9$$

Distribution factor,

$$K_{d1} = \frac{\sin\left(\frac{m\beta}{2}\right)}{m \sin\left(\frac{\beta}{2}\right)} = \frac{\sin\left(\frac{9 \times 20}{2}\right)}{9 \sin\left(\frac{20}{2}\right)} = 0.6398$$

Number of turns,

$$N_1 = 180 \times \frac{6}{2} = 540$$

For three-phase connection,

$$m = \frac{180}{20 \times 3} = 3$$

Distribution factor,

$$K_{d3} = \frac{\sin\left(\frac{m\beta}{2}\right)}{m \sin\left(\frac{\beta}{2}\right)} = \frac{\sin\left(\frac{3 \times 20}{2}\right)}{3 \sin\left(\frac{20}{2}\right)} = 0.9597$$

Turns per phase,

$$N_{ph} = 180 \times \frac{6}{2} \times \frac{1}{3} = 180$$

Using the emf equation,

$$E = 4.44K_d f N\phi$$

Hence,

$$\frac{E_{1\phi}}{E_{3\phi}} = \frac{V_1}{V_2} = \frac{4.44 \times 0.6398 \times f \times 540 \times \phi}{4.44 \times 0.9597 \times f \times 180 \times \phi}$$

$$\frac{V_1}{V_2} = 1.9999 \approx 2$$

$$\boxed{D}$$

Key Points

- Distribution factor:

$$K_d = \frac{\sin\left(\frac{m\beta}{2}\right)}{m \sin\left(\frac{\beta}{2}\right)}$$

- Generated emf is proportional to $K_d N$.
- For three-phase winding, turns per phase are one-third of the total turns.

Mistakes to be avoided

- Use the same slot angle β for both winding arrangements.
- Do not forget that m depends on the number of phases.

Q5.14.4 NAT | 2014 | 2M | Ans: 0.10 to 0.12 | ● ○ ○

Given,

$$P = 0.6\,\text{pu}$$

$$E_f = 1.3\,\text{pu}$$

$$V_t = 1\,\text{pu}$$

$$X_s = 1.1\,\text{pu}$$

Real power delivered by the generator is

$$P = \frac{E_f V_t}{X_s} \sin\delta$$

Substituting the given values,

$$0.6 = \frac{1.3 \times 1}{1.1} \sin\delta$$

$$\sin\delta = \frac{0.6 \times 1.1}{1.3} = 0.5076$$

$$\delta = 30.51^\circ$$

Reactive power delivered by the generator is

$$Q = \frac{V_t}{X_s}\left(E_f \cos\delta - V_t\right)$$

Substituting the values,

$$Q = \frac{1}{1.1}\left(1.3\cos 30.51^\circ - 1\right)$$

$$Q = \frac{1}{1.1}\left(1.1201 - 1\right)$$

$$Q = 0.109 \text{ pu}$$

$$\boxed{0.109}$$

Key Points

- For a cylindrical rotor synchronous generator with negligible armature resistance,

$$P = \frac{E_f V_t}{X_s}\sin\delta$$

- Reactive power delivered by the generator is

$$Q = \frac{V_t}{X_s}\left(E_f \cos\delta - V_t\right)$$

- Positive value of Q indicates reactive power supplied by the generator.

Q5.14.5 NAT | 2014 | 2M | Ans: 50 ●●○

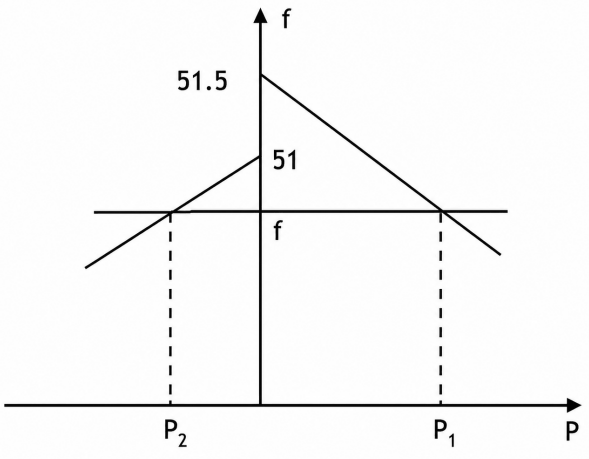


Fig. Solution Q5.14.5

Given,

$$P_1 + P_2 = 2.5\,\text{MW}$$

Let the steady-state frequency be f Hz.

For generator 1,

$$\frac{51.5 - f}{P_1} = 1$$

$$f = 51.5 - P_1$$

Similarly, for generator 2 having the same droop,

$$f = 51 - P_2$$

Equating the two expressions,

$$51.5 - P_1 = 51 - P_2$$

$$P_1 - P_2 = 0.5\,\text{MW}$$

Solving,

$$P_1 + P_2 = 2.5\,\text{MW}$$

$$P_1 = 1.5\,\text{MW}, \qquad P_2 = 1\,\text{MW}$$

Therefore,

$$f = 51.5 - 1.5 = 50\,\text{Hz}$$

50

Key Points

- In parallel operation of generators, the steady-state frequency is common.
- For equal droop slopes, frequency drop is proportional to the load shared by each generator.

Mistakes to be avoided

- Do not assume equal load sharing when no-load frequencies are different.
- Use power in MW consistently while applying the droop equation.

Q5.14.6 MCQ 2014 1M Ans: D ● ○ ○

Hunting in a synchronous machine refers to the oscillation of the rotor about its equilibrium position due to sudden changes in load or system disturbances.

The damping torque required to suppress hunting is mainly provided by the damper (amortisseur) winding embedded in the rotor pole faces.

When hunting occurs, relative motion exists between the rotating magnetic field and the rotor. This induces currents in the damper winding, producing damping torque. Since the damper winding is located on the rotor, the damping action is due to rotor copper losses.

Therefore, hunting is predominantly damped by rotor copper losses.

D

Q5.14.7 NAT 2014 2M Ans: 33.47 ●●○

Given,

$$X_s = 0.8\,\text{pu}$$

$$P = 1\,\text{pu}$$

$$V_t = 1.1\,\text{pu}$$

At unity power factor,

$$\cos\phi = 1$$

Therefore,

$$I_a = \frac{P}{V_t \cos\phi} = \frac{1}{1.1} = 0.9091\,\text{pu}$$

For a synchronous generator,

$$E_f = V_t + jI_aX_s$$

$$E_f = 1.1 + j(0.9091)(0.8)$$

$$E_f = 1.1 + j0.7273$$

The angle of E_f with respect to V_t is

$$\delta = \tan^{-1}\left(\frac{0.7273}{1.1}\right)$$

$$\delta = 33.47^\circ$$

$$\boxed{33.47}$$

Q5.11.1 MCQ 2011 1M Ans: B ●●○

When a three-phase, salient pole synchronous motor is connected to an infinite bus and operated at no-load with normal excitation, the initial operating conditions are:

$$E_f \approx V_t, \quad \delta \approx 0^\circ$$

The armature current I_a is at its minimum absolute value (near zero). **Phase 1: Reducing Excitation from Normal to Zero** As the field excitation current I_f is gradually reduced to zero, the excitation voltage E_f decreases towards zero. To maintain the synchronous rotating magnetic field against the terminal voltage V_t, the motor behaves as an under-excited machine and must draw a lagging reactive magnetizing current from the infinite bus. Therefore, the armature current I_a **increases continuously**.

At exactly zero excitation ($E_f = 0$), the motor does not drop out of step because it is a salient pole machine. It continues to run at synchronous speed using **reluctance torque** alone:

$$P = \frac{V_t^2}{2}\left(\frac{1}{X_q} - \frac{1}{X_d}\right)\sin(2\delta)$$

To supply the tiny amount of active power required to overcome internal rotational and core losses, the power angle shifts to approximately $\delta \approx -45^\circ$.

Phase 2: Increasing Excitation in the Reverse Direction When the field current is reversed and gradually increased, it creates a magnetic field that directly opposes the stator's main field. Operating with a fully opposing negative field on the d-axis creates a highly unstable state.

To find a lower-energy stable operating configuration, the rotor physically shifts relative to the stator field by 180° electrical (it slips a pole pitch). Due to this structural inversion, the reversed field excitation now aligns with and **assists** the stator's magnetic field instead of opposing it. Since the reversed field is now helping to magnetize the machine, the armature no longer needs to draw massive magnetizing current from the bus. Consequently, the armature current I_a **decreases steeply**.

Thus, the armature current first increases and then decreases steeply.

B

Q5.11.2 MCQ 2011 2M Ans: B ●●○

Given,

$$S = V_{pu} I_{pu} \cos\phi$$

Rated KVA = 1 pu, rated voltage = 1 pu, and power factor = 1.

Hence,

$$1 = 1 \times I_{pu} \times 1$$

$$I_{pu} = 1, \qquad \cos\phi = 1$$

$$\phi = 0^\circ$$

For a salient-pole synchronous machine,

$$\tan\psi = \frac{V\sin\phi + I_a X_q}{V\cos\phi + I_a R_a}$$

Substituting

$$V = 1, \quad I_a = 1, \quad X_q = 1, \quad R_a = 0, \quad \phi = 0^\circ$$

$$\tan\psi = \frac{0 + 1\times 1}{1\times 1 + 1\times 0} = 1$$

$$\psi = 45^\circ$$

Also,

$$\psi = \delta + \phi$$

Since,

$$\phi = 0^\circ$$

$$\delta = \psi = 45^\circ$$

Therefore, the correct answer is

B

Q5.09.1 MCQ 2009 1M Ans: D ● ○ ○

For the same field excitation of 20 A,

$$E_{oc} = 2000\,\text{V}$$

and the short-circuit current is

$$I_{sc} = 400\,\text{A}$$

Hence, the synchronous impedance is

$$Z_s = \frac{E_{oc}}{I_{sc}} = \frac{2000}{400} = 5\,\Omega$$

The internal voltage drop at a load current of 200 A is

$$\Delta V = I_a Z_s$$

$$\Delta V = 200 \times 5 = 1000\,\text{V}$$

Since the machine is operating in the linear region and armature resistance is neglected,
Therefore, the correct option is

D

Q5.08.1 MCQ 2008 2M Ans: D ● ● ○

Given,

$$\overline{V}_t = 1\angle 0^\circ\,\text{pu}$$

$$\overline{I}_a = 0.6\angle 0^\circ\,\text{pu}$$

Since the power factor is unity, current and terminal voltage are in phase.
Applying KVL for a synchronous motor,

$$\overline{E} = \overline{V}_t - jI_aX_s$$

Substituting $X_s = 1\,\text{pu}$,

$$\overline{E} = 1 - j(0.6)(1)$$

$$\overline{E} = 1 - j0.6$$

Magnitude of emf:

$$|\overline{E}| = \sqrt{1^2 + 0.6^2} = 1.166 \approx 1.17\,\text{pu}$$

Angle of emf:

$$\delta = \tan^{-1}\left(\frac{0.6}{1}\right) = 30.96^\circ$$

Hence,

$$\overline{E} = 1.167\angle -30.96^\circ \text{ pu}$$

Hence, the correct option is

D

Key Points

- At unity power factor, $\overline{I}_a$ and $\overline{V}_t$ are in phase.
- For a synchronous motor,
$$\overline{E} = \overline{V}_t - jI_aX_s$$
when armature resistance is neglected.
- The load angle is obtained from the phasor angle of the internal emf with respect to the terminal voltage.

Q5.08.2 MCQ 2008 2M Ans: A

For a synchronous motor,

$$E_f = \left[(V\cos\phi - I_aR_a)^2 + (V\sin\phi - I_aX_s)^2\right]^{1/2}$$

Given,

$$R_a = 0$$

Hence,

$$E_f = \left[V^2\cos^2\phi + V^2\sin^2\phi + I_a^2X_s^2 - 2VI_aX_s\sin\phi\right]^{1/2}$$

Using

$$V = 1\ pu, \quad X_s = 1\ pu$$

and the same excitation as the previous case,

$$E_f = 1.17\ pu$$

The armature current increases by 20%,

$$I_a = 1.2(I_a)_{\text{initial}} = 1.2 \times 0.6 = 0.72\ pu$$

Substituting,

$$1.17 = \left[1^2 + (0.72)^2 - 2(1)(0.72)(1)\sin\phi\right]^{1/2}$$

Squaring both sides,

$$(1.17)^2 = 1 + (0.72)^2 - 1.44\sin\phi$$

$$1.3689 = 1 + 0.5184 - 1.44\sin\phi$$

$$-1.44\sin\phi = -0.1495$$

$$\sin\phi = 0.1038$$

$$\phi = 5.96^\circ$$

Therefore,

$$pf = \cos\phi = 0.995$$

Hence, the correct answer is

$$\boxed{A}$$

Key Points

- For a synchronous motor at lagging power factor

$$E_f = \sqrt{(V\cos\phi - I_a R_a)^2 + (V\sin\phi - I_a X_s)^2}$$

- If excitation is unchanged, E_f remains constant.
- Increasing armature current while keeping excitation constant changes the operating power factor.

Q5.08.3 MCQ | 2008 | 1M | Ans: D

In AC machines, the generated emf is given by

$$E = 4.44\, f\, \Phi\, T\, k_w$$

where

$$k_w = k_p k_d$$

is the winding factor.
For a distributed winding,

$$k_d < 1$$

and for a short-pitched (chorded) winding,

$$k_p < 1$$

Therefore,

$$k_w = k_p k_d < 1$$

which reduces the generated emf compared to a concentrated full-pitch winding.

Also, both distribution factor and pitch factor reduce the magnitude of harmonic components more severely than the fundamental component. Hence, harmonic content in the generated emf decreases.

Hence, the correct option is

$$\boxed{D}$$

Key Points

- Winding factor:

$$k_w = k_p k_d$$

- Distributed windings reduce harmonic content by spreading conductors in multiple slots.

- Short-pitching eliminates or suppresses selected harmonics and improves waveform quality.
- Since $k_p < 1$ and $k_d < 1$, the fundamental emf is also reduced.

Q5.07.1 MCQ | 2007 | 2M | Ans: B

Neglecting armature resistance,

$$\overline{E}_f = \overline{V}_t - j\overline{I}_a X_s$$

For a synchronous motor,

$$P = \frac{E_f V_t}{X_s} \sin\delta$$

When E_f is constant and P is reduced, δ decreases.
From the phasor diagram, when

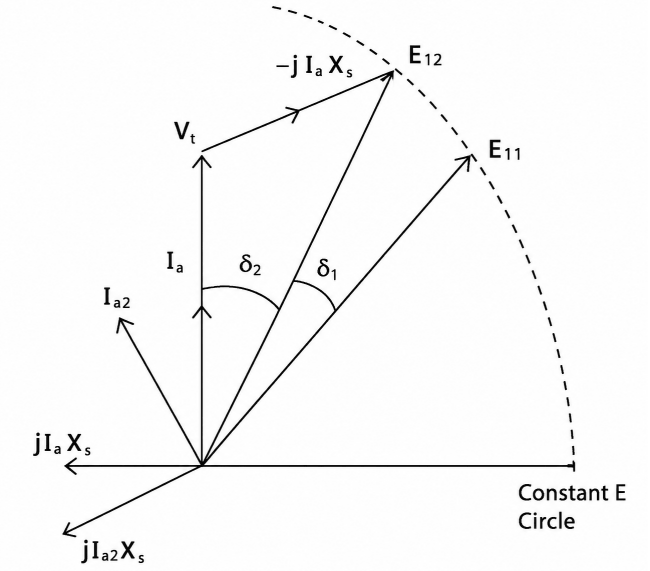


Fig. Solution Q5.07.1

$$\delta_2 < \delta_1$$

the armature current $\overline{I}_{a2}$ becomes leading. Hence, the motor operates at leading power factor.
This can also be verified from the input reactive power expression,

$$Q = \frac{V_t}{X_s}\left(V_t - E_f \cos\delta\right)$$

At unity power factor,

$$Q = 0$$

$$V_t = E_f \cos\delta$$

But as δ decreases,

$$V_t < E_f \cos\delta$$

Hence,

$$Q < 0$$

Therefore, the synchronous motor operates at leading power factor.

B

Key Points

- For a synchronous motor,
$$P = \frac{E_f V_t}{X_s} \sin\delta$$
- If excitation is constant, E_f remains constant.
- Reduction in load decreases δ.
- Negative input reactive power indicates leading power factor operation.

Mistakes to be avoided

- Do not assume that power factor remains unity when load changes.
- Do not use the generator phasor equation for a synchronous motor.
- Remember that $Q < 0$ represents leading power factor for motor operation.

Q5.07.2 MCQ 2007 2M Ans: C ●○○

The short-circuit ratio (SCR) is defined as

$$\text{SCR} = \frac{I_f \text{ (rated } V_{oc})}{I_f \text{ (rated } I_{sc})}$$

Substituting the given values,

$$\text{SCR} = \frac{15}{10} = 1.5$$

For a cylindrical-rotor synchronous machine, the synchronous reactance in per unit is approximately

$$X_s(\text{pu}) = \frac{1}{\text{SCR}}$$

Hence,

$$X_s(\text{pu}) = \frac{1}{1.5} = 0.666$$

Hence, the correct option is

C

Key Points

- Short-circuit ratio is the ratio of field current required for rated open-circuit voltage to that required for rated short-circuit current.

- For synchronous machines,
$$X_s(\text{pu}) \approx \frac{1}{\text{SCR}}$$
- Higher SCR implies lower synchronous reactance and better voltage regulation.

Q5.07.3 MCQ | 2007 | 2M | Ans: C

Initially, a load of 12 MW is connected at 50 Hz. When an additional load of 3.5 MW is connected, the frequency change due to the 5% governor droop is

$$\frac{\Delta f}{f} = -\frac{\Delta P}{P_R} \times \text{droop}$$

where $P_R = 15$ MW is the maximum continuous power limit of the prime mover. Hence,

$$\Delta f = -\frac{3.5}{15} \times 0.05 \times 50$$

$$\Delta f = -0.583 \text{ Hz}$$

Therefore, the initially settled frequency would be

$$f = 50 - 0.583$$

$$f = 49.417 \text{ Hz}$$

However, the total load is

$$P_{\text{total}} = 12 + 3.5 = 15.5 \text{ MW},$$

which exceeds the maximum continuous power limit of 15 MW. Hence, the prime mover cannot sustain the load and the rotor decelerates. Consequently, the frequency continues to decrease until it falls below 48 Hz, causing one of the 4 MW loads to trip.

The remaining load is, therefore,

$$P_L = 15.5 - 4 = 11.5 \text{ MW}.$$

The change in load from the initial 12 MW is

$$\Delta P = 11.5 - 12 = -0.5 \text{ MW}.$$

Hence,

$$\frac{\Delta f}{50} = -\frac{-0.5}{15} \times 0.05$$

$$\Delta f = 0.083 \text{ Hz}$$

Thus, the final frequency is

$$f = 50 + 0.083$$

$$\boxed{C}$$

Key Points

- Governor droop characteristic:

$$\frac{\Delta f}{f} = -\frac{\Delta P}{P_{\text{rated}}} \times \text{droop}$$

- An increase in load causes a reduction in frequency.
- Under-frequency relays disconnect loads when frequency falls below the preset threshold.
- Rotor decelerates when electrical power exceeds mechanical input power.

Mistakes to be avoided

- Do not neglect the negative sign in the droop equation.
- Do not stop at 49.417 Hz; check whether the load exceeds the prime-mover limit.
- Remember that under-frequency load shedding changes the final connected load.

Q5.06.1 MCQ | 2006 | 2M | Ans: B | ● ● ○

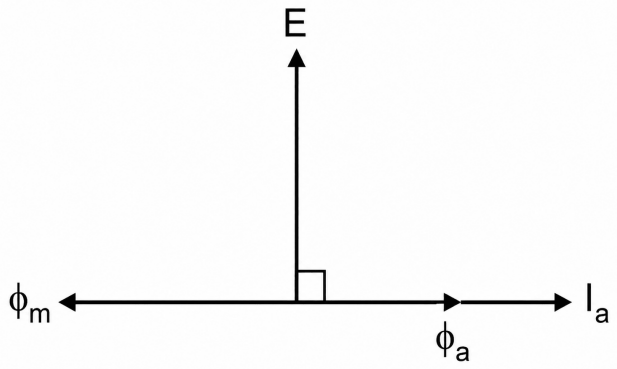


Fig. Solution Q5.06.1

For a synchronous generator supplying a zero power factor lagging load, the armature current lags the terminal voltage by 90°.

In this condition, the armature reaction MMF acts directly opposite to the main field MMF produced by the rotor.

Hence, the resultant air-gap flux decreases.

Thus, the correct option is

$\boxed{B}$

Key Points

- For a synchronous generator:
  - Unity power factor load → cross-magnetizing armature reaction.
  - Zero power factor lagging load → demagnetizing armature reaction.
  - Zero power factor leading load → magnetizing armature reaction.
- Demagnetizing armature reaction reduces the resultant air-gap flux and generated EMF.

Mistakes to be avoided

- Do not confuse generator and motor armature reaction effects.
- At unity power factor, armature reaction is cross-magnetizing, not demagnetizing.
- For a lagging load on a generator, armature reaction weakens the main field.

Q5.06.2 MCQ | 2006 | 2M | Ans: A | ● ● ●

At 50% load,

$$P = \frac{5000}{2} = 2500\,\text{W}$$

The armature current at unity power factor is

$$I_a = \frac{P}{\sqrt{3}V_L \cos\phi}$$

$$I_a = \frac{2500}{\sqrt{3} \times 400 \times 1} = 3.608\,\text{A}$$

The induced emf is

$$\overline{E}_f = \overline{V}_t + jI_aX_s$$

$$\overline{E}_f = \frac{400}{\sqrt{3}}\angle 0^\circ + j(3.608)(10)$$

$$\overline{E}_f = 233.74\angle 8.879^\circ\,\text{V}$$

If the excitation is increased by 1%,

$$E_{\text{new}} = 233.74 \times 1.01 = 236.0774\,\text{V}$$

For a synchronous generator,

$$|E| = \sqrt{(V\cos\phi + I_aR_a)^2 + (V\sin\phi + I_aX_s)^2}$$

Squaring,

$$|E|^2 = |V|^2 + I_a^2\left(R_a^2 + X_s^2\right) + 2I_aX_sV\sin\phi + 2I_aR_aV\cos\phi$$

At unity power factor,

$$\cos\phi = 1, \qquad \sin\phi = 0$$

Also,

$$R_a = 0, \qquad X_s = 10$$

Hence,

$$(236.0774)^2 = \left(\frac{400}{\sqrt{3}}\right)^2 + I_a^2(10)^2$$

$$I_a = 4.8981\,\text{A}$$

Rated current at unity power factor is

$$I_{\text{rated}} = \frac{5000}{\sqrt{3} \times 400 \times 1} = 7.2168 \text{ A}$$

Therefore,

$$\text{New current in percent} = \frac{4.8981}{7.2168} \times 100 = 67.9\%$$

Hence, the correct option is

A

Q5.06.3 MCQ | 2006 | 2M | Ans: C | ● ● ○

Slots per pole:

$$\frac{48}{4} = 12$$

Slot angle,

$$\beta = \frac{180^\circ}{12} = 15^\circ$$

Slots per pole per phase,

$$m = \frac{12}{3} = 4$$

Distribution factor,

$$K_d = \frac{\sin\left(\frac{m\beta}{2}\right)}{m \sin\left(\frac{\beta}{2}\right)}$$

$$K_d = \frac{\sin\left(\frac{4 \times 15}{2}\right)}{4 \sin\left(\frac{15}{2}\right)} = 0.9576$$

Pitch factor,

$$K_p = \cos\frac{\epsilon}{2} = \cos 18^\circ = 0.951$$

Winding factor,

$$K_w = K_p K_d = 0.951 \times 0.9576 = 0.91$$

In a double-layer winding,

$$\text{Number of coils} = \text{Number of slots} = 48$$

Total turns,

$$N = 48 \times 10 = 480$$

Turns per phase,

$$N_{ph} = \frac{480}{3} = 160$$

Induced emf per phase,

$$E_{ph} = 4.44 f N_{ph} \phi K_w$$

$$E_{ph} = 4.44 \times 50 \times 160 \times 0.025 \times 0.91 = 808.08\ \text{V}$$

Line voltage,

$$E_L = \sqrt{3}\, E_{ph}$$

$$E_L = \sqrt{3} \times 808.08 = 1400\ \text{V}$$

Therefore,

$$\boxed{C}$$

Mistakes to be avoided

- In double-layer winding, number of coils equals number of slots.
- Use turns per phase, not total turns, in the emf equation.
- Convert phase voltage to line voltage using $\sqrt{3}$ for star connection.
- Do not confuse slot angle with chording angle.

Q5.06.4 MCQ | 2006 | 2M | Ans: C

Slots per pole per phase:

$$m = \frac{12}{2} = 6$$

Distribution factor,

$$K_d = \frac{\sin\left(\frac{m\beta}{2}\right)}{m \sin\left(\frac{\beta}{2}\right)}$$

$$K_d = \frac{\sin\left(\frac{6 \times 15^\circ}{2}\right)}{6 \sin\left(\frac{15^\circ}{2}\right)}$$

$$K_d = 0.9028$$

Pitch factor,

$$K_p = \cos\frac{\epsilon}{2} = \cos 18^\circ = 0.951$$

Winding factor,

$$K_w = K_p K_d = 0.951 \times 0.9028 = 0.8585$$

Number of turns per phase:

$$N_{ph} = \frac{480}{2} = 240$$

Induced emf per phase:

$$E_{ph} = 4.44 f N_{ph} \phi K_w$$

$$E_{ph} = 4.44 \times 50 \times 240 \times 0.025 \times 0.8585 = 1143.522\,\text{V}$$

Line-to-line voltage:

$$E_L = \sqrt{2} \times 1143.522 = 1617.18\,\text{V}$$

Hence, the correct option is

C

Q5.06.5 MCQ | 2006 | 2M | Ans: A | ● ○ ○

For the 5th harmonic, the pitch factor is

$$K_{p5} = \cos\left(\frac{5\epsilon}{2}\right)$$

Given,

$$\epsilon = 36^\circ$$

Therefore,

$$K_{p5} = \cos\left(\frac{5 \times 36^\circ}{2}\right)$$

$$K_{p5} = \cos 90^\circ = 0$$

Since the induced emf of a harmonic is directly proportional to its pitch factor,

$$E_5 \propto K_{p5}$$

$$E_5 = 0$$

Hence, the correct option is

A

Key Points

- Short-pitching can be used to eliminate specific harmonics.
- If $K_{pn} = 0$, the corresponding harmonic emf is completely eliminated.

Q5.05.1 MCQ | 2005 | 2M | Ans: C | ● ○ ○

In salient-pole synchronous machines, the air gap along the d-axis is smaller than that along the q-axis.

Therefore, reluctance along the d-axis is less, which results in

$$X_d > X_q$$

A synchronous motor is not self-starting because the rotor cannot continuously follow the rotating magnetic field at synchronous speed. The average starting torque is zero.

Damper bars are provided to produce induction motor action during starting and help the machine accelerate close to synchronous speed.

The short-circuit ratio (SCR) is defined as

$$\text{SCR} = \frac{I_f \text{ (rated } V_{oc})}{I_f \text{ (rated } I_{sc})}$$

and is the reciprocal of per-unit synchronous reactance.

The V-curve of a synchronous motor represents the variation of armature current with field excitation at constant output power.

Hence, the third statement is false.

Therefore, the correct option is

C

Q5.05.2 MCQ | 2005 | 1M | Ans: B | ● ○ ○

Given,

$$S = 1000\,\text{kVA}, \qquad V_L = 6.6\,\text{kV}, \qquad X_s = 20\,\Omega$$

At full load and unity power factor,

$$I_a = \frac{S}{\sqrt{3}V_L}$$

$$I_a = \frac{1000 \times 10^3}{\sqrt{3} \times 6600} = 87.48\,\text{A}$$

Phase voltage is

$$V_{ph} = \frac{6600}{\sqrt{3}} = 3810.5\,\text{V}$$

For a synchronous generator,

$$\overline{E}_{ph} = \overline{V}_{ph} + jI_aX_s$$

$$E_{ph} = \sqrt{V_{ph}^2 + (I_aX_s)^2}$$

$$E_{ph} = \sqrt{(3810.5)^2 + (87.48 \times 20)^2} = 4193\,\text{V}$$

Line-to-line induced emf is

$$E_L = \sqrt{3}E_{ph}$$

$$E_L = \sqrt{3} \times 4193 = 7262\,\text{V} \approx 7.2\,\text{kV}$$

Therefore,

B

Q5.05.3 MCQ 2005 1M Ans: C ● ○ ○

From the previous solution,

$$V_{ph} = 3810.5\,\text{V}$$

and

$$I_a X_s = 87.48 \times 20 = 1749.6\,\text{V}$$

The power angle is

$$\delta = \tan^{-1}\left(\frac{I_a X_s}{V_{ph}}\right)$$

$$\delta = \tan^{-1}\left(\frac{1749.6}{3810.5}\right)$$

$$\delta = 24.6^\circ$$

Therefore,

C

Q5.04.1 MCQ 2004 2M Ans: A ● ● ○

When the excitation of two alternators operating in parallel is unequal, a synchronizing current (circulating current) flows between the machines.

The synchronizing current per phase is given by

$$I_{sy} = \frac{|E_1 - E_2|}{Z_{s1} + Z_{s2}}$$

Since both induced emfs are in phase, only their magnitudes differ.

$$E_1 = \frac{3300}{\sqrt{3}}\,\text{V}$$

$$E_2 = \frac{3200}{\sqrt{3}}\,\text{V}$$

Given,

$$Z_{s1} = Z_{s2} = j1.7\,\Omega$$

Hence,

$$I_{sy} = \frac{\frac{3300}{\sqrt{3}} - \frac{3200}{\sqrt{3}}}{2 \times 1.7}$$

$$I_{sy} = 16.98\,\text{A}$$

Hence, the correct option is

A

Key Points

- Unequal excitation of parallel alternators causes circulating current to flow between the machines.
- Circulating current flows even when there is no external load change.
- The synchronizing current depends on the emf difference and the sum of synchronous impedances.
- If the emfs are in phase, only the magnitude difference is considered.

Q5.04.2 MCQ | 2004 | 2M | Ans: C

Given,

$$P_{shaft} = 9\,\text{kW}$$

Total losses are

$$P_{loss} = P_{core} + P_{fw} + P_{cu}$$

Given,

$$P_{core} = 0.8\,\text{kW}$$

$$P_{fw} = 2\,\text{kW}$$

Since armature resistance is negligible,

$$P_{cu} \approx 0$$

Therefore,

$$P_{loss} = 0.8 + 2 = 2.8\,\text{kW}$$

Input power drawn by the motor is

$$P_{in} = P_{shaft} + P_{loss}$$

$$P_{in} = 9 + 2.8 = 11.8\,\text{kW}$$

For a three-phase motor,

$$P_{in} = \sqrt{3}\,V_L I_L \cos\phi$$

Hence,

$$I_L = \frac{P_{in}}{\sqrt{3}\,V_L \cos\phi}$$

Substituting,

$$I_L = \frac{11.8 \times 10^3}{\sqrt{3} \times 400 \times 0.8}$$

$$I_L = 21.29\,\text{A}$$

Hence, the correct option is

$$\boxed{\text{C}}$$

Q5.04.3 MCQ 2004 2M Ans: B ● ○ ○

For a three-phase system, the real power output is

$$P = \sqrt{3}\, V_L I_L \cos\phi$$

Given,

$$P = 500\,\text{MW} = 500 \times 10^6\,\text{W}$$

$$V_L = 21.5\,\text{kV} = 21.5 \times 10^3\,\text{V}$$

$$\cos\phi = 0.85$$

Therefore,

$$I_L = \frac{P}{\sqrt{3}\, V_L \cos\phi}$$

$$I_L = \frac{500 \times 10^6}{\sqrt{3} \times 21.5 \times 10^3 \times 0.85}$$

$$I_L = 15.79 \times 10^3\,\text{A}$$

$$I_L = 15.79\,\text{kA}$$

Hence, the correct option is

B

Q5.04.4 MCQ 2004 2M Ans: D ● ○ ○

The synchronous speed of an alternator is given by

$$N_s = \frac{120f}{P}$$

Given,

$$N_s = 250\,\text{rpm}$$

$$f = 50\,\text{Hz}$$

Therefore,

$$P = \frac{120f}{N_s}$$

$$P = \frac{120 \times 50}{250} = 24$$

Therefore, the correct option is

D

Q5.03.1 MCQ 2003 2M Ans: A ● ● ○

The coil distribution is as shown in the figure.

							a1'	a2'
a1	a2	a3	b1	b2	b3	c1	c2	c3
a3'	b1'	b2'	b3'	c1'	c2'	c3'	a4'	a5'
a4	a5	a6	b4	b5	b6	c4	c5	c6
—	—	—	—	—	—	—	—	—
—	—	—	—	—	—	—	—	—

Fig. Solution Q5.03.1

Looking at the second row, out of every 9 slots, 3 slots contain coils of the same phase and 6 slots contain coils of different phases.

Therefore, the fraction of slots having coils of different phases is

$$\frac{6}{9}$$

For a total of 36 slots,

$$\text{Number of slots having coils of different phases} = 36 \times \frac{6}{9}$$

$$= 24$$

Therefore, the correct option is

$$\boxed{\text{A}}$$

Key Points

- In a double-layer winding, the phase distribution can be determined directly from the slot assignment table.
- The required number of slots is obtained by multiplying the total slots by the fraction of slots occupied by coils of different phases.
- Here, 6 out of every 9 slots contain coils belonging to different phases.

Q5.03.2 MCQ 2003 1M Ans: D ● ● ○

A capacitor connected across the partly inductive load supplies the required leading reactive current locally.

Hence, the inductive reactive current drawn from the synchronous generator is nullified and the generator operates at unity power factor.

The real power demand of the load is unchanged. Therefore, the mechanical input power supplied by the engine remains unchanged.

Hence, fuel input is left unaltered.

Since the reactive current supplied by the generator is reduced, the armature reaction becomes less demagnetizing. Therefore, to maintain the same terminal voltage, the field current must be reduced.

Thus,

I_f has to be reduced

and

fuel input is left unaltered

Therefore, the correct option is

D

Key Points

- Capacitor compensation supplies reactive power locally.
- Field current controls reactive power and terminal voltage.
- Fuel input controls real power output.
- Reactive power compensation does not change real power demand.

Mistakes to be avoided

- Do not reduce fuel input because active power demand remains unchanged.
- Do not increase field current after reducing lagging reactive current.
- Do not confuse reactive power compensation with real power reduction.

Q5.03.3 MCQ 2003 1M Ans: A

The upper curve represents the open-circuit voltage (induced emf) of an alternator, while the lower curve represents the terminal voltage corresponding to a zero power factor lagging load.

For a zero power factor lagging load,

$$\overline{E}_f = \overline{V}_t + \overline{I}_a \angle(-90^\circ)\,(jX_s)$$

Since

$$\overline{I}_a \angle(-90^\circ)\,(jX_s) = I_a X_s$$

we get

$$E_f = V_t + I_a X_s$$

Therefore,

$$E_f - V_t = I_a X_s$$

Hence, the difference between the induced emf and the terminal voltage is proportional to the synchronous reactance.

Therefore, the correct option is

A

Key Points

- At zero power factor lagging, armature current lags terminal voltage by 90°.
- The voltage drop between the OCC and ZPF characteristics is used in the Potier and synchronous impedance methods.

Q5.02.1 MCQ | 2002 | 2M | Ans: B | ● ● ○

The air-gap flux due to field excitation alone is

$$\phi_f = 25\,\text{mWb}$$

After connecting the stator supply while keeping the field excitation unchanged, the air-gap flux reduces to

$$\phi_r = 20\,\text{mWb}$$

Thus, the armature reaction opposes the main field and cancels an additional

$$25 - 20 = 5\,\text{mWb}$$

of flux.

Since the resultant flux decreases, the armature reaction is **demagnetizing**. In a synchronous motor, demagnetizing armature reaction corresponds to a **leading power factor** current.

Therefore, the motor draws leading current from the supply.

D

Q5.02.2 NAT | 2002 | 2M | Ans: 242.92, 35.38, 0.9845 lag | ● ○ ○

Given:

$$V_L = 415\,\text{V}, \qquad X_s = 20\,\Omega, \qquad I_a = 2\,\text{A}$$

For constant excitation and variable load, the induced emf remains constant.

Per-phase voltage:

$$V_t = \frac{415}{\sqrt{3}} = 239.60\,\text{V}$$

Induced emf is

$$E_f = V_t - jI_aX_s$$

$$E_f = \frac{415}{\sqrt{3}} - j(20 \times 2)$$

$$E_f = 239.60 - j40$$

$$E_f = 242.92\angle -9.47^\circ\,\text{V}$$

When load is increased, excitation remains unchanged, so

$$E_f = \sqrt{(V_t \cos\phi)^2 + (V_t \sin\phi + I_aX_s)^2}$$

Substituting,

$$242.92^2 = \left(\frac{415}{\sqrt{3}}\right)^2 - 2\left(\frac{415}{\sqrt{3}}\right)(100)\sin\phi + (50 \times 2)^2$$

$$\sin\phi = 0.175$$

$$\phi = 10.09^\circ$$

Since ϕ is positive, the assumed lagging power factor is correct.

$$\cos\phi = 0.9845 \text{ lag}$$

Since the machine is lossless,

$$P_{out} = P_{in}$$

$$P = \sqrt{3}\, V_L I_L \cos\phi$$

$$P = \sqrt{3} \times 415 \times 50 \times 0.9845$$

$$P = 35.38\,\text{kW}$$

$$\boxed{242.92,\ 35.38,\ 0.9845 \text{ lag}}$$

Q5.01.1 MCQ | 2001 | 1M | Ans: B | ● ○ ○

To eliminate the n^{th} harmonic from the generated emf of an alternator, the coil-span factor for that harmonic must be zero.

The pitch factor is

$$k_{pn} = \cos\left(\frac{n\alpha}{2}\right)$$

where α is the short-pitch angle.

For elimination of the 5^{th} harmonic,

$$k_{p5} = 0$$

$$\cos\left(\frac{5\alpha}{2}\right) = 0$$

$$\frac{5\alpha}{2} = 90^\circ$$

$$\alpha = \frac{180^\circ}{5}$$

$$\alpha = 36^\circ$$

Therefore, the correct option is

$$\boxed{\text{B}}$$

Q5.01.2 MCQ | 2001 | 1M | Ans: C ● ● ○

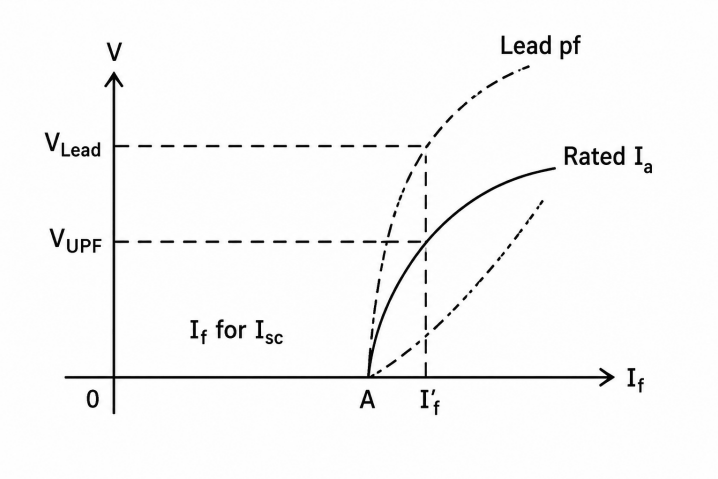


Fig. Solution Q5.01.2

At short-circuit condition,

$$V_t = 0$$

To produce rated armature current under short-circuit, a non-zero field current is required. Since this field current is the same for leading, lagging and unity power factor loads, all OCC curves must originate from the same point corresponding to the short-circuit field current.

Hence, curves A and B are eliminated.

Now consider a leading power factor load. For constant armature current and excitation,

$$V_t = E_f - jI_aX_s$$

For a leading power factor load,

$$I_a = I_a\angle 90^\circ$$

Therefore,

$$V_t = E_f - (jI_a)\,(jX_s)$$

$$V_t = E_f + I_aX_s$$

Thus, the reactance drop adds to the generated emf, resulting in a higher terminal voltage for the same field current.

Hence, the upper curve corresponds to leading power factor operation, while the lower curve corresponds to lagging power factor operation.

C

Key Points

- At short circuit, $V_t = 0$, but a finite field current is required to produce rated armature current.
- Leading power factor load produces a voltage rise, whereas lagging power factor load produces a voltage drop.
- Hence, for the same excitation, the terminal voltage is maximum for leading power factor loading.

Q5.01.3 MCQ | 2001 | 2M | Ans: D | ● ● ○

For zero voltage regulation,

$$\cos(\theta + \phi) = -\frac{|I_a||Z_s|}{2|V_t|}$$

Substituting the given values,

$$\cos(\theta + \phi) = -\frac{20 \times 10}{2 \times \left(\frac{440}{\sqrt{3}}\right)} = -0.3936$$

$$\theta + \phi = 113.178^\circ$$

Since the synchronous impedance consists only of reactance,

$$\theta = 90^\circ$$

Therefore,

$$\phi = 113.178^\circ - 90^\circ = 23.178^\circ$$

$$\cos\phi = \cos(23.178^\circ) = 0.92$$

Therefore, the correct option is

D

Key Points

- For zero voltage regulation of a synchronous generator,

$$\cos(\theta + \phi) = -\frac{|I_a||Z_s|}{2|V_t|}$$

- If armature resistance is neglected, the synchronous impedance is purely reactive and

$$\theta = 90^\circ.$$

- Zero voltage regulation occurs at a leading power factor load.

Q5.01.4 NAT | 2001 | 5M | Ans: 91.80 to 91.85 | ● ● ●

When excitation is OFF for the synchronous motor, the driving motor supplies only the friction and windage losses.

$$P_{\text{friction \& windage}} = 800\,\text{W}$$

When the armature is short-circuited and rated armature current flows, the input power corresponds to friction and windage losses plus full-load copper losses.

$$P_{\text{friction \& windage}} + P_{\text{cu}} = 2500\,\text{W}$$

$$P_{\text{cu}} = 2500 - 800 = 1700\,\text{W}$$

With excitation ON and armature open-circuited, the driving motor supplies friction and windage losses and iron losses.

$$P_{\text{friction \& windage}} + P_{\text{iron}} = 1800\,\text{W}$$

$$P_{\text{iron}} = 1800 - 800 = 1000\,\text{W}$$

At 50% load, the armature current is half of the rated current. Copper loss varies as the square of the current:

$$P_{\text{cu, 50\%}} = (0.5)^2 \times 1700 = 425\,\text{W}$$

Total losses at 50% load,

$$P_{\text{loss}} = 800 + 1000 + 425 = 2225\,\text{W}$$

Output at 50% load,

$$P_{\text{out}} = 0.5 \times 50 = 25\,\text{kW}$$

Efficiency,

$$\eta = \frac{P_{\text{out}}}{P_{\text{out}} + P_{\text{loss}}} = \frac{25000}{25000 + 2225} = \frac{25000}{27225}$$

$$\eta = 0.9183 = 91.83\%$$

$$\boxed{91.83\%}$$

At half-load, the output power is

$$P_o = 50 \times 0.5 \times 10^3 = 25\,000\,\text{W}$$

Since copper loss varies as the square of load current,

$$P_{\text{cu, half-load}} = \left(\frac{1}{2}\right)^2 P_{\text{cu}} = \frac{1700}{4} = 425\,\text{W}$$

Total losses at half-load are

$$P_{\text{loss}} = P_{\text{friction \& windage}} + P_{\text{iron}} + P_{\text{cu, half-load}}$$

$$P_{\text{loss}} = 800 + 1000 + 425 = 2225\,\text{W}$$

Hence,

$$\eta = \frac{P_o}{P_o + P_{\text{loss}}}$$

$$\eta = \frac{25\,000}{25\,000 + 2225} = 0.9182$$

$$\eta = 91.82\%$$

$$\boxed{91.82}$$

Q5.01.5 NAT | 2001 | 5M | Ans: A:35.32, 26.49, B:44.6, 33.45 | ● ○ ○

Rating of each alternator:

$$S = 100\,\text{MVA}, \qquad \cos\phi = 0.8$$

Initially,

$$P_1 = P_2 = S\cos\phi = 100 \times 0.8 = 80\,\text{MW}$$

$$Q_1 = Q_2 = S\sin\phi = 100 \times 0.6 = 60\,\text{MVAR}$$

Assuming a base of 100 MVA,

$$V = 1\,\text{p.u.}, \qquad I_a = 1\,\text{p.u.}$$

Since direct-axis and quadrature-axis reactances are equal, the machine is cylindrical rotor with

$$X_s = 0.8\,\text{p.u.}, \qquad R_a = 0$$

Initially,

$$E_{fA} = E_{fB} = \sqrt{(V\cos\phi + I_a R_a)^2 + (V\sin\phi + I_a X_s)^2}$$

$$E_{fA} = E_{fB} = \sqrt{(1 \times 0.8)^2 + (1 \times 0.6 + 0.8)^2}$$

$$E_{fA} = E_{fB} = 1.612\,\text{p.u.}$$

Assuming linear magnetization characteristics, the change in field current is equal to the change in excitation emf.
Generator A excitation is reduced by 5%,

$$E'_{fA} = 0.95 \times 1.612 = 1.532\,\text{p.u.}$$

Generator B excitation is increased by 5%,

$$E'_{fB} = 1.05 \times 1.612 = 1.693\,\text{p.u.}$$

Assuming negligible change in power factor,

$$1.532 = \sqrt{(V\cos\phi)^2 + (V\sin\phi + I'_a X_s)^2}$$

$$I'_a = 0.883\,\text{p.u.}$$

Hence, the modified active and reactive powers for generator A are

$$P'_A = 0.883 \times 40 = 35.32\,\text{MW}$$

$$Q'_A = 0.883 \times 30 = 26.49\,\text{MVAR}$$

Similarly, for generator B,

$$1.693 = \sqrt{(V\cos\phi)^2 + (V\sin\phi + I'_a X_s)^2}$$

$$I'_a = 1.115\,\text{p.u.}$$

Therefore,

$$P'_B = 1.115 \times 40 = 44.6\,\text{MW}$$

$$Q'_B = 1.115 \times 30 = 33.45\,\text{MVAR}$$

$$\boxed{A : 35.32, 26.49, B : 44.6, 33.45}$$

Key Points

- Under linear magnetization, percentage change in excitation emf equals percentage change in field current.
- Active and reactive power sharing are proportional to armature current when the power factor remains approximately constant.

Q5.00.1 MCQ | 2000 | 1M | Ans: B | ● ○ ○

The phase sequence of a three-phase alternator depends upon:

- Direction of rotation of the rotor.
- Direction of cutting of armature conductors by the rotating magnetic field.

If the field current is reversed while keeping the direction of rotation unchanged, all phase voltages reverse their polarities simultaneously, but the phase sequence remains unchanged.

However, if the direction of rotation is reversed while keeping the field current unchanged, the order in which the phase voltages attain their maximum values reverses.

For example,

$$R \rightarrow Y \rightarrow B$$

becomes

$$R \rightarrow B \rightarrow Y$$

Hence, the phase sequence reverses only when the direction of rotation is reversed.

Therefore, the correct option is

$$\boxed{\text{B}}$$

Key Points

- Phase sequence depends on the direction of rotation of the magnetic field.
- Reversing the field current changes the polarity of induced emf but does not change the phase sequence.

Mistakes to be avoided

- Do not confuse reversal of polarity with reversal of phase sequence.
- Doubling the number of poles changes frequency according to

$$f = \frac{PN}{120}$$

but does not reverse phase sequence.

- Remember that all three phase voltages reverse together when field current is reversed.

Q5.00.2 MCQ | 2000 | 1M | Ans: B

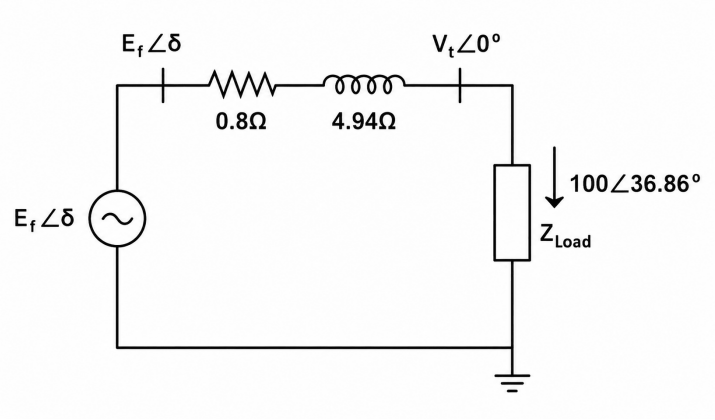


Fig. Solution Q5.00.2

From the per-phase KVL equation of an alternator,

$$E_f \angle \delta = V_t \angle 0° + I_a \angle \phi \times Z_s \angle \theta$$

where

$$\phi = \cos^{-1}(0.8) = 36.86°$$

For leading power factor,

$$I_a = 100 \angle 36.86°$$

and

$$Z_s = 0.8 + j4.94$$

Hence,

$$E_f \angle \delta = 2000 \angle 0° + 100 \angle 36.86° (0.8 + j4.94)$$

$$E_f \angle \delta = 1822.316 \angle 14.07°$$

Voltage regulation is

$$\%VR = \frac{|E_f| - |V_t|}{|V_t|} \times 100$$

$$\%VR = \frac{1822.316 - 2000}{2000} \times 100$$

$$\%VR = -8.884\%$$

Hence, the correct option is

B

Q5.00.3 NAT | 2000 | 5M | Ans: 35.056,26.29 | ● ● ○

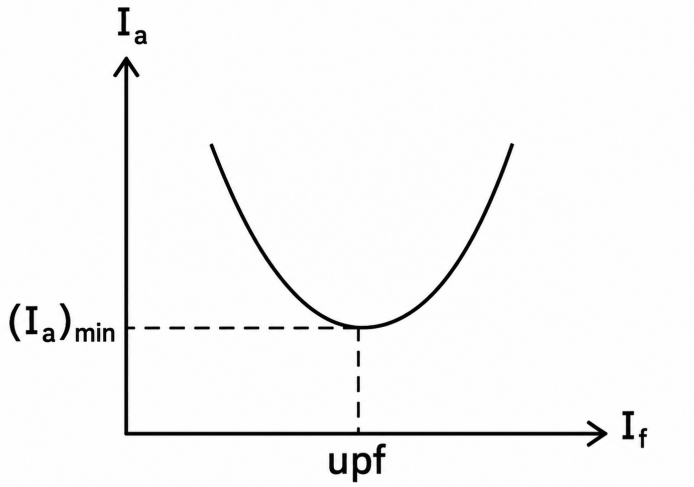


Fig. Solution Q5.00.3

From the V-curve of a synchronous motor, the minimum armature current occurs at unity power factor.

$$(I_a)_{\min} = 8.8\,\text{A}$$

At unity power factor,

$$P = \sqrt{3}\,V_L I_a \cos\phi$$

Since losses are neglected,

$$P_{\text{in}} = P_{\text{out}}$$

$$P = \sqrt{3} \times 2300 \times 8.8 \times 1$$

$$P = 35.056\,\text{kW}$$

For obtaining a leading power factor of 0.8, the synchronous motor must be overexcited. Since the load remains unchanged, the output power remains constant.

$$\sqrt{3}\,V_L I_{a1} \cos\phi_1 = \sqrt{3}\,V_L I_{a2} \cos\phi_2$$

$$I_{a2} = I_{a1}\left(\frac{\cos\phi_1}{\cos\phi_2}\right)$$

$$I_{a2} = 8.8\left(\frac{1}{0.8}\right) = 11\,\text{A}$$

Hence, the field current should be increased until the line ammeter reads 11 A.

Reactive power delivered to the supply under this condition is

$$Q = \sqrt{3}\,V_L I_{a2} \sin\phi_2$$

For 0.8 leading power factor,

$$\sin\phi_2 = \sqrt{1 - 0.8^2} = 0.6$$

Therefore,

$$Q = \sqrt{3} \times 2300 \times 11 \times 0.6$$

$$Q = 26.29\,\text{kVAR}$$

$$\boxed{P = 35.056, 26.29}$$

Q5.99.1 MCQ 1999 1M Ans: C ●●○

Higher synchronous reactance in modern alternators is preferred because it limits the current during transient disturbances and short-circuit conditions.

For a sudden disturbance,

$$I_{\text{transient}} \propto \frac{E}{X_s}$$

Hence, if synchronous reactance X_s is increased, the transient current decreases.

$$X_s \uparrow \quad \Rightarrow \quad I_{\text{transient}} \downarrow$$

Therefore, higher synchronous reactance helps in reducing transient currents.

$$\boxed{\text{C}}$$

Key Points

- Higher synchronous reactance limits current during transient conditions..
- Higher voltage regulation is a disadvantage, not the reason for preference.

Mistakes to be avoided

- Do not select sub-transient current unless sub-transient reactance X_d'' is specifically involved.
- Do not select higher voltage regulation; it is not an advantage.
- Harmonic currents are reduced mainly by winding design, not by increasing X_s.

Q5.99.2 MCQ 1999 2M Ans: B ●●○

The generated emf of an alternator is proportional to

$$E \propto k_d k_p N_{ph}$$

Since the armature conductors are the same in both cases, the current carrying capability remains unchanged.

For a three-phase winding with 60° phase spread,

$$k_{d1} = \frac{\sin\left(\frac{\pi}{2}\right)}{\frac{\pi}{2}} = \frac{2}{\pi}$$

For a single-phase winding with 180° phase spread,

$$k_{d2} = \frac{\sin\left(\frac{\pi}{2}\right)}{\frac{\pi}{2}} = \frac{2}{\pi}$$

Using the winding factors obtained in the given solution,

$$k_{d1} = \frac{3}{\pi}, \qquad k_{d2} = \frac{2}{\pi}$$

Since

$$(\text{kVA})_{\text{rating}} \propto E \propto k_d$$

$$\frac{(\text{kVA})_1}{(\text{kVA})_2} = \frac{k_{d1}}{k_{d2}} = \frac{\frac{3}{\pi}}{\frac{2}{\pi}} = \frac{3}{2}$$

Given

$$(\text{kVA})_1 = 300\,\text{kVA}$$

$$\frac{300}{(\text{kVA})_2} = \frac{3}{2}$$

$$(\text{kVA})_2 = 200\,\text{kVA}$$

Therefore,

$$\boxed{B}$$

Key Points

- Generated emf is proportional to the distribution factor k_d.
- With the same armature conductors, current carrying capability remains unchanged.
- Hence, machine kVA rating is directly proportional to generated emf.
- Distribution factor depends on the phase spread of the winding.

Q5.99.3 NAT | 1999 | 5M | Ans: 0.8556 | ● ● ○

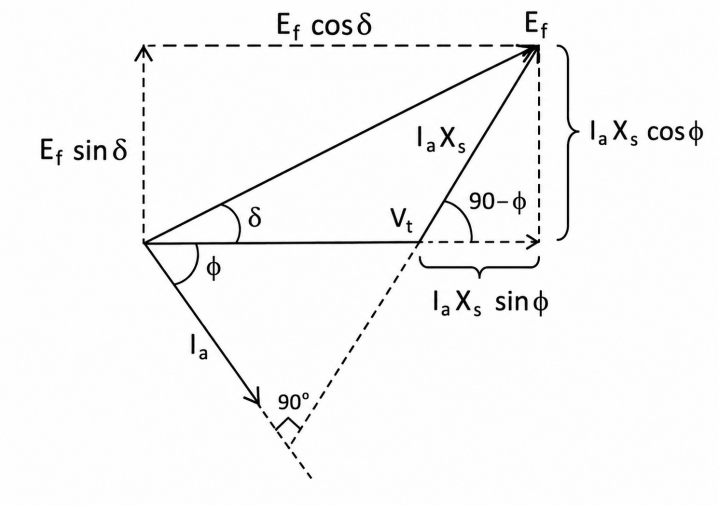


Fig. Solution Q5.99.3

Since

$$|E_f| > |V_t|$$

the alternator is over-excited.
For an over-excited alternator, the load power factor is lagging.

$$|E_f| = 1.25|V_t|$$

From the per-phase phasor equation of an alternator,

$$E_f \angle\delta = V_t \angle 0^\circ + (I_a \angle -\phi)\, X_s \angle 90^\circ$$

$$E_f \angle\delta = V_t \angle 0^\circ + I_a X_s \angle (90^\circ - \phi)$$

Resolving into horizontal and vertical components,

$$E_f \sin\delta = I_a X_s \cos\phi \tag{1}$$

$$E_f \cos\delta = V_t + I_a X_s \sin\phi \tag{2}$$

Squaring and adding (1) and (2),

$$E_f^2 = (I_a X_s)^2 + V_t^2 + 2V_t I_a X_s \sin\phi$$

Given,

$$E_f = 1.25 \times \frac{11}{\sqrt{3}} \times 10^3 \text{ V}$$

$$V_t = \frac{11}{\sqrt{3}} \times 10^3 \text{ V}$$

$$I_a X_s = 500 \times 5 = 2500 \text{ V}$$

Substituting,

$$\left(1.25 \times \frac{11}{\sqrt{3}} \times 10^3\right)^2 = (2500)^2 + \left(\frac{11}{\sqrt{3}} \times 10^3\right)^2 + 2\left(\frac{11}{\sqrt{3}} \times 10^3\right)(2500)\sin\phi$$

$$\sin\phi = 0.5176$$

$$\phi = 31.17^\circ$$

$$\cos\phi = 0.8556$$

Hence,

$$\boxed{0.8556}$$

Q5.98.1 MCQ | 1998 | 1M | Ans: A | ● ○ ○

A synchronous generator connected to an infinite bus and operating under over-excited condition supplies reactive power to the system.

For a synchronous generator,

$$Q = \frac{EV}{X_s}\cos\delta - \frac{V^2}{X_s}$$

When excitation is increased,

$$E > V$$

and the generator delivers positive reactive power ($Q > 0$) to the bus.

A source supplying reactive power behaves as a capacitor from the system point of view.

$$\text{Over-excited Generator} \implies Q > 0 \implies \text{Capacitive Behaviour}$$

Hence, the machine acts as a capacitor.

$$\boxed{\text{A}}$$

Key Points

- An over-excited synchronous machine supplies reactive power to the system.
- A source supplying reactive power is equivalent to a capacitive element.
- An under-excited synchronous machine absorbs reactive power and behaves like an inductor.
- Infinite bus maintains constant voltage and frequency irrespective of generator excitation changes.

Q5.98.2 MCQ | 1998 | 2M | Ans: D | ● ● ○

For a distributed winding, the distribution factor is

$$k_d = \frac{\sin\left(\frac{m\gamma}{2}\right)}{m\sin\left(\frac{\gamma}{2}\right)}$$

For small values of γ,

$$\sin\left(\frac{\gamma}{2}\right) \approx \frac{\gamma}{2}$$

Hence,

$$k_d \approx \frac{\sin\left(\frac{m\gamma}{2}\right)}{\frac{m\gamma}{2}}$$

For a 1-phase alternator,

$$m\gamma = \pi$$

(phase spread = 180°)
Therefore,

$$k_d = \frac{\sin\left(\frac{\pi}{2}\right)}{\frac{\pi}{2}} = \frac{2}{\pi}$$

The induced emf is proportional to the distribution factor.

$$E \propto k_d$$

Thus,

$$E = \frac{2}{\pi}(2T) = 1.273\,T \text{ V}$$

Hence,

D

Key Points

- Distribution factor:

$$k_d = \frac{\sin\left(\frac{m\gamma}{2}\right)}{m\sin\left(\frac{\gamma}{2}\right)}$$

- For a single-phase winding, the phase spread is 180° ($m\gamma = \pi$).

Q5.97.1 NAT | 1997 | 2M | Ans: 180, -10 | ● ○ ○

Given,

$$V_t = 200\text{ V}$$

$$I_a = 10\text{ A}$$

$$Z_s = 2\angle 90^\circ\ \Omega$$

Since,

$$\theta \rightarrow \text{Impedance angle}$$

$\phi \rightarrow$ Load power factor angle

Applying the alternator phasor equation,

$$E_f \angle\delta = V_t \angle 0^\circ + I_a \angle\phi \; Z_s \angle\theta$$

Substituting the given values,

$$E_f \angle\delta = 200 + (10\angle 90^\circ)\,(2\angle 90^\circ)$$

$$E_f \angle\delta = 200 + 20\angle 180^\circ$$

$$E_f \angle\delta = 200 - 20$$

$$E_f \angle\delta = 180\angle 0^\circ$$

Hence,

$$|E_f| = 180\,\text{V}$$

Voltage regulation is

$$\text{VR} = \frac{|E_f| - |V_t|}{|V_t|} \times 100$$

$$\text{VR} = \frac{180 - 200}{200} \times 100$$

$$\text{VR} = -10\%$$

Therefore,

$$\boxed{180,\ -10}$$

PrepFusion

Q5.96.1 MCQ | 1996 | 1M | Ans: B | ● ○ ○

When the field winding of a cylindrical-rotor synchronous motor is short-circuited, no DC excitation is available.

Hence, the motor cannot develop synchronous torque at standstill and therefore cannot start as a synchronous motor.

However, the short-circuited field winding acts like a squirrel-cage rotor due to induced currents.

Therefore, the machine develops induction motor torque and starts as an induction motor.

As the field winding remains short-circuited,

$$I_f = 0$$

and no excitation emf is produced.

$$E_f = 0$$

Since synchronous torque is proportional to excitation,

$$T_s \propto E_f$$

$$T_s = 0$$

Thus, after starting, the machine continues to operate only as an induction motor and cannot pull into synchronism.
Hence, it starts but does not run at synchronous speed.

B

Q5.96.2 MCQ 1996 1M Ans: B ●○○

The synchronous reactance of a synchronous motor is obtained from the ratio of open-circuit voltage to short-circuit current for the same field current.

$$X_s = \frac{E_{oc}}{I_{sc}}$$

At low values of field current, the magnetic circuit is unsaturated and X_s is nearly constant. As field current increases, saturation occurs, due to which the increase in E_{oc} is less than proportional.

Therefore,

$$I_f \uparrow \quad \Rightarrow \quad X_s \downarrow$$

Hence, the curve corresponding to synchronous reactance versus field current is the decreasing curve.

B

Key Points

- Synchronous reactance is given by
$$X_s = \frac{E_{oc}}{I_{sc}}.$$
- Due to saturation, synchronous reactance decreases with increase in field current.
- The decreasing characteristic in the given figure is curve II.

Q5.96.3 MCQ 1996 1M Ans: C ●○○

During hunting of a synchronous motor, the rotor oscillates about its equilibrium position.

Due to these oscillations, relative motion exists between the rotor damper bars and the rotating magnetic field. Hence, currents are induced in the damper bars.

These induced currents produce a damping torque which opposes the oscillations.

Hence, the correct option is

C

Key Points

- Hunting means oscillation of the rotor about synchronism.
- Damper bars reduce hunting by producing damping torque.
- The torque developed by damper bars opposes rotor oscillations.

Q5.95.1 MCQ | 1995 | 1M | Ans: C

"Internal power factor angle" ψ in alternator and motor is the angle between E_f and I_a.

The phasor diagram of a synchronous motor operating at leading power factor is shown below.

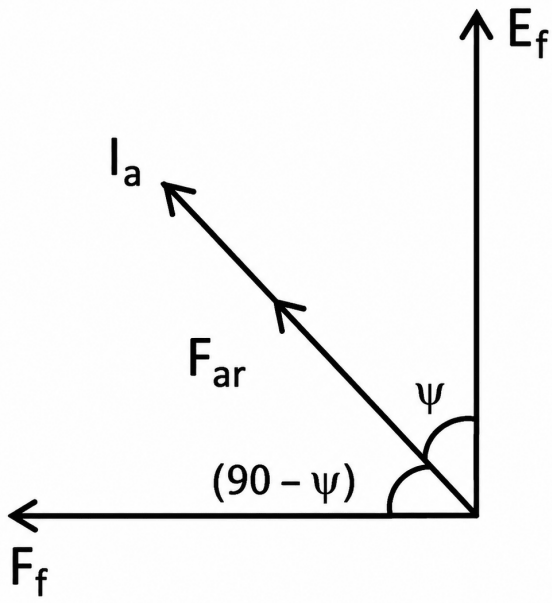


Fig. Solution Q5.95.1

As can be seen from the phasor diagram, the angle between the excitation emf and armature current (or mmf) is ψ.

Since the angle between excitation emf and field mmf is 90°, the angle between armature mmf and field mmf is

$$(90^\circ - \psi)$$

Hence, the correct option is

C

Q5.95.2 NAT | 1995 | 1M | Ans: 0.844

Given:

$$\text{Number of stator slots} = 36$$

$$\text{Phase spread} = m\beta = 120^\circ$$

Slots per pole per phase,

$$m = \frac{36}{8 \times 3} = 1.5$$

Since m is not an integer, it is a fractional slot winding.

$$m = \frac{\text{slots/ph}}{\text{pole}} = \frac{36/3}{8} = \frac{4 \times 3}{4 \times 2} = \frac{Ks}{Kp}$$

Here, K is the HCF and

$$s = 3$$

For a fractional slot winding, the distribution factor is

$$k_d = \frac{\sin\left(\frac{m\beta}{2}\right)}{s\sin\left(\frac{m\beta}{2s}\right)}$$

Substituting,

$$k_d = \frac{\sin\left(\frac{120^\circ}{2}\right)}{3\sin\left(\frac{120^\circ}{2\times 3}\right)}$$

$$k_d = \frac{\sin 60^\circ}{3\sin 20^\circ}$$

$$k_d = \frac{0.8660}{3\times 0.3420}$$

$$k_d = 0.844$$

$$\boxed{0.844}$$

Key Points

- For fractional slot windings, slots per pole per phase (m) is non-integer.
- Distribution factor for fractional slot winding is

$$k_d = \frac{\sin\left(\frac{m\beta}{2}\right)}{s\sin\left(\frac{m\beta}{2s}\right)}$$

  where s is obtained from the fractional-slot grouping.
- Distribution factor is always less than unity.

Q5.95.3 NAT | 1995 | 5M | Ans: 2.73 | ●●○

Given parameters are

$$V_L = 415\,\text{V}, \quad E_L = 400\,\text{V}, \quad P_{\text{sh}} = 15\,\text{kW}, \quad \cos\phi = 0.9 \text{ (leading)}, \quad R_a = 0.5\,\Omega.$$

Calculation of the armature current I_a:

The input power balance, accounting for the leading power factor and armature copper losses, is

$$\sqrt{3}V_L I_a \cos\phi + 3I_a^2 R_a = P_{\text{sh}}$$

Substituting the given values,

$$(\sqrt{3} \times 415 \times I_a \times 0.9) + (3I_a^2 \times 0.5) = 15000$$

$$646.92I_a + 1.5I_a^2 = 15000$$

Rearranging,

$$1.5I_a^2 + 646.92I_a - 15000 = 0$$

Solving for the positive value of I_a,

$$I_a = \frac{-646.92 + \sqrt{(646.92)^2 - 4(1.5)(-15000)}}{2(1.5)}$$

$$I_a = \frac{-646.92 + 713.10}{3} = 22.06\text{A}$$

Phase-based vector analysis for X_s:

Assuming a star-connected system, the per-phase terminal voltage is

$$V_t = \frac{415}{\sqrt{3}} \approx 239.60\,\text{V}$$

The per-phase induced EMF is

$$E_f = \frac{400}{\sqrt{3}} \approx 230.94\,\text{V}$$

For a power factor of 0.9 leading,

$$\sin\phi = \sqrt{1 - (0.9)^2} \approx 0.4359$$

For a leading power factor alternator,

$$E_f^2 = (V_t \cos\phi + I_a R_a)^2 + (V_t \sin\phi - I_a X_s)^2$$

Substituting the values,

$$(230.94)^2 = (239.60 \times 0.9 + 22.06 \times 0.5)^2 + (239.60 \times 0.4359 - 22.06X_s)^2$$

$$53333.28 = (215.64 + 11.03)^2 + (104.44 - 22.06X_s)^2$$

$$53333.28 = (226.67)^2 + (104.44 - 22.06X_s)^2$$

$$53333.28 = 51379.29 + (104.44 - 22.06X_s)^2$$

Therefore,

$$(104.44 - 22.06X_s)^2 = 1953.99$$

Taking the positive square root for the standard inductive solution,

$$104.44 - 22.06X_s = \sqrt{1953.99} \approx 44.20$$

$$22.06X_s = 60.24$$

$$X_s = \frac{60.24}{22.06}$$

$$\boxed{2.73}$$

Key Points

- For a star-connected alternator, the phase voltage is $V_t = V_L/\sqrt{3}$.
- The input power includes both useful shaft power and armature copper losses:

$$P_{\text{in}} = P_{\text{sh}} + 3I_a^2 R_a.$$

- For a leading power factor alternator, the phasor relation is

$$E_f^2 = (V_t \cos\phi + I_a R_a)^2 + (V_t \sin\phi - I_a X_s)^2.$$

- The synchronous reactance obtained is the per-phase value.

Mistakes to be avoided

- Do not use the line voltage directly in the per-phase phasor equation for a star-connected alternator.
- Do not omit the armature copper loss $3I_a^2 R_a$ from the input power balance.
- For a leading power factor, use the appropriate sign of the $I_a X_s$ term.
- In the quadratic formula, the discriminant contains $(646.92)^2$, not $(646.92)^3$.
- Select the physically meaningful positive value of X_s.

Q5.94.1 MCQ 1994 1M Ans: A

Match each item with the machine or circuit element it belongs to.

- **(a) Commutation → (S) Inter pole.** In a dc machine the current in a coil must reverse while the coil is short-circuited by a brush. Interpoles (commutating poles) placed between the main poles produce the emf that makes this reversal sparkless.
- **(b) V-curves → (U) Synchronous motor.** V-curves are plots of armature current against field current at constant load and constant terminal voltage. They are drawn for a synchronous motor, and the minimum of each curve is at unity power factor.
- **(c) Freewheeling diode → (Q) Inductive load.** A freewheeling diode is connected across the load of a rectifier so that the current of an inductive load has a path to continue when the source voltage reverses.
- **(d) Overlap → (T) Source inductance.** In a phase-controlled rectifier the source inductance prevents the current from changing instantly from one thyristor to the next, so two devices conduct together for the overlap angle.

Hence the correct matching is $a - S,\ b - U,\ c - Q,\ d - T$, which is

$$\boxed{\text{A}}$$

Q5.93.1 MCQ | 1993 | 2M | Ans: A | ● ● ●

For a short-circuit test on an alternator with negligible armature resistance,

$$I_{sc} = \frac{E_g}{\sqrt{R_a^2 + X_s^2}} \approx \frac{E_g}{X_s}$$

The generated emf is proportional to speed and field flux:

$$E_g \propto N\phi$$

Assuming operation in the unsaturated region,

$$\phi \propto I_f$$

Hence,

$$E_g \propto NI_f$$

Also, synchronous reactance is

$$X_s = 2\pi f L_s$$

Since frequency is proportional to speed,

$$X_s \propto f \propto N$$

Substituting in the short-circuit current expression,

$$I_{sc} \propto \frac{NI_f}{N}$$

$$I_{sc} \propto I_f$$

Thus, the short-circuit current is independent of speed and depends only on the field current.

Therefore, when the speed is reduced to half, the field current required to maintain the rated armature current remains unchanged.

$$I_f = I_{f1}$$

Hence, the correct answer is

A

Key Points

- During short-circuit test, R_a is negligible compared to X_s.
- Generated emf:
$$E_g \propto N\phi \propto NI_f$$
- Synchronous reactance:
$$X_s \propto N$$
- Therefore,
$$I_{sc} \propto \frac{NI_f}{N} = I_f$$
- Short-circuit characteristic (SCC) is independent of speed.

Q5.93.2 MSQ 1993 2M Ans: C,D ● ● ○

A synchronous motor working at 0.8 lagging power factor is under-excited. At a fixed mechanical load and terminal voltage, the active component of the armature current stays approximately constant.

If the excitation or field current is increased, the lagging reactive current drawn from the AC supply decreases. Hence, the power factor increases towards unity.

At unity power factor, the armature current becomes minimum.

If the field current is increased further, the motor becomes over-excited and starts operating at a leading power factor. The power factor therefore decreases again and the armature current increases.

This is represented by the V-curves and inverted V-curves of the synchronous motor. Checking the options:

- **(A)** The power factor does not decrease first; it rises to unity. Not correct.
- **(B)** The armature current does not increase first; it falls to a minimum at unity power factor. Not correct.
- **(C)** The power factor increases up to unity (a certain value of field current) and thereafter decreases (leading). Correct.
- **(D)** The armature current decreases up to its minimum (at unity power factor) and thereafter increases. Correct.

(C) and (D)

Q5.92.1 MCQ 1992 1M Ans: A ● ○ ○

The torque angle δ of a synchronous machine is defined as the space angle between the rotor magnetic field axis and the stator resultant magnetic field axis.

The rotor mmf wave is produced by the DC field winding, whereas the stator mmf wave is produced by the three-phase armature currents.

$$T_e \propto F_r F_s \sin\delta$$

where

δ = space angle between rotor mmf wave and stator mmf wave

Therefore, the correct option is

A

Q5.92.2 MCQ 1992 5M Ans: A ● ● ○

(a) For non-zero leading power factor and constant excitation, armature reaction is partly magnetizing and partly cross-magnetizing. Hence, terminal voltage V_t increases with load current.

$$(a) \rightarrow (Q)$$

(b) For zero power factor leading load, armature reaction is completely magnetizing.

$$(b) \rightarrow (P)$$

(c) For constant terminal voltage and zero power factor leading current, excitation must be reduced.

$$(c) \rightarrow (S)$$

(d) For non-zero leading power factor load, excitation must be reduced initially and increased beyond rated current.

$$(d) \rightarrow (R)$$

Therefore,

$$\boxed{A}$$

Q5.91.1 MCQ | 1991 | 1M | Ans: C ● ○ ○

A synchronous motor is not self-starting because the average starting torque developed by the main field system is zero.

Damper winding (amortisseur winding) is provided on the rotor pole faces. At starting, it acts like the squirrel-cage winding of an induction motor and develops induction motor torque.

Thus, the motor starts as an induction motor and accelerates close to synchronous speed, after which the DC field is excited and the rotor pulls into synchronism.

Therefore, the primary purpose of providing damper winding is to provide starting torque.

$$\boxed{C}$$

Q5.91.2 NAT | 1991 | 5M | Ans: 0.069, 1.038, 85.54, 705.07 ● ● ●

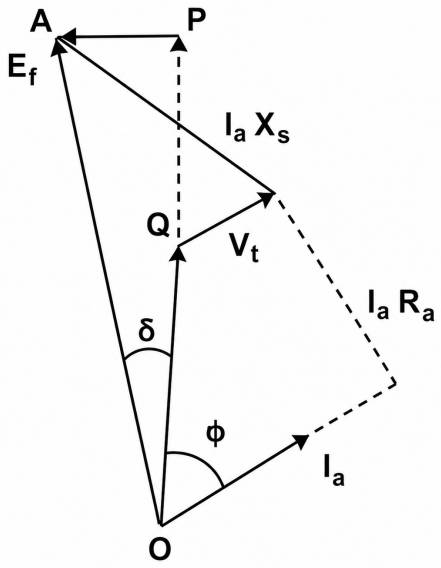


Fig. Solution Q5.91.2

Given:

$$S = 10\,\text{kVA}, \qquad V_L = 380\,\text{V}, \qquad R_s = 1\,\Omega, \qquad X_s = 15\,\Omega$$

$$P_o = 8\,\text{kW}, \qquad \cos\phi = 0.8$$

Since the alternator is working at lagging power factor, it is overexcited.

$$E_f \cos\delta > V_t$$

Base impedance is

$$Z_{\text{base}} = \frac{(V_{\text{base}})^2}{(S_{\text{base}})}$$

$$Z_{\text{base}} = \frac{(380)^2}{10 \times 10^3} = 14.44\,\Omega$$

Therefore,

$$R_s(\text{p.u.}) = \frac{1}{14.44} = 0.069$$

$$X_s(\text{p.u.}) = \frac{15}{14.44} = 1.03878$$

The output power is

$$P = \sqrt{3} V_L I_L \cos\phi$$

$$8000 = \sqrt{3} \times 380 \times I_L \times 0.8$$

$$I_L = 15.2\,\text{A}$$

For lagging power factor,

$$I_L = 15.2\angle(-36.87^\circ)$$

The generated emf per phase is

$$\vec{E}_f = \vec{V}_t + \vec{I}_a \vec{Z}_s$$

$$E_f\angle\delta = \frac{380}{\sqrt{3}}\angle 0^\circ + 15.2\angle(-36.87^\circ)(1 + j15)$$

$$E_f\angle\delta = 407.075\angle 25.19^\circ\,\text{V}$$

Voltage regulation is

$$\%VR = \frac{E_f - V_t}{V_t} \times 100$$

$$\%VR = \frac{407.075 - \dfrac{380}{\sqrt{3}}}{\dfrac{380}{\sqrt{3}}} \times 100$$

$$\%VR = 85.54\%$$

If load is suddenly disconnected,

$$|V_t| = |E_f|$$

$$V_t = 407.075\,\text{V per phase}$$

$$V_L = \sqrt{3} V_t$$

$$V_L = \sqrt{3} \times 407.075$$

$$V_L = 705.07\,\text{V}$$

$$\boxed{0.069, 1.038, 85.54, 705.07}$$

Key Points

- For a star-connected alternator,
$$V_{ph} = \frac{V_L}{\sqrt{3}}$$
- Base impedance is
$$Z_{\text{base}} = \frac{V_{\text{base}}^2}{S_{\text{base}}}$$
- For an alternator,
$$\vec{E}_f = \vec{V}_t + \vec{I}_a \vec{Z}_s$$
- When load is suddenly disconnected, terminal voltage becomes equal to generated emf.

PrepFusion

SOLUTIONS

VI

Miscellaneous topics

Topics

1. Single Phase/Universal Motors
2. Special machines

Unit Snapshot*

Stream: EE	**Questions:** 25	**MCQ:** 23	**MSQ:** 0	**NAT:** 2
1M: 14	**2M:** 11	**Descriptive:** 0	**Avg Weightage** ~1M	**Range:** 0M(2014)–2M(2018)

**See preface for how Avg Weightage and Range are calculated.*

PrepFusion

Q6.23.1 MCQ | 2023 | 1M | Ans: B | ● ○ ○

Synchronous speed is

$$N_s = \frac{120f}{P}$$

$$N_s = \frac{120 \times 50}{10} = 600\,\text{rpm}$$

Since rotor rotates opposite to backward field,

$$s = \frac{N_s + N_r}{N_s}$$

$$s = \frac{600 + 540}{600} = 1.9$$

Rotor induced frequency due to backward field is

$$f_r = sf = 1.9 \times 50 = 95\,\text{Hz}$$

$\boxed{B}$

Key Points

- For backward rotating field, effective relative speed is $N_s + N_r$.
- Rotor frequency is given by $f_r = sf$.

Q6.22.1 MCQ | 2022 | 1M | Ans: D | ● ○ ○

Among single-phase induction motors, the capacitor start capacitor run motor has the best running performance because the auxiliary winding remains permanently connected through a running capacitor.

This provides a nearly two-phase balanced operation, resulting in:

- Higher power factor
- Better efficiency
- Smoother torque
- Lower current for the same output

Hence, it has the maximum power factor during steady-state running condition.

$\boxed{D}$

Q6.18.1 NAT | 2018 | 2M | Ans: 0.106 | ● ● ○

At near synchronous speed,

$$s \approx 0$$

Hence, the rotor branch impedance becomes very large and the equivalent circuit is

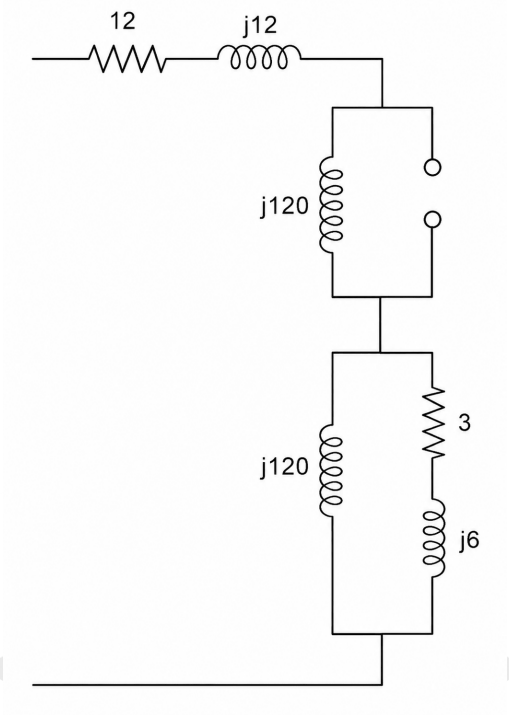


Fig. Solution Q6.18.1

$$Z_{eq} = (12 + j12) + j120 + \frac{(j120)(3 + j6)}{3 + j126}$$

$$Z_{eq} = 12 + j132 + \frac{(-720 + j360)(3 - j126)}{3^2 + 126^2}$$

$$Z_{eq} = 12 + j132 + \frac{43200 + j91800}{15885}$$

$$Z_{eq} = 12 + j132 + 2.7195 + j5.779$$

$$Z_{eq} = 14.7195 + j137.779$$

Therefore,

$$\phi = \tan^{-1}\left(\frac{137.779}{14.7195}\right) = 83.9^\circ$$

Power factor,

$$\text{pf} = \cos\phi = \cos(83.9^\circ) = 0.1062$$

$$\boxed{0.106}$$

Q6.17.1 NAT 2017 2M Ans: 149.5 ● ● ○

The condition for maximum starting torque is,

$$\theta_a = \frac{90 - \theta_m}{2}$$

$$\theta_m = \tan^{-1}\left(\frac{15.75}{12.50}\right) = 51.56^\circ$$

$$\theta_a = \tan^{-1}\left(\frac{X_c - 12.75}{24.50}\right)$$

Hence,

$$\frac{X_c - 12.75}{24.50} = \tan\left(\frac{90 - 51.56}{2}\right) = 0.3486$$

$$X_c = 21.29\,\Omega$$

$$C = \frac{1}{\omega X_c} = \frac{1}{2\pi \times 50 \times 21.29} = 149.5\,\mu\text{F}$$

$$\boxed{149.5}$$

Key Points

- Maximum starting torque in a capacitor-start induction motor occurs when $\theta_a = \frac{90 - \theta_m}{2}$.
- The auxiliary winding current should lead the main winding current by an optimum angle for maximum starting torque.

Q6.16.1 MCQ 2016 1M Ans: C ● ○ ○

In a single-phase capacitor-run induction motor, the direction of rotation depends on the phase relationship between the main winding and the auxiliary winding.

Reversing the connections of only one winding reverses the phase sequence and hence reverses the rotating magnetic field.

Interchanging the terminals of the auxiliary winding reverses the direction of rotation.

Interchanging the AC supply terminals does not change the direction because both windings are affected simultaneously.

If both windings are interchanged, their relative phase relationship remains unchanged, so the direction does not reverse.

$$\boxed{C}$$

Q6.14.1 MCQ 2014 1M Ans: C ● ○ ○

If the centrifugal switch works correctly, the starting capacitor is disconnected when the motor speed reaches approximately

$$0.7N_s$$

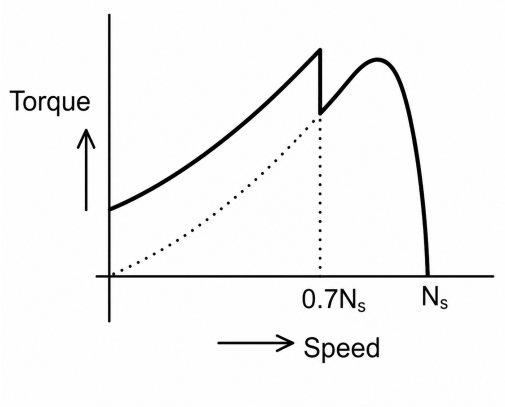


Fig. Solution Q6.14.1

After disconnection of the starting capacitor, the motor continues to operate with only the running winding, and the torque-speed characteristic changes accordingly.

If the switch fails to disconnect the starting capacitor, then the capacitor remains permanently connected and effectively acts as a run capacitor. In that case, the torque-speed characteristic follows the curve corresponding to continuous capacitor operation.

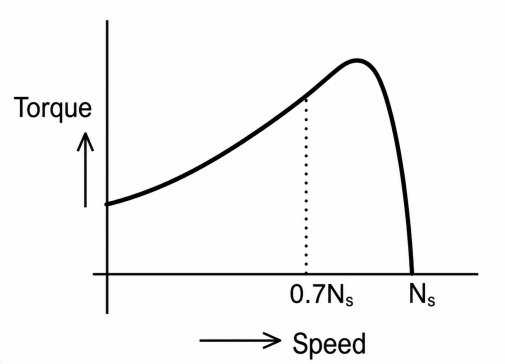


Fig. Solution Q6.14.1

The correct characteristic is therefore the one shown in option (c).

$\boxed{C}$

Q6.12.1 MCQ 2012 1M Ans: B ●○○

Balanced two-phase currents must have equal magnitudes and a phase difference of 90°.

Given:

$$i_1(t) = I_m \sin(\omega t - \phi_1)$$

$$i_2(t) = I_m \cos(\omega t - \phi_2) = I_m \sin\left(\omega t - \phi_2 + \frac{\pi}{2}\right)$$

For balance, i_2 must lead i_1 by $\frac{\pi}{2}$:

$$-\phi_2 + \frac{\pi}{2} = -\phi_1 + \frac{\pi}{2}$$

Hence,

$$\phi_1 = \phi_2$$

Hence the correct option is

$$\boxed{B}$$

Key Points

For a balanced two-phase system, the two phase currents must have equal magnitudes and be displaced by 90° electrical.

Mistakes to be avoided

Do not treat $\cos\theta$ as being in phase with $\sin\theta$. Use $\cos\theta = \sin(\theta + \frac{\pi}{2})$ before comparing phase angles.

Q6.09.1 MCQ 2009 2M Ans: C ●●○

Assume the supply voltage as the reference phasor.

Current in the auxiliary winding is

$$\bar{I}_a = \frac{V\angle 0^\circ}{Z_a\angle\theta_a} = \frac{V}{Z_a}\angle(-\theta_a)$$

Hence, the phase angle of auxiliary winding current is

$$-\theta_a = -\tan^{-1}\left(\frac{\omega L_a}{R_a}\right)$$

$$-\theta_a = -\tan^{-1}\left(\frac{2\pi \times 50 \times \frac{10}{\pi}}{1}\right) = -\tan^{-1}(1000) = -89.9427^\circ$$

Current in the main winding is

$$\bar{I}_m = \frac{V\angle 0^\circ}{Z_m\angle\theta_m} = \frac{V}{Z_m}\angle(-\theta_m)$$

Hence, the phase angle of main winding current is

$$-\theta_m = -\tan^{-1}\left(\frac{\omega L_m}{R_m}\right)$$

$$-\theta_m = -\tan^{-1}\left(\frac{2\pi \times 50 \times \frac{0.1}{\pi}}{0.1}\right) = -\tan^{-1}(100) = -89.427^\circ$$

Therefore,

$$\delta = \theta_m - \theta_a = -0.5156^\circ$$

Since

$$T \propto \tan\delta$$

and δ is very small, the developed torque is very small and insufficient to rotate the motor.

Hence, the correct answer is

$$\boxed{C}$$

Key Points

- Starting torque in a single-phase induction motor depends on the phase difference between main and auxiliary winding currents.
- Current phase angle is given by $-\tan^{-1}\left(\frac{\omega L}{R}\right)$.
- A larger phase displacement between winding currents produces higher starting torque.
- If the phase difference is nearly zero, the motor develops negligible starting torque.

Q6.08.1 MCQ 2008 2M Ans: A ●○○

Synchronous speed is

$$N_s = \frac{120f}{P}$$

$$N_s = \frac{120 \times 50}{4} = 1500\,\text{rpm}$$

Forward slip is

$$s = \frac{N_s - N_r}{N_s}$$

$$s = \frac{1500 - 1425}{1500} = 0.05$$

For the backward rotating field, slip is

$$s_b = 2 - s$$

$$s_b = 2 - 0.05 = 1.95$$

In the backward branch of a single-phase induction motor equivalent circuit, the effective rotor resistance is

$$R_b = \frac{r_2}{2s_b}$$

$$R_b = \frac{7.8}{2(1.95)}$$

$$R_b = 2\,\Omega$$

$\boxed{A}$

Key Points

- Forward slip is $s = \frac{N_s - N_r}{N_s}$.
- Backward slip in a single-phase induction motor is $s_b = 2 - s$.
- In double revolving field theory, rotor resistance in each branch is divided by 2.
- Effective rotor resistance in the backward branch is $\frac{r_2}{2(2-s)}$.

Q6.08.2 MCQ 2008 1M Ans: D ●○○

Detent torque is the torque present in a stepper motor when the phase windings are not excited. It is produced due to the interaction between the permanent magnet rotor and the toothed stator structure.

Detent torque represents the maximum torque that must be applied to move the rotor away from its preferred resting position when the windings are unexcited.

Hence,

$$\text{Detent Torque} = \text{Maximum static torque with phase winding unexcited}$$

Therefore, the correct answer is

D

Key Points

- Detent torque exists only in permanent-magnet and hybrid stepper motors.
- It is present even when no current flows in the phase windings.
- Detent torque is caused by rotor permanent magnets interacting with stator teeth.
- Holding torque is measured with the phase windings excited, whereas detent torque is measured with the windings unexcited.

Q6.07.1 MCQ 2007 2M Ans: B ●●○

For a multi-stack variable reluctance stepper motor, the step angle is given by

$$\beta = \frac{360^\circ}{mP_r}$$

where

- m = number of stacks
- P_r = number of rotor poles

Given,

$$m = 3, \qquad P_r = 20$$

Substituting,

$$\beta = \frac{360^\circ}{3 \times 20}$$

$$\beta = 6^\circ$$

Therefore, the correct option is

B

Key Points

- For a multi-stack variable reluctance stepper motor, the step angle is given by $\beta = \frac{360^\circ}{mP_r}$.
- The step angle decreases as the number of stacks or rotor poles increases.
- Smaller step angle provides higher positioning resolution.

Mistakes to be avoided

- Do not use the permanent-magnet/hybrid stepper motor formula $\beta = \frac{(P_s - P_r)}{P_s P_r} \times 360^\circ$ for a multi-stack variable reluctance motor.
- Do not confuse the number of stacks with the number of phases.
- Ensure rotor poles per stack are used in the calculation.

Q6.06.1 MCQ | 2006 | 1M | Ans: B | ● ○ ○

For a single-phase capacitor-start induction motor, the direction of rotation is reversed by reversing the connections of either the main winding or the auxiliary winding (but not both simultaneously).

Reversing only the main winding terminals reverses the phase relationship between the main and auxiliary winding currents, thereby reversing the rotating magnetic field and the direction of rotation.

Hence, statement (B) is correct.

$\boxed{B}$

Q6.05.1 MCQ | 2005 | 2M | Ans: B | ● ○ ○

In a single-phase induction motor, the total developed torque is the difference between the forward-field torque and the backward-field torque.

$$T_{\text{total}} = T_f - T_b$$

where,

T_f = Torque due to forward field

T_b = Torque due to backward field

Hence,

$$T_{\text{total}} = \frac{I_2'^2 R_2'}{2\omega_s}\left(\frac{1}{s} - \frac{1}{2-s}\right)$$

Thus,

$$T_{\text{total}} \propto R_2'$$

Therefore, increasing the rotor resistance increases the total starting torque.

A higher starting torque produces a larger accelerating torque, enabling the motor to reach its operating speed more quickly.

Hence, a high-resistance rotor is used to obtain quick acceleration.

$\boxed{B}$

Key Points

- In a single-phase induction motor, total torque is the difference between forward and backward field torques.
- Rotor resistance increases the starting torque.
- Higher starting torque results in higher accelerating torque.
- High-resistance rotors are preferred where quick acceleration is required.

Q6.04.1 MCQ 2004 1M Ans: D ● ○ ○

Among the single-phase induction motors, the capacitor-run motor has a capacitor permanently connected in series with the auxiliary winding during both starting and running conditions.

The running capacitor improves the phase displacement between the main and auxiliary winding currents, resulting in a nearly two-phase operation.

Therefore, the motor operates with

- higher power factor,
- smoother torque,
- better efficiency.

Hence, the capacitor-run motor has the highest power factor at full load among the given options.

D

Q6.04.2 MCQ 2004 2M Ans: A ● ● ○

For a capacitor-start induction motor, the auxiliary winding current must lead the main winding current by 90° for maximum starting torque.

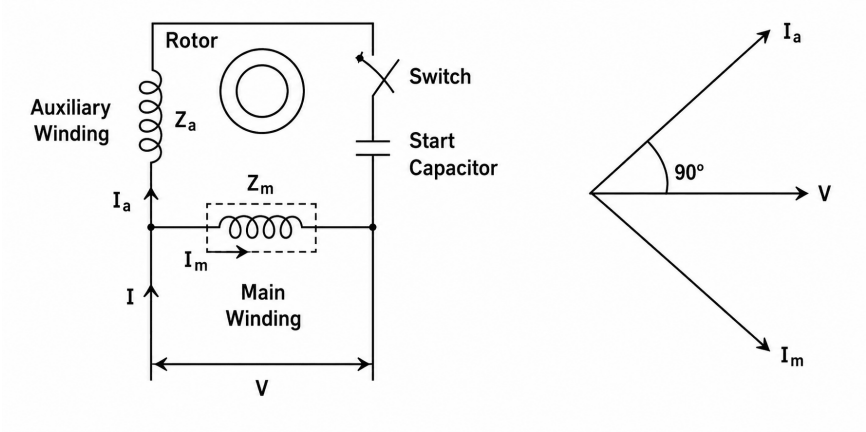


Fig. Solution Q6.04.2

The impedance of the main winding is

$$Z_m = (6 + j4)\,\Omega$$

Hence,

$$I_m = \frac{V}{Z_m} = \frac{230\angle 0^\circ}{6 + j4} = 31.89\angle -33.7^\circ \text{ A}$$

Therefore, the auxiliary winding current should lead by

$$90^\circ - 33.7^\circ = 56.3^\circ$$

The auxiliary winding impedance is

$$Z_a = 8 + j6 - \frac{j}{\omega C}$$

Hence,

$$\tan^{-1}\left(\frac{\frac{1}{\omega C}-6}{8}\right)=56.3^\circ$$

Substituting $\omega = 2\pi \times 50$ rad/s,

$$\frac{\frac{1}{\omega C}-6}{8}=\tan 56.3^\circ$$

$$\frac{1}{\omega C}=6+8\tan 56.3^\circ$$

$$C=\frac{1}{2\pi(50)\,(6+8\tan 56.3^\circ)}$$

$$C=176.84\,\mu\text{F}$$

$\boxed{A}$

Key Points

- Maximum starting torque in a capacitor-start motor is obtained when the main and auxiliary winding currents are in quadrature.
- Main winding current lags due to inductive impedance.
- The capacitor is selected so that the auxiliary winding current leads by the required angle.
- Capacitive reactance is $X_C = \frac{1}{\omega C}$.

Q6.04.3 MCQ | 2004 | 2M | Ans: B | ● ● ● | CLICK HERE FOR VIDEO SOLUTION

For a stepper motor, the speed is given by

$$N=\frac{\beta f}{360}$$

where

- β = step angle in degrees
- f = stepping rate in steps/second

Given,

$$\beta = 1.8^\circ$$

$$f = 100\ \text{steps/s}$$

Substituting,

$$N = \frac{1.8 \times 100}{360}$$

$$N = 0.5\,\text{rps}$$

Converting into rpm,

$$N = 0.5 \times 60$$

$$N = 30\,\text{rpm}$$

Hence, the rotational speed of the motor is

$$\boxed{B}$$

Key Points

- Stepper motor speed is directly proportional to both step angle and stepping frequency.
- Speed in revolutions per second is given by $N = \frac{\beta f}{360}$.
- To convert rps into rpm, multiply by 60.

Mistakes to be avoided

- Do not forget to convert rps into rpm by multiplying by 60.
- Use step angle in degrees while applying the speed formula.
- Do not confuse stepping rate (steps/s) with rotational speed (rpm).

Q6.04.4 MCQ 2004 1M Ans: C

For a stepper motor, the torque ratings satisfy

$$T_{\text{pull-out}} > T_{\text{holding}} > T_{\text{pull-in}} > T_{\text{detent}}$$

where

- Detent torque is the torque present when the winding is unexcited.
- Holding torque is the maximum static torque with the winding excited.
- Pull-in torque is the maximum load torque at which the motor can start, stop, or reverse without losing steps.
- Pull-out torque is the maximum torque that the motor can deliver while running synchronously.

Hence, the highest numerical value is the pull-out torque.

$$\boxed{C}$$

Key Points

- Pull-out torque is the maximum operating torque of a stepper motor.
- Holding torque is measured under energized standstill conditions.
- Detent torque exists even when the motor winding is not energized.

Mistakes to be avoided

- Do not confuse holding torque with pull-out torque.
- Detent torque is always smaller than holding torque.
- Pull-in torque is not the maximum torque capability of the motor.

Q6.04.5 MCQ 2004 2M Ans: D ● ○ ○

Reluctance torque depends on the variation of inductance with rotor position.

$$T_r = \frac{1}{2} i^2 \frac{dL}{d\theta}$$

If the inductance is independent of rotor position, then

$$\frac{dL}{d\theta} = 0$$

Hence, reluctance is also not varying with rotor position. This means the machine has a cylindrical rotor structure with uniform reluctance and uniform air-gap.

Therefore, reluctance torque should be zero.

Reluctance torque is produced only when the rotor tends to align itself along the path of minimum reluctance.

$$\boxed{D}$$

Q6.03.1 MCQ 2003 2M Ans: B ● ○ ○

A single-phase induction motor produces a pulsating magnetic field. According to the double revolving field theory, this pulsating field can be resolved into two rotating magnetic fields of equal magnitude rotating in opposite directions.

At synchronous speed,

$$N_r = N_s$$

Hence, the slip with respect to the forward rotating field is

$$s_f = \frac{N_s - N_r}{N_s} = 0$$

Therefore, the rotor frequency due to the forward field is

$$f_{rf} = s_f f = 0$$

and the forward-field rotor current becomes zero.

For the backward rotating field,

$$s_b = 2 - s_f = 2$$

Thus, the backward field still induces rotor current of frequency

$$f_{rb} = s_b f = 2f$$

Hence, rotor current is not zero even at synchronous speed.

$$\boxed{B}$$

Q6.03.2 MCQ 2003 1M Ans: C ● ○ ○

Match the applications with the most suitable motor types.

- Food mixers require high speed operation and compact size. Universal motors are commonly used in mixers, vacuum cleaners, and portable tools.

$$P \rightarrow 3$$

- Cassette tape recorders require compact, efficient drives with good speed control. Permanent magnet DC motors are commonly used.

$$Q \rightarrow 1$$

- Domestic water pumps are generally low-rating appliances operated from single-phase supply.

$$R \rightarrow 2$$

- Escalators require high starting torque and are usually driven by three-phase induction motors.

$$S \rightarrow 4$$

Therefore, the correct matching is

$$P \rightarrow 3, \quad Q \rightarrow 1, \quad R \rightarrow 2, \quad S \rightarrow 4$$

C

Q6.00.1 MCQ 2000 1M Ans: B ● ○ ○

For a variable reluctance stepper motor,

$$\beta = \frac{N_r - N_s}{N_s N_r} \times 360^\circ$$

Given,

$$\beta = 1.8^\circ, \qquad N_s = 40$$

$$1.8 = \frac{N_r - 40}{40N_r} \times 360$$

$$\frac{1.8}{360} = \frac{N_r - 40}{40N_r}$$

$$\frac{1}{200} = \frac{N_r - 40}{40N_r}$$

$$40N_r = 200N_r - 8000$$

$$160N_r = 8000$$

$$N_r = 50$$

Therefore, the correct option is

B

Key Points

- Step angle of a variable reluctance stepper motor is

$$\beta = \frac{N_r - N_s}{N_s N_r} \times 360^\circ.$$

- Here, N_s is the total number of stator teeth.
- Rotor teeth must be more than stator teeth for the given options.

Q6.99.1 MCQ 1999 1M Ans: C ● ○ ○

A single-phase induction motor is not self-starting because the pulsating magnetic field produces zero starting torque.

To obtain starting torque, a phase difference must be created between the currents in the main and auxiliary windings so that an approximately rotating magnetic field is produced.

When the main and auxiliary windings are identical, this phase shift is obtained by connecting a capacitor in series with the auxiliary winding.

Hence, the motor develops a starting torque.

$\boxed{C}$

Q6.96.1 MCQ 1996 1M Ans: A ● ○ ○

In a split-phase induction motor, the main winding is highly inductive while the auxiliary winding is made more resistive.

Because of the difference in the winding impedances, the currents in the two windings are displaced in phase.

The phase difference obtained is approximately

$$25^\circ \text{ to } 30^\circ$$

Therefore, the correct option is

$\boxed{A}$

Key Points

- Split-phase motors use resistance splitting to obtain starting torque.
- Auxiliary winding current leads the main winding current.
- Typical phase displacement is about 25°–30°.

Q6.96.2 MCQ 1996 1M Ans: C ● ○ ○

A repulsion motor develops torque due to the interaction between stator flux and rotor currents induced through the commutator arrangement.

The motor possesses

- Very high starting torque
- High starting current
- Speed decreasing rapidly with load

These characteristics closely resemble those of a DC series motor.

Therefore, the correct option is

C

Q6.96.3 MCQ 1996 2M Ans: D ● ○ ○

Computer printers require precise angular positioning and accurate incremental motion.

A stepper motor rotates through a fixed angle for every input pulse and provides excellent position control without requiring feedback.

Therefore, stepper motors are widely used in

- Computer printers
- Plotters
- CNC positioning systems
- Robotics

Therefore, the correct option is

D

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